

FINAL REPORT

**A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114
IN CYNOMOLGUS MONKEYS**

**BATTELLE STUDY NUMBER: 52447
CALVERT STUDY NUMBER: 1259SL35.001**

TESTING FACILITY

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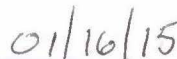
JANUARY 2015

SIGNATURE PAGE



Michael A. Hawk, M.S., D.A.B.T.
Study Director

Date

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Date

COMPLIANCE STATEMENT

This study was conducted in compliance with the current version of the United States Food and Drug Administration's (FDA) Good Laboratory Practice (GLP) Regulations, 21 CFR Part 58 and was listed on Battelle's master schedule of regulated studies. Except for analyses performed by the Sponsor and/or designee, all records required to reconstruct the study including all electronic physiologic data and the final report were maintained under the direction of Battelle, Columbus, Ohio according to standard operating procedures (SOPs).

This study was conducted in an Association for Assessment and Accreditation of Laboratory Animal Care International (AAALAC)-accredited facility in compliance with the National Research Council (NRC) Guide for Care and Use of Laboratory Animals, and the U.S. Animal Welfare Act, as amended.

All portions of this study performed at Battelle adhered to the study protocol and any amendments, as well as to applicable Battelle facility SOPs.

The portions of the study conducted by the Sponsor and Sponsor-designated test site were conducted in compliance with the current version of the U.S. FDA's GLP Regulations, 21 CFR Part 58. The bulk drug substance (LT3114) used in this study was placed on stability. The 6-month stability study report will be submitted as part of the IND.

The data presented accurately reflect the results of this study. The protocol, amendments to the protocol, and deviations to the protocol are presented in [Appendix A](#).



Michael A. Hawk, M.S., D.A.B.T.
Study Director



Date

QUALITY ASSURANCE STATEMENT

The study was inspected by the Quality Assurance Unit (QAU). Reports were submitted to the Study Director and Management as follows.

Phase Inspected	Date Inspected	Date Reported to Study Director and Management
Protocol review	9/2/14	9/2/14
Blood collection	9/8/14	9/9/14
Clinical pathology sample analysis	9/8/14	9/9/14
Protocol amendment 1 review	9/9/14	9/9/14
Dose preparation	9/17/14	9/18/14
Analysis sample collection	9/17/14	9/18/14
Body weights	9/18/14	9/18/14
Clinical observations	9/18/14	9/18/14
Test article administration - intravenous	9/18/14	9/18/14
Telemetry data collection	9/18/14	9/18/14
Review of off-site QA records	10/8/14	10/8/14
Review of off-site QA records	11/3/14	11/3/14
Protocol amendment 2 review	11/5/14	11/5/14
Audit formulation preparation data	11/10/14	11/10/14
Audit study file	11/19/14	11/19/14
Audit electrocardiograms	12/3/14	12/3/14
Audit statistical analysis report	12/3/14	12/3/14
Audit draft final report	12/3/14	12/3/14
Review of off-site QA records	1/5/15	1/5/15
Audit final report	1/15/15	1/15/15


Quality Assurance Unit

SUMMARY

The purpose of this study was to evaluate the cardiovascular effects of LT3114 following a single slow bolus injection in Cynomolgus monkeys. A crossover design was used where the monkeys were administered once with vehicle, followed three days later by an administration of one of two dose levels of LT3114, via an approximate 3-minute slow bolus injection.

Blood samples were collected for clinical pathology prior to Day 1 (for health screen purposes only). Male Cynomolgus monkeys (n=8) previously surgically implanted with radio-telemetry devices, were dosed once via an approximately 3 minute slow bolus injection with vehicle or LT3114 at target doses of 0, 10, and 50 mg/kg administered at a volume of 2 mL/kg in a crossover design. The parameters evaluated included: cardiovascular function [e.g., systolic, diastolic, and mean blood pressures, heart rate, electrocardiogram (ECG)], body temperature, clinical observations, body weights, and blood collection for LT3114 analysis. Telemetry data were collected and analyzed statistically both in restraint chairs (approximately 0 to 120 minutes post-dose) and unrestrained in their respective home cage (approximately 3 to 24 hours post-dose). Data were analyzed for statistically significant differences as compared to the vehicle data.

The cardiovascular analysis demonstrated that systolic, diastolic, and mean blood pressure data were not remarkably different from vehicle for both the in-box and in-cage data. Heart rate values were not remarkably different from vehicle for the in-box and in-cage data, however, there was a large standard error for the 50 mg/kg LT3114 group for the in-box data following dosing which was attributed to Animal 203 having an increased (by approximately 100 BPM) heart rate. This increase in heart rate was directly associated with the events noted during dose administration. This increase in heart rate attributed to large standard errors for the RR interval, as well as the corrected QT (QTcB) values. Additionally, there was an increasing trend for QTcB for the 50 mg/kg LT3114 group approximately 20 to 50 minutes following dosing, which can be directly associated with the increase in heart rate, thus causing the Bazett's formula to over correct QT for heart rate during that time period. All the other

ECG parameters (PR, QRS, and QT) for the in-box and in-cage data were not remarkably different from vehicle. Body temperature for both the 10 and 50 mg/kg LT3114 group for the in-cage data revealed sporadic significant increases and decreases between 3.5 and 24 hours post-dose as compared to vehicle. These sporadic changes in body temperature did not occur during the first two hours following dosing (in-box data), were small in magnitude (maximum change was 0.7°C), and could not be directly related to the test article.

Qualitative assessment of the ECG waveforms determined that there were no changes in the ECGs attributable to the test article.

There were no clinical observations or changes in body weight noted in this study that were attributable to the test article.

Toxicokinetic (TK) data revealed a dose response relationship and confirmed exposure.

In conclusion, the 50 mg/kg LT3114 dose level was considered no observable adverse effect level (NOAEL) for this study.

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LIST OF PARTICIPATING PERSONNEL

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Program/Project Manager	Monroe, P., Ph.D., D.A.B.T.
Study Cardiologist	Hamlin, R., D.V.M., Ph.D., Diplomate A.C.V.I.M., Cardiology
Principal Investigator, Bioanalytical Analysis	Calvarese, M., B.S.
Principal Investigator, Dose Formulation Analysis	Johnson, V., Ph.D.
Study Monitor	Modi, M., Ph.D.
Sponsor Representative Contact	Fogleman, J., Ph.D.

1.0 INTRODUCTION

The purpose of this study was to evaluate the cardiovascular effects of LT3114 following a single slow bolus injection in Cynomolgus monkeys. LT3114 is a monoclonal antibody under evaluation as a possible treatment for neuropathic pain in humans.

The following parameters were evaluated: systolic, diastolic, and mean blood pressures, heart rate, ECG, body temperature, clinical observations, body weights, and determination of LT3114 concentrations in plasma.

2.0 ROUTE AND DURATION OF ADMINISTRATION

Each non-human primate (NHP) was dosed once via an approximately 3 minute slow bolus injection with vehicle (Day 1) or one of two levels (10 or 50 mg/kg) of LT3114 (Day 4) at a volume of 2 mL/kg using a simple crossover design.

3.0 TESTING FACILITY AND PERSONNEL

3.1 Testing Facility

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4.3 Sponsor's Project Representative

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Telephone: (570) 585-2308
Email: jill.fogleman@calvertlabs.com

5.0 MATERIALS AND METHODS

5.1 Test System

5.1.1 Test System Justification

Considerable scientific documentation exists for the Cynomolgus monkey as a predictive animal model for humans. An extensive biochemical and physiological data base for the Cynomolgus monkey is available. This is an accepted non-rodent species for use in toxicology studies of drugs intended for human use. At this time, studies in laboratory animals are required to support regulatory submissions. The number of monkeys is considered to be the minimum number necessary to yield meaningful results.

5.1.2 Source and Arrival

Eight male Cynomolgus monkeys were required for this study. A sufficient number of in-house animals were supplied by Covance Research Products, Inc. (Alice, TX) to provide the required number of healthy animals for testing. The monkeys were transferred from the Battelle holding colony on September 5, 2014. The animals were 4 to 6 years of age and weighed 4.4 to 6.5 kg on Day 1.

5.1.3 Quarantine and Animal Care

Study animals were quarantined and accepted in accordance with facility SOPs at Battelle. A staff veterinarian verified that each animal was healthy based on a routine

pre-study clinical pathology screen, physical examination and observations. A staff veterinary clinical pathologist reviewed the clinical pathology data. Animals not suitable for health reasons were excluded from the study.

All animal housing and environmental conditions followed testing facility SOPs. Monkeys were individually housed in stainless steel cages. All monkeys were identified by tattoo and cage card.

Each monkey was provided with a Certified Primate Diet 5048 (Purina Lab Diet), throughout the study twice daily except during designated fasting periods and once daily on dosing days. Analytical reports of each feed lot of Purina Lab Diet were provided by the manufacturer. Analytical reports were reviewed according to facility SOP to ensure acceptable standards, and freedom from levels of contaminants that may interfere with the purpose or conduct of the study. Analytical results were retained in facility records. Water was provided to each cage *ad libitum* via an automatic watering system. The water source was from the municipal city supply of the City of Columbus, Ohio and West Jefferson, Ohio, which conforms to the Ohio Environmental Protection Agency (EPA) water standards. There were no known contaminants in the food or water that would interfere with the purpose or outcome of a study.

Environmental controls were set to maintain temperature and relative humidity in the room within the ranges specified by facility SOPs. Cage size and animal care conformed to the current AAALAC recommendations, current requirements stated in the “[Guide for the Care and Use of Laboratory Animals](#)” (NRC, 2011), the U.S. Department of Agriculture through the Animal Welfare Act (Public Law 99-198, as amended), and to testing facility SOPs.

5.1.4 Group Assignment and Identification

Prior to Day 1, at least 12 hours of hemodynamic, core body temperature, and electrocardiographic waveform data were collected for each animal. The Study Director reviewed all pre-test data for the monkeys and deemed them acceptable for use on study. One monkey (C59031) was excluded from the study due to a low telemetry battery life. The

remaining eight monkeys, deemed acceptable for use on study, were assigned to dose groups on September 12, 2014 using a computer software program [Xybion Next Generation PATH/TOX SYSTEM (Version 1.7.2)]. Following group assignment, each monkey was identified with an individual identification (ID) number unique within the study.

5.2 Test Article and Vehicle

5.2.1 Test Article

The bulk test article, LT3114, Lot No. 103948 was used to formulate doses to be used on the study. The test article was manufactured by Gallus Biopharmaceuticals, LLC. and supplied to Battelle from Calvert Laboratories Inc. at a concentration of 32.1 mg/mL (Lot No. GA121459). All formulated test article was received at Battelle on September 3, 2014 and was stored in a refrigerator unit set to maintain 2 to 8°C. A Certificate of Analysis (CoA) of the bulk test article is presented in [Appendix B](#). The remaining bulk test article was shipped back to Calvert Laboratories Inc. on January 15, 2015.

5.2.2 Vehicle

The vehicle for this study was 300 mM Trehalose, 51.4 mM Na₂HPO₄, 24.3 mM Citric acid, pH:5.0. The vehicle (Lot No. 104011-001) was manufactured by Gallus Biopharmaceuticals, LLC., supplied to Battelle from Calvert Laboratories Inc. and was stored refrigerated (2 to 8°C). The vehicle as provided was spiked with polysorbate 80 solution. A Certificate of Compliance for the vehicle is presented in [Appendix B](#).

5.2.3 Reserve Samples of Test Article and Vehicle

Reserve samples for archiving were not required on this study because the duration was less than 4 weeks.

5.2.4 Disposition of the Formulated Test Article and Vehicle

Any unused formulated test article and vehicle was discarded on January 15, 2015.

5.2.5 Formulation Preparation

Calvert Laboratories provided Battelle LT3114 dose formulation at a concentration of 32.1 mg/mL and vehicle to be used on study. The 32.1 mg/mL formulation was used to dilute with the vehicle provided for the two concentrations (5 and 25 mg/mL) used on study. Dose formulations were prepared a day prior to dosing and stored in a refrigerator set to maintain 2 to 8°C. On the day of dosing, all dose formulations were removed from the refrigerator at least 1 hour prior to the start of dose administration.

5.2.6 Dose Formulation Analysis

Dose aliquots (6X approximately 1.0 mL) were collected from the top, middle, and bottom of the dosing formulations and vehicle. These samples were labeled “for concentration analysis” and stored frozen (-30 to -15°C) until shipped. One set of samples was shipped on dry ice after each day of dosing to Dr. Johnson at Burleson Research Technologies, Inc. in Morrisville, NC. Prior to shipment of samples, the Study Monitor and the Principal Investigator were informed by email of the shipment date and specifications (see [Protocol Deviation in Appendix A](#)).

Backup samples were discarded on January 15, 2015.

The formulation analysis report is located in [Appendix B](#).

5.2.7 Justification of Dose Levels

The doses for this study were selected based on a dose range finding study performed in Cynomolgus monkeys ([Calvert Study No. 0437LS35.001](#)). The monkeys tolerated the test article when administered as an intravenous injection over 3 minutes as three consecutive doses (one dose per week) at 10, 50, and 150 mg/kg. For this cardiovascular study, the route of test article administration was a slow intravenous (IV) injection (approximately 3 minutes) in order to achieve sufficient exposure.

6.0 EXPERIMENTAL DESIGN

6.1 Dose Administration

Each monkey was conditioned to a restraint chair and isolation box prior to Day 1 (the first day of dose administration). The monkeys were acclimated for 6 days to accept up to 4 consecutive hours of chair restraint and to the isolation box.

On the day of dosing, all eight NHPs were administered either vehicle or test article at the following target doses:

Experimental Design

Animal Numbers/Sex	Treatment	Study Day	LT3114 Dose Level (mg/kg)	LT3114 Dose Concentration (mg/mL)	Dose Volume (mL/kg) ^a
101-104/M and 201-204/M	Vehicle	1	0	0	2
101-104/M	LT3114 (Low-Dose)	4	10	5	2
201-204/M	LT3114 (High-Dose)	4	50	25	2

a. Actual dose volume was determined by individual NHP body weight. The volume determined was delivered over an ~3 minute slow bolus injection (see [Protocol Deviation in Appendix A](#)).

Each monkey was dosed once via an approximately 3 minute slow bolus injection with vehicle or LT3114 at target doses of 0, 10, and 50 mg/kg administered at a volume of 2 mL/kg.

6.2 Telemetry Data Collection

All monkeys were surgically implanted with a Data Sciences International radiotelemetry transmitter, which has systemic arterial blood pressure, heart rate, core body temperature, and ECG data collection capabilities. Telemetry data were collected on each dose administration day. Each monkey's home cage and isolation box was equipped with a Data Sciences International telemetry receiver. The transmitters, receivers, data matrices, ambient pressure monitors, cabling, and computers (which run the Dataquest A.R.T. data acquisition and analysis software), are all components of the PhysioTel Telemetry System.

The Dataquest A.R.T. telemetry software collected systemic arterial blood pressures (systolic, diastolic, and mean), heart rate, core body temperature, and ECG data continuously (except when the monkeys were out of range of the receivers for study-driven activities) at least 24 hours before the first dose administration through at least 7 days (see [Protocol Deviation in Appendix A](#)) following the last dose administration. Additionally, at least 12 hours of telemetry data was collected for each animal on a day prior to selection for assessing cardiovascular, core body temperature, and ECG status. Telemetry data was only collected when the monkeys were in their home cages or isolation boxes. A detailed account of the telemetry data collection procedure was included in the study file.

6.3 Clinical Observations

Observations for mortality and moribundity were performed twice daily in accordance with testing facility SOPs. Prior to Day 1, clinical observations were performed twice daily and recorded on Individual Animal History Records. Beginning on Day 1 and continuing through the last day of in-life (Day 11), clinical observations were performed at least once daily and entered into the Xybion Next Generation PATH/TOX System. On each day of dose administration, clinical observations were performed a minimum of once prior to dosing and approximately 4 to 5 hours following dosing. Clinical observations were recorded as required to document indications of toxic and/or pharmacologic effects following the start of the dose administrations. All clinical abnormalities were recorded.

6.4 Body Weights

The individual body weight of each monkey was recorded prior to animal selection and on each day of dose administration (Days 1 and 4) and on Day 11. Body weights were used for the calculation of dose volumes for each day of dose administration.

6.5 Blood Collection for LT3114

Approximately 2.0 mL of whole blood was obtained from an appropriate vein and the blood placed into tubes containing di-potassium ethylene diamine tetraacetic acid (K₂EDTA). The blood specimens were obtained approximately 2 hours following the start of dosing.

Exact collection times were documented. The blood specimens were placed on wet ice until centrifugation, which was performed within 60 minutes from collection (see [Protocol Deviation in Appendix A](#)). The specimens were centrifuged for 10 minutes in a refrigerated centrifuge (set to 4°C) at 2700 rpm. The resultant plasma was transferred into an appropriately labeled cryovial and placed on dry ice until stored in a freezer set to maintain -85 to -60°C (see [Protocol Deviation in Appendix A](#)) until shipment to QPS, LLC for analysis.

After the final sample was collected, all samples were shipped on dry ice via an overnight courier to the following address for analysis:

Melissa Calvarese
QPS, LLC
One Innovation Way, Suite 200
Newark, DE 19711
Telephone: (302) 369-5196
Fax: (302) 737-3759
Email: melissa.calvarese@qps.com

The principal investigator for bioanalytical analysis and study Sponsor were notified prior to shipment as to the shipment details. On the day of shipment, the tracking number, sample count, and carrier name were provided via email, along with sample collection and processing records.

Plasma analysis for LT3114 was conducted by QPS, LLC.

The principal investigator for bioanalytical analysis provided a bioanalytical report from the data generated to the Study Director for inclusion in the final report.

6.6 Disposition of Animals

After completion of the in-life study phase, all study and replacement animals were transferred to Battelle's Laboratory Animal Resources (LAR) holding pool.

7.0 DATA ANALYSIS

7.1 Hemodynamic and Core Body Temperature Data

Group mean average and individual animal plots were produced for the systemic arterial blood pressures (systolic, diastolic, and mean), heart rate, and core body temperature.

7.2 Electrocardiographic Assessment

Periodic ECG waveform tracings (approximately 60 seconds) were generated using ECGAUTO software from EMKA Technologies (Paris, France) at pre-dose and at approximately 0.5, 1, 2, 4, 8, 12, 24, and 48 hours post-dosing for all dosing days (Days 1 and 4). Some of the ECG strips produced at these time points varied from the targeted time point due to the presence of unreadable ECG waveforms. The ECG waveform strips were qualitatively assessed for rhythm and morphology alterations by the Study Cardiologist.

ECG intervals (PR, QRS, QT, and RR) were measured and analyzed continuously and averaged for each 5 minute time period from 24 hours pre-dose through at least 24 hours post-dose for all dosing days (Days 1 and 4). The QT interval was normalized for changes in heart rate (RR interval) by conversion to the corrected QT (QTc) interval using Bazett's formula $QTc = QT/(RR)^{1/2}$ (Bazett, 1920).

7.3 TK Analyses

TK samples were analyzed for concentration of LT3114 in plasma and mean concentrations were reported. A description of the analysis method used and bioanalytical report are provided in [Appendix G](#).

8.0 STATISTICAL ANALYSIS

Telemetry data were electronically transferred to a study statistician for statistical analyses. Statistical analyses were performed separately for each parameter and for in-cage and in-box data. Significance levels were less than or equal to 0.05 unless specified otherwise.

The hemodynamic and ECG interval end points data collected during a predetermined time period prior to dosing on each dosing day were averaged for each animal under both in-cage and in-box conditions, resulting in separate in-cage and in-box "baseline" averages for the animal. The baseline period for the in-box data was approximately 30-minute period prior to each dosing when the animal was in the isolation box. The baseline period for the home cage was approximately 2 hours prior to dosing and prior to moving the animal to the isolation box.

For each dose administration on Day 1 and Day 4, the animal's in-cage and in-box data collected post-dosing were divided into specified time intervals (for example, 10-minute for the in-box data and 30-minute for the in-cage data) during the 2 hours of collecting in-box data and the 22 hours of collecting in-cage data (3 to 24 hours post-dosing), and the data collected within each interval were averaged. For each animal and parameter, the appropriate baseline average (either in-cage or in-box) was subtracted from the post-dosing data collected on the same day and within that same environment, resulting in "baseline-adjusted" in-cage and in-box data, respectively.

For both the unadjusted and baseline-adjusted post-dosing averages, means and the standard error of each mean (SEM) were calculated across animals by dosing day, test condition (in-cage or in-box), and time interval and were presented in tables for each end point. Means (and accompanying standard errors) of the baseline averages were calculated across animals for each and test condition (in-cage or in-box) and dosing day and were presented in tables for each end point.

The baseline-adjusted post-dosing averages were statistically analyzed using a repeated measures mixed effect analysis of variance (ANOVA) model. The model included dose level (i.e., vehicle, low-dose, and high-dose), time after dose and the interaction of time after dose with dose level as fixed effects, and animal as a random effect, when appropriate. The ANOVA model took into account correlation in the post-dosing averages that was present due to monitoring the same animals over multiple time intervals following dosing using either an AR(1) covariance structure or a sp(pow) covariance structure depending on whether or not the time intervals were equally spaced. F-tests were conducted within the ANOVA to test for significant differences among the dose levels in the baseline-adjusted averages at each specified time interval following dosing. To control the positive discovery rate of these tests over the time intervals during a post-dosing period less than or equal to 5 percent, the significance level of each test was adjusted using the Benjamini and Hochberg method ([Benjamini, 1995](#)) and conclusions of significant differences among dose levels at specific time intervals were made based on these adjusted significance levels. When significant differences among dose levels were concluded at a given post-dosing time interval, t-tests

were performed within the ANOVA to compare each of the two test article dose levels with the vehicle level at that time interval. A Bonferroni correction ([Miller, 1981](#)) was applied to the significance levels of these comparisons in order to control the overall error rate associated with comparisons to vehicle at a given time interval to less than or equal to 5 percent. Details on specific statistical methods were documented in the study file.

9.0 COMPUTER SYSTEMS FOR DATA MANAGEMENT AND ANALYSIS

The Honeywell Excel Building Supervisor (Version 1.7) was used for the capture, storage, and summarization of all appropriate animal room temperature and percent relative humidity data. The T-Track System (Version 1.0.0) was used for the capture, storage, and summarization of all temperatures of the storage units. The Xybion Next Generation PATH/TOX SYSTEM (Version 1.7.2) was used to maintain all test article inventory information, for randomization (animal number assignment), and for the capture, storage, and summarization of all appropriate animal-derived in-life data. The PhysioTel Telemetry System with Dataquest A.R.T. software (Version 3.1) and the ECGAUTO software (Version 3.3) were used for the capture, storage, and summarization of all appropriate animal-derived telemetry data. All statistical summaries and analyses were performed using Stata Version 13 (Release 13.1).

10.0 RESULTS

10.1 Concentration, Homogeneity, and Stability Analysis

The LT3114 formulation concentration analysis results are presented in [Appendix B](#). Dose formulation samples were provided for the vehicle and two dose concentrations, 5.0 mg/mL and 25.0 mg/mL. Dose formulation samples were collected from each dosing solution and the vehicle from the top, middle, and bottom strata of the LT3114 dose formulations and from the middle stratum of the vehicle. Analyses demonstrated the absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 5.5 to 7.3 mg/mL, and 27.9 to 33.1 mg/mL for the 5.0, and 25.0 mg/mL dosing solutions, respectively. All LT3114 concentrations met acceptance criteria for accuracy and precision, with the exception of the samples collected from the top and middle strata of the 5.0 mg/mL dose formulation and from

the bottom stratum of the 25.0 mg/mL dose formulation. The concentration of these three samples were high and did not meet acceptance criteria for accuracy but met the acceptance criteria for precision. LT3114 concentrations from the top, middle, and bottom fractions of the dose formulations had a CV% of 15.5% and 9.7% demonstrating acceptable homogeneity of the 5.0 and 25.0 mg/mL dose formulations.

10.2 Mortality

There were no deaths during this study. All monkeys were transferred to Battelle's LAR holding pool following the end of in-life procedures.

10.3 Cardiovascular, Body Temperature, and ECG Data

Cardiovascular, body temperature, and ECG data are presented in [Appendix D](#). Individual animal cardiovascular, body temperature, and ECG data are presented in [Appendix E](#). Descriptive statistics (means and SEM) of unadjusted and baseline-adjusted averages were calculated for each day and dose level at each time, presented in tables, and displayed in figures. If significance was found among dose levels at a given time point (marked "yes" on tables), individual dose levels (10 and 50 mg/kg) were tested for significance compared to vehicle (marked * if the p-value was at or below 0.05).

10.3.1 IV Bolus Dosing In-Box Data

In-box baseline adjusted group mean cardiovascular and body temperature data are presented in [Tables 1 through 10](#) and [Figures 1 through 4](#) in [Appendix D](#). In-box unadjusted group mean cardiovascular and body temperature data are presented in [Figures 9 through 12](#) in [Appendix D](#). All values listed below are baseline adjusted and are based on the statistically significant maximum change from vehicle.

Systolic, diastolic, and mean blood pressures ([Appendix D, Tables 1 through 3, Figure 1](#)) were not remarkably different from vehicle. Heart rate was not remarkably different from vehicle ([Appendix D, Table 4, Figure 2](#)), however, there was a large standard error for the 50 mg/kg dose group which was attributed to Animal 203 having an increased (approximately 100 BPM) heart rate following dosing ([Figure E-26](#)). This increase in heart

rate was caused by the dose administration, during which the butterfly needle came out of the vessel and the animal had to be re-handled in order to replace the needle in the same leg to complete the dosing event. Body temperature ([Appendix D, Table 5, Figure 2](#)) were not remarkably different from vehicle. There was a large standard error for the 50 mg/kg dose group for RR (inverse of heart rate) and QTcB as well as an increasing trend for QTcB from 20 to 50 minutes post-dose ([Appendix D, Tables 6 and 10, Figure 4](#)) which was attributed to Animal 203 having an increased heart rate following dosing. As stated above, this increase in heart rate was due to the complications of dosing. This increase in heart rate lead to an over correction for QT using the Bazett's correction method QTcB [$QTc = QT/(RR)^{1/2}$] and is not considered test article-related. All other ECG parameters (PR, QRS, and QT) ([Appendix D, Tables 7 through 9, Figures 3 and 4](#)) were not remarkably different from vehicle.

10.3.2 IV Bolus Dosing In-Cage Data

In-cage baseline adjusted group mean cardiovascular, body temperature, and ECG data are presented in [Tables 11 through 20](#) and [Figures 5 through 8](#) in [Appendix D](#). In-cage unadjusted group mean cardiovascular and body temperature data are presented in [Figures 13 through 16](#). All values listed below are baseline adjusted and are based on the statistically significant maximum change from vehicle.

Systolic, diastolic, mean blood pressures, and heart rate ([Appendix D, Tables 11 through 14, Figures 5 and 6](#)) were not remarkably different from vehicle. Body temperature for the 50 mg/kg dose group revealed sporadic significant increases between 3.5 and 24 hours post-dose as compared to vehicle. The maximum increase (0.7°C) occurred at 3.5 hours post-dose ([Appendix D, Table 15 and Figure 6](#)). Body temperature for the 10 mg/kg dose group revealed sporadic statistically significant decreases and one statistically significant increase between 12.5 and 22.5 hours post-dose as compared to vehicle. The maximum decrease (0.5°C) occurred between 15 and 16 hours post-dose and the one significant increase (0.4°C) occurred at 22.5 hours post-dose ([Appendix D, Table 15 and Figure 6](#)). These sporadic changes in body temperature were not considered physiologically relevant nor could be directly related to the test article. All the ECG parameters (RR, PR, QRS, QT, and QTcB)

([Appendix D](#), [Tables 16](#) through [20](#), [Figures 7](#) and [8](#)) were not remarkably different from vehicle.

10.4 Qualitative Evaluation of the ECG Waveforms

Qualitative evaluation of the ECG waveforms is presented in [Appendix F](#). With few exceptions when muscle tremor or motion artifacts occurs, all ECGs are of excellent quality for interpretation. As usual with serial ECGs, there are frequent changes in both heart rates and in configurations of component of deflections (in particular ST-T complexes).

There were no ECG findings attributable to the test article at either dose level for this study.

10.5 Clinical Observations

Individual animal clinical observations are presented in [Appendix C](#), [Table C-1](#).

There were no test article-related clinical observations for this study.

10.6 Body Weights

Individual animal body weights are presented in [Appendix C](#), [Table C-2](#). There were no changes in body weight that were considered test article-related.

10.7 Plasma Analysis for LT3114

Post-dose animal TK concentrations (ng/mL) of LT3114 were assessed using the validated enzyme-linked immunosorbent assay (ELISA) method (QPS Project No. 344-1403). The below quantifiable limit (BQL) was 100 ng/mL for LT3114. Concentrations (ng/mL) of LT3114 in individual monkey K₂EDTA are presented in the [table](#) below. The bioanalytical report including the individual animal TK data can be found in [Appendix G](#).

Concentration (ng/mL) of LT3114 in Individual Monkey K₂EDTA Plasma Samples

Subject	Treatment Description	Day Nominal	Hour Nominal	Concentration (ng/mL)
101	10mg/kg	1	2	BQL<(100)
101	10mg/kg	4	2	313000
102	10mg/kg	1	2	BQL<(100)
102	10mg/kg	4	2	332000
103	10mg/kg	1	2	BQL<(100)
103	10mg/kg	4	2	371000
104	10mg/kg	1	2	BQL<(100)
104	10mg/kg	4	2	291000
201	50mg/kg	1	2	BQL<(100)
201	50mg/kg	4	2	1650000
202	50mg/kg	1	2	BQL<(100)
202	50mg/kg	4	2	1590000
203	50mg/kg	1	2	BQL<(100)
203	50mg/kg	4	2	1450000
204	50mg/kg	1	2	BQL<(100)
204	50mg/kg	4	2	2210000

BQL = Below quantifiable limit.

11.0 DISCUSSION/CONCLUSION

The cardiovascular analysis demonstrated that systolic, diastolic, and mean blood pressure data were not remarkably different from vehicle for both the in-box and in-cage data. Heart rate values were not remarkably different from vehicle for the in-box and in-cage data, however, there was a large standard error for the 50 mg/kg LT3114 group for the in-box data following dosing which was attributed to Animal 203 having an increased (by approximately 100 BPM) heart rate. This increase in heart rate was directly associated with the events noted during dose administration. This increase in heart rate attributed to large standard errors for the RR interval as well as the corrected QT (QTcB) values. Additionally, there was an increasing trend for QTcB for the 50 mg/kg LT3114 group approximately 20 to 50 minutes following dosing, which can be directly associated with the increase in heart rate, thus causing the Bazett's formula to over correct QT for heart rate during that time period. All the other ECG parameters (PR, QRS, and QT) for the in-box and in-cage data were not remarkably different from vehicle. Body temperature for both the 10 and 50 mg/kg LT3114 group for the in-cage data revealed sporadic significant increases and decreases between 3.5 and 24 hours post-dose as compared to vehicle. These sporadic changes in body temperature did not occur during the first 2 hours following dosing (in-box data), were small in magnitude (maximum change was 0.7°C), and could not be directly related to the test article.

Qualitative assessment of the ECG waveforms determined that there were no changes in the ECGs attributable to the test article.

There were no clinical observations or changes in body weight noted in this study that were attributable to the test article.

TK data revealed a dose response relationship and confirmed exposure.

In conclusion, the 50 mg/kg LT3114 dose level was considered the NOAEL for this study.

12.0 STORAGE AND RETENTION OF RECORDS AND MATERIALS

Except for analyses performed by the Sponsor or Sponsor's designated laboratory, all records required to reconstruct the study and the final report are being maintained under the direction of Battelle for 3 years following the completion of in-life. At the completion of the 3 years, the Sponsor will be contacted to determine where the records will be shipped.

13.0 REFERENCES

A Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. B*, 57:289-300.

Guide for the Care and Use of Laboratory Animals. (2011). National Academy Press, Washington, D.C. Current Version.

Bazett, H.C. (1920). An analysis of the time-relations of electrocardiograms. *Heart*, 57:353-370.

Miller, R.G., Jr. (1981). *Simultaneous Statistical Inference*, Springer Verlag, NY.

Neter, J., Kutner, M., Nachtsheim, C., and Wasserman, W. (1996). *Applied Linear Statistical Models*, Fourth Edition, Chicago:Irwin.

APPENDIX A

PROTOCOL, PROTOCOL AMENDMENTS, AND PROTOCOL DEVIATIONS

STUDY PROTOCOL

A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114 IN CYNOMOLGUS MONKEYS

**TESTING FACILITY:
BATTELLE
505 KING AVENUE
COLUMBUS, OH 43201-2696**

**SPONSOR REPRESENTATIVE:
CALVERT LABORATORIES INC.
130 DISCOVERY DRIVE
SCOTT TOWNSHIP, PA 18447**

Battelle Study No.: 52447
Sponsor Study No.: 1259SL35.001
Preparation Date: September 2, 2014
Page 2 of 16

SIGNATURE PAGE

APPROVED, BATTELLE:



Michael A. Hawk, M.S., D.A.B.T.
Study Director9/3/14

Date

Barney R. Sparrow, Ph.D., D.A.B.T.
Manager, Toxicology09/03/14

Date

APPROVED, CALVERT LABORATORIES, INC.:



Jill L. Fogleman, Ph.D.
Sponsor Representative
Calvert Laboratories, Inc.

Date

APPROVED, MWM Consulting Group Inc. for Lpath Inc.:



Marlene W. Modi, Ph.D.
Sponsor Study Monitor

Date

① SEE FOLLOWING PAGE FOR SPONSOR REPRESENTATIVES AND SPONSOR STUDY MONITOR'S SIGNATURES. MAH 9/4/14

The final version of the protocol was approved by the study monitor for Study Director signature via:

Fax _____ E-mail ☒ Telephone _____ on 9/3/14 1 MAH
(Check appropriate method) Date Initials

To the best of our knowledge, this study does not unnecessarily duplicate any previous experiments. There are no known alternatives to the use of animals in this type of study.

Battelle Study No.: 52447
Sponsor Study No.: 1259SL35.001
Preparation Date: September 2, 2014
Page 2 of 16

SIGNATURE PAGE

APPROVED, BATTELLE:

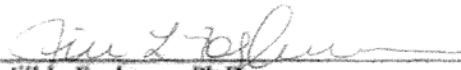
Michael A. Hawk, M.S., D.A.B.T.
Study Director

Date

Barney R. Sparrow, Ph.D., D.A.B.T.
Manager, Toxicology

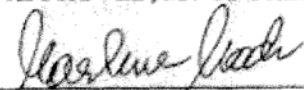
Date

APPROVED, CALVERT LABORATORIES, INC.:


Jill L. Fogleman, Ph.D.
Sponsor Representative
Calvert Laboratories, Inc.

3 Sep 2014
Date

APPROVED, MWM Consulting Group Inc. for Lpath Inc.:


Marlene W. Modi, Ph.D.
Sponsor Study Monitor

02 September 2014
Date

The final version of the protocol was approved by the study monitor for Study Director signature via:

Fax _____ E-mail _____ Telephone _____ on _____ / _____
(Check appropriate method) Date Initials

To the best of our knowledge, this study does not unnecessarily duplicate any previous experiments. There are no known alternatives to the use of animals in this type of study.

1.0 PURPOSE OF STUDY

The purpose of this study is to evaluate the cardiovascular effects of LT3114 following a single slow bolus injection in Cynomolgus Monkeys.

2.0 REGULATORY COMPLIANCE

This study will be conducted in compliance with the current version of the Food and Drug Administration's (FDA) Good Laboratory Practice regulations, 21 CFR Part 58, for the conduct of non-clinical studies. All records required to reconstruct the study including all electronic records and the final report will be maintained under the direction of Battelle, Columbus, OH according to facility Standard Operating Procedures except for records by the Sponsor and/or designee.

All portions of this study will adhere to the study protocol and any amendments, as well as to applicable Battelle facility Standard Operating Procedures.

This study will be conducted in an AAALAC-accredited facility in compliance with the ILAR/NRC Guide for Care and Use of Laboratory Animals, and the U.S. Animal Welfare Act, as amended.

3.0 ROUTE AND DURATION OF ADMINISTRATION

Male Cynomolgus monkeys will be administered once with vehicle, followed three days later by an administration of one of two dose levels of LT3114, via an approximate 3 minute slow bolus injection.

4.0 TESTING FACILITY AND STUDY PERSONNEL**4.1 Testing Facility**

Battelle
505 King Avenue
Columbus, Ohio, 43201-2696

4.2 Study Personnel

Study Director: Michael A. Hawk, M.S., D.A.B.T.
Telephone (614) 424-7740
Telefax (614) 458-7740
Email: hawkm@battelle.org

Study Coordinator: Shawnda V. Armentrout, A.S., R.V.T., L.A.T.G.
Telephone (614) 424-4726
Telefax (614) 458-4726
Email: armentrouts@battelle.org

Study Management: Barney R. Sparrow, Ph.D., D.A.B.T.
Study Statistician: Brandon J. Wood, M.S.
Study Physiologist: Craig R. Hassler, Ph.D.
Study Cardiologist: Robert L. Hamlin, D.V.M., Ph.D., D.A.C.V.I.M.
Telemetry Scientist: Thomas M. Vinci, B.S.
Program Director: Phil Monroe, Ph.D., D.A.B.T.

5.0 SPONSOR AND SPONSOR REPRESENTATIVE

5.1 Sponsor

Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

Study Monitor Marlene W. Modi, Ph.D.
MWM Consulting Group, Inc. for Lpath Inc.
Telephone: (732) 274-9562
Mobile: (732) 822-6101
E-mail: mwmconsultinggroup@gmail.com

5.2 Sponsor Representative

Calvert Laboratories, Inc.
130 Discovery Drive
Scott Township, PA 18447

Sponsor Representative Contact: Jill L. Fogleman, Ph.D.
Telephone: (570) 585-2308
Telefax: (570) 586-3450
Email: jill.fogleman@calvertlabs.com

6.0 Principal Investigator(s)

Bioanalytical analysis: Melissa Calvarese
QPS, LLC
One Innovation way, Suite 200
Newark, DE 19711

Telephone: (302) 369-5196
Telefax: (302) 737-3759
Email: melissa.calvarese@qps.com

Dose formulation analysis: Victor J. Johnson, Ph.D.
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
Tel: (919) 719-2500
Fax: (919) 719-2505
E-mail: vjohnson@brt-labs.com

7.0 PROPOSED STUDY SCHEDULE

Proposed dates for the following study events are listed below. The actual dates will be documented in the study file.

Study Initiation Date:	Week of September 1, 2014
First Day of Dosing:	Week of September 15, 2014
In-life Completion Date:	Week of September 22, 2014
Submission of Draft Report:	Within 8 weeks following final in-life procedures.
Submission of Final Report:	Within 20 working days following receipt of Sponsor's final comments.

8.0 TEST ARTICLE AND VEHICLE

The Supplier/Manufacturer will provide sufficient quantities of the test article and vehicle to conduct this study. A log of receipt and use of the test article will be maintained. Receipt of vehicle will be maintained. Copies of the Material Safety Data Sheets (MSDS) will also be maintained. The bulk test article identity, strength, purity and composition, along with the Certificate of Analysis (C of A) or equivalent, will be the responsibility of the Supplier/Manufacturer and will be supplied to Battelle for inclusion in the study file and final report. Methods of synthesis of each test article will be maintained by the Supplier/Manufacturer.

8.1 Transmission of Test Article and Vehicle

Signed receipts, shipping slips, and any associated documentation (if provided) will be retained to verify transfer of materials. The test article shipped from the Sponsor will be sent to:

Battelle
Attention: Mr. Kevin A. Carrico
TOXBC Test Article Custodian

Chemistry Technical Center
651 West Fifth Avenue
Columbus, OH 43201-2696
Telephone: (614) 424-5361
Fax: (614) 424-5221
Email: carrico@battelle.org

8.2 Test Article Identification

Test Article:	LT3114
Lot/Batch No.:	103948
Purity:	To be documented in the final report
Retest Date:	December 2014
Supplier:	Gallus Biopharmaceuticals, LLC
Storage Conditions (bulk):	2 to 8°C, protected from light
Storage Conditions (formulated):	2 to 8°C, protected from light

8.3 Vehicle Identification

Vehicle:	300 mM Trehalose, 51.4 mM Na ₂ HPO ₄ , 24.3 mM Citric acid, pH: 5.09
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2 ml of 10% polysorbate 80 to be added to each 1 liter bag of vehicle/diluent prior to use to achieve a final concentration of 0.02% polysorbate 80.

Lot/Batch No.:	To be documented in raw data notebooks and the final report
Expiration Date:	March 2015
Supplier:	To be documented in raw data notebooks and the final report
Storage Conditions:	2 to 8°C, protected from light

8.4 Reserve Samples

Since this study is less than 28 days in length, no reserve samples will be collected.

8.5 Disposition of Bulk Test Article, Dose Formulations and/or Vehicle

The bulk test article and/or vehicle will be returned to the Sponsor or its designee, disposed of or kept for future use on additional studies at the Sponsor's request. Residual dose formulations may be disposed of after the end of in-life with permission from the Sponsor.

8.6 Dose Preparation

The Sponsor will provide diluent formulation already spiked with the polysorbate 80 solution. This diluent may then be used on multiple studies for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations.

The Sponsor will provide the test article already formulated at 32.1 mg/ml in sterile containers. Diluent will also be provided in separate sterile containers. Diluent and test article need to come to room temperature before the formulation is prepared. **Do not shake the formulation.** When necessary depending on the dose level, doses will be prepared by dilution using the provided 32.1 mg/ml and diluent solutions under aseptic conditions. Before dosing, visual inspection of the formulation should be conducted against a black background for any irregularities.

Test article formulations will be prepared on the day prior to dosing and stored in a refrigerator (set to maintain 2 to 8°C) until use. On the day of dosing, all dose formulations will be removed from the refrigerator at least 1 hour prior to the start of dose administration. After the end of in-life and with Sponsor's permission the residual dose formulations may be discarded.

8.7 Dose Formulation Analysis

Aliquots (6 X ~1.0 mL) will be collected from the top, middle and bottom of the dosing formulations and vehicle. These samples will be labeled "for concentration analysis" and stored frozen (-30 to -15°C) until shipped. One set of samples will be shipped on dry ice after each day of dosing. Backup samples will be maintained at Battelle and either shipped for analysis or discarded after Sponsor approval.

Dose formulation analysis will be performed by Victor J. Johnson, Ph.D. The report regarding the test article analysis will be sent to Battelle for inclusion in the final report. Shipments will be sent to:

Victor J. Johnson, Ph.D.
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
Tel: (919) 719-2500
Fax: (919) 719-2505
E-mail: vjohnson@brt-labs.com

Prior to shipment of samples, the Study Monitor and the Principal Investigator will be informed by email of the shipment date and specifications.

8.8 Vehicle and Test Article Administration

The intended route of administration to humans is intravenous. The route of test article administration will be intravenous in order to get sufficient exposure for evaluation of potential cardiovascular effects of LT3114.

Vehicle and test article will be administered via a slow intravenous injection (~3 minutes) by an ambulatory pump. Details of the dosing procedure will be documented in the study file.

On each day of dosing, each animal selected for dosing will be removed from its home cage and placed in a restraint chair. The animals will then be placed in an isolation box and allowed to calm down to baseline CV data levels. Following the baseline data collection, each animal will be administered either the vehicle or test article intravenously (depending on study day). All animals will be acclimated to the restraint chair and isolation box for approximately 6 days. All acclimation procedures will be documented in the study file.

8.9 Justification of Dose Level Selection

The doses for this study were selected based on a dose range finding study performed in Cynomolgus Monkeys (Calvert Study #0437LS35.001). The monkeys tolerated the test article when administered as an intravenous injection over 3 minutes as three consecutive doses (one dose per week) at 10, 50 and 150 mg/kg. For this cardiovascular study, the route of test article administration will be a slow intravenous injection (~3 minutes) in order to achieve sufficient exposure.

9.0 TEST SYSTEM

Species:	Monkey (non-naïve)
Breed	Cynomolgus (<i>Macaca fascicularis</i>)
Supplier:	Covance Research Products, Inc.
Age of Animals at Start of Dosing:	2 to 7 years
Sex:	Male
Body Weight Range at Start of Dosing:	3 to 7 kg
Number of Animals on Study:	Eight male monkeys (plus one monkey will be available as a replacement animal, if needed).

9.1 Test system justification

Considerable scientific documentation exists for the Cynomolgus monkey as a predictive animal model for humans. An extensive biochemical and physiological data base for the Cynomolgus monkey is available. This is an accepted non-rodent species for use in toxicology studies of drugs intended for human use. At this time, studies in laboratory animals are required to support regulatory submissions. The number of monkeys is considered to be the minimum number necessary to yield meaningful results.

10.0 ANIMAL CARE, HOUSING AND ENVIRONMENTAL CONDITIONS

General procedures for animal care and housing will meet current AAALAC recommendations, current requirements stated in the *ILAR Guide for Care and Use of Laboratory Animals* (National Research Council, as revised) and current requirements as stated by the U.S. Department of Agriculture through the Animal Welfare Act, as amended.

10.1 Quarantine and Acceptance

Study animals were quarantined and accepted in accordance with facility Standard Operating Procedures (SOPs) at Battelle, West Jefferson, Ohio. A staff veterinarian will verify that each animal is healthy based upon a routine pre-study clinical pathology screen, physical examination and observations. A staff veterinary clinical pathologist will review the clinical pathology data. Animals not suitable for health reasons will be excluded from the study.

10.2 Animal Housing and Environmental Conditions

Animals will be housed individually in stainless steel cages during the study period. General procedures for animal care and housing will meet current AAALAC standards, current requirements stated in the NRC "Guide for Care and Use of Laboratory Animals" (National Research Council) and the U.S. Department of Agriculture through the United States Animal Welfare Act, as amended. The animal room temperature, relative humidity, light/dark cycle, and air changes will be maintained in accordance with facility and or Test site SOPs.

All animal housing and environmental conditions will follow facility and Test site SOPs. Non Human Primates (NHPs) will be singly housed at least 7 days prior to Day 1 to mimic testing conditions.

To avoid interference with cardiovascular data collection, cage changes may need to occur on a non-routine schedule, not to be delayed more than 1 week.

10.3 Feed

A certified primate diet (Purina Lab Diet – 5048) will be used. Animals will have access to their daily ration except during the fasting or restraint periods (e.g., physical exams, during dosing and blood collection periods). Certified analysis of each lot used during the study will be provided by the manufacturer and maintained under the direction of Battelle. It is not anticipated that there will be any unacceptable levels of known contaminants in the food that would affect the outcome of this study.

On dosing days, NHPs will be fed after the toxicokinetic blood collection is completed. The amount of food provided will be as described in facility and Test site SOPs.

10.4 Water

Fresh water from the Columbus, Ohio Municipal Water Supply will be used. The water will be provided ad libitum to the animals via an automatic watering system, except during the restraint periods (e.g., during dosing and blood collection periods). The water supply is periodically monitored by bacterial and chemical analyses. Results of these analyses are maintained under the direction of Battelle and copies of the most recent water analysis will be maintained in the Battelle facility records. It is not anticipated that there will be any unacceptable levels of known contaminants in the water that would affect the outcome of this study.

11.0 EXPERIMENTAL DESIGN**11.1 Dose Administration**

The NHPs will be conditioned to the isolation boxes and restraint chairs approximately 4 hours prior to the start of dosing (with the exception of any NHPs that are removed early due to poor temperament or questionable cardiovascular data). The NHPs will be progressively acclimated according to testing facility SOP or a modified version of this procedure prior to study start. On the day of dosing, all 8 NHPs will be administered either vehicle or test article at the following target doses:

Animal Numbers/Sex	Treatment	Study Day	LT3114 Dose Level (mg/kg)	LT3114 Dose Concentration (mg/mL)	Dose Volume (mL/kg)*
101-104/M and 201-204/M	Vehicle	1	0	0	2
101-104/M	LT3114 (Low-Dose)	4	10	5	2
201-204/M	LT3114 (High-Dose)	4	50	25	2

*Actual dose volume will be determined by individual NHP body weight. The volume determined will be delivered over a ~3 minute slow bolus injection.

Each NHP will be dosed once via a ~3 minute slow bolus injection with vehicle or LT3114 at target doses of 0, 10 and 50 mg/kg administered at a volume of 2 mL/kg.

11.2 Surgical Procedure for Implantation of Radiotelemetry Transmitters Used for Dosing

Eight male NHPs (plus one replacement animal) were previously surgically implanted with Data Sciences International radiotelemetry transmitters, which have ECG and core body temperature capabilities. A telemetry unit was placed into abdomen of each NHP and ECG leads were placed beneath the skeletal musculature in a configuration emulating a standard Lead II electrocardiogram.

11.3 Telemetry Data Collection

Each animal's home cage and isolation box will be equipped with a Data Sciences International telemetry receiver. The transmitters, receivers, data matrices, ambient pressure monitors, cabling, and computers which run the Dataquest A.R.T. data acquisition and analysis software are all components of the PhysioTel Telemetry System. The Dataquest A.R.T. telemetry software will collect systemic arterial blood pressures (systolic, diastolic, and mean), heart rate, core body temperature, and ECG data continuously (except when the animals are out of range of the receivers for study-driven activities) at least 24 hours before the first dosing event through at least 7 days following the last dosing event.

Prior to each dose administration, each animal will be chair-restrained and allowed baseline values to become consistent with the experimental environment. Following this initial calm down period, baseline data will be collected for at least 30 minutes. Monitoring will continue for approximately 2 hours ($\pm$ 30 minutes) following dosing. Following each in-box cardiovascular monitoring period, the animals will be returned to their home cages and in-cage telemetry data collection will be resumed.

Telemetry data will only be collected when the animals are in their isolation boxes or home cages. A detailed account of the telemetry data collection procedure will be included in the study file.

11.4 Animal Selection and Identification

The Study Director will determine which NHPs will be placed on study. This selection will be based upon each NHP's physical exam, clinical observations and preliminary evaluation of at least 12 hours of each NHP's cardiovascular data. If all of the NHPs' parameters are acceptable, then the NHPs to be used for dosing will be randomly selected using the Xybion PATH/TOX System.

NHPs will be uniquely identified by both tattoo and cage card. The NHPs selected for dosing will be randomly assigned a study specific identification number (using the Xybion PATH/TOX System), which will be documented on their cage card.

11.5 Clinical Observations

Observations for morbidity and mortality will be performed twice in accordance with facility Standard Operating Procedures. Prior to Day 1, clinical observations will be performed daily and recorded on the Individual Animal History Records. Beginning on Day 1, detailed clinical observations will be performed at least once daily and entered into the Xybion PATH/TOX System. On each day of dosing, detailed clinical observations will be performed a minimum of once prior to and following dosing.

11.6 Body Weights

The individual body weight of each NHP will be recorded prior to animal selection and on Study Days 1 and 4 (prior to dosing for dose volume determinations) and on Day 11.

11.7 Blood Collection for LT3114

A blood sample (approximately 2.0 mL) will be collected from each dosed NHP on Days 1 and 4 at approximately 2 hours from the start of dosing. All collection times will be recorded. All blood samples will be collected into tubes containing K₂EDTA. Within 60 minutes from collection, the collection tubes will be centrifuged to obtain plasma. The plasma will be transferred into appropriately labeled tubes and placed on dry ice until stored in a freezer set to maintain -80 to -60°C until shipment to the Sponsor's laboratory for analysis.

After the final sample is collected, all samples will be shipped on dry ice via an overnight courier to the following address for analysis:

Melissa Calvarese
QPS, LLC
One Innovation way, Suite 200

Newark, DE 19711
Telephone: (302) 369-5196
Telefax: (302) 737-3759
Email: melissa.calvarese@qps.com

Prior to shipment, the above-mentioned person and the study sponsor will be informed by the Study Monitor of shipment details. Shipments will be performed only at the beginning of the week. The results of analysis performed will be included in the final study report.

Analyses of plasma samples for LT3114 concentration will be performed by QPS. Complete details on the methods and results will be provided as a bioanalytical sub-report which will be included in the Final Report.

11.8 Necropsy

A gross necropsy will be performed under the supervision of a veterinary pathologist (when possible) on all NHPs that die during the study or are euthanized for any other reason (moribund as per Battelle SOP ARFII-003 or other), after notification of the Sponsor. Tissues will be collected according to a standard primate tissue list and they will be fixed in 10 percent neutral-buffered formalin. The Sponsor will determine if any of the tissues will be processed for histopathological evaluation or discarded.

11.9 Disposition of Animals

After completion of the in-life study phases, all surviving animals will be transferred to Battelle's Laboratory Animal Resources (LAR) holding pool or a subsequent study.

12.0 DATA ANALYSIS

12.1 Data Analysis

11.1.1 Hemodynamic and Core Body Temperature Assessments

Group mean average and individual animal plots will be produced for the systemic arterial blood pressures (systolic, diastolic, and mean), heart rate, and core body temperature.

11.1.2 Electrocardiographic Assessment

Periodic ECG waveform tracings (approximately 60 seconds) will be generated using ECGAUTO software from EMKA Technologies (Paris, France) at predose and at approximately 0.5, 1, 2, 4, 8, 12, 24 and 48 hours post-dosing on each dosing day (Days 1 and 4). Some of the ECG strips produced at these time points may vary from the targeted time point due to the presence of unreadable ECG waveforms. The ECG waveform strips may be qualitatively assessed for rhythm and morphology alterations by the Study Cardiologist.

ECG intervals (PR, QRS, QT and RR) will be measured and analyzed continuously and averaged for each 5 minute time period from 24 hours predose through at least 24 hours post-dose on each dosing day (Days 1 and 4). The QT interval will be normalized for changes in heart rate (RR interval) by conversion to the corrected QT (QTc) interval using Bazett's formula $QTc = QT/(RR)^{1/2}$ (Bazett, 1920).

11.1.3 Statistical Analyses

Telemetry data will be electronically transferred to a study statistician for analyses. Statistical analyses will be performed separately for each parameter and for in-cage and in-box data. Significance levels will be set to at 0.05 unless specified otherwise.

For each of the hemodynamic and ECG interval end points data collected during a predetermined time period prior to dosing on each dosing day will be averaged for each animal under both in-cage and in-box conditions on each dosing day, resulting in separate in-cage and in-box "baseline" averages for the animal. The baseline period for the in-box data will be the ~30-minute period prior to each dosing when the animal is in the isolation box. The baseline period for the home cage will be ~2 hours prior to dosing and prior to moving the animal to the isolation box.

For each dose administration on Day 1 and Day 4, the animal's in-cage and in-box data collected post-dosing will be divided into specified time intervals (for example, 10-minute for the in-box data and 30-minute for the in-cage data) during the 2 hours of collecting in-box data and the 22 hours of collecting in-cage data (3 to 24 hours post-dosing), and the data collected within each interval will be averaged. For each animal and parameter, the appropriate baseline average (either in-cage or in-box) will be subtracted from the post-dosing data collected on the same day and within that same environment, resulting in "baseline-adjusted" in-cage and in-box data, respectively.

For both the unadjusted and baseline-adjusted post-dosing averages, means and the standard error of each mean (SEM) will be calculated across animals by dosing day, test condition (in-cage or in-box), and time interval and will be presented in tables for each end point. Means (and accompanying standard errors) of the baseline averages will be calculated across animals for each and test condition (in-cage or in-box) and dosing day and will be presented in tables for each end point.

The baseline-adjusted post-dosing averages will be statistically analyzed using a repeated measures mixed effect ANOVA model. The model is expected to include dose level (i.e., vehicle, low-dose and high dose), time after dose and the interaction of time after dose with dose level as fixed effects, and animal as a random effect when appropriate. The ANOVA model will take into account correlation in the post-dosing averages that is present due to monitoring the same animals over multiple time intervals following a dosing using either an AR(1) covariance structure or a sp(pow) covariance structure depending on whether or not the time intervals are equally spaced. F-tests will be conducted within the ANOVA to test for significant differences among the dose

levels in the baseline-adjusted averages at each specified time interval following dosing. To control the positive discovery rate of these tests over the time intervals during a post-dosing period to no more than 5 percent, the significance level of each test will be adjusted using the Benjamini and Hochberg method (Benjamini, 1995) and conclusions of significant differences among dose levels at specific time intervals will be made based on these adjusted significance levels. When significant differences among dose levels are concluded at a given post-dosing time interval, t-tests will be performed within the ANOVA to compare each of the two test article dose levels with the vehicle level at that time interval. A Bonferroni correction (Miller, 1981) will be applied to the significance levels of these comparisons in order to control the overall error rate associated with comparisons to vehicle at a given time interval to less than or equal to 5 percent. Details on specific statistical methods will be documented in the study file.

13.0 REPORTING

An audited draft report of this study will be submitted to the Sponsor. The Sponsor shall submit comments, if any, on the draft report to the Study Director. Battelle will submit a final report to the Sponsor following receipt of the Sponsor's final comments. The final report will be provided to the Sponsor as a searchable and hyperlinked PDF file.

14.0 COMPUTER SYSTEMS FOR DATA MANAGEMENT AND ANALYSIS

Computer System Name	Version Number	Manufacturer	Data Type
Dataquest A.R.T.	3.1	Data Sciences International	Physiological Signals
ECGAUTO	3.3	EMKA Technologies	ECG
Excel Building Supervisor	1.7	Honeywell	Animal Facility Environmental
Next Generation PATH/TOX SYSTEM	1.7.2	Xybion Medical Systems Corporation	Animal Toxicology, Pathology and Pharmacy Data Management
Stata Release	13	StataCorp	Statistical Analysis
T-Track	1.0.0	Battelle	Environmental Storage

15.0 STORAGE OF STUDY MATERIALS AND RECORDS RETENTION

Except for analyses performed by the sponsor or sponsor's designated laboratory, all records required to reconstruct the study and the final report will be maintained under the direction of Battelle for 3 years following the completion of in-life. At the completion of the 3 years, the sponsor shall be contacted to determine where the records will be shipped. Any raw data or documentation generated from analyses performed by the Sponsor or Sponsor's designated laboratory will be archived by the Sponsor or Sponsor's designated laboratory.

16.0 REFERENCES

Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. B*, **57**:289-300.

Guide for the Care and Use of Laboratory Animals. (2011). National Academy Press, Washington, D.C. Current Version.

Bazett, H.C. (1920). An analysis of the time-relations of electrocardiograms. *Heart.*, **57**: pp.353-370.

Miller, R.G., Jr. (1981). *Simultaneous Statistical Inference*, Springer Verlag, NY.

Neter, J., Kutner, M., Nachtsheim, C., and Wasserman, W. (1996). *Applied Linear Statistical Models*, Fourth Edition, Chicago:Irwin.

Protocol Amendment Number 1
Battelle Study Number: 52447
Sponsor Study Number: 1259SL35.001
Page 1 of 4**PROTOCOL AMENDMENT NUMBER 1****Battelle Study Number: 52447****A SINGLE INTRAVENOUS INFUSION STUDY OF LT3114
IN CYNOMOLGUS MONKEYS****1. Part to be Changed: Page 3, Section 2.0, Regulatory Compliance****Effective Date: September 9, 2014**

Change: All portions of this study will adhere to the study protocol and any amendments, as well as to applicable Battelle facility Standard Operating Procedures.

This study will be conducted in an AAALAC-accredited facility in compliance with the ILAR/NRC Guide for Care and Use of Laboratory Animals, and the U.S. Animal Welfare Act, as amended.

To: All portions of this study **conducted at Battelle** will adhere to the study protocol and any amendments, as well as to applicable Battelle facility Standard Operating Procedures.

All portions of this study conducted at Burleson Research Technologies, Inc. (BRT) will adhere to the study protocol and any amendments, as well as to applicable BRT facility Standard Operating Procedures.

This study will be conducted in an AAALAC-accredited facility in compliance with the ILAR/NRC Guide for Care and Use of Laboratory Animals, and the U.S. Animal Welfare Act, as amended.

Justification: Regulatory compliance was changed to specify facilities.

2. Part to be Changed: Page 7, Section 8.7, Dose Formulation Analysis**Effective Date: September 9, 2014**

- Change:** Dose formulation analysis will be performed by Victor J. Johnson, Ph.D. The report regarding the test article analysis will be sent to Battelle for inclusion in the final report.
- To:** Dose formulation analysis will be performed by **BRT under the direction of** Victor J. Johnson, Ph.D. **Analysis will be performed using a validated method (BRT 20140629-A).** The report regarding the test article analysis will be sent to Battelle for inclusion in the final report.
- Justification:** Change in wording requested by Principal Investigator and validated method added.

3. Part to be Changed: Page 12, Section 11.5, Clinical Observations**Effective Date: September 9, 2014**

- Change:** Beginning on Day 1, detailed clinical observations will be performed at least once daily and entered into the Xybion PATH/TOX System. On each day of dosing, detailed clinical observations will be performed a minimum of once prior to and following dosing.
- To:** Beginning on Day 1, **clinical observations** will be performed at least once daily and entered into the Xybion PATH/TOX System. On each day of dosing, **clinical observations** will be performed a minimum of once prior to and following dosing.
- Justification:** Change in wording from detailed clinical observations to clinical observations. Detailed clinical observations would entail the technician to physically handle the animals and on a cardiovascular telemetry study animal handling will greatly affect cardiovascular data.

4. Part to be Changed: Page 13, Section 11.7, Blood Collection for LT3114**Effective Date: September 9, 2014**

Change: Prior to shipment, the above-mentioned person and the study sponsor will be informed by the Study Monitor of shipment details

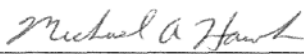
To: **Prior to shipment, the above-mentioned person and the Study Sponsor will be informed of shipment details.**

Justification: The Study Monitor will not inform the Principal Investigator and the Study Sponsor but someone at Battelle will notify them of the shipment details.

5. Part to be Added: Page 15, Section 15.0, Storage of Study Materials and Records Retention**Effective Date: September 9, 2014**

Add the following: All work products, including raw data and final study phase report, generated by BRT will be archived at BRT for a period of 2 year, unless other arrangements are made with the Sponsor. After completion of the 2 year period, all study records archived at BRT will be returned to the Sponsor. The Battelle Study Director will be informed of the final disposition of the BRT study records.

Justification: Wording added to this section to clarify how BRT will store study materials and retain records.

Protocol Amendment Number 1
Battelle Study Number: 52447
Sponsor Study Number: 1259SL35.001
Page 4 of 4**SIGNATURE PAGE**This protocol amendment was approved by the Sponsor's Study Monitor on 9/9/14 / MAH.
Date Initials**APPROVED, BATTELLE:**

Michael A. Hawk, M.S., D.A.B.T.
Study Director9/10/14

Date

Barney Sparrow, Ph.D., D.A.B.T.
Manager09/11/14

Date**APPROVED, CALVERT LABORATORIES, INC.:**

Jill L. Fogleman, Ph.D.
Sponsor Representative
Calvert Laboratories, Inc.

Date**APPROVED, MWM Consulting Group Inc. for
Lpath Inc.:**

Marlene W. Modi, Ph.D.
Sponsor Study Monitor

Date

① SEE FOLLOWING PAGE FOR SPONSOR REPRESENTATIVE'S AND SPONSOR STUDY
MONITOR'S SIGNATURES. MAH 9/10/14

Protocol Amendment Number 1
Battelle Study Number: 52447
Sponsor Study Number: 1259SL35.001
Page 4 of 4**SIGNATURE PAGE**This protocol amendment was approved by the Sponsor's Study Monitor on ____/____/____.
Date Initials**APPROVED, BATTELLE:**_____
Michael A. Hawk, M.S., D.A.B.T.
Study Director_____
Date_____
Barney Sparrow, Ph.D., D.A.B.T.
Manager_____
Date**APPROVED, CALVERT LABORATORIES, INC.:**_____
Jill L. Fogleman, Ph.D.
Sponsor Representative
Calvert Laboratories, Inc._____
10 Sep 2014

Date

**APPROVED, MWM Consulting Group Inc. for
Lpath Inc.:**_____
Marlene W. Modi

09 September 2014

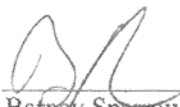
Marlene W. Modi, Ph.D.
Sponsor Study Monitor_____
Date

PROTOCOL AMENDMENT NUMBER 2**Battelle Study Number: 52447****A SINGLE INTRAVENOUS INFUSION STUDY OF LT3114
IN CYNOMOLGUS MONKEYS****1. Part to be Changed: Page 6, Section 8.2, Test Article Identification****Effective Date: November 3, 2014**

Change:	Supplier:	Gallus Biopharmaceuticals, LLC
To:	Manufacturer: Supplier	Gallus Biopharmaceuticals, LLC Calvert Laboratories, Inc.
Justification:	Clarification of Manufacturer and Supplier for the test article.	

Protocol Amendment Number 2
Battelle Study Number: 52447
Sponsor Study Number: 1259SL35.001
Page 2 of 2**SIGNATURE PAGE**This protocol amendment was approved by the Sponsor's Study Monitor on 11/5/14 / MAH.
Date Initials**APPROVED, BATTELLE:**

Michael A. Hawk, M.S., D.A.B.T.
Study Director11/6/14

Date

Barney Sparrow, Ph.D., D.A.B.T.
Manager11/06/14

Date**APPROVED, CALVERT LABORATORIES, INC.:**

Jill L. Fogleman, Ph.D.
Sponsor Representative
Calvert Laboratories, Inc.

Date**APPROVED, MWM Consulting Group Inc. for
Lpath Inc.:**

Marlene W. Modi, Ph.D.
Sponsor Study Monitor

Date

①: SEE FOLLOWING PAGE FOR SPONSOR'S APPROVAL.

MAH 11/6/14

Protocol Amendment Number 2
Battelle Study Number: 52447
Sponsor Study Number: 1259SL35.001
Page 2 of 2**SIGNATURE PAGE**This protocol amendment was approved by the Sponsor's Study Monitor on ____/____/____.
Date Initials**APPROVED, BATTELLE:**_____
Michael A. Hawk, M.S., D.A.B.T.
Study Director_____
Date_____
Barney Sparrow, Ph.D., D.A.B.T.
Manager_____
Date**APPROVED, CALVERT LABORATORIES, INC.:**_____
Jill L. Fogleman, Ph.D.
Sponsor Representative
Calvert Laboratories, Inc._____
5 NOV 2014
Date**APPROVED, MWM Consulting Group Inc. for
Lpath Inc.:**_____
Marlene W. Modi, Ph.D.
Sponsor Study Monitor_____
05 November 2014
Date

Battelle Deviation Report	
Deviation No. (Assigned by QAU): DR-757-KA	
Type of Deviation (check one)	
☒ Study	☐ Facility
Study Number: 52447 Study Title: A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114 IN CYNOMOLGUS MONKEYS	
Deviation Classification (check one)	
☐ GLP ☒ Protocol ☐ SOP/Method: Number and Version ☐ Validation/Qualification ☐ Other	
Date(s) of Deviation: 09/15/14-09/23/14	
Description of Deviation Protocol page 12 section 11.7 states: The plasma will be transferred into appropriately labeled tubes and placed on dry ice until stored in a freezer set to maintain -80 to -60°C until shipment to the Sponsor's laboratory for analysis. The freezer settings (where the samples were stored) are set to maintain -85 to -60°C.	
Cause of Deviation: Incorrect range of freezer setting in the protocol.	
Deviation Form Prepared by/Date: S. Armentrout 11/04/14 110414 Jt	
Corrective Action: None.	
Impact of Deviation: NONE ALL SAMPLES WERE ANALYZED AND RESULTS WERE ADEQUATE FOR THIS STUDY.	
Deviation Corrective Action/Impact Approved by/Date: (Study Director and Contributing Scientist/Principal Investigator as applicable or Responsible Individual) Michael A. Hand 11/4/14	
Deviation Reviewed and Corrective Action Accepted by/Date: (Supervisor, Supervisor Representative, or Manager) BT 11/4/14	
Deviation Reviewed and Registered by QAU/Date: Kathleen E. Reed 11/5/14	

Battelle Deviation Report	
Deviation No. (Assigned by QAU): DR-723-KA	
Type of Deviation (check one)	
☒ Study	☐ Facility
Study Number: 52447 Study Title: A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114 IN CYNOMOLGUS MONKEYS	
Deviation Classification (check one)	
☐ GLP ☒ Protocol ☐ SOP/Method: Number and Version ☐ Validation/Qualification ☐ Other	
Date(s) of Deviation: 09/18/2014	
Description of Deviation Protocol section 11.7 states: Within 60 minutes from collection, the collection tubes will be centrifuged to obtain plasma. On 09/18/14, animal 101 was bled at 1150a and the sample was not centrifuged until 1252p.	
Cause of Deviation: Technician oversight <i>AP 092614</i>	
Deviation Form Prepared by/Date: S. Armentrout 09/23/14 <i>0924148T</i>	
Corrective Action: None.	
Impact of Deviation: Impact will be evaluated after analysis of data. <i>after reviewing the TK data, there was no apparent impact to the data. mmt 11/19/14</i>	
Deviation Corrective Action/Impact Approved by/Date: (Study Director and Contributing Scientist/Principal Investigator, as applicable or Responsible Individual) <i>Michael A. Hawk 9/24/14</i>	
Deviation Reviewed and Corrective Action Accepted by/Date: (Supervisor, Supervisor Representative, or Manager) <i>Tw/10/14/14</i>	
Deviation Reviewed and Registered by QAU/Date: <i>Kathleen E. Reed 11/7/14</i>	

Battelle Deviation Report	
Deviation No. (Assigned by QAU): DR-715-KA	
Type of Deviation (check one)	
☒ Study	☐ Facility
Study Number: 52447 Study Title: A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114 IN CYNOMOLGUS MONKEYS	
Deviation Classification (check one)	
☐ GLP ☒ Protocol ☐ SOP/Method: Number and Version ☐ Validation/Qualification ☐ Other	
Date(s) of Deviation: 09/15/2014	
Description of Deviation Protocol section 8.7 (Dose Formulation Analysis) states: Prior to shipment of samples, the Study Monitor and the Principal Investigator will be informed by email of the shipment date and specifications. On September 15, 2014, prior to shipment of samples, an email notification was sent to the Principal Investigator with the shipment date and specifications but was not sent to the study monitor.	
Cause of Deviation: Inadvertently did not notify the Study Monitor. <i>R 11/10/14</i>	
Deviation Form Prepared by/Date: S. Armentrout 09/16/14 <i>11/10/14</i>	
Corrective Action: An email was sent out the following day informing the Study Monitor of the sample shipment.	
Impact of Deviation: None.	
Deviation Corrective Action/Impact Approved by/Date: (Study Director and Contributing Scientist/Principal Investigator as applicable or Responsible Individual) <i>Michael A. Havel 11/10/14</i>	
Deviation Reviewed and Corrective Action Accepted by/Date: (Supervisor, Supervisor Representative, or Manager) <i>Natalie L. Smith 10 Nov 2014</i>	
Deviation Reviewed and Registered by QAU/Date: <i>Kathleen E. Reed 11/10/14</i>	

Battelle Deviation Report	
Deviation No. (Assigned by QAU): DR-777-KA	
Type of Deviation (check one)	
☒ Study	☐ Facility
Study Number: 52447 Study Title: A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114 IN CYNOMOLGUS MONKEYS	
Deviation Classification (check one)	
☐ GLP ☒ Protocol ☐ SOP/Method: Number and Version ☐ Validation/Qualification ☐ Other	
Date(s) of Deviation: 09/24/14	
Description of Deviation Protocol section 11.3 states: The Dataquest A.R.T. telemetry software will collect systemic arterial blood pressures (systolic, diastolic, and mean), heart rate, core body temperature, and ECG data continuously (except when the animals are out of range of the receivers for study-driven activities) at least 24 hours before the first dosing event through at least 7 days following the last dosing event. On 09/24/14 there was an issue with the telemetry hardware and data collection stopped at 12:18p, 6 days following the last dosing event for animal 202.	
Cause of Deviation: Disconnected cable.	
Deviation Form Prepared by/Date: S. Armentrout 11/19/14	
Corrective Action: None.	
Impact of Deviation: Minimal. No data collected during the last ~22 hours of the study. This set of data will not be analyzed according to protocol. To prevent prolonged disturbance of the remaining animals on study, the cable was not re-connected.	
Deviation Corrective Action/Impact Approved by/Date: (Study Director and Contributing Scientist/Principal Investigator as applicable or Responsible Individual) <i>Michael Hawk</i> 11/19/14	
Deviation Reviewed and Corrective Action Accepted by/Date: (Supervisor, Supervisor Representative, or Manager) <i>[Signature]</i> 11/19/14	
Deviation Reviewed and Registered by QAU/Date: <i>Kathleen E. Reed</i> 11/20/14	

Battelle Deviation Report	
Deviation No. (Assigned by QAU): DR-788-KA	
Type of Deviation (check one)	
☒ Study	☐ Facility
Study Number: 52447 Study Title: A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114 IN CYNOMOLGUS MONKEYS	
Deviation Classification (check one)	
☐ GLP ☒ Protocol ☐ SOP/Method: Number and Version ☐ Validation/Qualification ☐ Other	
Date(s) of Deviation: 09/03/14	
Description of Deviation Protocol page 8 section 8.8 states: Vehicle and test article will be administered via a slow intravenous injection (~3 minutes) by an ambulatory pump. Details of the dosing procedure will be documented in the study file. Vehicle and test article was administered via a slow intravenous injection (~3minutes) but not by ambulatory pump.	
Cause of Deviation: Typographical error in the protocol.	
Deviation Form Prepared by/Date: S. Armentrout 12/04/14	
Corrective Action: None.	
Impact of Deviation: None. All doses were manually delivered by a slow intravenous injection.	
Deviation Corrective Action/Impact Approved by/Date: (Study Director and Contributing Scientist/Principal Investigator as applicable or Responsible Individual) Michael Hurl 12/4/14	
Deviation Reviewed and Corrective Action Accepted by/Date: (Supervisor, Supervisor Representative, or Manager) Roy Bannister 12/4/14	
Deviation Reviewed and Registered by QAU/Date: Heather E. Reed 12/4/14	

APPENDIX B

**CERTIFICATE OF ANALYSIS, CERTIFICATE OF COMPLIANCE, STABILITY
AND FORMULATION ANALYSIS REPORT**

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
		PAGE	1 of 2

Name	Part Number	Lot Number
Lpathomab-DS, 30 mg/mL	TBD	GA121459
Formulation Buffer		Storage Temperature
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0		2 to 8°C
Production Date	Test Date	Retest Date
JUN 2014	JUL 2014	DEC 2014
Intended Use	GLP-conforming Toxicity Testing	
Manufacturer	Manufacturer Part Number	Manufacturer Lot Number
Gallus-MO, St. Louis, MO	TBD	Gallus-MO 103948

Attribute	Test	Specification	Results
	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution	Particulate free, clear, slightly yellow solution
Identity	pH	5.0 ± 0.2	5.1
	CGE (Non-reduced)	Major band at 150 kDa	Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Conforms
	iCE	Comparable to RS	Conforms
		Minor Peak at pI ~8.3	15%
Major Peak at pI ~8.4		60%	
Minor Peak at pI ~8.6		13%	
	Faint Peak at pI ~8.7	4%	
Potency	API Concentration by OD at 280 nm	30 ± 3 mg/mL	32.1 mg/mL
	LPA Binding	100% ±40% of RS	89%

Confidential

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
			PAGE 2 of 2

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected

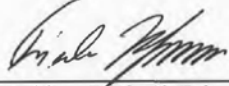
Attribute	Test	Specification	Results
Safety	Residual Protein-A	≤ 10 ppm	<1 ppm
	Residual DNA	≤ 5 pg/mg	<0.05 pg/mg
	CHO HCP	≤ 100 ppm	15 ppm
	Endotoxin, USP <85>	Report	<0.06 EU/mL ¹
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL	None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

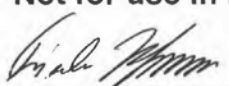
Quality Assurance:


Acting Head of QA - Frieder K. Hofmann, PhD

Date: 22AUG2014

¹ At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

Confidential

Lpath Inc.		Certificate of Compliance			
TITLE	24.3 mM Citrate, 51.4 mM Sodium Phosphate Dibasic, 300 mM D-Trehalose, pH 5.0 Buffer				
					PAGE 1 of 1
Manufacturer			Lot Number		
Gallus Biopharmaceuticals, St. Louis, MO			104011-001		
Production Date		Expiry Dates¹		Intended Use	
JUN 2014	Storage: 15 to 30°C; SEP 2014	Storage: 2 to 8°C; MAR 2015	Storage: -25 to -15°C; JUN 2016	Storage: -90°C to -60°C; JUN 2019	GLP-conforming toxicity testing and Lpathomab sample diluent
Attribute	Specification			Results	
Appearance	Particulate free, clear, colorless			Conforms	
pH	4.9 to 5.1			5.1	
Sterility	Confirmed integrity of sterilizing membrane filter			Conforms	
JUDGMENT					
Based on a comprehensive review of the executed production batch record, this buffer is approved for use in GLP-conforming toxicity testing and as sample diluent for Lpathomab.					
Caution – Not for use in humans					
Quality Assurance:				Date: 18JUL2014	
Acting Head of QA - Frieder K. Hofmann, PhD					

¹ Buffer not stored < -60°C or used for toxicity testing, requires aseptic handling.

Lpath Certificate of Analysis														
Stability Assessment of Lpathomab-DS, lot 103948, stored at 5 ± 3°C					Formulation Age									
Assay		Specification			Time 0		1 Month		2 Months		3 Months			
Appearance		Particulate-free, clear, colorless to slightly yellowish and slightly opalescent solution.			Conforms		Conforms		Conforms		Conforms			
pH		4.8 to 5.2			5.2		5.1		5.1		5.1			
ICE		Comparable to Reference Standard			Conforms		Conforms		Conforms		Conforms			
		Report pI and relative peak area of Major Isoform and total relative peak area of acidic and basic variants			pl		pl		pl		pl			
					Area		Area		Area		Area			
					Acidic		Acidic		Acidic		Acidic			
		8.2			53%		8.2		53%		8.2		52%	
		Basic			24%		Basic		25%		Basic		25%	
CGE-NR		IgG + IgG-NG ≥ 90%			98.4%		97.9%		98.8%		97.6%			
		IgG-NG			2.1%		1.8%		1.5%		2.2%			
		2H-1L			0.3%		0.4%		0.4%		0.4%			
		2H			0%		0%		0%		0%			
		HL			0%		0%		0%		0%			
		HC			0%		0%		0%		0%			
		LC			0.4%		0.4%		0.5%		0.6%			
CGE-R		HC + HC-NG + LC ≥ 90%			98.1%		95.5%		99.1%		98.1%			
		HC			66%		63%		67%		66%			
		HC-NG			1.1%		1.5%		1.1%		1.1%			
		LC			31%		31%		31%		31%			
qSE-HPLC		API concentration: 29 to 35 mg/mL			33.2		33.5%		34.3		34.5			
		Purity: ≥ 90%			99.2		98.6%		98.3%		98.2			
		Aggregates: ≥ 5%			0.8%		1.4%		1.7%		1.8			
		No single impurity > %			0%		0%		0%		0%			
LPA-binding ELISA		60% to 140% of reference standard activity			125%		106%		92%		136%			
Particulate		≤ 6,000 Particles ≥ 10 µm per vial			51						51			
		≤ 600 Particles ≥ 25 µm per vial			2						0			
Judgment														
Lot 103948 of Lpathomab-DS was placed in a validated 5 ± 3°C stability chamber at Lpath's qualified contract laboratory, Eurofins Lancaster Laboratories in Lancaster, PA, and tested monthly for three months with the stability-indicating assays appearance, pH, isoelectric focusing by capillary electrophoresis (ICE), non-reduced and reduced capillary gel electrophoresis (CGE-NR and CGE-R), quantitative size-exclusion high performance liquid chromatography (qSE-HPLC), LPA-binding ELISA and subvisible particulate matter. The obtained results and underlying raw data were reviewed and verified by Lpath's Quality Department. The results only varied within each test-specific accuracy tolerance and indicate that the lot exhibited excellent product stability under the storage conditions.														
Department		Title		Name			Signature			Date				
Quality Assurance		Head of Product Quality		Frieder K Hofmann, PhD						22DEC2014				

STUDY PHASE REPORT

STUDY TITLE: A Single Bolus Injection Study of LT3114 in Cynomolgus Monkeys

REPORT TITLE: Determination of LT3114 in Dose Formulation Samples

BRT Reference No.: BRT 20140827

Author: Victor J. Johnson, PhD (BRT)

Dose Formulation Analysis Laboratory: Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560

Testing Facility: Battelle
505 King Avenue
Columbus, OH 43201-2696

Study Number: 52447

Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

Sponsor Reference Number: 1259SL35.001

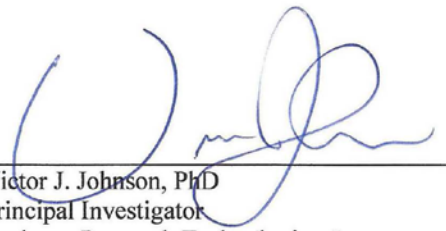
Report Date: 05 January 2015



COMPLIANCE STATEMENT

This study was performed according to the study protocol, applicable BRT SOPs, and in compliance with the following regulations:

- Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58)



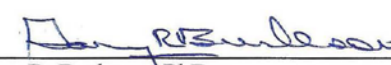
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USA

05 Jan 15

Date

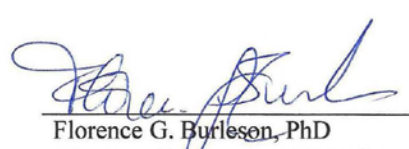
APPROVALS



Victor J. Johnson, PhD
Principal Investigator
Burleson Research Technologies, Inc.05 Jan 15
Date

Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.05 Jan 2015
Date

Report Review:



Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.05 Jan 15
Date

QUALITY ASSURANCE STATEMENT

Study Phase Reference Number: BRT 20140827

Study Phase Title: Determination of LT3114 in Dose Formulation Samples

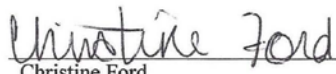
Testing Site: Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560

Principal Investigator: Victor J. Johnson, PhD

Phase Inspected	Date of Inspection	Date Report Sent to BRT Principal Investigator/Management	Date Report Sent to Battelle Study Director/Management
Substrate Addition Audit	09 Dec 14	09 Dec 14	09 Dec 14
Raw Data Audit	18 Dec 14	18 Dec 14	19 Dec 14
Report Audit	18 Dec 14	18 Dec 14	19 Dec 14

The final report and data record audit were performed by the BRT Quality Assurance Unit.

To the best of my knowledge, this final report accurately describes the study methods and procedures used, and the reported results accurately reflect the raw data.


Christine Ford
Quality Assurance Manager

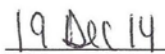

Date

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STUDY INFORMATION**Determination of LT3114 in Dose Formulation Samples**

Study Title:	A Single Bolus Injection Study of LT3114 in Cynomolgus Monkeys
Testing Facility:	Battelle
Testing Facility Study No:	52447
Study Director:	Michael Hawk, M.S., D.A.B.T. (Battelle)
Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121
Sponsor Reference Number:	1259SL35.001
Test Site (Dose Formulation Analysis):	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Test Site Reference No.:	BRT 20140827
Principal Investigator:	Victor J. Johnson, PhD (BRT)
Analyst:	Kristen Fisher (BRT)
Study Initiation Date:	03 Sep 14
Sample Receipt Date:	16 Sep 14 – Vehicle (top, middle, bottom) 19 Sep 14 – LT3114 at 5.0 mg/mL (top, middle, bottom) 19 Sep 14 – LT3114 at 25.0 mg/mL (top, middle, bottom)
Experimental Start Date:	09 Dec 14
Experimental Completion Date:	16 Dec 14
Study Phase Completion Date:	05 Jan 15

SUMMARY

The purpose of this study phase was to determine the concentration of LT3114 in dose formulation samples generated by Battelle Study No. 52447: “A Single Bolus Injection Study of LT3114 in Cynomolgus Monkeys”.

Dose formulation samples were provided for the vehicle and two dose concentrations, 5.0 mg/mL and 25.0 mg/mL. Dose formulation samples were collected from each dosing solution and the vehicle from the top, middle, and bottom strata of the LT3114 dose formulations and from the middle stratum of the vehicle. Analyses demonstrated the absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 5.5 – 7.3 mg/mL, and 27.9 – 33.1 mg/mL for the 5.0, and 25.0 mg/mL dosing solutions, respectively. All LT3114 concentrations met acceptance criteria for accuracy and precision, with the exception of the samples collected from the top and middle strata of the 5.0 mg/mL dose formulation and from the bottom stratum of the 25.0 mg/mL dose formulation. The concentration of these three samples were high and did not meet acceptance criteria for accuracy but met the acceptance criteria for precision. LT3114 concentrations from the top, middle, and bottom fractions of the dose formulations had a CV% of 15.5% and 9.7% demonstrating acceptable homogeneity of the 5.0 and 25.0 mg/mL dose formulations.

INTRODUCTION

The objective of this study phase was to determine the concentration of LT3114 in dose formulation samples in support of the Battelle study 52447 entitled “A Single Bolus Injection Study of LT3114 in Cynomolgus Monkeys”.

STUDY DESIGN**Dose Formulation Samples**

Dose formulations were prepared by Battelle on 15 Sep 14 (vehicle) and 17 Sep 14 (LT3114 5.0mg/mL and 25.0 mg/mL dose formulations) according to the study protocol, using LT3114 in a citrate buffer formulation supplied by the Sponsor. Aliquots (6x1 mL) were collected on 15 Sep 14 (vehicle) and 17 Sep 14 (LT3114 dose formulations) from the top, middle, and bottom of the vehicle and each dosing solution. The samples were stored frozen (-30 to -15°C) until single aliquots from each sample were shipped to BRT overnight on dry ice and received on 16 Sep 14 (vehicle) and 19 Sep 14 (LT3114 dose formulations) (Table 1). Samples were logged into BRT inventory and stored frozen in a freezer set to maintain -20°C until analyzed for LT3114 concentration. Samples were returned a freezer set to maintain -20°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director per protocol. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 1: Dose formulation samples received at BRT on 16 Sep 14 (vehicle) and 19 Sep 14 (LT3114 dose formulations).

Preparation Date	Date of Receipt at BRT	Aliquots	LT3114 Concentration (mg/mL)	Location	Volume (mL)
15 Sep 14	16 Sep 14	1	Vehicle	Top	1
15 Sep 14	16 Sep 14	1	Vehicle	Middle	1
15 Sep 14	16 Sep 14	1	Vehicle	Bottom	1
17 Sep 14	19 Sep 14	1	5.0	Top	1
17 Sep 14	19 Sep 14	1	5.0	Middle	1
17 Sep 14	19 Sep 14	1	5.0	Bottom	1
17 Sep 14	19 Sep 14	1	25.0	Top	1
17 Sep 14	19 Sep 14	1	25.0	Middle	1
17 Sep 14	19 Sep 14	1	25.0	Bottom	1

METHODOLOGY

Dose Formulation Samples

Dose formulation samples (Vehicle plus 2 dose levels with 3 samples each from top, middle, and bottom of the preparations = 9 total samples) were received in good condition on frozen dry ice on 16 Sep 14 (vehicle) and 19 Sep 14 (LT3114 dose formulations). Upon receipt at BRT, all samples were stored frozen in a freezer set to maintain -20°C. The samples were analyzed for LT3114 concentrations at BRT.

LT3114 ELISA Method

Dose formulation samples were analyzed for LT3114 concentration using a validated ELISA method (BRT 20140629A). The ELISA method is provided in [Appendix 1](#).

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. Standards, quality control (QC) samples, delipidized bovine serum (DBS) assay buffer, and dose formulation samples were added to the coated plates and incubated to allow for capture of the LT3114. A horseradish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a Spectramax 340 microplate spectrophotometer (Molecular Devices) using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Data and Statistical Analysis

Data analysis was performed using SoftMax[®]Pro v2.2.1 (Molecular Devices) software. Calibration curves were generated by plotting the mean OD₄₅₀ values (y-axis) versus concentration (x-axis) and fitting the curve using the 4-parameter logistic function. Computer printouts included OD₄₅₀ values (raw data), the calibration curve, the standard deviation, the coefficient of variation between replicates, and the interpolated values of the unknown test samples. Concentrations of LT3114 in test samples were obtained by multiplying the mean interpolated value (result), from duplicate wells, by the sample dilution factor. Results were reported to one decimal place.

Calculations, including means, standard deviations, and coefficients of variation, for determining accuracy and precision were performed using Microsoft Excel 2013.

Acceptance Criteria (per method validation BRT 20140629A)*Calibration Curve Acceptance Criteria:*

- Calibration curve range: 3.9 to 500 ng/mL
- Accuracy of standards: 20% of nominal (25% at LLOQ and ULOQ)
- Precision of the standards within 20%
- At least 75% of the standards (minimum of 6 points) met these criteria. Failed standards were removed from the analysis, if present.

QC Sample Acceptance Criteria:

- QC1-300 ng/mL; QC2-150 ng/mL; QC3-5 ng/mL
- Accuracy of QC samples: 20% of nominal (25% at 5 ng/mL and 300 ng/mL)
- Precision of the QCs were within 20% (25% at 5 ng/mL and 300 ng/mL)
- At least 67% of all QC samples met these criteria. In addition, at least 50% of the QC samples at each level met these criteria.
- Total error ($|\%RE| + \%CV$) of each QC was $\leq 30\%$ ($\leq 40\%$ at 5 ng/mL and 300 ng/mL)

Study Dose Formulation Sample Acceptance Criteria:

- Precision of the replicates was within 20% (25% at LLOQ and ULOQ)
- Sample concentrations were acceptable when accuracy was within 20% of nominal

Study Dose Formulation Sample Reanalysis:

- Samples were reanalyzed if the assay is rejected due to failure of the standards or QCs to meet acceptance criteria provided above.
- Samples were reanalyzed if the mean OD450 for the replicates is outside the range of OD450 values for the calibration curve. If applicable, the sample was diluted appropriately and reanalyzed. This did not apply to samples diluted at the MRD that were below the range of the standard curve.
- For samples that were within the range of the standard curve, samples were reanalyzed if the precision of the replicates was not acceptable.

RESULTS AND CONCLUSIONS

Dose Formulation Sample Analysis

LT3114 concentrations for the triplicate samples (top, middle, and bottom) of the vehicle and each dose formulation are provided in [Table 2](#). Analysis confirmed the absence of LT3114 in the vehicle preparations. The concentrations of LT3114 in the 5.0 mg/mL dose formulations were 7.3, 7.3, and 5.5 mg/mL and 28.3, 27.9, and 33.1 mg/mL for the 25.0 mg/mL dose formulation. Dose formulation samples collected from the bottom stratum of the 5.0 mg/mL dose formulation and the top and middle strata of the 25 mg/mL dose formulation met acceptance criteria for accuracy and precision. The concentrations of LT3114 in samples collected from the top and middle strata of the 5.0 mg/mL dose formulation and the bottom stratum of the 25.0 mg/mL dose formulation were high and did not meet acceptance criteria for accuracy but met acceptance criteria for precision. The samples that did not meet acceptance criteria for accuracy were repeated. The repeat analysis confirmed the high results ([Table 3](#)).

The homogeneity of each dose formulation was assessed by determining the CV% for the top, middle, and bottom samples collected from each formulation. The CV% was 15.5% and 9.7% for the 5.0 and 25.0 mg/mL dose formulations, respectively, demonstrating acceptable homogeneity for all dose formulations.

All dose formulation samples were analyzed within the frozen storage stability for LT3114 in a citrate buffer formulation established in the method validation (BRT 20140629A).

DOCUMENTATION AND RETENTION OF DATA

Description of Circumstances Affecting Data Quality or Integrity

BRT is not aware of any circumstances surrounding this study phase that would adversely affect data quality or integrity.

Maintenance of Study Records

All raw data, protocol, amendments, study records, and the final study phase report will be archived at the Test Site (BRT) for a period of at least 2 years from the date that the final study phase report is issued unless alternative arrangements are made with the Sponsor. After that time, or as per contract agreement, the Sponsor will be notified and the data and report will be returned to the Sponsor, unless alternative arrangements are made. All raw data not specific to this study (e.g., instrument logs, CVs, etc.) will be archived at BRT.

Deviations from the Study Protocol

- There was a technical error in preparation of the calibration curve for the assay run performed on 09 Dec 14 that resulted in failure of the QC samples to meet acceptance criteria. The assay run was rejected and the analysis was repeated. There was no impact of these deviation on the study phase since the analysis was successfully repeated.

TABLES

Table 1: Dose formulation samples received at BRT on 16 Sep 14 (vehicle) and 19 Sep 14 (LT3114 dose formulations).	8
Table 2: LT3114 concentrations in dose formulation samples.	14
Table 3: Repeat analyses of dose formulations that did not show acceptable accuracy...	15

Table 2: LT3114 concentrations in dose formulation samples.

Sampling Dates	Dose Formulation	Location	LT3114 (mg/mL)	% of Target	Mean LT3114 (mg/mL)	% of Target	CV%
15-Sep-14	Vehicle	Top	BLQ	N/A	BLQ	N/A	N/A
	Vehicle	Middle	BLQ	N/A			
	Vehicle	Bottom	BLQ	N/A			
17-Sep-14	LT3114 Low-Dose: 5.0 mg/ml	Top	7.3	146	6.7	134	15.5
	LT3114 Low-Dose: 5.0 mg/ml	Middle	7.3	146			
	LT3114 Low-Dose: 5.0 mg/ml	Bottom	5.5	110			
	LT3114 High-Dose: 25.0 mg/ml	Top	28.3	113	29.8	119	9.7
	LT3114 High-Dose: 25.0 mg/ml	Middle	27.9	112			
	LT3114 High-Dose: 25.0 mg/ml	Bottom	33.1	132			

BLQ – Below Lower Limit of Quantification (3.9 ng/mL); N/A – Not applicable.

Gray highlight indicates that concentration did not meet acceptance criteria for accuracy (within 20% of target).

Table 3: Repeat analyses of dose formulations that did not show acceptable accuracy.

Sampling Dates	Dose Formulation	Location	LT3114 (mg/mL)		
			Assay Run 1 (12 Dec 14)	Assay Run 2 (16 Dec 14)	Mean
17-Sep-14	LT3114 Low-Dose: 5.0 mg/ml	Top	7.3	7.2	7.3
	LT3114 Low-Dose: 5.0 mg/ml	Middle	6.3	8.3	7.3
	LT3114 High-Dose: 25.0 mg/ml	Bottom	32.5	33.6	33.1

Gray highlight indicates that concentration did not meet acceptance criteria for accuracy (within 20% of target).

APPENDIX 1: ELISA METHOD FOR QUANTIFICATION OF LT3114.**Materials and Equipment**

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Citrate buffer provided by Lpath
- LT3114 Citrate Formulated Drug Substance (Provided by Sponsor)
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS), assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H₂SO₄ (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)
 - 1.1 LPA-Conjugated-BSA coating material were kept frozen until use.
 - 1.2 After dilution, LPA-Conjugated-BSA coating material was stored at 2-8° C for up to 4 months.
2. Peroxidase conjugated-anti-human IgG secondary antibody
 - 2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%).
 - 2.2 Store at -20° C for up to 1 year.
3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)
 - 3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly.
 - 3.2 Store at 2-8°C for up to 3 months.
4. Blocking Solution 1xPBS, 1% BSA 1, 0.1% Tween 20
 - 4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X).
 - 4.2 Mix thoroughly and sterile filter.
 - 4.3 Store at 2-8°C for up to 1 month.

Preparation of Quality Control Samples

Prepare QC samples in citrate buffer using Human Reference Anti-LPA IgG LT3114 at 300, 150, and 10 ng/mL. Aliquot and store at $\leq -70^{\circ}\text{C}$.

ELISA Method

1. Coating Plates
 - 1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at $\leq -70^{\circ}\text{C}$. Thaw the aliquot at room temperature.
 - 1.2 Dilute LPA-Conjugated-BSA coating material to 2.5 $\mu\text{g}/\text{ml}$ in 0.05 M Carbonate Buffer.
 - 1.3 Coat a 96-well high-binding microtiter plate(s) with 100 $\mu\text{L}/\text{well}$ coating solution.
 - 1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.
2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).
3. Blocking Plates
 - 3.1 Wash coated plate(s) 4 times with wash solution.

- 3.2 Dispense 150 µL/well of blocking solution to each well and cover the plate(s) with plate sealers.
 - 3.3 Incubate plate(s) at 37°C for 1 hour ± 5 minutes.
4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)
 - 4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)
 - 4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.
 - 4.3 Prepare Human Reference Anti-LPA IgG 8 point standard curve by diluting in DBS to achieve desired range (3.9 – 500 ng/ml).
 - 4.4 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 4.5 Add samples (controls, calibrators and unknowns) at 100 µL/well.
5. Add samples to appropriate wells
 - 5.1 Dispense 100 µl of appropriate sample to designated wells in duplicate. Add 100 µl of DBS to assay blank wells.
 - 5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
6. Add HRP-goat-anti-human secondary antibody solution.
 - 6.1 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody
 - 6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.
 - 6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.
 - 6.3 Dispense 100 µl of diluted HRP-goat-anti-human secondary antibody to each well.
 - 6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
7. Substrate Addition
 - 7.1 Wash coated plate(s) 4 times with wash solution.
 - 7.2 Dispense 100 µl **COLD** TMB solution to each well.
 - 7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.

8. Stop Solution Addition
 - 8.1 Dispense 100 µl 1.0 M H₂SO₄ solution to each well.
 - 8.2 Measure absorbance at 450 nm on microplate reader.

APPENDIX C
IN-LIFE DATA

Table C-1. Clinical Observations

Animal ID	Dose Level	Dose Day	Observations
101	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
102	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
103	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
104	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
201	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
202	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
203	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
204	0 mg/kg	1	No Abnormal Clinical Observations Days 1-3
Animal ID	Dose Level	Dose Day	Observations
101	10 mg/kg	4	No Abnormal Clinical Observations Days 4-11
102	10 mg/kg	4	No Abnormal Clinical Observations Days 4-11
103	10 mg/kg	4	No Abnormal Clinical Observations Days 4-11
104	10 mg/kg	4	No Abnormal Clinical Observations Days 4-11
201	50 mg/kg	4	No Abnormal Clinical Observations Days 4-11
202	50 mg/kg	4	No Abnormal Clinical Observations Days 4-11
203	50 mg/kg	4	No Abnormal Clinical Observations Days 4-11
204	50 mg/kg	4	No Abnormal Clinical Observations Days 4-11

Table C-2. Individual Animal Body Weight (kg) Data

Group	Animal ID	Day			
		-3	1	4	11
Vehicle or LT3114 (Low-Dose)	101	5.4	5.4	5.4	5.4
Vehicle or LT3114 (Low-Dose)	102	4.4	4.4	4.4	4.5
Vehicle or LT3114 (Low-Dose)	103	4.6	4.6	4.6	4.6
Vehicle or LT3114 (Low-Dose)	104	6.4	6.5	6.4	6.5
Vehicle or LT3114 (High-Dose)	201	4.7	4.7	4.7	4.7
Vehicle or LT3114 (High-Dose)	202	4.5	4.5	4.6	4.6
Vehicle or LT3114 (High-Dose)	203	6.1	6.0	6.1	6.1
Vehicle or LT3114 (High-Dose)	204	6.5	6.5	6.6	6.6

APPENDIX D
STATISTICAL ANALYSIS REPORT

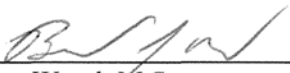
Battelle*The Business of Innovation*

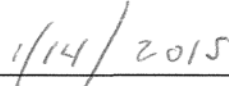
STATISTICAL ANALYSIS REPORT**A SINGLE SLOW BOLUS INJECTION STUDY OF LT3114
IN CYNOMOLGUS MONKEYS**

Battelle Study No. 52447

January 14, 2015

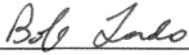
Prepared By:

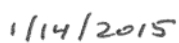


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1.0 OBJECTIVES

The objective of this study was to evaluate the cardiovascular effects of LT3114 when given a single slow bolus injection to Cynomolgus monkeys. The study's statistical objectives, which were based on this overall objective, were as follows:

- For a given set of cardiovascular, body temperature, and electrocardiogram (ECG) parameters, determine whether differences between the test article and the vehicle existed on a 10-minute interval basis during a 2-hour in-box period immediately following dosing of the test article or the vehicle.
- For a given set of cardiovascular, body temperature, and ECG parameters, determine whether differences between the test article and the vehicle existed on a 30-minute interval basis during the period from 3 to 24 hours following dosing of the test article or the vehicle, when the animals were in their home cages.

The statistical methods used to address these statistical objectives, as well as an overview of the study design and the data parameters considered in the statistical analysis, are presented in this report. All statistical analysis results and summaries of data presented in this report were executed using Version 13.1 of STATA.

2.0 STUDY DESIGN AND STUDY DATA

2.1 Design for Dosing Regimen

The non-human primates (NHPs) will be administered either test article or vehicle at the following target doses:

Animal Numbers/Sex	Treatment	Study Day	LT3114 Dose Level (mg/kg)	LT3114 Dose Concentration (mg/mL)	Dose Volume (mL/kg) ^a
101-104/M and 201-204/M	Vehicle	1	0	0	2
101-104/M	LT3114 (Low-Dose)	4	10	5	2
201-204/M	LT3114 (High-Dose)	4	50	25	2

a. Actual dose volume was determined by individual NHP body weight. The volume determined was delivered over an ~3 minute slow bolus injection.

Each NHP will be dosed once via an approximately 3 minute slow bolus injection with vehicle or LT3114 at target doses of 0, 10, and 50 mg/kg administered at a volume of 2 mL/kg. Details of the dosing procedure were documented in the study file.

2.2 Telemetry Data Collection

Each animal's home cage and isolation box were equipped with a Data Sciences International telemetry receiver. The transmitters, receivers, data matrices, ambient pressure monitors, cabling, and computers which run the Dataquest A.R.T. data acquisition and analysis software are all components of the PhysioTel Telemetry System. The Dataquest A.R.T. telemetry software collected systemic arterial blood pressures (systolic, diastolic, and mean), heart rate, core body temperature, and ECG (RR, PR, QRS, and QT) data continuously (except when the animals were out of range of the receivers for study-driven activities) at least 24 hours before each dosing event through at least 7 days following the last dosing event. Telemetry data were only collected when the animals were in their home cages or isolation boxes. A detailed account of the telemetry data collection procedure was included in the study file.

The QT interval was normalized for changes in heart rate for each animal by conversion to the corrected values using the formula: Bazett [$QTcB = QT/(RR)^{1/2}$] (Bazett, 1920).

2.3 Average and Baseline-Adjusted Average Data

There were either four or eight animals for each calculated average; however, the number of animals' data included in any particular average at a specific time point may be less than four or eight if any of the animals' data was missing. This could be due to an animal being out of range of the telemetry receiver (i.e., no data collected), data was screened due to outliers or that the derived data was not interpretable at a specific time point.

2.3.1 Telemetry Data

Telemetry data were screened for outlier values and were set to missing if data were in the following ranges:

1. SBP < 30 mmHg or SBP > 220 mmHg, DBP < 15 mmHg or DBP > 200 mmHg, MBP < 20 mmHg or MBP > 200 mmHg, HR < 40 bpm or HR > 300 bpm, or BT < 35°C or BT > 40°C. If data were missing for any of the above parameters in a given data record, then no data from that record were included in the data summary for these parameters.
2. RR ≤ 0 msec, PR ≤ 0 msec, QRS ≤ 0 msec, QT ≤ 0 msec, or QTcB ≤ 0 msec.

For each parameter and study animal, "in-cage" data within the 2-hour baseline period (4:00 AM to 6:00 AM) prior to each dosing, were averaged to obtain the baseline average for each dosing. "In-cage" data from 3 to 24 hours following each dosing were averaged into forty-two 30-minute averages. The "in-cage" baseline

average data were then subtracted from each dosing day's post-dosing 30-minute averages, resulting in "in-cage baseline-adjusted" post-dosing average data. These "baseline-adjusted" averages were the primary endpoints for the analysis on the in-cage data.

For each parameter and study animal, "in-box" data within the 30-minute baseline period immediately prior to each dosing, were averaged to obtain the baseline average for each dosing. "In-box" data from 0 to 120 minutes following each dosing were averaged into 10-minute averages. The "in-box" baseline average data were then subtracted from each dosing day's post-dosing 10-minute averages, resulting in "in-box baseline-adjusted" post-dosing average data. These "baseline-adjusted" averages were the primary endpoints for the analysis on the in-box data.

3.0 STATISTICAL METHODS

The statistical analyses were applied separately for "in-cage" and "in-box" data for each of the hemodynamic, body temperature and ECG endpoints ("parameters"). All data were analyzed using Version 13.1 of STATA.

For both the unadjusted and baseline-adjusted post-dosing averages, means and the standard errors of mean (SEM) were calculated across animals by time interval and dose level and were presented in tables and displayed in figures for each parameter. Also included in the tables are the means and SEMs for the baseline data, as calculated across animals by dose level separately for the cage and box conditions. The baseline-adjusted post-dosing averages were analyzed using an analysis of variance (ANOVA) model (Neter, 1996). The ANOVA included dose, day, time, and the interaction of dose and time as fixed effects, and animal as random effect. The model took into account the correlations in the post-dosing averages that were present due to monitoring the same animals over multiple time intervals following a given dosing through the use of an AR(1) covariance structure. F-tests were conducted within the ANOVA to test for significant differences among the dose levels in the baseline-adjusted averages at each specified time interval following dosing. To control the false discovery rate for these tests across all time intervals to less than or equal to 5 percent, the significance level of each test was adjusted using the Benjamini and Hochberg method (Benjamini, 1995) and conclusions of significant differences among dose levels at specific time intervals were made based on these adjusted significance levels. When significant differences among dose levels were concluded at a given post-dosing time interval, t-tests were performed within the ANOVA to compare each test article dose level with the vehicle level at that time interval. A Bonferroni correction (Miller, 1981) was applied to the significance levels of these comparisons in order to control the overall error rate associated with comparisons to vehicle at a given time interval to less than or equal to 5 percent.

4.0 STATISTICAL ANALYSIS RESULTS AND CONCLUSIONS

Descriptive statistics (means and SEM) of unadjusted and baseline-adjusted averages for hemodynamic, body temperature, and ECG endpoints were calculated for each day and dose level at each time and presented in tables (along with analysis results) and displayed in figures as follows:

- Tables 1 through 10 - in-box
- Tables 11 through 20 - in-cage
- Figures 1 through 4 - in-box - baseline-adjusted
- Figures 5 through 8 - in-cage - baseline-adjusted
- Figures 9 through 12 - in-box - unadjusted
- Figures 13 through 16 - in-cage - unadjusted

The results of the statistical analyses were also reported in the above tables. The significant dose effects at a particular time point were indicated by a “Yes”.

Pairwise comparisons of test article dose levels to the vehicle at the given time points were performed if the dose effects were significant at a given time point and the dose levels differed significantly from the vehicle were marked with an “*”.

5.0 REFERENCES

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Miller, R.G., Jr. (1981). *Simultaneous Statistical Inference*, Springer Verlag, NY.

Neter, J., Kutner, M., Nachtsheim, C. and Wasserman, W. (1996). *Applied Linear Statistical Models*, Fourth Edition, Chicago: Irwin.

Table 1. In Box, Systolic Blood Pressure (mmHg), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	127.7 (3.5)	130.3 (7.2)	134.8 (4.4)				
10	145.2 (4.3)	151.5 (5.6)	151.2 (6.7)	17.5 (3.3)	21.3 (3.3)	16.4 (4.2)	
20	131.0 (2.8)	139.4 (7.5)	139.4 (8.0)	3.3 (2.4)	9.1 (0.6)	4.6 (5.5)	
30	131.0 (3.2)	135.8 (6.8)	134.6 (10.1)	3.2 (2.3)	5.6 (1.1)	-0.2 (7.5)	
40	127.4 (2.4)	131.5 (6.9)	129.7 (10.5)	-0.4 (2.8)	1.2 (2.1)	-5.1 (8.5)	
50	126.4 (3.7)	132.0 (7.0)	133.1 (9.3)	-1.3 (1.8)	1.7 (3.0)	-1.7 (7.7)	
60	125.4 (3.7)	132.0 (7.6)	128.5 (7.6)	-2.3 (2.0)	1.7 (2.3)	-6.2 (5.2)	
70	130.9 (3.0)	134.3 (8.3)	129.5 (7.6)	3.2 (1.6)	4.0 (1.6)	-5.3 (4.1)	
80	127.5 (2.9)	137.6 (10.6)	129.6 (8.0)	-0.3 (1.8)	7.3 (4.5)	-5.2 (3.9)	
90	128.7 (3.9)	137.0 (10.4)	130.2 (5.9)	1.0 (1.6)	6.7 (4.5)	-4.6 (2.0)	
100	129.0 (2.6)	134.8 (10.7)	130.3 (6.1)	1.3 (1.7)	4.5 (4.6)	-4.4 (2.6)	
110	131.3 (2.4)	134.2 (9.4)	132.5 (5.7)	3.6 (2.8)	3.9 (2.2)	-2.3 (3.5)	
120	137.5 (2.4)	137.8 (7.2)	135.2 (4.5)	9.8 (2.8)	7.5 (0.8)	0.5 (2.9)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 2. In Box, Diastolic Blood Pressure (mmHg), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	87.6 (2.4)	92.0 (7.4)	93.5 (3.4)				
10	103.8 (3.3)	110.2 (5.5)	110.6 (6.1)	16.2 (2.9)	18.2 (2.0)	17.1 (4.8)	
20	90.8 (2.3)	99.9 (8.1)	99.0 (5.9)	3.2 (1.8)	7.9 (0.9)	5.5 (4.8)	
30	90.2 (2.7)	96.6 (8.4)	94.6 (8.0)	2.5 (1.9)	4.6 (1.2)	1.2 (7.0)	
40	87.9 (2.4)	93.6 (8.4)	90.5 (8.7)	0.2 (2.0)	1.5 (1.3)	-2.9 (8.1)	
50	87.1 (2.9)	93.8 (8.4)	93.4 (6.8)	-0.5 (1.4)	1.8 (2.0)	-0.1 (6.3)	
60	86.3 (3.0)	94.5 (8.0)	90.1 (5.9)	-1.4 (1.5)	2.5 (1.8)	-3.4 (4.7)	
70	90.8 (2.4)	96.0 (8.0)	90.4 (6.3)	3.1 (1.2)	4.0 (1.2)	-3.1 (4.2)	
80	88.1 (2.4)	98.3 (9.3)	90.5 (6.3)	0.4 (1.4)	6.3 (3.6)	-3.0 (3.9)	
90	88.2 (2.9)	97.6 (9.0)	90.7 (4.5)	0.5 (1.2)	5.6 (3.5)	-2.8 (1.6)	
100	88.1 (2.1)	95.8 (9.3)	90.6 (4.9)	0.4 (1.4)	3.7 (3.7)	-2.9 (2.5)	
110	89.8 (2.3)	94.4 (9.7)	92.1 (4.3)	2.1 (2.1)	2.3 (2.3)	-1.4 (2.4)	
120	95.2 (2.5)	97.5 (8.2)	94.8 (3.1)	7.6 (2.0)	5.4 (1.4)	1.3 (1.9)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 3. In Box, Mean Arterial Blood Pressure (mmHg), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	107.1 (2.7)	110.2 (7.1)	113.7 (3.3)				
10	124.3 (3.8)	130.2 (5.0)	130.7 (6.3)	17.2 (2.9)	20.1 (2.5)	17.1 (4.5)	
20	110.6 (2.4)	118.8 (7.6)	119.2 (6.5)	3.5 (2.1)	8.6 (0.8)	5.5 (5.2)	
30	110.2 (3.0)	115.4 (7.5)	114.6 (8.6)	3.1 (2.2)	5.2 (1.0)	0.9 (7.3)	
40	107.2 (2.2)	111.5 (7.5)	110.0 (9.2)	0.1 (2.2)	1.4 (1.6)	-3.6 (8.4)	
50	106.1 (3.1)	112.0 (7.6)	113.1 (7.5)	-0.9 (1.5)	1.8 (2.5)	-0.6 (6.9)	
60	105.2 (3.1)	112.2 (7.7)	109.2 (6.2)	-1.9 (1.7)	2.1 (2.0)	-4.4 (5.0)	
70	110.3 (2.6)	114.1 (8.0)	109.8 (6.4)	3.2 (1.3)	3.9 (1.4)	-3.8 (4.2)	
80	107.0 (2.5)	116.9 (9.7)	109.7 (6.6)	-0.0 (1.6)	6.7 (4.0)	-3.9 (4.0)	
90	107.7 (3.2)	116.2 (9.5)	110.1 (4.6)	0.7 (1.4)	6.1 (4.0)	-3.6 (1.8)	
100	107.7 (2.2)	114.2 (9.7)	110.1 (4.9)	0.6 (1.5)	4.0 (4.0)	-3.6 (2.6)	
110	109.7 (2.1)	113.3 (9.3)	112.0 (4.6)	2.6 (2.4)	3.1 (2.3)	-1.7 (3.0)	
120	115.5 (2.3)	116.6 (7.6)	114.7 (3.3)	8.4 (2.3)	6.4 (1.2)	1.0 (2.4)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 4. In Box, Heart Rate (BPM), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages and Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	132.1 (10.9)	135.5 (8.4)	141.3 (23.3)				
10	198.3 (5.6)	217.5 (2.5)	206.0 (9.5)	66.2 (7.6)	82.1 (5.9)	64.7 (23.1)	
20	155.8 (9.4)	172.2 (10.7)	170.9 (17.7)	23.8 (6.0)	36.7 (5.1)	29.6 (19.7)	
30	151.9 (10.5)	161.5 (9.5)	161.1 (19.3)	19.8 (7.9)	26.0 (3.6)	19.8 (26.6)	
40	146.4 (10.9)	146.8 (9.2)	150.0 (18.5)	14.4 (7.7)	11.3 (3.8)	8.7 (29.5)	
50	141.2 (10.4)	147.5 (7.1)	153.8 (18.4)	9.1 (4.6)	12.0 (4.9)	12.5 (23.1)	
60	137.5 (9.6)	150.9 (14.1)	142.0 (15.5)	5.4 (6.2)	15.4 (8.7)	0.7 (18.2)	
70	148.3 (12.0)	159.6 (14.1)	140.3 (17.4)	16.2 (5.7)	24.2 (6.1)	-1.0 (13.3)	
80	140.2 (11.2)	163.9 (21.0)	137.7 (17.3)	8.1 (4.7)	28.4 (13.4)	-3.7 (12.0)	
90	136.9 (11.5)	166.4 (20.6)	138.5 (19.6)	4.8 (5.2)	31.0 (13.7)	-2.8 (4.6)	
100	135.5 (9.3)	156.2 (20.7)	140.6 (19.3)	3.5 (5.8)	20.8 (13.9)	-0.7 (5.9)	
110	143.3 (11.8)	153.4 (15.3)	146.8 (26.7)	11.2 (8.2)	18.0 (9.5)	5.5 (5.0)	
120	165.2 (13.7)	162.3 (9.2)	153.9 (21.5)	33.2 (11.1)	26.9 (4.7)	12.6 (3.2)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 5. In Box, Body Temperature (°C), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	38.0 (0.2)	38.1 (0.3)	37.7 (0.3)				
10	37.9 (0.2)	38.1 (0.3)	37.9 (0.3)	-0.0 (0.1)	0.0 (0.1)	0.2 (0.1)	
20	37.9 (0.2)	38.1 (0.3)	37.9 (0.2)	-0.1 (0.1)	-0.0 (0.1)	0.2 (0.2)	
30	37.9 (0.2)	38.1 (0.4)	37.9 (0.2)	-0.0 (0.0)	-0.0 (0.1)	0.1 (0.2)	
40	37.9 (0.2)	38.1 (0.3)	37.8 (0.2)	-0.0 (0.0)	-0.0 (0.1)	0.1 (0.2)	
50	37.9 (0.2)	38.1 (0.3)	37.8 (0.2)	-0.0 (0.0)	0.0 (0.1)	0.0 (0.2)	
60	37.9 (0.2)	38.1 (0.3)	37.8 (0.2)	-0.1 (0.0)	0.0 (0.1)	0.1 (0.2)	
70	37.9 (0.2)	38.1 (0.3)	37.8 (0.2)	-0.1 (0.0)	-0.0 (0.1)	0.1 (0.2)	
80	37.9 (0.2)	38.1 (0.3)	37.8 (0.2)	-0.0 (0.0)	-0.0 (0.1)	0.0 (0.1)	
90	37.9 (0.2)	38.1 (0.3)	37.8 (0.3)	-0.0 (0.1)	0.0 (0.1)	0.0 (0.1)	
100	37.9 (0.2)	38.2 (0.3)	37.8 (0.3)	-0.0 (0.1)	0.0 (0.1)	0.1 (0.1)	
110	38.0 (0.2)	38.2 (0.3)	37.8 (0.3)	0.0 (0.1)	0.1 (0.1)	0.1 (0.1)	
120	38.0 (0.2)	38.1 (0.3)	37.9 (0.3)	0.0 (0.1)	0.0 (0.1)	0.2 (0.1)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a $0.05/2=0.025$ level and the significant dose levels were marked with an "*".

Table 6. In Box, RR Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	489.2 (41.3)	450.8 (29.5)	475.7 (77.4)				
10	349.5 (16.3)	313.7 (18.0)	337.9 (20.2)	-139.7 (33.4)	-137.1 (19.8)	-134.7 (109.8)	
20	398.7 (24.3)	358.1 (23.5)	377.4 (44.7)	-90.5 (27.0)	-92.7 (12.1)	-95.2 (97.0)	
30	413.6 (26.2)	379.9 (24.0)	416.5 (41.6)	-75.6 (26.2)	-70.9 (9.8)	-56.1 (91.7)	
40	432.5 (31.5)	417.0 (29.5)	446.3 (47.1)	-56.7 (26.6)	-33.8 (10.6)	-94.3 (70.7)	
50	443.1 (35.5)	414.5 (22.3)	433.6 (34.3)	-30.4 (10.8)	-36.3 (15.2)	-42.0 (54.8)	
60	474.3 (41.9)	417.9 (39.2)	497.8 (82.8)	-14.9 (10.9)	-32.9 (17.3)	22.1 (36.7)	
70	445.0 (36.0)	404.6 (36.1)	489.3 (70.5)	-44.2 (11.1)	-46.2 (11.3)	13.6 (28.4)	
80	463.0 (34.6)	409.5 (45.8)	492.3 (65.1)	-26.2 (10.1)	-41.3 (16.4)	16.7 (31.7)	
90	475.9 (44.8)	395.5 (42.0)	500.1 (86.0)	-13.3 (15.4)	-55.3 (13.9)	24.5 (19.3)	
100	468.3 (37.9)	420.3 (50.8)	481.4 (73.0)	-20.9 (15.9)	-30.4 (22.0)	5.8 (13.3)	
110	451.8 (39.6)	412.7 (43.7)	478.2 (85.3)	-37.4 (18.9)	-38.1 (19.7)	2.5 (14.9)	
120	438.8 (37.2)	406.5 (31.6)	463.6 (73.3)	-50.4 (16.9)	-44.3 (13.5)	-12.1 (11.0)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "**".

Table 7. In Box, PR Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	82.6 (5.1)	83.2 (3.6)	73.9 (6.3)				
10	76.7 (3.7)	78.9 (5.5)	67.5 (6.3)	-5.9 (2.8)	-4.3 (3.3)	-3.6 (1.6)	
20	80.6 (4.4)	83.3 (3.6)	69.2 (9.0)	-2.0 (1.0)	0.0 (2.5)	-2.0 (1.1)	
30	80.8 (4.9)	84.2 (2.9)	71.2 (10.8)	-1.8 (0.9)	1.0 (3.0)	0.1 (2.8)	
40	82.0 (5.7)	84.7 (3.4)	75.5 (9.9)	-0.6 (0.8)	1.4 (2.4)	-3.1 (7.1)	
50	77.5 (4.9)	84.1 (3.3)	73.1 (6.3)	-2.0 (0.5)	0.9 (2.1)	-0.9 (3.9)	
60	80.9 (5.1)	82.6 (3.7)	73.4 (8.1)	-1.7 (0.6)	-0.6 (1.9)	-0.6 (3.9)	
70	79.3 (4.2)	82.8 (3.1)	74.4 (8.4)	-3.3 (1.5)	-0.5 (2.7)	0.4 (3.4)	
80	81.1 (5.2)	82.9 (3.8)	74.3 (8.4)	-1.5 (0.5)	-0.3 (2.3)	0.4 (3.3)	
90	79.8 (4.6)	82.8 (3.2)	73.8 (6.8)	-2.8 (1.0)	-0.5 (2.6)	-0.1 (1.8)	
100	80.2 (4.2)	83.4 (3.4)	75.0 (7.0)	-2.4 (1.7)	0.2 (2.0)	1.0 (2.1)	
110	80.4 (4.0)	84.6 (3.7)	74.2 (7.6)	-2.3 (1.9)	1.4 (4.0)	0.2 (2.0)	
120	80.5 (4.2)	84.2 (3.3)	73.4 (7.2)	-2.1 (1.9)	1.0 (3.5)	-0.6 (1.3)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 8. In Box, QRS Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	59.4 (2.1)	63.5 (1.9)	56.3 (2.3)				
10	59.6 (2.1)	61.3 (2.4)	53.7 (3.3)	0.2 (0.9)	-2.2 (0.8)	-0.7 (1.5)	
20	60.7 (2.1)	62.8 (2.7)	55.4 (3.6)	1.3 (0.7)	-0.7 (1.0)	0.9 (1.6)	
30	60.3 (2.1)	62.9 (3.1)	54.9 (4.5)	0.9 (0.7)	-0.6 (1.4)	0.5 (2.5)	
40	60.2 (2.2)	62.3 (2.6)	60.6 (2.7)	0.9 (0.8)	-1.2 (1.2)	2.4 (0.9)	
50	59.7 (2.4)	62.5 (2.6)	57.4 (3.2)	0.1 (0.5)	-1.0 (1.1)	1.1 (0.9)	
60	59.0 (2.4)	62.7 (2.5)	55.8 (3.4)	-0.3 (0.5)	-0.8 (1.0)	-0.5 (1.1)	
70	59.5 (2.0)	62.3 (2.8)	56.7 (2.9)	0.1 (0.4)	-1.1 (1.4)	0.4 (1.1)	
80	59.0 (2.3)	62.3 (2.7)	56.3 (3.2)	-0.4 (0.4)	-1.2 (1.3)	-0.0 (1.2)	
90	58.9 (2.3)	62.3 (2.8)	56.9 (2.2)	-0.5 (0.3)	-1.1 (1.3)	0.6 (0.7)	
100	58.9 (2.0)	62.5 (2.7)	56.7 (2.1)	-0.4 (0.3)	-0.9 (1.3)	0.5 (0.6)	
110	59.0 (2.0)	62.4 (3.0)	56.1 (2.8)	-0.3 (0.4)	-1.1 (1.4)	-0.2 (0.7)	
120	59.3 (2.2)	62.3 (2.9)	56.7 (2.4)	-0.1 (0.5)	-1.2 (1.3)	0.4 (0.4)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "**".

Table 9. In Box, QT Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	251.1 (16.7)	238.9 (12.4)	249.1 (33.5)				
10	225.2 (15.3)	206.2 (5.3)	220.2 (32.7)	-25.9 (5.5)	-32.7 (8.5)	-23.6 (15.8)	
20	231.9 (13.7)	216.2 (9.9)	236.1 (35.9)	-19.2 (4.9)	-22.8 (4.1)	-7.8 (10.9)	
30	233.0 (13.9)	219.8 (11.4)	244.2 (33.9)	-18.2 (6.2)	-19.2 (1.2)	0.3 (13.0)	
40	235.5 (14.2)	227.4 (11.2)	258.6 (33.6)	-15.6 (5.2)	-11.5 (1.4)	-13.4 (12.6)	
50	240.1 (16.6)	227.3 (10.5)	251.7 (30.3)	-11.1 (4.1)	-11.7 (2.2)	2.6 (10.6)	
60	244.0 (15.1)	226.8 (10.5)	246.3 (24.8)	-7.1 (3.7)	-12.2 (4.3)	-2.8 (10.6)	
70	237.2 (15.4)	223.8 (10.7)	251.1 (28.3)	-13.9 (3.6)	-15.2 (4.1)	2.0 (7.8)	
80	242.8 (15.6)	225.2 (15.4)	253.6 (28.9)	-8.3 (3.3)	-13.7 (4.0)	4.5 (8.1)	
90	246.1 (17.5)	222.9 (15.4)	247.9 (30.6)	-5.0 (3.4)	-16.1 (3.3)	-1.2 (4.0)	
100	245.6 (17.9)	230.3 (17.1)	250.3 (31.0)	-5.6 (4.6)	-8.6 (4.9)	1.3 (4.5)	
110	242.2 (18.5)	228.7 (16.2)	246.4 (35.1)	-9.0 (5.9)	-10.3 (4.9)	-2.7 (4.1)	
120	238.8 (17.8)	227.1 (14.5)	244.2 (33.4)	-12.3 (5.6)	-11.8 (3.4)	-4.8 (2.6)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "**".

Table 10. In Box, QTcB Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (minutes)	Means (Standard Error) of Unadjusted 10-Minute Averages			Means (Standard Error) of Baseline-Adjusted 10-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	361.8 (14.1)	357.4 (14.5)	363.8 (22.9)				
10	381.7 (23.4)	369.8 (9.5)	381.2 (62.1)	19.9 (11.6)	12.4 (5.2)	22.9 (31.3)	
20	368.7 (18.6)	362.7 (15.4)	387.5 (58.1)	6.9 (5.6)	5.3 (3.7)	29.2 (27.3)	
30	364.1 (17.5)	357.7 (17.0)	381.5 (51.0)	2.3 (3.9)	0.3 (3.2)	23.2 (19.9)	
40	361.0 (19.3)	354.0 (17.6)	390.4 (49.8)	-0.8 (6.0)	-3.4 (3.9)	18.8 (21.5)	
50	362.8 (15.0)	354.7 (17.7)	382.7 (35.7)	-4.3 (1.9)	-2.7 (3.5)	18.9 (14.5)	
60	357.9 (12.6)	353.6 (15.9)	355.0 (18.0)	-3.8 (2.4)	-3.8 (2.3)	-8.8 (9.6)	
70	358.4 (14.1)	354.1 (13.3)	364.0 (23.5)	-3.4 (1.3)	-3.3 (2.7)	0.1 (1.7)	
80	359.1 (14.3)	354.5 (16.4)	365.3 (25.6)	-2.7 (1.2)	-2.9 (2.4)	1.5 (3.7)	
90	358.9 (12.5)	356.2 (14.8)	353.8 (18.0)	-2.9 (2.9)	-1.2 (2.4)	-10.0 (6.0)	
100	360.1 (14.8)	357.6 (15.6)	363.9 (22.1)	-1.7 (1.7)	0.3 (2.5)	0.0 (1.1)	
110	361.8 (15.1)	358.1 (16.8)	359.3 (23.7)	-0.0 (2.2)	0.7 (3.5)	-4.5 (3.4)	
120	362.7 (15.6)	357.8 (18.3)	361.9 (26.0)	0.9 (2.5)	0.4 (4.3)	-1.9 (4.1)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 11. In Cage, Systolic Blood Pressure (mmHg), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	116.7 (4.4)	114.6 (6.7)	119.6 (9.8)				
3.5	125.3 (2.7)	129.5 (7.8)	125.2 (3.8)	8.6 (4.1)	14.9 (1.7)	5.6 (7.0)	
4	127.9 (4.1)	120.1 (8.4)	137.4 (3.6)	11.1 (5.1)	5.5 (2.3)	17.8 (7.8)	
4.5	123.7 (2.3)	118.4 (5.1)	133.2 (8.2)	7.0 (3.7)	3.7 (5.1)	13.6 (14.5)	
5	118.3 (3.1)	116.7 (6.9)	125.1 (3.9)	1.6 (4.8)	2.1 (5.0)	5.5 (8.7)	
5.5	114.0 (3.3)	109.6 (6.0)	123.2 (4.2)	-2.7 (4.1)	-5.0 (2.7)	3.6 (6.8)	
6	116.0 (3.8)	109.6 (5.3)	120.3 (5.1)	-0.7 (4.1)	-5.0 (1.8)	0.7 (6.1)	
6.5	115.4 (4.2)	107.3 (5.1)	117.8 (5.5)	-1.4 (4.1)	-7.4 (1.9)	-1.8 (5.4)	
7	109.6 (3.0)	106.9 (4.9)	114.2 (4.5)	-7.1 (2.7)	-7.7 (2.2)	-5.4 (5.5)	
7.5	107.4 (3.5)	102.6 (6.0)	110.8 (7.0)	-9.3 (2.4)	-12.0 (1.6)	-8.8 (3.3)	
8	106.5 (3.2)	102.3 (6.2)	110.2 (7.4)	-10.2 (2.5)	-12.3 (2.1)	-9.4 (2.8)	
8.5	107.2 (2.9)	104.0 (5.8)	111.7 (6.5)	-9.5 (3.3)	-10.7 (2.3)	-7.9 (3.8)	
9	106.5 (3.0)	105.5 (5.0)	111.2 (7.3)	-10.3 (2.5)	-9.1 (3.0)	-8.4 (3.9)	
9.5	107.6 (3.1)	105.6 (6.0)	110.5 (7.9)	-9.1 (2.8)	-9.0 (1.7)	-9.1 (3.7)	
10	108.4 (2.6)	105.9 (6.2)	111.8 (7.7)	-8.3 (2.9)	-8.7 (1.9)	-7.8 (4.3)	
10.5	107.9 (2.7)	106.7 (7.5)	111.7 (6.9)	-8.9 (2.9)	-8.0 (1.4)	-7.9 (3.9)	
11	107.8 (2.9)	110.1 (6.9)	109.0 (7.4)	-9.0 (2.6)	-4.5 (1.8)	-10.6 (3.1)	
11.5	108.5 (3.2)	109.5 (8.8)	109.1 (8.3)	-8.2 (2.7)	-5.1 (2.3)	-10.5 (3.2)	
12	108.5 (2.9)	109.8 (7.5)	111.7 (9.4)	-8.3 (2.3)	-4.8 (1.5)	-7.9 (3.4)	
12.5	109.2 (3.4)	109.1 (8.2)	111.4 (9.0)	-7.5 (2.3)	-5.5 (2.2)	-8.2 (3.9)	
13	108.7 (3.8)	109.9 (7.8)	109.2 (6.8)	-8.0 (2.3)	-4.7 (1.2)	-10.4 (3.6)	
13.5	108.6 (3.5)	110.5 (7.5)	109.1 (6.5)	-8.1 (2.4)	-4.2 (1.7)	-10.5 (3.7)	
14	109.6 (3.6)	110.8 (8.0)	109.7 (7.4)	-7.2 (1.8)	-3.9 (1.4)	-9.9 (3.0)	
14.5	109.6 (3.3)	112.4 (6.4)	111.8 (7.1)	-7.1 (2.3)	-2.3 (2.5)	-7.8 (3.4)	
15	109.4 (3.1)	112.4 (6.6)	110.4 (6.6)	-7.3 (2.3)	-2.2 (1.9)	-9.2 (3.9)	
15.5	108.8 (2.8)	112.5 (6.9)	111.1 (7.6)	-7.9 (2.5)	-2.2 (2.4)	-8.5 (3.5)	
16	110.2 (3.4)	113.6 (6.7)	111.9 (7.0)	-6.6 (1.9)	-1.1 (1.5)	-7.7 (3.3)	
16.5	110.9 (3.2)	115.0 (7.4)	110.5 (6.5)	-5.8 (2.3)	0.4 (2.6)	-9.1 (3.7)	
17	111.4 (3.0)	116.7 (7.0)	115.6 (11.3)	-5.3 (2.4)	2.1 (1.1)	-4.0 (4.8)	
17.5	113.1 (3.3)	116.3 (7.6)	119.5 (13.7)	-3.6 (2.2)	1.7 (1.6)	-0.1 (7.0)	
18	113.7 (4.0)	115.2 (7.2)	119.1 (11.8)	-3.0 (1.8)	0.5 (1.8)	-0.5 (5.4)	
18.5	118.9 (4.7)	114.2 (8.4)	120.3 (10.0)	2.2 (1.7)	-0.4 (1.7)	0.7 (0.6)	
19	120.0 (4.7)	122.5 (7.3)	122.7 (8.0)	3.2 (1.4)	7.9 (1.9)	3.1 (2.5)	
19.5	114.8 (4.5)	122.0 (9.0)	118.0 (8.4)	-2.0 (1.6)	7.4 (2.5)	-1.6 (4.3)	
20	119.2 (4.8)	118.0 (8.9)	121.2 (9.4)	2.5 (1.3)	3.4 (2.5)	1.6 (1.8)	
20.5	120.4 (4.7)	121.4 (8.2)	118.7 (9.2)	3.7 (2.0)	6.8 (1.6)	-0.9 (5.3)	
21	119.4 (4.9)	122.5 (9.4)	120.0 (9.3)	2.7 (2.1)	7.9 (3.0)	0.4 (5.2)	
21.5	123.0 (4.7)	123.1 (9.7)	123.2 (9.8)	6.2 (2.4)	8.5 (3.1)	3.6 (4.6)	
22	126.9 (3.8)	123.6 (8.2)	129.4 (6.7)	10.2 (2.2)	9.0 (1.7)	9.8 (4.9)	
22.5	125.7 (4.0)	128.8 (10.1)	127.0 (5.7)	8.9 (3.9)	14.2 (3.5)	7.4 (5.0)	
23	118.0 (3.5)	120.9 (9.1)	121.3 (6.3)	1.3 (3.8)	6.2 (4.5)	1.7 (6.0)	
23.5	113.2 (3.6)	117.4 (8.6)	116.8 (6.7)	-3.5 (2.2)	2.8 (4.0)	-2.8 (3.7)	
24	112.7 (4.0)	117.5 (9.4)	115.0 (6.2)	-4.0 (1.9)	2.9 (4.2)	-4.6 (4.1)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "a".

Table 12. In Cage, Diastolic Blood Pressure (mmHg), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	79.8 (3.3)	80.2 (6.6)	81.3 (7.2)				
3.5	88.8 (2.2)	93.8 (7.9)	89.4 (3.5)	9.0 (3.4)	13.6 (2.3)	8.1 (5.1)	
4	90.1 (2.9)	88.0 (8.1)	99.5 (1.3)	10.3 (4.0)	7.8 (2.8)	18.1 (6.5)	
4.5	87.4 (1.9)	85.8 (5.0)	96.0 (7.5)	7.6 (3.0)	5.6 (4.0)	14.7 (11.9)	
5	83.6 (2.2)	84.4 (6.3)	89.6 (2.9)	3.8 (3.8)	4.2 (4.0)	8.3 (6.1)	
5.5	79.9 (2.3)	79.5 (5.9)	87.7 (3.9)	0.1 (3.2)	-0.7 (2.5)	6.4 (4.1)	
6	80.5 (2.6)	79.2 (5.2)	83.9 (4.8)	0.7 (3.1)	-1.0 (1.7)	2.6 (3.7)	
6.5	80.1 (3.0)	76.2 (5.0)	81.4 (5.3)	0.3 (3.3)	-4.0 (1.8)	0.1 (3.3)	
7	74.6 (2.2)	76.1 (4.4)	78.3 (4.1)	-5.2 (2.1)	-4.1 (2.3)	-3.0 (3.3)	
7.5	73.3 (2.8)	72.3 (5.7)	75.4 (6.6)	-6.4 (2.3)	-7.9 (1.9)	-5.9 (1.9)	
8	72.6 (2.5)	71.5 (5.7)	75.0 (6.8)	-7.2 (2.4)	-8.7 (2.3)	-6.3 (1.7)	
8.5	73.2 (2.4)	72.4 (5.3)	76.5 (5.7)	-6.6 (2.8)	-7.8 (2.5)	-4.8 (2.6)	
9	72.1 (2.6)	72.7 (4.9)	75.5 (6.3)	-7.7 (2.0)	-7.5 (2.6)	-5.8 (2.8)	
9.5	73.5 (2.5)	72.9 (5.7)	74.9 (7.2)	-6.3 (2.4)	-7.3 (1.8)	-6.5 (2.7)	
10	73.2 (2.2)	73.5 (6.0)	75.8 (7.0)	-6.6 (2.0)	-6.7 (1.4)	-5.6 (3.1)	
10.5	73.2 (2.1)	74.3 (6.8)	76.1 (5.8)	-6.6 (2.1)	-5.9 (1.1)	-5.2 (2.1)	
11	73.3 (2.5)	76.6 (6.5)	73.0 (6.8)	-6.5 (2.0)	-3.6 (1.2)	-8.3 (2.0)	
11.5	74.0 (2.6)	76.6 (8.0)	73.3 (7.9)	-5.8 (2.2)	-3.6 (1.7)	-8.1 (2.1)	
12	73.8 (2.2)	76.9 (6.9)	75.6 (8.2)	-6.0 (2.1)	-3.3 (1.1)	-5.7 (2.5)	
12.5	74.4 (2.6)	77.2 (7.2)	75.1 (7.7)	-5.4 (2.0)	-3.0 (1.6)	-6.3 (2.7)	
13	74.4 (2.8)	77.8 (7.3)	73.1 (5.7)	-5.4 (2.0)	-2.4 (0.9)	-8.2 (2.0)	
13.5	73.9 (2.4)	78.0 (7.1)	74.0 (6.1)	-5.9 (2.1)	-2.2 (1.4)	-7.3 (2.3)	
14	75.0 (2.4)	78.8 (7.5)	74.7 (6.6)	-4.8 (1.8)	-1.4 (1.0)	-6.6 (1.4)	
14.5	74.7 (2.4)	79.4 (6.4)	76.9 (5.9)	-5.1 (2.0)	-0.8 (1.3)	-4.5 (1.6)	
15	74.8 (2.4)	80.0 (6.4)	75.5 (5.7)	-5.0 (2.0)	-0.1 (1.1)	-5.8 (2.0)	
15.5	74.1 (2.5)	80.0 (6.9)	75.8 (6.5)	-5.7 (2.1)	-0.2 (1.7)	-5.5 (1.7)	
16	76.3 (2.9)	81.2 (7.1)	76.3 (5.7)	-3.5 (1.5)	1.0 (0.9)	-5.1 (1.6)	
16.5	76.1 (2.4)	82.9 (7.6)	75.3 (5.6)	-3.7 (1.7)	2.7 (1.8)	-6.0 (2.0)	
17	77.3 (2.4)	84.2 (7.5)	79.6 (10.0)	-2.5 (1.7)	4.0 (1.0)	-1.7 (3.8)	
17.5	77.9 (2.9)	84.8 (8.2)	83.7 (11.3)	-1.9 (1.7)	4.6 (1.8)	2.3 (5.1)	
18	79.0 (3.3)	83.3 (7.6)	82.6 (9.4)	-0.8 (1.4)	3.1 (1.4)	1.3 (3.7)	
18.5	82.3 (3.2)	82.3 (8.4)	83.3 (7.4)	2.5 (1.1)	2.1 (2.1)	1.9 (0.7)	
19	81.8 (4.0)	87.7 (8.1)	84.4 (6.5)	2.0 (1.5)	7.5 (2.2)	3.0 (0.9)	
19.5	78.5 (4.0)	87.6 (8.9)	80.3 (8.2)	-1.3 (1.7)	7.4 (2.4)	-1.0 (3.2)	
20	83.4 (3.7)	85.4 (8.7)	82.8 (8.3)	3.6 (1.0)	5.2 (2.5)	1.5 (1.6)	
20.5	82.7 (3.6)	88.1 (7.8)	79.5 (8.9)	2.9 (1.4)	7.9 (1.3)	-1.8 (4.3)	
21	81.6 (3.8)	88.5 (8.6)	80.9 (8.7)	1.8 (1.8)	8.3 (2.2)	-0.5 (4.2)	
21.5	84.5 (3.3)	88.7 (9.2)	83.2 (8.7)	4.7 (1.7)	8.5 (2.7)	1.8 (3.7)	
22	88.1 (2.6)	88.7 (8.0)	89.8 (6.1)	8.3 (1.8)	8.5 (1.6)	8.5 (3.3)	
22.5	88.3 (2.6)	92.8 (9.2)	89.7 (4.8)	8.5 (3.4)	12.6 (2.6)	8.3 (3.3)	
23	81.5 (2.4)	87.0 (8.4)	84.6 (5.7)	1.7 (3.4)	6.8 (3.4)	3.2 (4.4)	
23.5	78.3 (2.3)	83.4 (7.8)	81.2 (6.5)	-1.5 (2.0)	3.3 (2.9)	-0.1 (1.6)	
24	77.4 (2.8)	83.9 (8.2)	79.4 (6.1)	-2.3 (2.0)	3.7 (3.3)	-2.0 (2.3)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "a".

Table 13. In Cage, Mean Arterial Blood Pressure (mmHg), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	96.6 (3.7)	95.7 (6.4)	98.9 (8.2)				
3.5	106.4 (2.5)	110.9 (7.9)	107.0 (3.4)	9.9 (3.7)	15.2 (2.1)	8.1 (6.3)	
4	108.1 (3.5)	103.1 (8.3)	118.0 (1.8)	11.5 (4.6)	7.4 (2.8)	19.1 (7.4)	
4.5	104.3 (2.1)	100.8 (4.9)	113.7 (8.2)	7.8 (3.4)	5.1 (4.4)	14.8 (13.5)	
5	99.5 (2.6)	99.1 (6.6)	106.3 (3.1)	2.9 (4.4)	3.4 (4.6)	7.4 (7.6)	
5.5	95.1 (2.7)	92.8 (5.9)	104.3 (3.7)	-1.4 (3.6)	-2.9 (2.5)	5.4 (5.6)	
6	96.6 (3.1)	92.5 (5.2)	100.7 (4.6)	-0.0 (3.6)	-3.2 (1.6)	1.7 (4.8)	
6.5	96.0 (3.5)	89.7 (5.0)	98.0 (5.0)	-0.6 (3.6)	-6.0 (1.7)	-1.0 (4.2)	
7	90.2 (2.5)	89.4 (4.6)	94.4 (3.9)	-6.3 (2.3)	-6.3 (2.1)	-4.5 (4.3)	
7.5	88.6 (3.0)	85.7 (5.7)	91.4 (6.6)	-7.9 (2.3)	-10.0 (1.7)	-7.6 (2.1)	
8	87.9 (2.8)	85.4 (5.5)	90.9 (6.9)	-8.6 (2.4)	-10.3 (2.1)	-8.0 (1.7)	
8.5	88.6 (2.6)	86.7 (5.1)	92.6 (5.7)	-7.9 (3.0)	-9.0 (2.4)	-6.3 (2.9)	
9	87.7 (2.7)	87.3 (4.7)	91.9 (6.6)	-8.9 (2.1)	-8.4 (2.9)	-7.1 (2.9)	
9.5	89.0 (2.7)	87.6 (5.5)	90.9 (7.5)	-7.5 (2.6)	-8.1 (1.7)	-8.0 (2.8)	
10	89.2 (2.3)	88.2 (5.7)	91.9 (7.1)	-7.4 (2.4)	-7.5 (1.6)	-7.0 (3.5)	
10.5	88.8 (2.3)	89.0 (6.8)	92.3 (6.1)	-7.7 (2.5)	-6.7 (1.0)	-6.7 (2.6)	
11	88.8 (2.6)	91.9 (6.3)	89.2 (6.8)	-7.8 (2.3)	-3.8 (1.3)	-9.7 (2.0)	
11.5	89.6 (2.8)	91.7 (8.0)	89.4 (8.0)	-6.9 (2.4)	-4.0 (1.8)	-9.5 (2.4)	
12	89.5 (2.3)	92.1 (6.8)	92.1 (8.7)	-7.1 (2.2)	-3.6 (1.1)	-6.8 (2.6)	
12.5	90.4 (2.9)	91.9 (7.2)	91.7 (8.3)	-6.2 (2.3)	-3.8 (1.7)	-7.2 (3.1)	
13	90.3 (3.1)	92.6 (7.1)	89.4 (6.0)	-6.3 (2.0)	-3.1 (0.9)	-9.5 (2.7)	
13.5	89.8 (2.7)	93.2 (6.8)	90.2 (6.2)	-6.7 (2.3)	-2.6 (1.5)	-8.7 (2.6)	
14	91.0 (2.8)	93.7 (7.4)	90.8 (6.9)	-5.6 (1.9)	-2.0 (1.0)	-8.2 (1.9)	
14.5	90.8 (2.7)	94.4 (6.1)	93.2 (6.3)	-5.8 (2.1)	-1.3 (2.0)	-5.7 (2.6)	
15	90.8 (2.6)	94.8 (6.3)	92.0 (6.0)	-5.8 (2.2)	-0.9 (1.5)	-6.9 (3.0)	
15.5	90.1 (2.5)	94.9 (6.7)	92.6 (6.9)	-6.4 (2.3)	-0.8 (2.1)	-6.3 (2.5)	
16	91.8 (2.9)	96.0 (6.7)	93.2 (6.1)	-4.8 (1.6)	0.3 (1.2)	-5.7 (2.4)	
16.5	92.1 (2.7)	97.8 (7.2)	91.6 (6.0)	-4.4 (2.0)	2.1 (2.2)	-7.3 (2.6)	
17	92.6 (2.5)	99.2 (6.9)	96.3 (10.6)	-4.0 (2.0)	3.5 (0.8)	-2.6 (4.1)	
17.5	93.8 (2.9)	98.8 (7.5)	99.6 (12.5)	-2.8 (2.0)	3.1 (1.5)	0.7 (5.9)	
18	94.8 (3.5)	97.7 (7.2)	99.1 (10.6)	-1.8 (1.7)	2.0 (1.6)	0.2 (4.3)	
18.5	99.3 (3.8)	96.5 (8.1)	100.3 (8.6)	2.7 (1.5)	0.8 (1.9)	1.3 (0.5)	
19	99.6 (4.2)	103.6 (7.2)	102.7 (6.8)	3.0 (1.4)	7.9 (1.8)	3.7 (1.7)	
19.5	94.8 (4.2)	103.5 (8.8)	97.8 (8.2)	-1.8 (1.8)	7.8 (2.5)	-1.1 (3.6)	
20	99.7 (4.2)	100.0 (8.7)	101.1 (8.5)	3.1 (1.2)	4.3 (2.7)	2.2 (1.1)	
20.5	100.2 (4.0)	103.3 (7.9)	97.9 (8.9)	3.6 (1.7)	7.6 (1.6)	-1.0 (4.5)	
21	99.0 (4.2)	103.9 (8.8)	99.1 (8.9)	2.4 (1.9)	8.2 (2.8)	0.1 (4.6)	
21.5	102.4 (3.8)	104.3 (9.3)	101.9 (9.0)	5.9 (2.0)	8.6 (3.0)	2.9 (3.9)	
22	106.4 (3.1)	104.4 (7.9)	108.8 (6.2)	9.9 (2.0)	8.7 (1.6)	9.8 (4.2)	
22.5	105.8 (3.2)	109.5 (9.4)	107.5 (4.7)	9.2 (3.8)	13.8 (3.1)	8.6 (4.3)	
23	98.0 (2.9)	102.6 (8.5)	101.6 (5.6)	1.4 (3.9)	6.9 (3.9)	2.6 (5.3)	
23.5	94.0 (2.7)	98.6 (8.0)	97.4 (6.3)	-2.5 (2.1)	2.9 (3.5)	-1.5 (2.5)	
24	93.2 (3.1)	98.8 (8.6)	95.4 (5.7)	-3.3 (1.9)	3.1 (3.8)	-3.5 (3.2)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "a".

Table 14. In Cage, Heart Rate (BPM), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages and Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	105.9 (4.8)	100.4 (9.4)	101.9 (2.2)				
3.5	132.1 (4.9)	157.2 (6.1)	140.1 (6.2)	26.2 (6.1)	56.8 (13.7)	38.2 (5.6)	
4	134.1 (4.1)	135.8 (3.1)	133.8 (7.2)	28.2 (4.9)	35.4 (8.2)	31.9 (7.0)	
4.5	128.0 (4.2)	129.7 (3.7)	129.1 (8.9)	22.1 (3.9)	29.3 (5.9)	27.2 (8.4)	
5	124.2 (4.2)	129.1 (3.8)	128.2 (5.2)	18.4 (3.2)	28.6 (9.0)	26.4 (3.9)	
5.5	120.0 (4.7)	118.9 (5.6)	123.1 (4.6)	14.2 (4.2)	18.4 (8.2)	21.2 (4.4)	
6	119.4 (5.4)	114.2 (7.3)	116.1 (7.3)	13.5 (4.0)	13.8 (3.2)	14.2 (6.5)	
6.5	118.9 (5.9)	107.6 (7.1)	114.6 (8.3)	13.0 (4.8)	7.2 (3.2)	12.7 (8.5)	
7	113.4 (6.3)	109.4 (8.1)	111.5 (7.8)	7.5 (4.5)	9.0 (2.3)	9.7 (7.5)	
7.5	111.5 (7.3)	104.3 (8.9)	107.5 (9.4)	5.6 (5.4)	3.9 (0.5)	5.6 (9.5)	
8	110.0 (7.6)	99.6 (11.0)	104.5 (9.2)	4.2 (6.5)	-0.9 (1.7)	2.6 (9.2)	
8.5	109.4 (7.4)	100.0 (12.3)	100.8 (8.5)	3.6 (5.3)	-0.5 (3.0)	-1.1 (8.1)	
9	106.8 (7.3)	99.4 (13.3)	99.3 (10.3)	0.9 (6.0)	-1.0 (4.0)	-2.6 (9.6)	
9.5	109.2 (8.2)	96.4 (11.8)	94.7 (8.1)	3.3 (6.0)	-4.0 (2.7)	-7.2 (7.8)	
10	105.5 (8.3)	98.2 (11.1)	96.8 (10.6)	-0.3 (6.4)	-2.3 (3.2)	-5.1 (10.0)	
10.5	104.5 (7.9)	98.8 (11.8)	95.3 (9.7)	-1.4 (6.1)	-1.6 (3.9)	-6.6 (8.7)	
11	104.9 (9.1)	95.7 (11.2)	90.3 (9.4)	-1.0 (7.9)	-4.7 (4.9)	-11.6 (8.4)	
11.5	105.0 (9.4)	96.2 (11.8)	87.6 (9.3)	-0.9 (8.2)	-4.2 (4.2)	-14.3 (9.0)	
12	102.6 (9.8)	96.3 (11.4)	88.1 (10.9)	-3.2 (8.6)	-4.2 (3.9)	-13.8 (10.2)	
12.5	103.0 (9.7)	98.1 (12.1)	88.7 (12.2)	-2.8 (9.1)	-2.4 (4.1)	-13.1 (11.9)	
13	104.1 (9.5)	95.1 (12.0)	82.4 (10.6)	-1.7 (9.0)	-5.3 (4.2)	-19.4 (9.6)	
13.5	102.2 (10.2)	94.0 (12.0)	85.7 (12.0)	-3.6 (9.2)	-6.4 (4.7)	-16.2 (11.5)	
14	104.2 (11.3)	94.0 (13.3)	84.0 (10.8)	-1.7 (10.2)	-6.4 (6.3)	-17.9 (9.9)	
14.5	105.5 (11.2)	93.8 (13.9)	85.4 (9.9)	-0.4 (10.4)	-6.6 (6.9)	-16.4 (8.8)	
15	104.2 (10.7)	95.7 (14.8)	83.7 (11.5)	-1.6 (9.7)	-4.7 (7.0)	-18.2 (10.0)	
15.5	100.8 (10.5)	91.8 (13.5)	81.9 (8.7)	-5.1 (9.3)	-8.6 (5.1)	-20.0 (7.4)	
16	102.6 (9.1)	92.6 (11.3)	82.2 (9.4)	-3.3 (7.3)	-7.8 (3.6)	-19.7 (7.9)	
16.5	99.6 (10.1)	94.5 (9.9)	79.7 (6.7)	-6.2 (8.8)	-5.9 (3.0)	-22.2 (5.0)	
17	102.3 (8.8)	96.3 (10.2)	80.9 (7.0)	-3.6 (7.4)	-4.1 (2.6)	-21.0 (6.0)	
17.5	99.6 (8.5)	98.7 (8.1)	86.8 (4.8)	-6.2 (7.2)	-1.8 (2.9)	-15.1 (4.7)	
18	97.9 (7.6)	96.5 (9.4)	83.9 (4.0)	-7.9 (6.8)	-3.9 (2.0)	-18.0 (3.1)	
18.5	108.9 (7.9)	93.7 (8.4)	99.7 (5.0)	3.1 (5.4)	-6.7 (4.1)	-2.2 (5.8)	
19	112.0 (8.8)	111.2 (11.5)	111.6 (10.8)	6.1 (5.1)	10.8 (5.4)	9.7 (11.6)	
19.5	100.3 (5.6)	111.1 (13.1)	97.5 (6.0)	-5.6 (6.1)	10.7 (7.3)	-4.4 (7.1)	
20	113.6 (5.4)	100.8 (9.9)	113.3 (2.8)	7.7 (4.4)	0.4 (3.7)	11.4 (4.8)	
20.5	116.2 (8.1)	119.8 (10.9)	104.2 (9.4)	10.4 (8.4)	19.4 (5.6)	2.4 (10.2)	
21	113.7 (7.1)	117.1 (14.4)	107.0 (10.6)	7.8 (8.0)	16.6 (10.5)	5.1 (11.0)	
21.5	128.0 (9.0)	115.3 (11.0)	107.4 (12.5)	22.1 (9.1)	14.9 (6.9)	5.5 (13.1)	
22	128.7 (7.2)	109.3 (10.3)	130.6 (12.2)	22.9 (7.7)	8.9 (6.1)	28.7 (11.5)	
22.5	118.6 (6.9)	120.0 (6.5)	128.3 (12.5)	12.8 (8.5)	19.5 (4.8)	26.4 (11.8)	
23	109.9 (5.4)	114.1 (5.9)	117.2 (9.0)	4.1 (6.9)	13.6 (3.9)	15.3 (7.5)	
23.5	107.6 (5.1)	116.0 (6.2)	114.0 (3.2)	1.8 (4.5)	15.5 (7.9)	12.1 (3.1)	
24	108.7 (5.0)	119.2 (4.2)	109.2 (4.7)	2.8 (4.8)	18.8 (9.1)	7.4 (3.2)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "a".

Table 15. In Cage, Body Temperature (°C), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	37.2 (0.1)	37.1 (0.3)	36.9 (0.1)				
3.5	37.6 (0.2)	38.2 (0.2)	38.0 (0.3)	0.4 (0.2)	1.0 (0.2)*	1.1 (0.3)*	Yes
4	37.7 (0.2)	38.0 (0.1)	37.4 (0.2)	0.5 (0.1)	0.8 (0.3)	0.5 (0.2)	
4.5	37.9 (0.1)	38.0 (0.2)	37.5 (0.3)	0.7 (0.1)	0.8 (0.1)	0.6 (0.3)	
5	37.8 (0.1)	37.9 (0.3)	37.6 (0.2)	0.6 (0.1)	0.8 (0.1)	0.7 (0.3)	
5.5	37.8 (0.1)	37.9 (0.2)	37.5 (0.3)	0.6 (0.1)	0.8 (0.1)	0.6 (0.3)	
6	37.8 (0.1)	38.1 (0.2)	37.6 (0.2)	0.6 (0.1)	0.9 (0.1)	0.7 (0.3)	
6.5	37.9 (0.1)	38.0 (0.2)	37.8 (0.1)	0.7 (0.1)	0.9 (0.2)	0.9 (0.2)	
7	37.9 (0.1)	38.1 (0.2)	37.7 (0.1)	0.7 (0.1)	0.9 (0.2)	0.8 (0.2)	
7.5	37.8 (0.1)	37.9 (0.2)	37.7 (0.1)	0.6 (0.1)	0.8 (0.2)	0.8 (0.2)	
8	37.6 (0.1)	37.6 (0.2)	37.6 (0.1)	0.4 (0.1)	0.4 (0.2)	0.7 (0.2)*	Yes
8.5	37.4 (0.1)	37.3 (0.2)	37.4 (0.1)	0.2 (0.1)	0.2 (0.2)	0.5 (0.1)*	Yes
9	37.3 (0.1)	37.1 (0.2)	37.2 (0.2)	0.1 (0.1)	-0.1 (0.2)	0.3 (0.2)	Yes
9.5	37.2 (0.1)	36.9 (0.3)	37.2 (0.1)	0.0 (0.1)	-0.2 (0.2)	0.3 (0.2)*	Yes
10	37.2 (0.1)	36.9 (0.2)	37.0 (0.1)	-0.0 (0.1)	-0.2 (0.2)	0.1 (0.2)	
10.5	37.1 (0.1)	36.9 (0.2)	37.0 (0.1)	-0.1 (0.1)	-0.3 (0.1)	0.1 (0.2)	Yes
11	37.1 (0.1)	36.9 (0.2)	37.1 (0.1)	-0.1 (0.1)	-0.3 (0.2)	0.2 (0.1)*	Yes
11.5	37.1 (0.1)	36.9 (0.2)	37.1 (0.1)	-0.1 (0.1)	-0.2 (0.2)	0.2 (0.1)	Yes
12	37.1 (0.1)	36.8 (0.2)	37.0 (0.0)	-0.1 (0.1)	-0.3 (0.1)	0.1 (0.1)	Yes
12.5	37.1 (0.1)	36.8 (0.2)	36.9 (0.1)	-0.1 (0.1)	-0.4 (0.2)*	0.0 (0.1)	Yes
13	37.0 (0.1)	36.8 (0.2)	36.9 (0.0)	-0.2 (0.1)	-0.3 (0.2)	-0.0 (0.1)	
13.5	37.0 (0.1)	36.8 (0.2)	36.9 (0.0)	-0.2 (0.1)	-0.4 (0.2)	0.0 (0.1)	Yes
14	37.1 (0.1)	36.8 (0.2)	36.9 (0.1)	-0.1 (0.1)	-0.4 (0.1)	-0.0 (0.1)	Yes
14.5	37.0 (0.1)	36.6 (0.3)	36.8 (0.1)	-0.2 (0.1)	-0.6 (0.1)*	-0.1 (0.1)	Yes
15	37.0 (0.1)	36.4 (0.3)	36.8 (0.1)	-0.2 (0.1)	-0.7 (0.1)*	-0.1 (0.1)	Yes
15.5	37.0 (0.1)	36.5 (0.3)	36.8 (0.1)	-0.2 (0.1)	-0.7 (0.1)*	-0.2 (0.1)	Yes
16	37.0 (0.1)	36.5 (0.3)	36.8 (0.1)	-0.2 (0.1)	-0.7 (0.1)*	-0.1 (0.1)	Yes
16.5	36.9 (0.1)	36.4 (0.3)	36.7 (0.1)	-0.3 (0.1)	-0.7 (0.1)*	-0.2 (0.0)	Yes
17	36.9 (0.1)	36.5 (0.3)	36.6 (0.2)	-0.3 (0.1)	-0.6 (0.0)*	-0.3 (0.1)	Yes
17.5	36.9 (0.1)	36.6 (0.3)	36.5 (0.2)	-0.3 (0.1)	-0.6 (0.0)	-0.4 (0.1)	
18	36.9 (0.1)	36.7 (0.3)	36.5 (0.1)	-0.3 (0.1)	-0.5 (0.1)	-0.4 (0.0)	
18.5	37.1 (0.1)	36.7 (0.4)	36.7 (0.1)	-0.1 (0.1)	-0.4 (0.1)	-0.2 (0.1)	
19	37.4 (0.1)	36.9 (0.4)	36.9 (0.2)	0.2 (0.1)	-0.2 (0.1)*	0.0 (0.1)	Yes
19.5	37.7 (0.1)	37.3 (0.4)	37.4 (0.2)	0.5 (0.1)	0.1 (0.1)*	0.5 (0.1)	Yes
20	37.9 (0.1)	37.6 (0.4)	37.6 (0.1)	0.7 (0.1)	0.5 (0.2)	0.7 (0.1)	
20.5	38.1 (0.1)	37.9 (0.3)	38.0 (0.1)	0.9 (0.1)	0.7 (0.2)	1.1 (0.1)	Yes
21	38.4 (0.1)	38.1 (0.3)	38.2 (0.1)	1.2 (0.1)	1.0 (0.2)	1.3 (0.1)	Yes
21.5	38.5 (0.1)	38.3 (0.3)	38.4 (0.1)	1.3 (0.1)	1.1 (0.1)	1.5 (0.1)	
22	38.6 (0.1)	38.4 (0.3)	38.4 (0.1)	1.4 (0.1)	1.2 (0.1)	1.5 (0.1)	
22.5	38.0 (0.2)	38.3 (0.1)	37.6 (0.2)	0.8 (0.2)	1.2 (0.2)*	0.7 (0.1)	Yes
23	37.7 (0.2)	38.1 (0.1)	37.4 (0.3)	0.5 (0.2)	0.9 (0.2)	0.5 (0.3)	
23.5	37.8 (0.2)	37.9 (0.2)	37.8 (0.1)	0.6 (0.1)	0.7 (0.1)	0.8 (0.2)	
24	37.8 (0.1)	38.0 (0.2)	37.9 (0.1)	0.6 (0.1)	0.9 (0.2)	1.0 (0.1)*	Yes

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 16. In Cage, RR Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	659.4 (42.5)	665.9 (83.0)	667.0 (34.2)				
3.5	472.3 (17.3)	404.5 (19.6)	440.2 (20.5)	-187.1 (32.5)	-261.5 (92.1)	-226.9 (17.8)	
4	465.3 (14.8)	449.0 (12.5)	466.2 (24.2)	-194.1 (32.0)	-216.9 (79.6)	-200.8 (14.0)	
4.5	483.6 (16.3)	478.1 (15.8)	482.1 (33.4)	-175.8 (27.6)	-187.8 (67.9)	-185.0 (17.1)	
5	492.1 (16.8)	477.9 (15.5)	490.4 (24.1)	-167.3 (27.9)	-188.0 (79.3)	-176.6 (20.7)	
5.5	512.8 (20.1)	514.2 (24.9)	508.7 (22.2)	-146.6 (30.3)	-151.7 (79.0)	-158.3 (22.0)	
6	519.4 (24.6)	549.4 (38.9)	530.9 (36.7)	-140.0 (25.1)	-116.5 (46.7)	-136.1 (30.6)	
6.5	522.5 (27.7)	570.5 (40.5)	540.5 (44.7)	-136.9 (27.0)	-95.4 (48.8)	-126.5 (38.8)	
7	560.4 (36.9)	570.2 (47.2)	554.8 (43.5)	-94.3 (30.2)	-95.7 (38.3)	-112.2 (37.1)	
7.5	562.0 (42.6)	598.8 (52.6)	565.7 (58.4)	-97.4 (38.1)	-67.1 (32.3)	-101.4 (52.8)	
8	573.5 (40.3)	638.0 (69.1)	593.4 (58.2)	-85.9 (34.1)	-28.0 (17.3)	-73.7 (51.4)	
8.5	582.6 (41.5)	644.0 (75.7)	622.8 (55.6)	-76.7 (32.1)	-22.0 (12.9)	-44.2 (47.0)	
9	592.2 (40.1)	655.9 (84.5)	649.3 (74.5)	-67.2 (27.6)	-10.1 (10.2)	-17.7 (71.1)	
9.5	598.5 (42.9)	668.7 (82.2)	661.4 (59.5)	-60.8 (24.6)	2.8 (12.4)	-5.7 (44.3)	
10	616.3 (43.7)	653.0 (74.5)	661.1 (71.1)	-43.1 (24.3)	-12.9 (18.8)	-5.9 (43.9)	
10.5	611.9 (44.6)	654.4 (79.4)	664.9 (68.9)	-47.5 (24.2)	-11.5 (23.7)	-2.2 (47.2)	
11	616.7 (48.5)	662.8 (75.3)	704.6 (67.7)	-42.7 (29.1)	-3.1 (31.5)	37.6 (43.1)	
11.5	622.5 (48.6)	665.7 (80.1)	722.8 (70.6)	-36.9 (27.9)	-0.2 (26.7)	55.7 (38.1)	
12	625.2 (56.9)	673.3 (80.0)	722.2 (84.0)	-34.2 (35.9)	7.4 (28.1)	55.2 (50.7)	
12.5	635.3 (60.7)	671.7 (88.2)	731.3 (98.1)	-24.0 (39.4)	5.8 (31.4)	64.2 (66.1)	
13	635.2 (55.4)	687.4 (87.4)	761.0 (102.2)	-24.2 (31.6)	21.5 (28.2)	94.0 (70.6)	
13.5	651.2 (66.9)	701.1 (85.4)	775.8 (99.0)	-8.2 (39.6)	35.1 (36.6)	108.8 (65.1)	
14	652.9 (71.5)	707.0 (99.3)	779.2 (90.0)	-6.5 (42.6)	41.0 (34.6)	112.2 (56.5)	
14.5	653.9 (72.1)	722.1 (102.5)	785.0 (87.2)	-5.4 (47.7)	56.2 (38.9)	117.9 (57.6)	
15	650.2 (70.0)	744.2 (115.6)	796.0 (99.1)	-9.2 (44.9)	78.3 (45.2)	128.9 (71.0)	
15.5	679.6 (71.7)	758.0 (116.9)	809.3 (84.7)	20.2 (45.7)	92.1 (41.8)	142.2 (57.8)	
16	681.4 (69.2)	776.0 (117.6)	802.5 (87.0)	22.0 (46.2)	110.1 (42.5)	135.4 (58.1)	
16.5	686.2 (74.2)	756.1 (111.7)	790.4 (77.4)	26.8 (49.7)	90.2 (36.7)	123.4 (45.5)	
17	705.5 (76.0)	742.1 (105.9)	796.8 (78.4)	46.1 (50.8)	76.2 (42.2)	129.7 (45.8)	
17.5	710.0 (69.8)	755.4 (111.2)	778.8 (89.4)	50.6 (45.3)	89.5 (36.2)	111.7 (61.7)	
18	711.2 (61.2)	752.2 (113.1)	786.6 (59.6)	51.8 (39.2)	86.3 (31.8)	119.6 (28.3)	
18.5	630.0 (62.3)	736.2 (98.0)	700.5 (62.5)	-29.4 (47.6)	70.3 (26.9)	33.5 (38.0)	
19	580.9 (53.3)	631.7 (107.9)	597.7 (72.4)	-78.5 (36.6)	-34.2 (35.1)	-69.4 (76.5)	
19.5	619.6 (43.4)	589.1 (87.6)	658.8 (40.8)	-39.8 (30.1)	-76.8 (27.8)	-8.3 (25.5)	
20	576.7 (39.2)	620.2 (85.6)	557.6 (20.8)	-82.7 (22.1)	-45.7 (23.2)	-109.4 (36.0)	
20.5	555.6 (40.2)	542.2 (41.7)	605.4 (51.8)	-103.8 (24.2)	-123.7 (42.6)	-61.7 (31.6)	
21	551.5 (34.7)	554.7 (56.8)	593.8 (54.1)	-107.9 (30.8)	-111.2 (28.7)	-73.2 (32.7)	
21.5	511.4 (31.7)	556.9 (48.1)	605.1 (59.9)	-148.0 (31.1)	-109.0 (56.2)	-61.9 (38.8)	
22	502.7 (30.0)	577.6 (56.3)	501.5 (47.3)	-156.7 (31.5)	-88.3 (57.1)	-165.5 (19.0)	
22.5	537.7 (30.4)	516.8 (29.7)	514.6 (46.6)	-121.7 (36.3)	-149.1 (61.4)	-152.5 (39.7)	
23	557.9 (22.0)	540.2 (30.4)	554.7 (35.5)	-101.5 (33.0)	-125.8 (53.9)	-112.3 (13.9)	
23.5	575.1 (27.1)	547.1 (31.5)	571.4 (22.9)	-84.3 (25.1)	-118.8 (72.8)	-95.7 (17.4)	
24	570.3 (25.0)	539.7 (28.5)	567.3 (25.4)	-89.0 (26.1)	-126.2 (75.8)	-99.7 (16.5)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "**".

Table 17. In Cage, PR Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	80.1 (4.9)	87.7 (8.7)	74.1 (5.2)				
3.5	80.5 (4.0)	81.4 (2.3)	72.8 (5.0)	0.4 (2.7)	-6.3 (7.3)	-1.3 (2.2)	
4	79.8 (4.5)	86.2 (3.4)	75.0 (6.2)	-0.3 (1.4)	-1.5 (5.9)	0.9 (3.0)	
4.5	81.3 (4.6)	86.6 (5.8)	73.9 (6.1)	1.2 (1.4)	-1.1 (2.9)	-0.2 (2.9)	
5	80.7 (5.1)	85.9 (5.8)	73.2 (5.4)	0.6 (1.4)	-1.8 (3.0)	-0.9 (1.5)	
5.5	79.9 (5.2)	87.7 (6.6)	73.8 (5.9)	-0.2 (1.0)	-0.0 (2.9)	-0.3 (1.7)	
6	79.8 (5.6)	87.0 (8.2)	73.8 (6.1)	-0.3 (1.0)	-0.7 (1.3)	-0.2 (1.8)	
6.5	79.1 (5.3)	89.2 (8.0)	73.8 (6.3)	-0.9 (1.1)	1.5 (1.1)	-0.3 (1.9)	
7	80.4 (5.8)	89.2 (7.9)	73.2 (5.7)	-1.4 (1.3)	1.5 (1.3)	-0.9 (1.8)	
7.5	76.7 (5.2)	89.5 (7.9)	70.0 (3.6)	-3.3 (1.5)	1.9 (2.4)	-4.1 (2.1)	
8	78.2 (5.4)	88.5 (7.7)	70.5 (3.9)	-1.9 (1.0)	0.8 (2.5)	-3.6 (1.5)	
8.5	78.8 (5.7)	86.4 (8.3)	72.3 (4.3)	-1.2 (1.2)	-1.2 (1.2)	-1.8 (1.0)	
9	78.4 (5.7)	86.5 (9.0)	72.7 (3.9)	-1.6 (1.1)	-1.2 (2.1)	-1.4 (2.1)	
9.5	78.7 (5.8)	87.6 (9.1)	73.6 (4.2)	-1.3 (1.8)	-0.1 (2.2)	-0.4 (1.4)	
10	79.1 (5.6)	87.7 (9.2)	75.2 (4.6)	-1.0 (2.1)	0.0 (2.0)	1.1 (1.7)	
10.5	78.8 (5.8)	88.0 (9.2)	74.0 (4.4)	-1.3 (2.0)	0.3 (2.5)	-0.1 (1.3)	
11	80.6 (5.7)	89.8 (9.1)	73.2 (5.2)	0.5 (2.2)	2.2 (2.2)	-0.9 (0.7)	
11.5	80.1 (6.0)	88.6 (9.3)	74.0 (5.2)	-0.0 (2.0)	0.9 (1.9)	-0.1 (0.6)	
12	79.9 (6.5)	90.1 (9.7)	75.4 (5.2)	-0.2 (2.1)	2.5 (1.7)	1.3 (1.4)	
12.5	80.9 (6.2)	90.2 (8.8)	76.0 (5.9)	0.9 (2.0)	2.6 (0.5)	1.9 (1.5)	
13	81.5 (6.0)	88.9 (10.1)	76.0 (6.5)	1.4 (1.5)	1.2 (1.4)	1.9 (2.1)	
13.5	81.6 (5.9)	88.7 (10.0)	76.6 (5.0)	1.6 (1.5)	1.1 (1.6)	2.6 (0.7)	
14	82.7 (6.6)	88.3 (9.9)	76.8 (5.0)	2.6 (2.1)	0.7 (1.3)	2.7 (0.9)	
14.5	82.1 (6.5)	89.2 (10.7)	77.9 (4.8)	2.1 (2.2)	1.5 (2.7)	3.8 (1.7)	
15	81.6 (6.3)	89.6 (11.0)	77.0 (6.2)	1.6 (1.9)	2.0 (2.8)	2.9 (1.6)	
15.5	81.6 (6.6)	89.5 (10.6)	78.5 (4.9)	1.6 (2.1)	1.8 (2.4)	4.4 (2.2)	
16	82.2 (6.2)	90.5 (10.6)	77.6 (5.0)	2.1 (2.1)	2.8 (2.3)	3.5 (1.8)	
16.5	81.2 (6.4)	90.8 (10.5)	76.1 (5.3)	1.2 (2.2)	3.1 (2.3)	2.0 (1.0)	
17	80.8 (6.4)	90.5 (10.5)	77.8 (6.4)	0.7 (2.0)	2.9 (2.5)	3.7 (1.9)	
17.5	81.5 (6.3)	90.9 (10.3)	78.1 (5.3)	1.5 (2.2)	3.2 (2.0)	4.0 (1.9)	
18	82.1 (5.8)	90.6 (9.9)	77.9 (5.2)	2.0 (1.6)	2.9 (1.6)	3.8 (1.6)	
18.5	80.9 (6.0)	88.8 (10.5)	77.2 (5.0)	0.9 (1.7)	1.1 (1.8)	3.1 (1.7)	
19	80.4 (5.5)	88.5 (10.4)	72.1 (4.9)	0.4 (1.7)	0.8 (2.0)	-1.9 (1.4)	
19.5	80.4 (4.9)	88.8 (10.0)	73.8 (5.5)	0.4 (0.7)	1.1 (1.5)	-0.3 (0.8)	
20	80.0 (4.5)	86.7 (8.8)	73.9 (5.4)	-0.0 (0.7)	-1.0 (1.0)	-0.2 (1.8)	
20.5	79.6 (3.8)	88.2 (8.8)	72.8 (5.7)	-0.5 (1.4)	0.5 (1.4)	-1.3 (1.2)	
21	80.1 (4.4)	85.7 (7.6)	73.3 (6.0)	0.1 (1.1)	-1.9 (1.9)	-0.8 (1.4)	
21.5	79.6 (4.6)	86.0 (7.5)	72.5 (6.1)	-0.5 (1.2)	-1.6 (1.7)	-1.6 (1.5)	
22	79.4 (4.2)	85.4 (7.9)	74.4 (5.9)	-0.6 (1.4)	-2.2 (1.4)	0.3 (2.0)	
22.5	83.1 (4.8)	88.6 (6.6)	75.2 (6.0)	3.1 (1.0)	1.0 (2.2)	1.1 (1.5)	
23	81.8 (4.7)	89.9 (6.1)	75.4 (6.1)	1.8 (1.7)	2.2 (2.8)	1.3 (1.5)	
23.5	80.7 (5.1)	88.7 (6.8)	74.5 (7.0)	0.6 (1.1)	1.0 (3.3)	0.4 (2.1)	
24	81.2 (5.2)	87.3 (6.1)	73.0 (6.1)	1.1 (1.4)	-0.3 (3.6)	-1.1 (1.7)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "a".

Table 18. In Cage, QRS Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	61.4 (1.6)	64.7 (2.5)	58.6 (1.1)				
3.5	60.0 (1.6)	62.6 (2.2)	57.0 (1.7)	-1.3 (1.2)	-2.1 (2.5)	-1.7 (1.5)	
4	58.9 (1.5)	62.0 (2.0)	58.2 (1.9)	-2.5 (1.0)	-2.7 (2.4)	-0.4 (1.7)	
4.5	59.8 (1.6)	61.7 (1.4)	57.2 (1.6)	-1.6 (1.3)	-2.9 (2.4)	-1.4 (1.6)	
5	59.1 (1.6)	62.1 (2.2)	55.9 (0.9)	-2.3 (1.1)	-2.5 (1.9)	-2.8 (0.6)	
5.5	58.8 (1.6)	63.0 (2.5)	56.9 (1.5)	-2.6 (1.2)	-1.6 (1.2)	-1.7 (1.4)	
6	58.9 (1.5)	63.6 (2.5)	56.5 (1.9)	-2.4 (0.8)	-1.0 (1.1)	-2.1 (1.8)	
6.5	59.2 (1.8)	63.8 (2.8)	57.0 (0.9)	-2.2 (0.6)	-0.8 (1.4)	-1.6 (1.2)	
7	61.3 (2.0)	64.4 (2.8)	57.8 (1.1)	-0.5 (1.0)	-0.3 (1.6)	-0.9 (1.4)	
7.5	62.1 (2.0)	64.8 (3.0)	58.7 (0.9)	0.7 (1.2)	0.1 (1.3)	0.1 (1.5)	
8	61.8 (2.1)	64.7 (3.1)	58.4 (1.0)	0.4 (1.2)	0.0 (1.3)	-0.3 (1.8)	
8.5	60.7 (2.1)	64.8 (3.2)	57.6 (0.8)	-0.6 (1.0)	0.2 (1.4)	-1.0 (1.6)	
9	62.2 (1.9)	64.7 (3.2)	57.5 (0.5)	0.8 (1.2)	0.0 (1.6)	-1.1 (1.3)	
9.5	60.7 (2.4)	65.3 (3.5)	57.4 (0.3)	-0.7 (1.1)	0.6 (2.0)	-1.2 (1.0)	
10	60.3 (2.3)	65.0 (3.4)	57.0 (1.1)	-1.0 (1.1)	0.3 (1.7)	-1.6 (0.8)	
10.5	60.9 (2.1)	64.9 (3.2)	57.5 (0.5)	-0.5 (0.9)	0.3 (1.8)	-1.2 (0.9)	
11	60.5 (2.1)	65.0 (3.1)	57.9 (0.2)	-0.9 (0.9)	0.3 (1.8)	-0.7 (1.0)	
11.5	60.4 (2.2)	65.4 (3.7)	58.0 (0.4)	-1.0 (1.0)	0.7 (2.1)	-0.7 (0.8)	
12	61.7 (2.1)	65.3 (3.4)	58.1 (1.3)	0.3 (0.9)	0.7 (1.9)	-0.6 (0.6)	
12.5	61.1 (2.5)	64.7 (3.1)	59.6 (2.3)	-0.3 (1.2)	-0.0 (1.3)	1.0 (1.2)	
13	59.9 (2.6)	65.2 (3.5)	60.0 (2.9)	-1.5 (1.2)	0.6 (1.6)	1.4 (1.8)	
13.5	60.0 (2.6)	64.9 (3.4)	58.5 (1.7)	-1.4 (1.2)	0.2 (1.7)	-0.2 (0.8)	
14	61.2 (2.4)	65.0 (3.7)	58.0 (1.9)	-0.2 (1.1)	0.3 (1.7)	-0.7 (0.9)	
14.5	61.4 (2.6)	64.8 (3.1)	59.0 (2.3)	0.1 (1.4)	0.2 (1.2)	0.4 (1.3)	
15	60.8 (2.4)	65.4 (2.6)	61.1 (2.1)	-0.6 (1.0)	0.7 (0.6)	2.5 (1.0)	
15.5	62.1 (1.8)	66.5 (2.4)	60.0 (1.8)	0.7 (0.7)	1.8 (0.4)	1.3 (0.7)	
16	61.1 (2.3)	67.3 (2.5)	59.5 (1.3)	-0.3 (1.1)	2.6 (0.3)	0.9 (0.5)	
16.5	61.8 (2.0)	66.1 (2.7)	59.5 (0.3)	0.4 (0.8)	1.4 (0.6)	0.9 (1.1)	
17	62.1 (2.0)	66.4 (2.7)	61.3 (1.8)	0.7 (0.9)	1.8 (0.5)	2.6 (0.7)	
17.5	61.1 (2.2)	67.0 (2.5)	58.8 (1.1)	-0.3 (0.9)	2.4 (0.6)	0.2 (0.4)	
18	61.8 (2.1)	66.6 (2.8)	59.1 (1.0)	0.4 (0.8)	1.9 (0.5)	0.5 (0.7)	
18.5	61.6 (1.8)	65.2 (3.2)	58.8 (1.3)	0.2 (0.9)	0.5 (1.0)	0.2 (0.6)	
19	61.5 (2.1)	65.2 (3.3)	58.8 (1.1)	0.1 (0.9)	0.5 (1.2)	0.1 (0.9)	
19.5	61.4 (1.9)	65.3 (3.1)	59.2 (0.9)	-0.0 (0.8)	0.6 (1.4)	0.5 (1.4)	
20	61.8 (2.0)	65.3 (2.8)	59.8 (1.4)	0.5 (0.7)	0.6 (0.8)	1.2 (1.7)	
20.5	61.5 (2.0)	65.4 (2.9)	57.6 (1.4)	0.1 (0.7)	0.8 (0.9)	-1.0 (1.3)	
21	61.9 (2.0)	65.5 (3.0)	58.4 (1.6)	0.5 (0.8)	0.8 (0.9)	-0.2 (1.5)	
21.5	61.7 (2.0)	65.4 (2.8)	58.9 (1.7)	0.3 (1.0)	0.7 (0.9)	0.3 (1.6)	
22	61.6 (1.8)	65.9 (2.8)	58.3 (2.0)	0.2 (0.8)	1.2 (0.8)	-0.3 (1.9)	
22.5	61.3 (1.7)	63.4 (2.6)	58.9 (1.7)	-0.1 (1.0)	-1.2 (1.6)	0.2 (1.7)	
23	61.5 (1.7)	64.1 (2.5)	56.9 (1.0)	0.1 (0.9)	-0.5 (1.6)	-1.7 (0.8)	
23.5	61.1 (1.8)	64.1 (2.9)	57.0 (0.9)	-0.3 (0.9)	-0.5 (1.0)	-1.6 (1.1)	
24	60.9 (2.1)	64.6 (2.9)	56.0 (0.6)	-0.5 (1.1)	-0.1 (1.0)	-2.6 (0.7)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "a".

Table 19. In Cage, QT Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	303.7 (15.8)	294.0 (18.1)	317.5 (20.6)				
3.5	243.8 (10.7)	227.6 (8.7)	242.9 (17.6)	-59.9 (10.0)	-66.4 (19.4)	-74.6 (10.1)	
4	240.9 (9.2)	234.5 (9.2)	250.0 (14.2)	-62.8 (9.6)	-59.5 (16.7)	-67.5 (7.9)	
4.5	247.3 (8.7)	239.9 (7.9)	253.3 (15.3)	-56.4 (8.6)	-54.1 (14.6)	-64.2 (7.3)	
5	250.2 (9.4)	238.9 (5.1)	255.4 (14.6)	-53.5 (7.7)	-55.1 (17.3)	-62.1 (7.1)	
5.5	254.6 (7.6)	245.8 (5.6)	260.4 (11.4)	-49.1 (8.8)	-48.2 (14.5)	-57.1 (9.7)	
6	257.2 (8.3)	259.5 (7.4)	267.2 (12.3)	-46.5 (8.0)	-34.5 (10.8)	-50.3 (8.4)	
6.5	260.4 (8.3)	266.3 (8.7)	274.0 (11.8)	-43.3 (8.0)	-27.7 (12.8)	-43.5 (9.9)	
7	266.5 (7.3)	265.3 (11.6)	287.2 (18.6)	-27.2 (8.4)	-28.7 (9.7)	-30.3 (7.7)	
7.5	280.4 (7.7)	279.7 (9.7)	300.7 (15.6)	-23.3 (10.8)	-14.4 (10.1)	-16.8 (10.9)	
8	284.3 (7.3)	293.6 (11.8)	306.5 (16.7)	-19.4 (11.0)	-0.5 (7.3)	-11.0 (11.3)	
8.5	288.6 (8.2)	295.9 (14.2)	315.0 (21.8)	-15.1 (9.9)	1.9 (7.0)	-2.6 (9.8)	
9	293.7 (8.8)	297.0 (16.6)	323.6 (29.2)	-10.0 (9.6)	3.0 (6.4)	6.1 (15.6)	
9.5	293.7 (11.1)	300.7 (16.0)	327.0 (21.6)	-10.0 (7.8)	6.6 (6.9)	9.5 (7.9)	
10	301.1 (12.9)	300.6 (15.5)	324.1 (22.9)	-2.6 (7.2)	6.6 (9.1)	6.6 (6.4)	
10.5	302.1 (12.7)	300.0 (16.0)	330.0 (26.7)	-1.6 (7.4)	6.0 (8.7)	12.5 (9.2)	
11	303.3 (13.1)	301.9 (16.9)	340.4 (29.1)	-0.4 (8.7)	7.9 (11.2)	22.9 (10.5)	
11.5	303.5 (13.9)	302.2 (17.0)	342.6 (21.9)	-0.2 (8.6)	8.2 (10.9)	25.1 (5.3)	
12	308.6 (15.5)	304.8 (17.1)	343.8 (25.8)	4.9 (9.3)	10.8 (9.8)	26.3 (10.6)	
12.5	311.8 (17.3)	303.8 (17.2)	343.3 (26.6)	8.1 (10.4)	9.8 (9.2)	25.8 (16.6)	
13	311.4 (16.4)	307.5 (18.5)	357.8 (30.8)	7.7 (9.8)	13.5 (10.6)	40.3 (14.6)	
13.5	316.7 (19.3)	308.7 (18.7)	357.1 (30.4)	13.0 (9.5)	14.7 (11.1)	39.6 (14.8)	
14	313.9 (20.9)	311.4 (19.8)	361.5 (30.6)	10.3 (10.8)	17.4 (12.7)	43.9 (13.3)	
14.5	315.7 (21.9)	315.4 (21.1)	361.7 (34.6)	12.1 (12.4)	21.4 (12.9)	44.2 (15.4)	
15	315.9 (21.1)	319.4 (23.0)	365.4 (38.2)	12.2 (12.3)	25.4 (12.9)	47.9 (18.6)	
15.5	320.2 (20.9)	320.8 (24.5)	370.0 (35.9)	16.5 (12.2)	26.8 (11.9)	52.5 (15.9)	
16	321.6 (20.6)	323.8 (24.1)	369.2 (36.7)	17.9 (12.7)	29.8 (11.7)	51.7 (16.5)	
16.5	323.0 (20.8)	319.7 (24.0)	366.7 (28.0)	19.3 (12.4)	25.7 (10.4)	49.2 (10.3)	
17	326.8 (20.3)	318.0 (23.3)	358.9 (23.6)	23.1 (12.8)	23.9 (11.1)	41.4 (12.5)	
17.5	329.3 (19.9)	322.5 (22.6)	353.2 (19.5)	25.6 (12.0)	28.5 (9.1)	35.7 (17.3)	
18	328.5 (18.4)	321.7 (21.9)	359.9 (20.0)	24.8 (10.6)	27.6 (7.2)	42.4 (7.6)	
18.5	303.1 (12.9)	318.5 (18.8)	332.1 (13.7)	-0.6 (12.2)	24.5 (8.9)	14.6 (11.7)	
19	290.0 (14.5)	293.4 (21.0)	297.3 (18.6)	-13.6 (11.2)	-0.6 (7.7)	-20.2 (19.9)	
19.5	303.0 (11.6)	281.6 (19.6)	315.1 (10.0)	-0.7 (10.4)	-12.4 (10.2)	-2.4 (10.9)	
20	289.2 (8.9)	286.4 (16.7)	286.1 (12.1)	-14.5 (8.6)	-7.6 (7.6)	-31.4 (12.7)	
20.5	280.5 (13.8)	271.8 (11.7)	291.5 (15.8)	-23.2 (6.1)	-22.2 (10.3)	-26.0 (11.4)	
21	279.7 (13.9)	271.0 (13.6)	287.3 (21.6)	-24.0 (9.9)	-23.0 (7.2)	-30.2 (16.1)	
21.5	266.8 (12.0)	270.4 (13.5)	289.9 (24.0)	-36.9 (10.6)	-23.6 (12.2)	-27.6 (20.3)	
22	262.4 (12.4)	275.1 (13.7)	270.4 (21.6)	-41.3 (10.0)	-18.9 (11.2)	-47.1 (11.3)	
22.5	273.5 (13.6)	262.0 (9.5)	272.3 (23.4)	-30.2 (10.0)	-32.0 (12.0)	-45.3 (12.3)	
23	279.8 (12.8)	266.5 (8.7)	275.4 (24.9)	-23.9 (10.5)	-27.5 (11.3)	-42.2 (10.7)	
23.5	283.2 (10.1)	265.6 (3.0)	276.6 (17.8)	-20.5 (8.9)	-28.4 (15.9)	-41.0 (9.1)	
24	281.5 (9.5)	263.1 (2.6)	277.8 (19.9)	-22.2 (8.6)	-30.9 (18.2)	-39.8 (9.9)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "*".

Table 20. In Cage, QTcB Interval (msec), Means (Standard Error) of Unadjusted and Baseline-Adjusted Averages Following Dosing with LT3114 or Vehicle

Time (hours)	Means (Standard Error) of Unadjusted 30-Minute Averages			Means (Standard Error) of Baseline-Adjusted 30-Minute Averages			Sig. Diff. Among Dose Levels? ^a
	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	0 mg/kg (n=8)	10 mg/kg (n=4)	50 mg/kg (n=4)	
Baseline	378.6 (12.7)	366.1 (5.5)	393.1 (23.5)				
3.5	356.2 (13.4)	359.1 (13.9)	366.2 (21.9)	-22.4 (6.2)	-7.1 (9.8)	-26.9 (10.0)	
4	354.5 (12.9)	350.9 (15.0)	367.5 (17.9)	-24.1 (4.3)	-15.2 (10.9)	-25.5 (8.3)	
4.5	356.7 (11.6)	348.3 (10.5)	366.3 (19.2)	-21.8 (4.4)	-17.8 (6.2)	-26.8 (10.4)	
5	357.7 (11.9)	346.9 (10.6)	366.1 (17.4)	-20.9 (3.8)	-19.2 (6.8)	-27.0 (11.4)	
5.5	357.5 (11.4)	344.1 (6.8)	366.9 (16.4)	-21.0 (5.5)	-22.0 (3.6)	-26.1 (11.1)	
6	359.2 (11.7)	352.1 (4.6)	369.3 (17.9)	-19.4 (4.4)	-14.0 (2.6)	-23.7 (10.0)	
6.5	362.9 (12.5)	354.3 (3.3)	376.1 (20.7)	-15.7 (5.2)	-11.8 (5.7)	-17.0 (7.0)	
7	358.6 (8.0)	353.1 (3.1)	389.0 (27.3)	-9.3 (6.3)	-13.1 (6.5)	-4.1 (7.6)	
7.5	378.3 (11.8)	363.5 (4.3)	404.2 (26.8)	-0.3 (5.7)	-2.6 (6.6)	11.1 (6.7)	
8	379.5 (10.6)	370.8 (5.9)	401.7 (22.7)	0.9 (5.5)	4.7 (6.0)	8.6 (5.5)	
8.5	382.9 (11.0)	372.6 (5.5)	402.1 (23.7)	4.3 (5.2)	6.4 (5.3)	9.1 (4.6)	
9	385.5 (11.3)	370.9 (4.6)	403.8 (22.5)	6.9 (5.1)	4.8 (5.3)	10.7 (3.7)	
9.5	383.4 (8.7)	371.3 (4.1)	405.8 (25.2)	4.8 (6.5)	5.2 (5.3)	12.8 (4.9)	
10	386.8 (11.5)	375.3 (2.7)	403.1 (25.3)	8.2 (5.1)	9.2 (6.3)	10.0 (4.5)	
10.5	390.1 (13.1)	374.8 (2.7)	408.4 (28.3)	11.5 (4.8)	8.7 (4.2)	15.3 (6.4)	
11	390.2 (12.3)	374.2 (1.2)	408.4 (27.8)	11.7 (5.3)	8.0 (5.5)	15.3 (5.8)	
11.5	389.0 (11.0)	374.0 (2.0)	407.2 (25.8)	10.4 (4.5)	7.9 (6.0)	14.2 (4.8)	
12	395.4 (14.2)	375.4 (1.3)	409.2 (25.7)	16.9 (5.6)	9.3 (4.4)	16.2 (3.4)	
12.5	395.9 (12.4)	375.4 (3.9)	406.6 (22.6)	17.3 (5.6)	9.3 (2.9)	13.6 (3.5)	
13	394.8 (9.8)	375.2 (1.6)	415.8 (32.0)	16.3 (5.5)	9.1 (5.1)	22.8 (9.0)	
13.5	397.5 (12.5)	372.9 (1.5)	409.6 (24.9)	18.9 (4.7)	6.7 (4.0)	16.5 (4.2)	
14	393.6 (13.1)	375.9 (3.8)	413.0 (23.7)	15.0 (5.2)	9.8 (6.9)	19.9 (4.0)	
14.5	395.2 (13.0)	376.7 (3.0)	410.4 (23.1)	16.7 (4.5)	10.6 (5.7)	17.3 (3.6)	
15	396.8 (12.4)	376.6 (4.2)	411.1 (23.2)	18.2 (4.2)	10.5 (6.0)	18.0 (2.1)	
15.5	393.2 (13.4)	374.1 (2.5)	412.1 (23.8)	14.7 (4.6)	8.0 (5.8)	19.0 (3.2)	
16	394.5 (12.8)	373.1 (2.4)	413.7 (25.7)	16.0 (5.3)	7.0 (5.3)	20.7 (3.3)	
16.5	395.8 (14.6)	372.8 (3.3)	417.0 (29.4)	17.2 (5.6)	6.7 (3.6)	23.9 (6.8)	
17	395.3 (13.2)	375.4 (3.3)	405.9 (18.8)	16.7 (5.5)	9.2 (2.9)	12.8 (7.8)	
17.5	396.3 (13.4)	377.0 (2.8)	405.9 (20.2)	17.8 (5.9)	10.9 (4.5)	12.9 (8.0)	
18	394.0 (12.8)	376.9 (3.3)	409.8 (21.3)	15.4 (5.8)	10.8 (4.8)	16.7 (4.7)	
18.5	388.5 (10.5)	377.0 (3.0)	403.1 (17.5)	10.0 (5.4)	10.9 (5.2)	10.0 (7.4)	
19	385.8 (11.8)	376.7 (5.4)	390.2 (18.9)	7.3 (4.0)	10.6 (3.3)	-2.9 (5.0)	
19.5	389.5 (12.0)	372.2 (7.9)	392.5 (18.7)	11.0 (6.6)	6.1 (5.5)	-0.6 (6.5)	
20	385.8 (9.5)	369.8 (6.3)	387.1 (22.7)	7.2 (6.3)	3.7 (4.0)	-6.0 (7.3)	
20.5	379.8 (12.2)	373.2 (9.1)	379.9 (26.5)	1.2 (3.0)	7.0 (7.3)	-13.2 (10.6)	
21	379.4 (13.7)	368.5 (9.6)	376.6 (27.0)	0.9 (4.8)	2.4 (6.4)	-16.5 (11.6)	
21.5	376.0 (11.8)	365.9 (10.2)	376.5 (27.8)	-2.5 (4.9)	-0.2 (6.7)	-16.6 (14.8)	
22	372.3 (10.7)	365.8 (8.8)	384.0 (21.5)	-6.3 (4.9)	-0.3 (5.4)	-9.0 (6.7)	
22.5	374.9 (12.2)	366.5 (7.5)	381.2 (21.3)	-3.7 (2.5)	0.4 (3.7)	-11.8 (5.1)	
23	376.6 (14.1)	364.6 (7.5)	371.8 (27.8)	-2.0 (4.6)	-1.5 (5.5)	-21.3 (10.2)	
23.5	376.2 (11.6)	362.6 (7.2)	368.9 (24.6)	-2.3 (4.8)	-3.5 (3.9)	-24.2 (12.0)	
24	375.6 (11.4)	361.7 (5.4)	371.7 (28.8)	-3.0 (4.8)	-4.4 (2.7)	-21.4 (13.3)	

a. Column indicates when F-tests for differences between dose levels in baseline-adjusted averages were significant at a given interval, as performed within a repeated measures ANOVA. At a given interval, significant dose level effects were concluded (marked with "Yes") when the adjusted p-value was at or below 0.05, where the adjustment was based on the Benjamini and Hochberg approach to controlling the false discovery rate to no higher than 0.05 across all intervals simultaneously. At a given time point, if significant dose level effects were concluded, comparisons of each test article dose level to the vehicle were performed at a 0.05/2=0.025 level and the significant dose levels were marked with an "**".

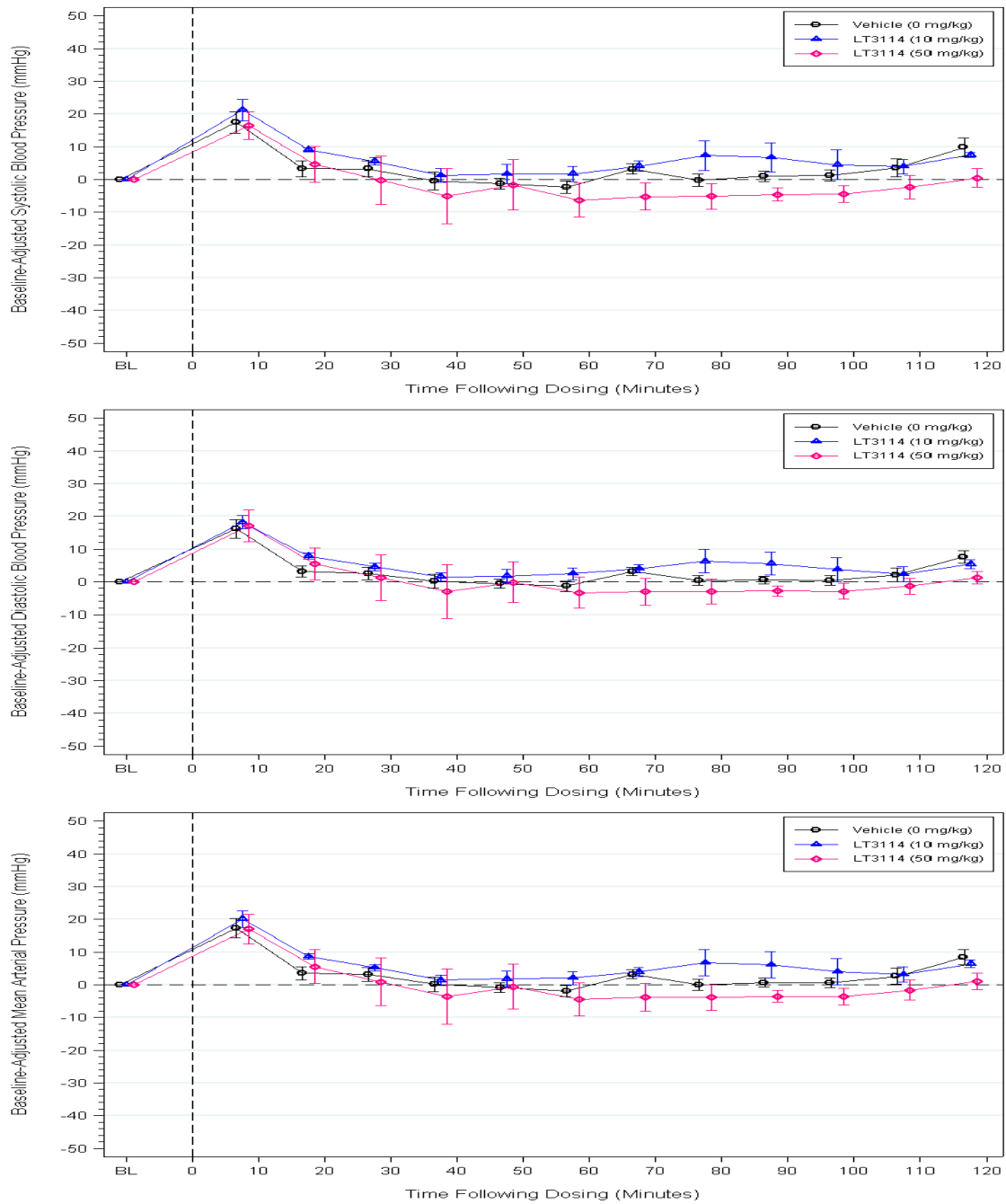


Figure 1. In Box, Blood Pressure Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

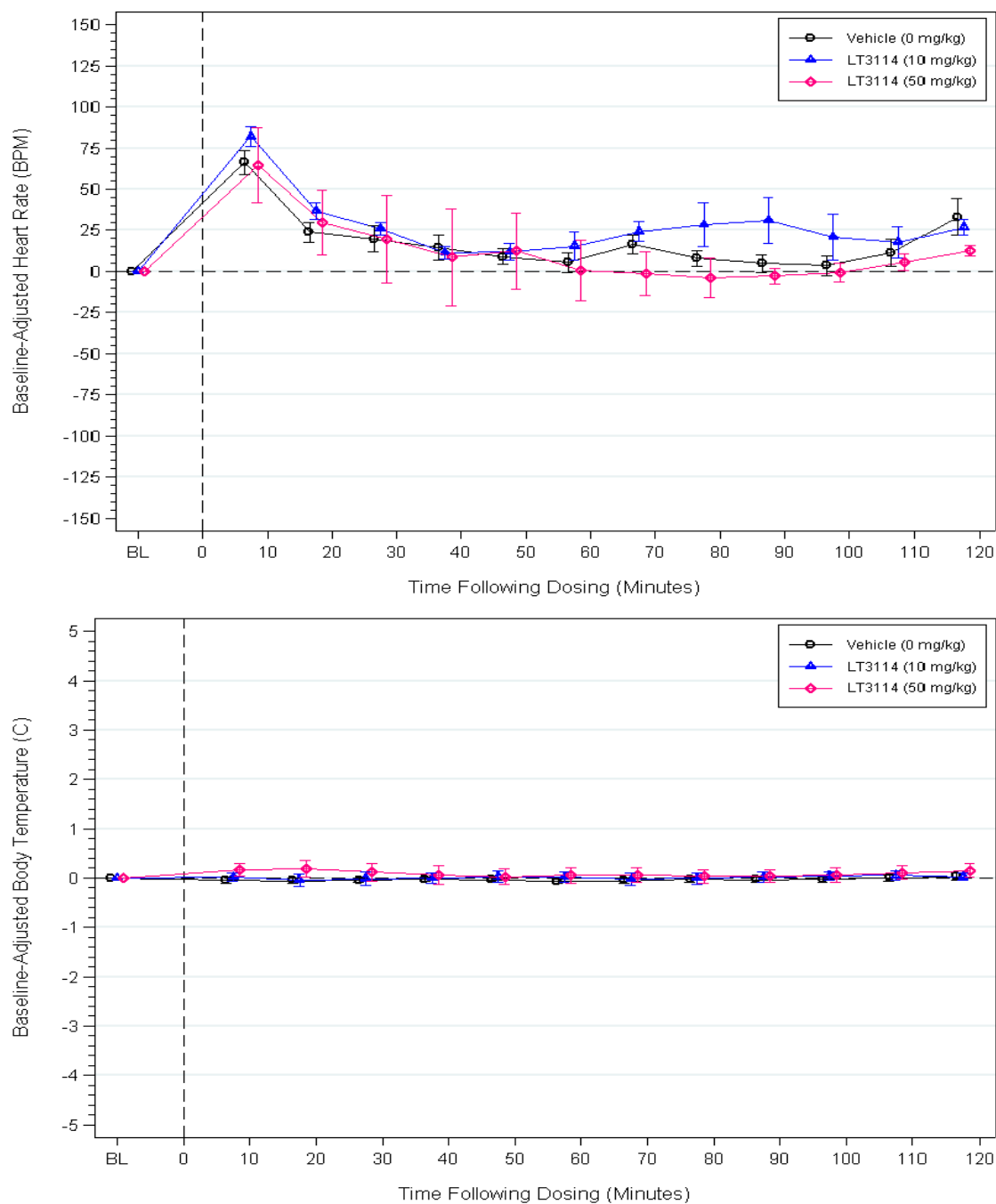


Figure 2. In Box, Heart Rate, and Body Temperature Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

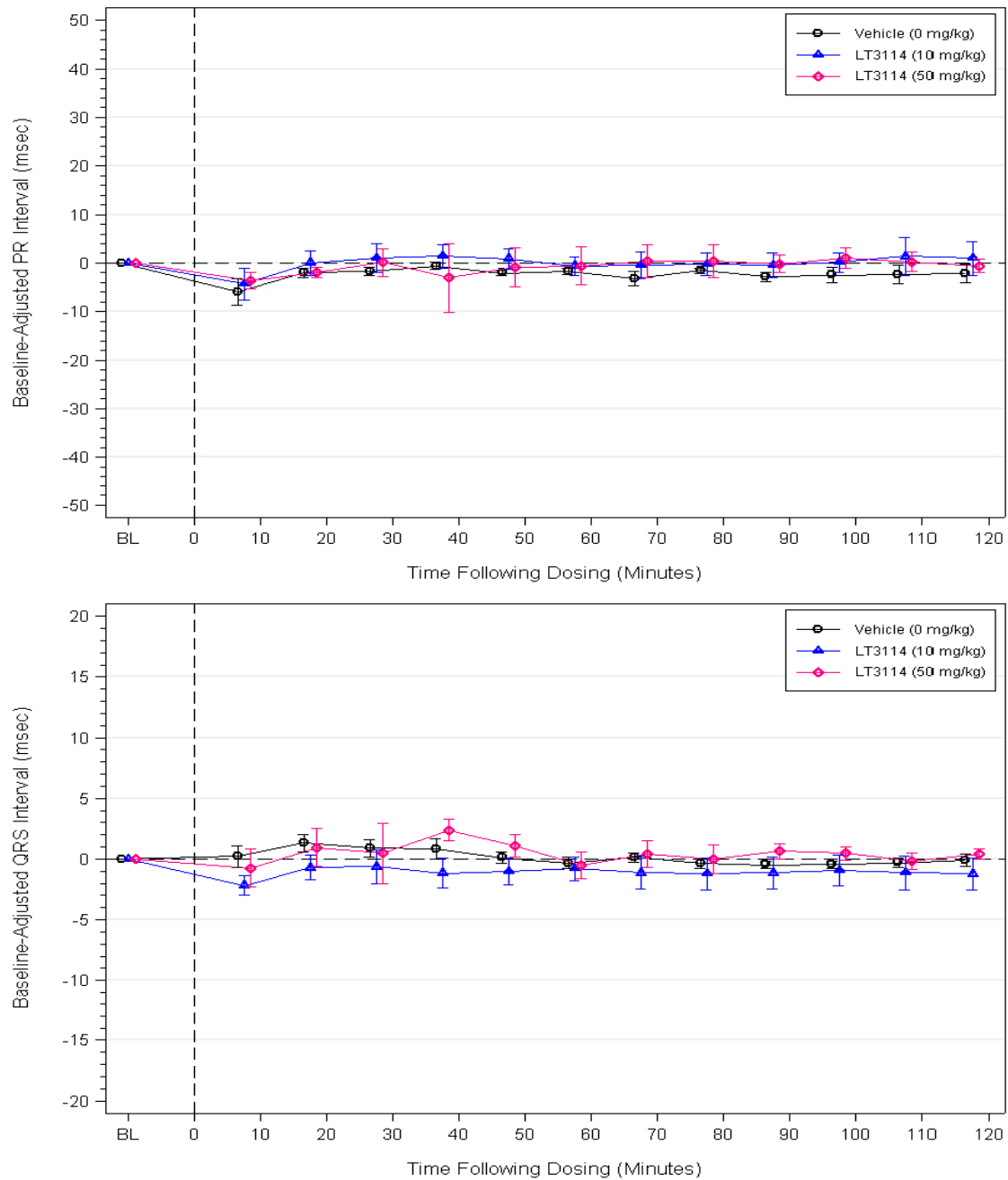


Figure 3. In Box, PR, and QRS Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

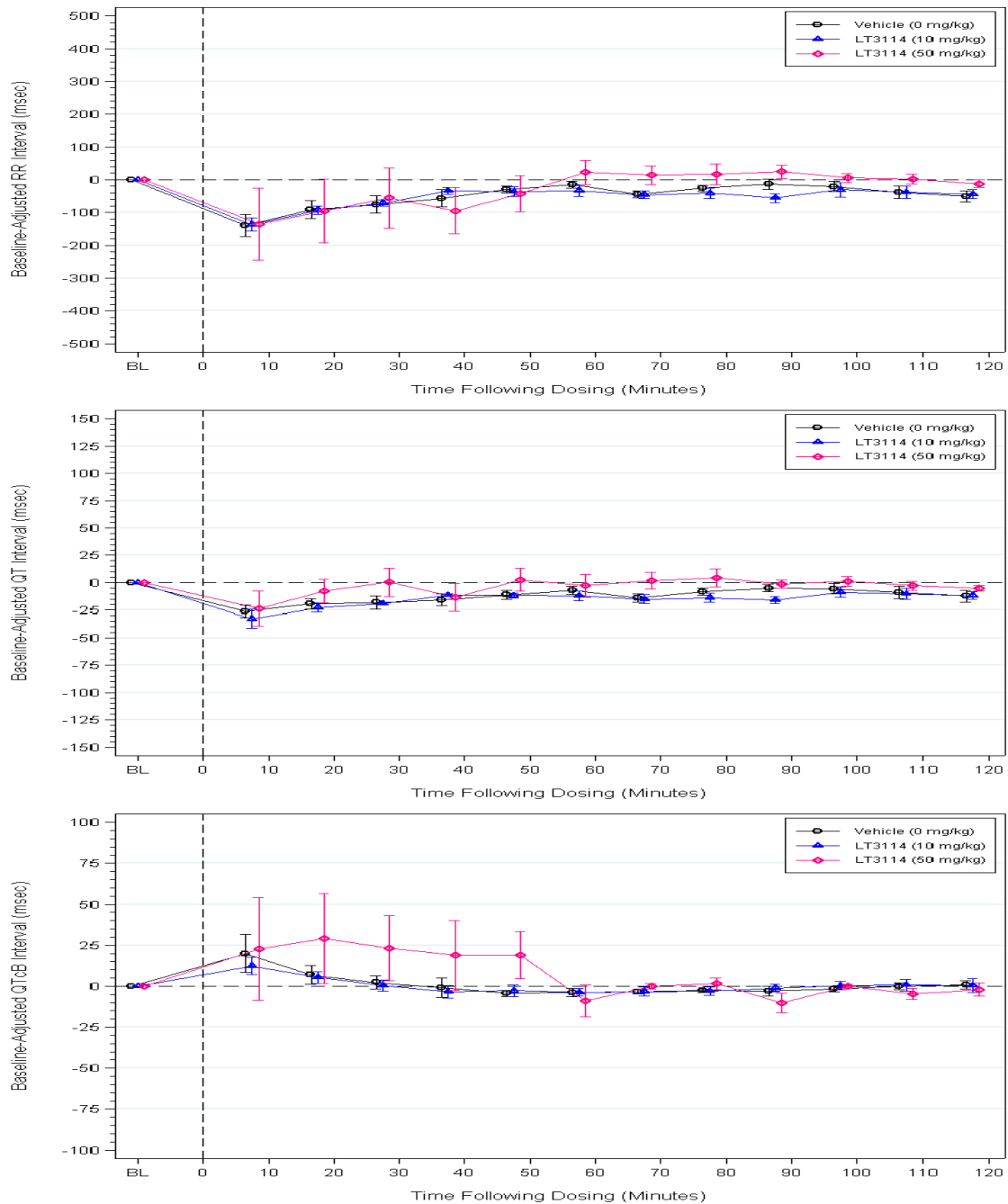


Figure 4. In Box, RR, QT, and QTcB Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

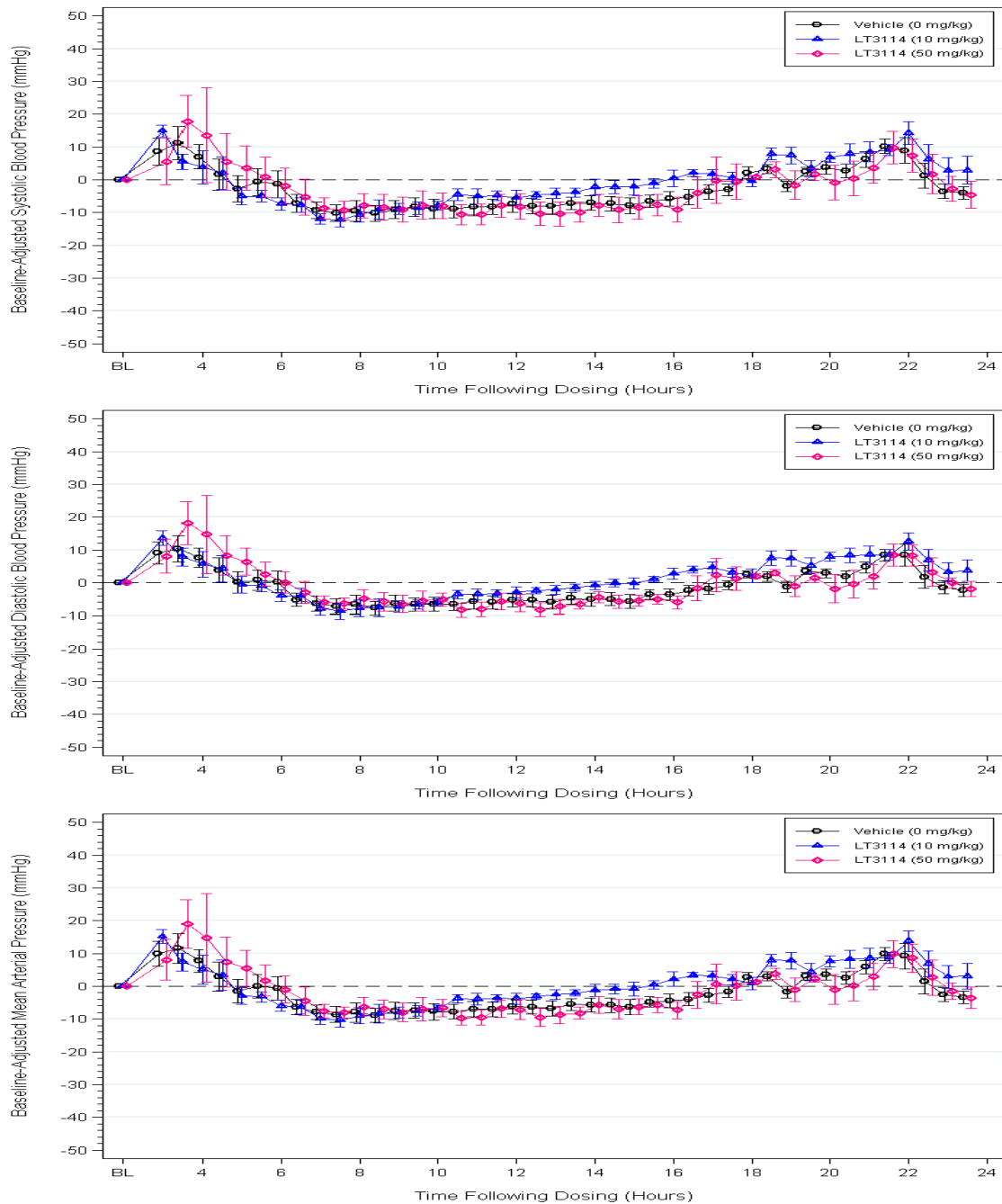


Figure 5. In Cage, Blood Pressure Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

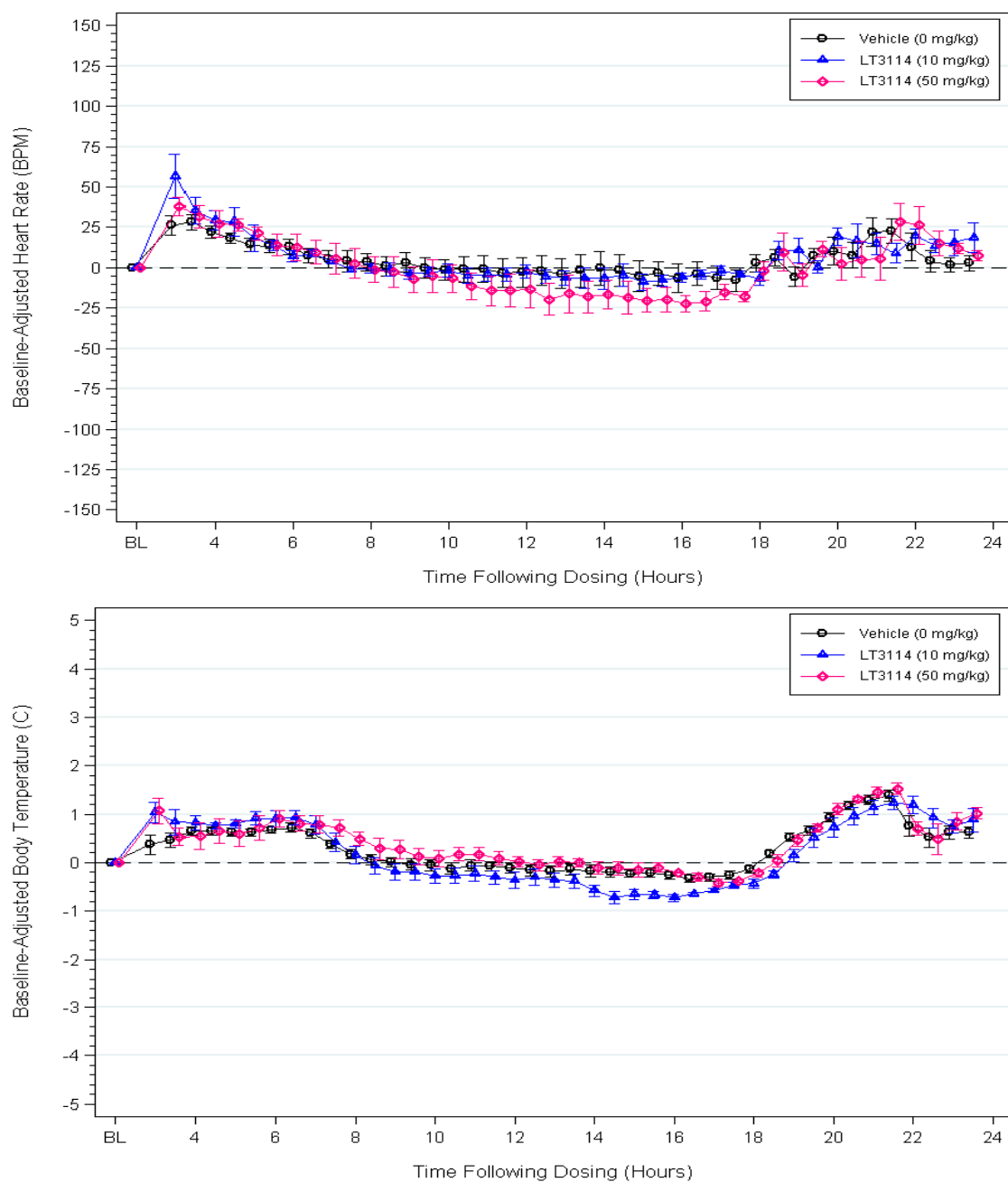


Figure 6. In Cage, Heart Rate, and Body Temperature Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

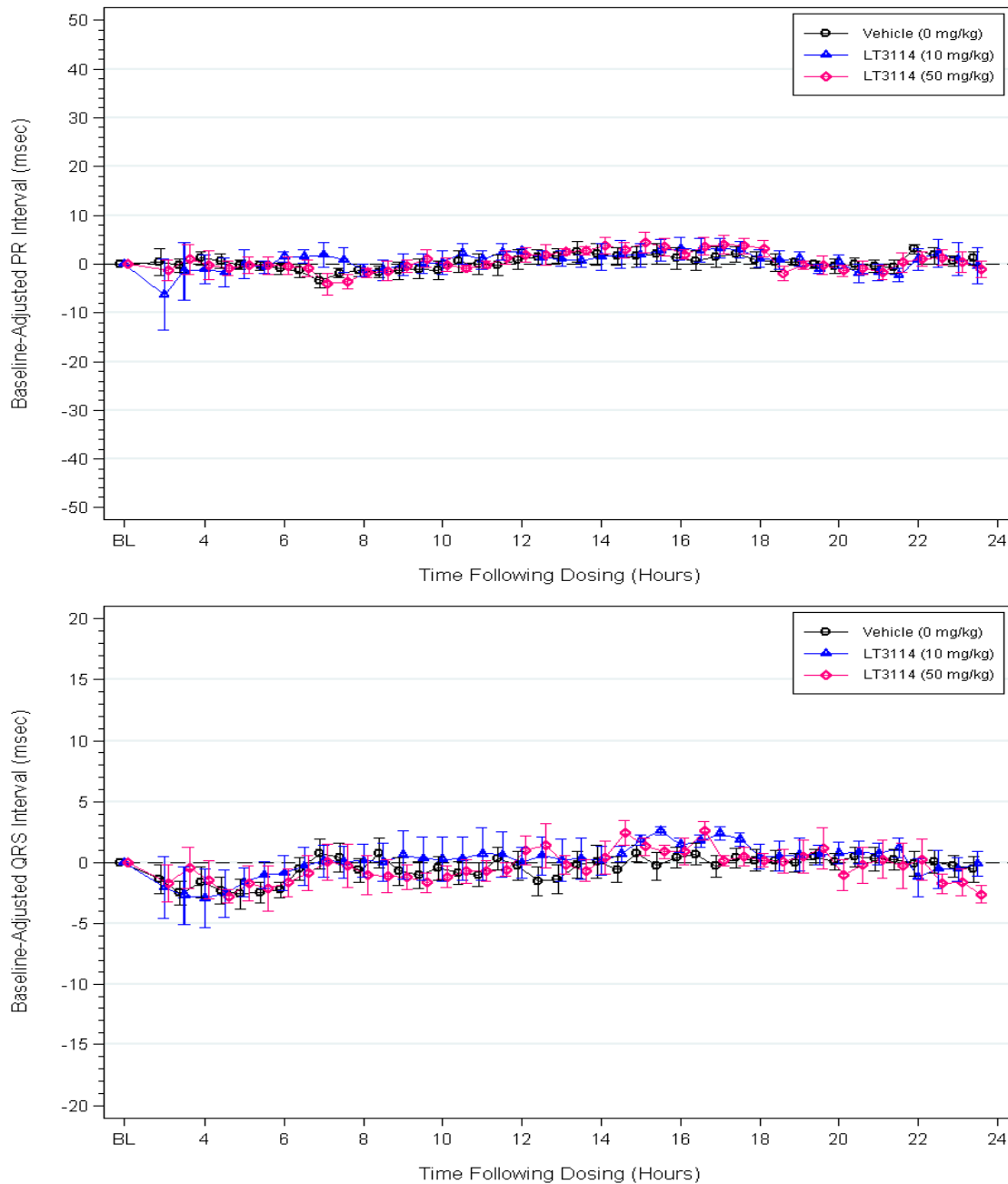


Figure 7. In Cage, PR, and QRS Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

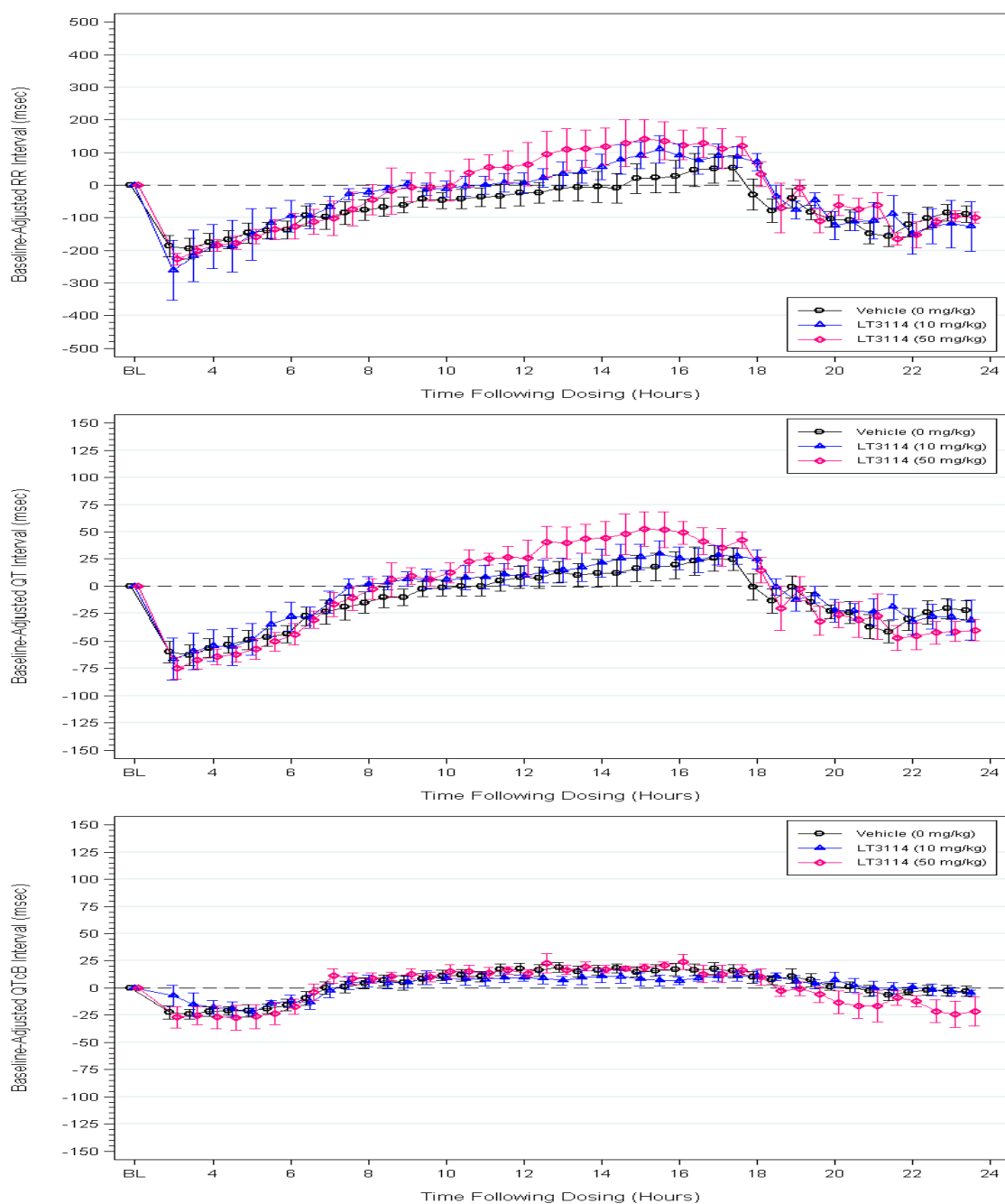


Figure 8. In Cage, RR, QT, and QTcB Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Baseline-Adjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

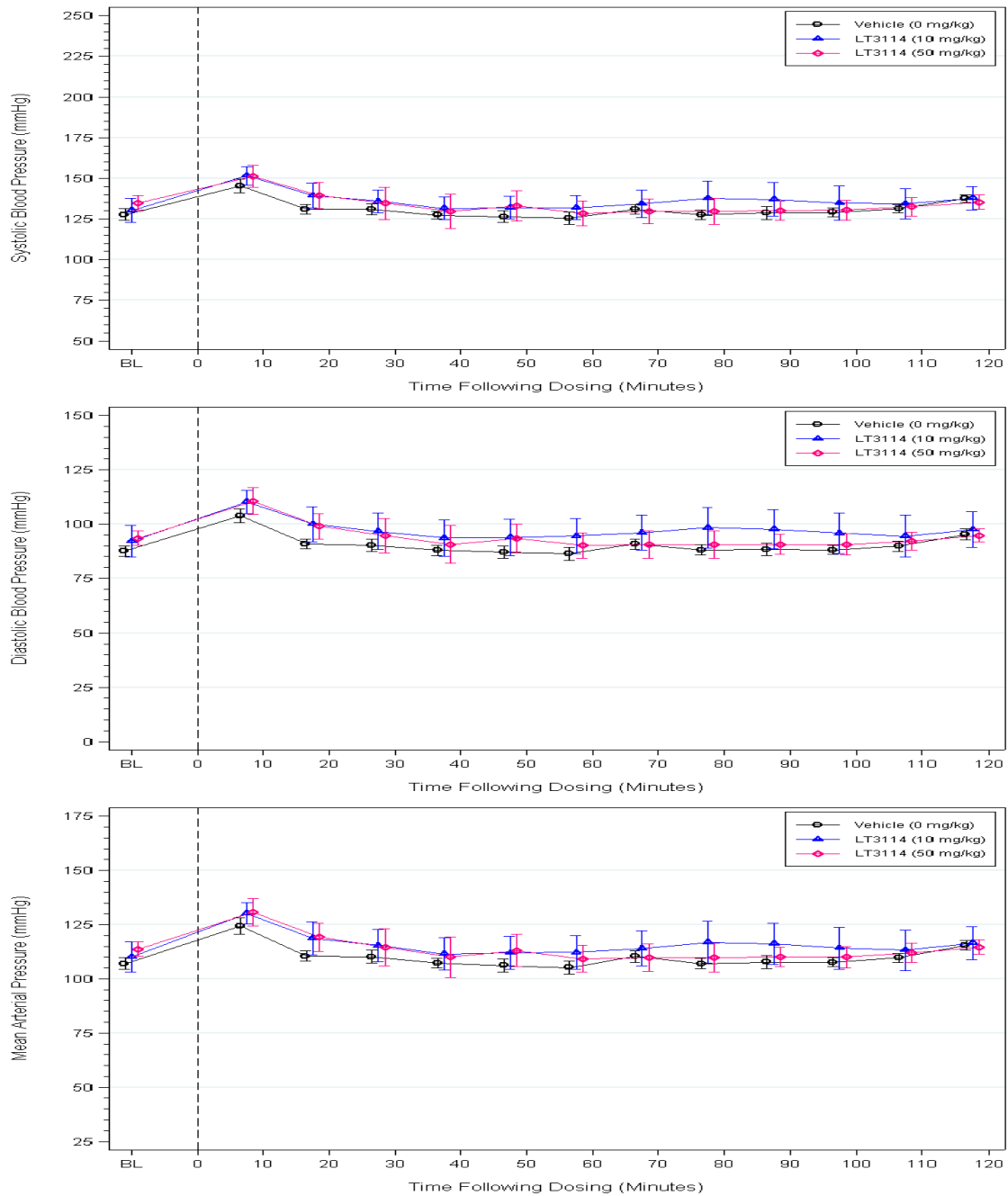


Figure 9. In Box, Blood Pressure Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

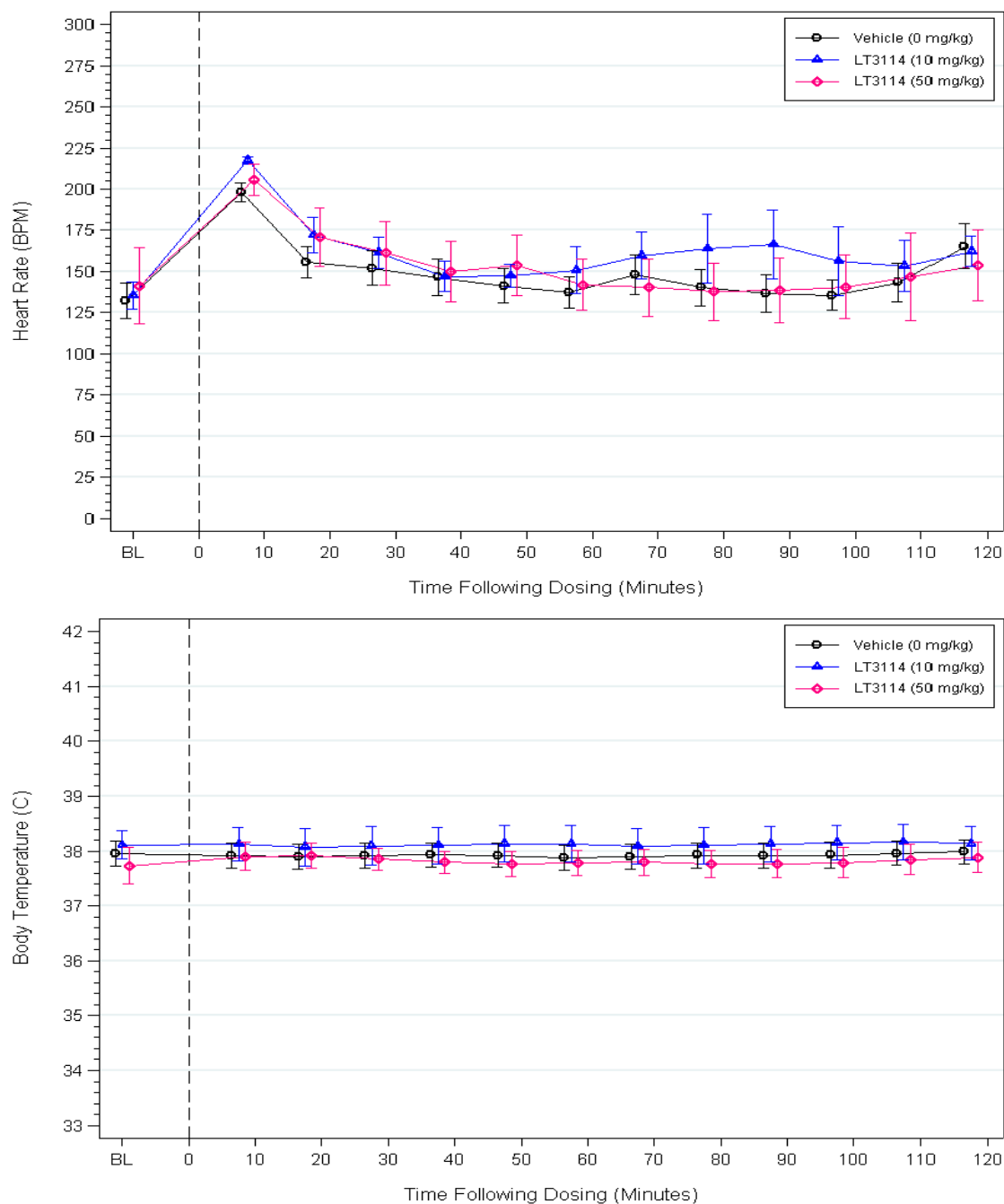


Figure 10. In Box, Heart Rate, and Body Temperature Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

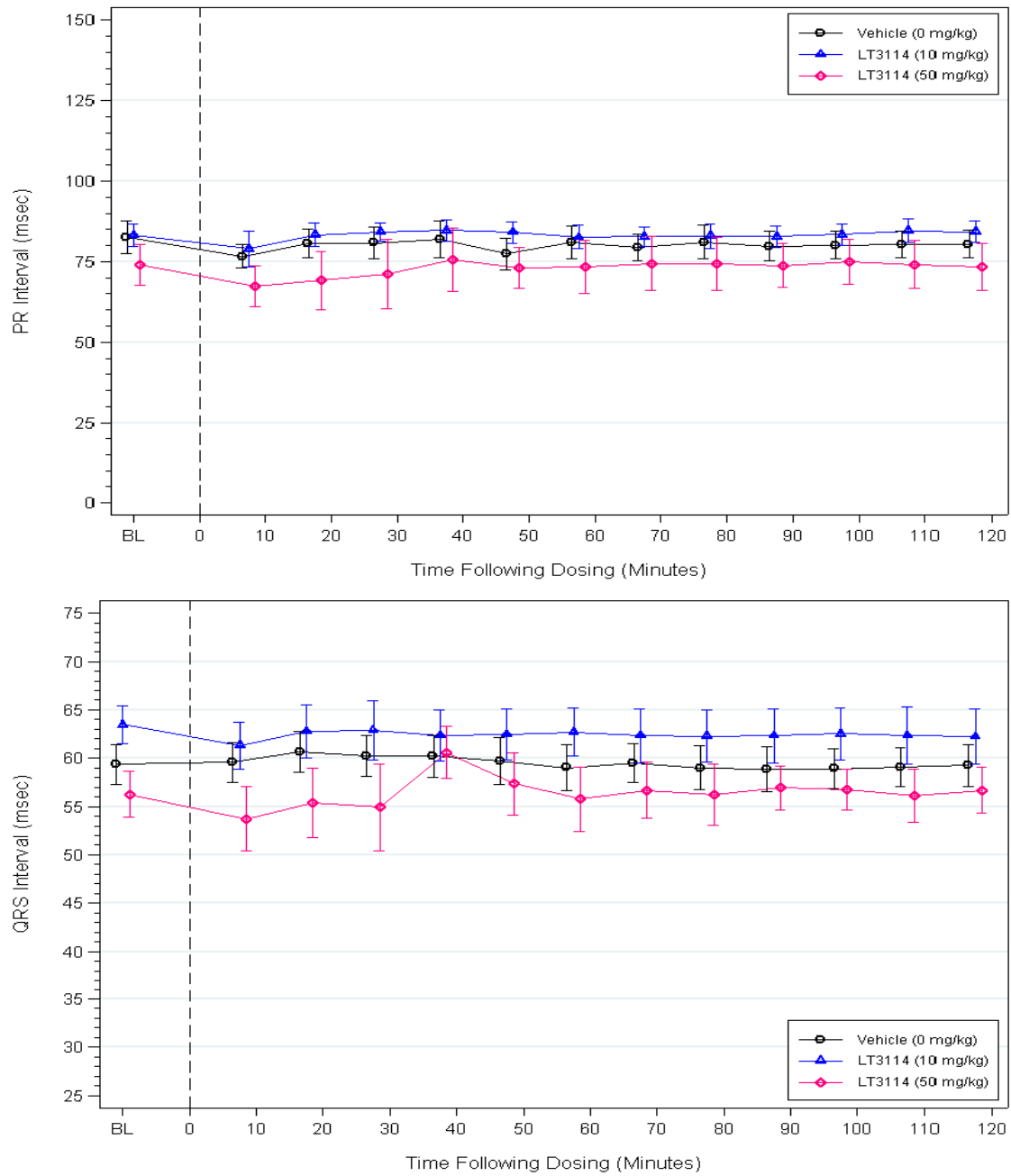


Figure 11. In Box, PR, and QRS Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

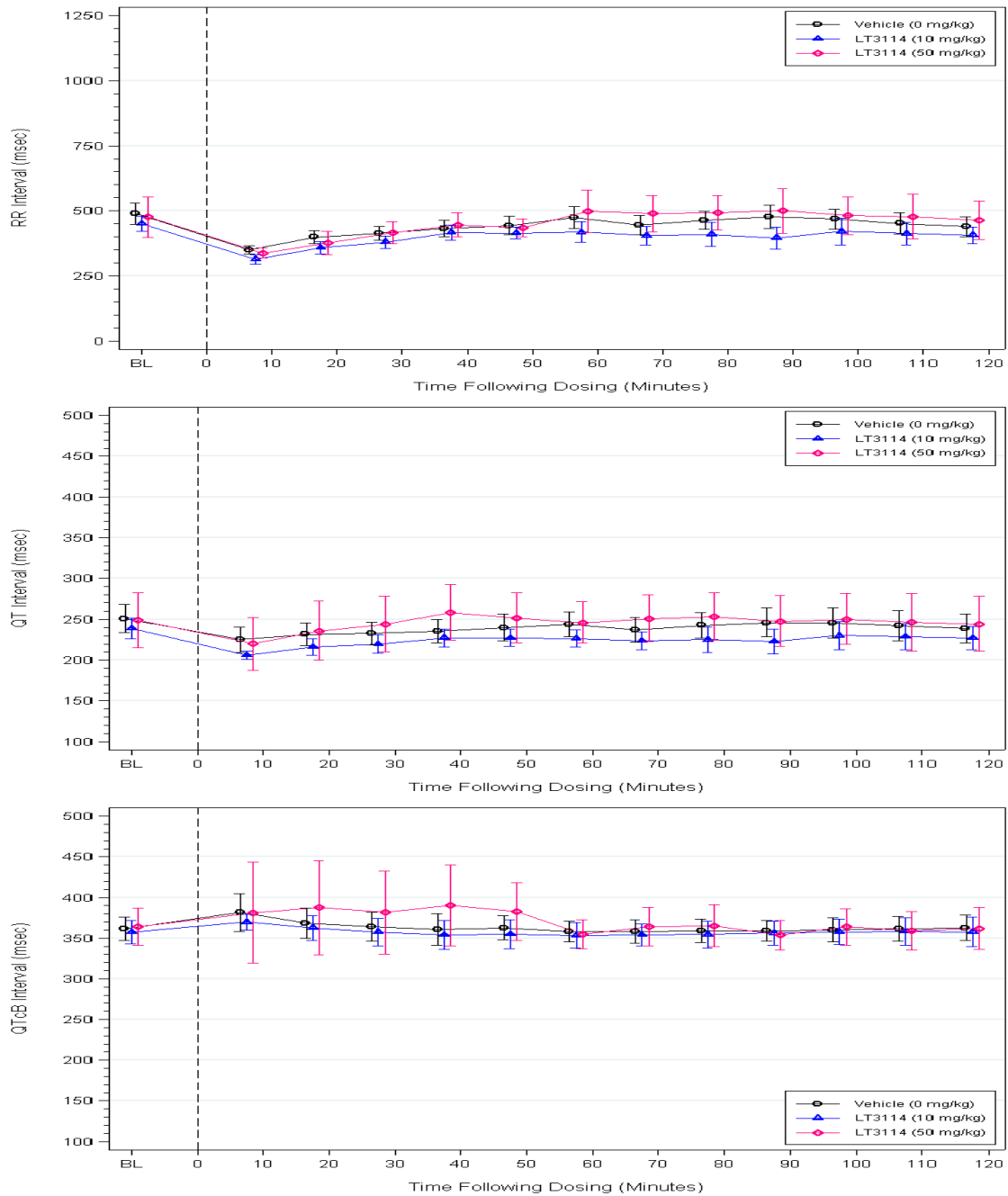


Figure 12. In Box, RR, QT, and QTcB Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 10-Minute Averages Following Dosing with LT3114 or Vehicle

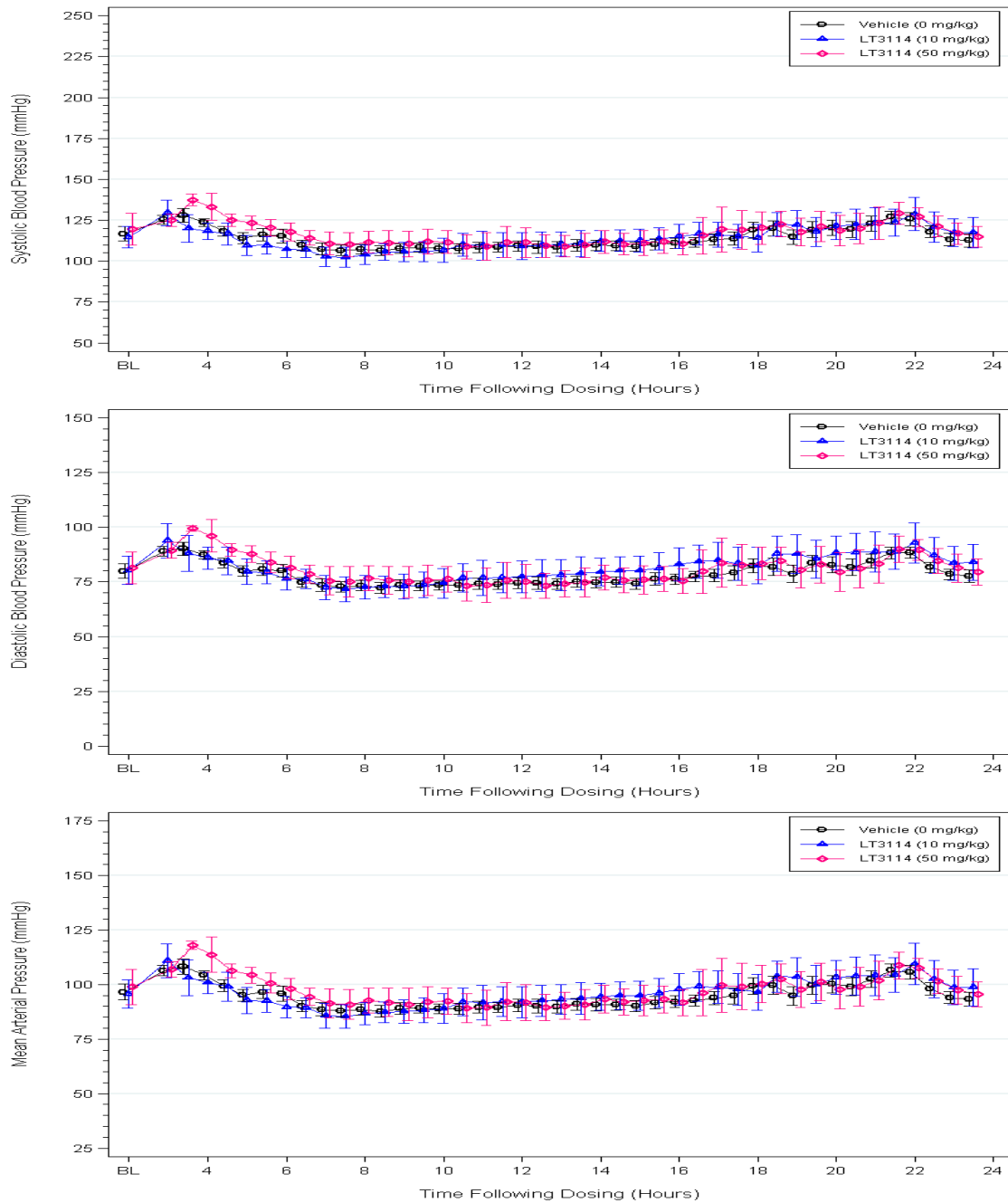


Figure 13. In Cage, Blood Pressure Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

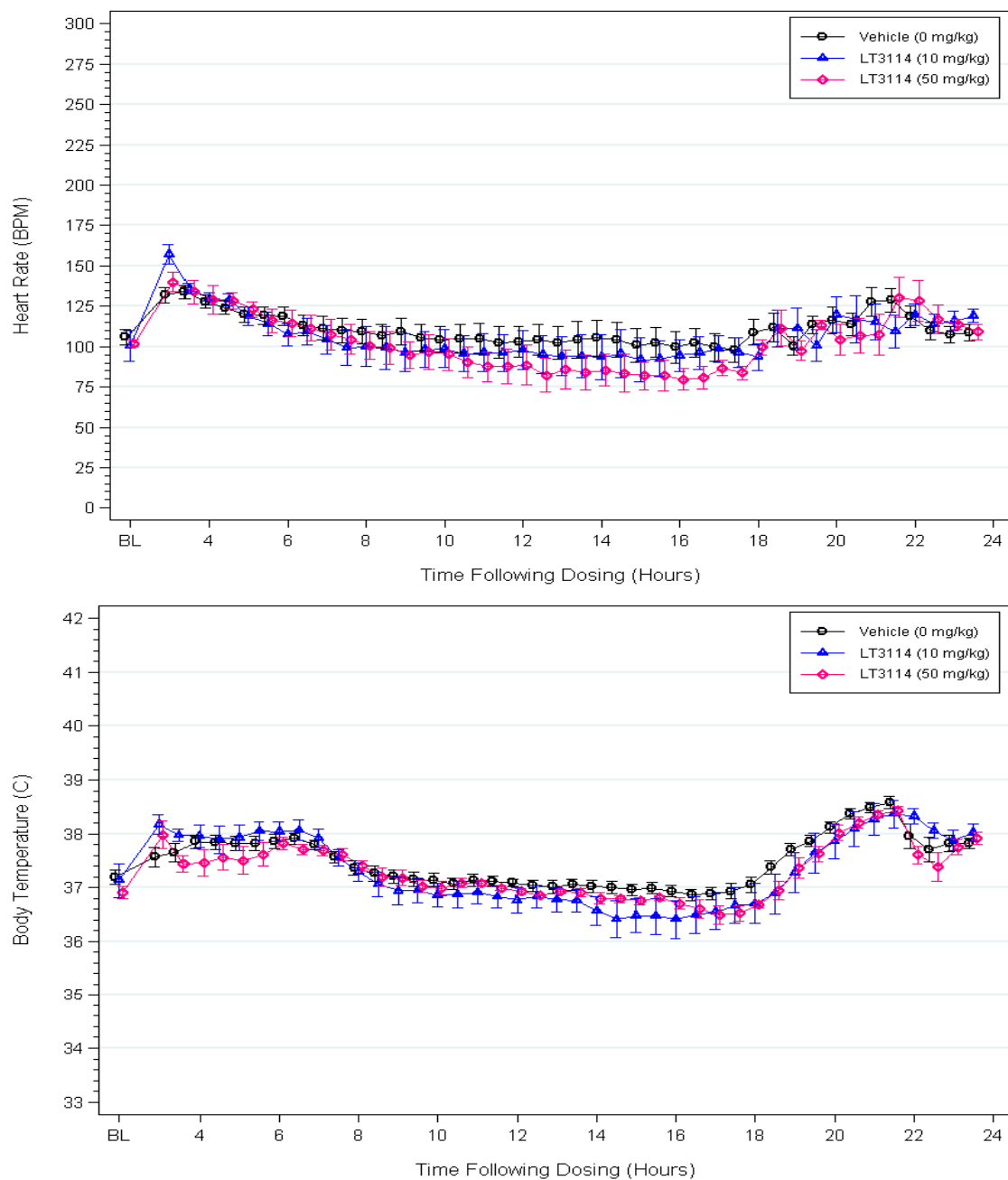


Figure 14. In Cage, Heart Rate, and Body Temperature Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

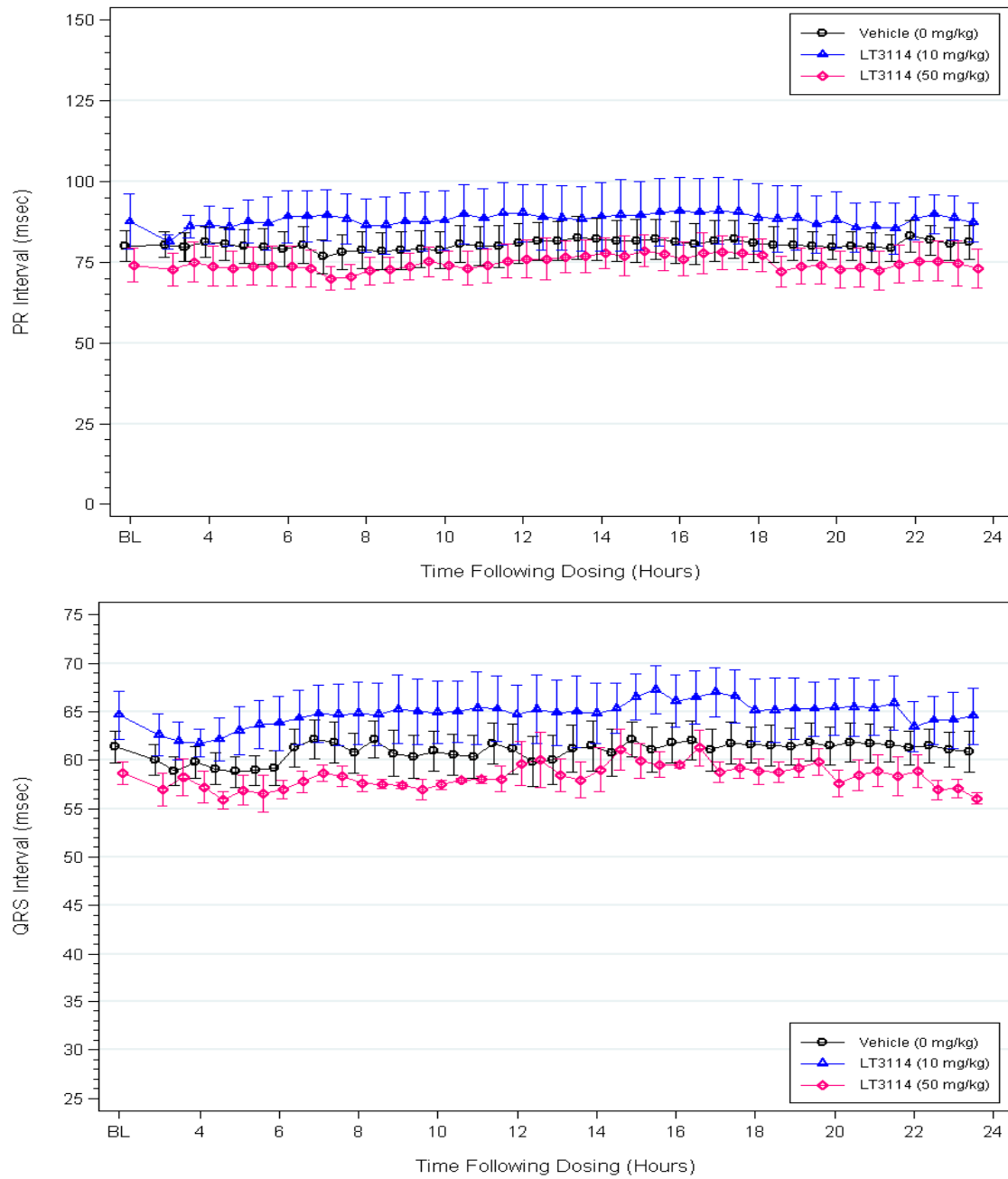


Figure 15. In Cage, PR, and QRS Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

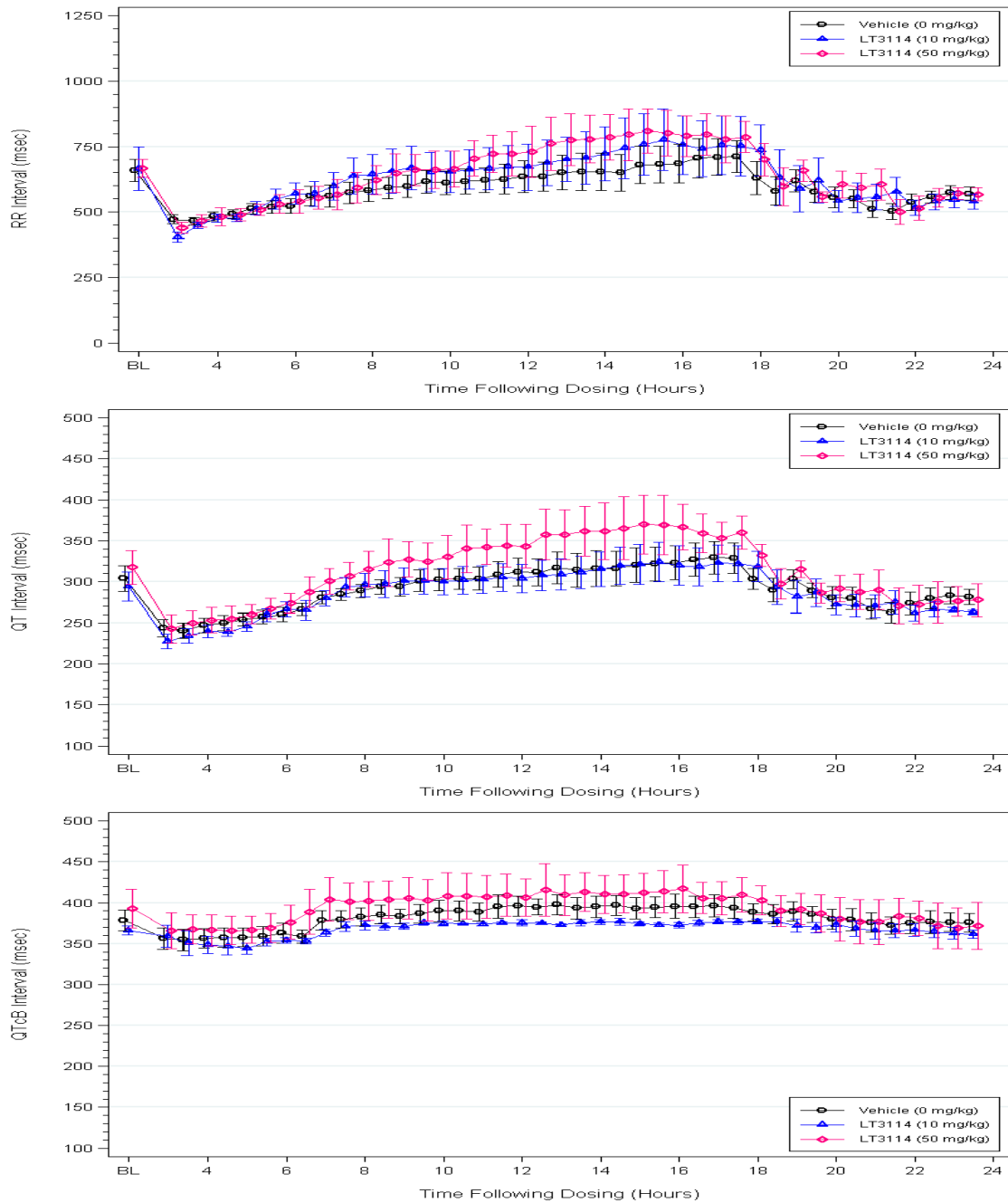


Figure 16. In Cage, RR, QT, and QTcB Interval Parameters, Dose Level Means (with $\pm$ Standard Error Bars) of Unadjusted 30-Minute Averages Following Dosing with LT3114 or Vehicle

APPENDIX E

INDIVIDUAL ANIMAL CARDIOVASCULAR AND ECG DATA

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
101	Chair	1	1	0	127	95	110	148	38	409	92	60	235	368
101	Chair	1	1	10	134	100	117	206	38	345	89	59	226	386
101	Chair	1	1	20	132	97	114	165	38	369	90	60	228	376
101	Chair	1	1	30	133	98	114	162	38	376	90	60	226	369
101	Chair	1	1	40	132	98	114	160	38	376	90	60	227	370
101	Chair	1	1	50	131	97	112	160	38	381	89	60	226	366
101	Chair	1	1	60	131	98	113	164	38	374	88	60	224	366
101	Chair	1	1	70	132	99	115	175	38	360	87	59	221	368
101	Chair	1	1	80	127	95	110	154	38	389	89	60	229	367
101	Chair	1	1	90	132	98	114	160	38	384	88	60	227	366
101	Chair	1	1	100	130	97	112	150	38	402	89	60	227	359
101	Chair	1	1	110	134	100	116	174	38	357	88	59	221	370
101	Chair	1	1	120	147	110	127	213	38	330	86	58	214	374
101	Cage	1	1	0	126	90	105	100	37	720	109	60	304	365
101	Cage	1	1	210	122	88	104	147	38	423	94	58	233	359
101	Cage	1	1	240	113	81	96	137	38	461	101	59	234	346
101	Cage	1	1	270	130	93	111	124	38	497	104	60	246	349
101	Cage	1	1	300	120	86	102	123	38	496	106	59	246	350
101	Cage	1	1	330	110	77	91	117	38	525	108	58	246	340
101	Cage	1	1	360	112	79	93	106	38	587	111	60	264	345
101	Cage	1	1	390	111	77	93	120	38	522	108	59	252	350
101	Cage	1	1	420	107	73	88	96	38	664	110	61	290	357
101	Cage	1	1	450	105	71	86	84	38	769	107	61	306	350
101	Cage	1	1	480	103	69	83	80	37	761	110	61	309	356
101	Cage	1	1	510	102	69	83	85	37	755	113	61	309	360
101	Cage	1	1	540	106	73	87	83	37	742	113	61	313	365
101	Cage	1	1	570	103	70	84	78	37	778	115	60	318	362
101	Cage	1	1	600	107	73	87	76	37	797	114	61	321	362
101	Cage	1	1	630	105	71	85	78	37	787	115	61	326	370
101	Cage	1	1	660	106	72	86	74	37	826	114	61	332	367
101	Cage	1	1	690	104	71	85	78	37	796	117	61	329	371
101	Cage	1	1	720	110	76	90	73	37	826	118	61	333	368
101	Cage	1	1	750	109	75	89	74	37	832	117	61	337	372
101	Cage	1	1	780	107	75	88	77	37	792	118	61	334	379
101	Cage	1	1	810	108	75	89	73	37	852	118	61	341	373
101	Cage	1	1	840	111	77	91	69	37	886	122	61	343	366
101	Cage	1	1	870	108	75	89	71	37	883	120	61	347	372
101	Cage	1	1	900	108	75	89	72	37	874	118	62	350	377
101	Cage	1	1	930	109	76	90	67	37	936	122	61	353	366
101	Cage	1	1	960	113	80	95	73	37	919	121	61	349	368
101	Cage	1	1	990	116	82	98	72	37	970	120	61	358	367
101	Cage	1	1	1020	114	81	96	75	36	967	119	61	361	371
101	Cage	1	1	1050	114	82	95	78	36	980	121	61	365	371

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
101	Cage	1	1	1080	119	87	100	82	37	932	118	61	349	365
101	Cage	1	1	1110	117	86	99	77	37	965	120	61	358	366
101	Cage	1	1	1140	129	92	108	85	37	760	118	60	326	380
101	Cage	1	1	1170	129	92	109	106	37	597	110	60	289	380
101	Cage	1	1	1200	129	93	109	106	38	601	108	60	286	374
101	Cage	1	1	1230	131	94	111	132	38	494	101	59	260	373
101	Cage	1	1	1260	131	94	111	122	38	518	105	59	268	374
101	Cage	1	1	1290	126	90	106	120	38	516	104	59	265	372
101	Cage	1	1	1320	135	95	113	129	38	487	101	58	257	371
101	Cage	1	1	1350	133	94	112	104	38	596	109	60	278	363
101	Cage	1	1	1380	122	83	100	110	38	593	106	59	276	364
101	Cage	1	1	1410	117	80	96	94	38	672	109	60	293	362
101	Cage	1	1	1440	120	83	100	120	38	539	109	60	267	367
101	Chair	2	4	0	146	113	128	139	38	436	93	60	240	363
101	Chair	2	4	10	159	126	141	219	38	321	88	58	211	373
101	Chair	2	4	20	157	123	139	184	38	339	88	59	220	378
101	Chair	2	4	30	152	121	136	175	38	349	87	59	219	371
101	Chair	2	4	40	149	118	132	156	38	387	90	60	229	369
101	Chair	2	4	50	152	119	134	157	38	385	90	60	226	365
101	Chair	2	4	60	149	117	132	157	38	385	89	60	226	365
101	Chair	2	4	70	151	118	133	167	38	369	87	59	222	366
101	Chair	2	4	80	152	120	135	159	38	380	89	60	225	366
101	Chair	2	4	90	151	118	133	169	38	364	87	60	223	369
101	Chair	2	4	100	149	117	132	156	38	389	89	60	230	369
101	Chair	2	4	110	154	121	137	182	38	347	86	59	219	373
101	Chair	2	4	120	154	121	136	179	38	353	86	59	220	371
101	Cage	2	4	0	134	100	114	81	36	877	114	60	335	362
101	Cage	2	4	210	151	117	133	175	38	357	86	58	219	367
101	Cage	2	4	240	144	112	127	138	38	442	95	59	238	359
101	Cage	2	4	270	128	99	112	120	37	511	104	60	249	350
101	Cage	2	4	300	133	101	116	133	37	471	103	59	238	348
101	Cage	2	4	330	126	96	110	122	37	502	107	59	247	350
101	Cage	2	4	360	125	95	108	102	38	635	111	61	275	346
101	Cage	2	4	390	122	91	105	97	38	642	113	60	275	345
101	Cage	2	4	420	120	89	102	90	38	690	113	61	290	350
101	Cage	2	4	450	119	88	101	86	37	718	113	61	297	351
101	Cage	2	4	480	118	87	100	77	37	797	111	61	316	355
101	Cage	2	4	510	119	87	100	73	37	826	111	61	323	357
101	Cage	2	4	540	118	86	99	71	37	863	112	61	331	357
101	Cage	2	4	570	122	89	103	70	37	877	114	61	335	359
101	Cage	2	4	600	122	90	104	74	37	844	114	61	336	368
101	Cage	2	4	630	128	94	108	73	37	860	114	61	339	368
101	Cage	2	4	660	128	95	109	72	37	850	116	61	341	371

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
101	Cage	2	4	690	135	100	115	70	37	867	116	61	341	368
101	Cage	2	4	720	130	97	111	72	36	875	119	62	346	373
101	Cage	2	4	750	131	98	112	70	36	903	117	61	347	367
101	Cage	2	4	780	132	99	113	70	36	905	119	61	350	370
101	Cage	2	4	810	131	99	113	71	36	897	118	61	350	373
101	Cage	2	4	840	134	101	115	70	36	948	118	61	354	366
101	Cage	2	4	870	129	97	111	68	36	954	120	61	358	368
101	Cage	2	4	900	130	98	112	70	36	998	122	62	363	365
101	Cage	2	4	930	131	99	113	68	36	1014	121	61	367	367
101	Cage	2	4	960	133	102	115	74	36	1042	122	62	372	366
101	Cage	2	4	990	135	105	118	78	36	1032	122	61	373	369
101	Cage	2	4	1020	136	106	119	80	36	965	121	61	366	377
101	Cage	2	4	1050	138	109	121	84	36	1000	121	61	369	371
101	Cage	2	4	1080	136	106	119	77	36	1028	120	61	370	368
101	Cage	2	4	1110	139	108	121	76	36	965	120	61	359	370
101	Cage	2	4	1140	144	112	125	78	36	941	119	61	353	367
101	Cage	2	4	1170	149	114	130	77	36	841	119	61	336	370
101	Cage	2	4	1200	145	111	126	75	37	851	113	61	333	365
101	Cage	2	4	1230	145	111	127	97	37	644	114	60	300	379
101	Cage	2	4	1260	149	114	129	92	37	686	108	60	301	369
101	Cage	2	4	1290	151	116	132	103	38	608	108	60	283	367
101	Cage	2	4	1320	147	112	128	101	38	625	109	60	286	365
101	Cage	2	4	1350	158	120	137	113	38	554	108	59	272	368
101	Cage	2	4	1380	146	111	127	101	38	606	108	60	285	369
101	Cage	2	4	1410	141	106	121	112	37	572	108	60	270	360
101	Cage	2	4	1440	143	108	123	114	38	577	105	60	266	354
102	Chair	1	1	0	132	91	112	183	38	352	73	70	230	388
102	Chair	1	1	10	142	102	123	214	38	278	67	68	206	392
102	Chair	1	1	20	130	88	109	179	38	339	75	71	229	392
102	Chair	1	1	30	130	86	108	166	38	362	73	71	238	395
102	Chair	1	1	40	128	85	107	159	38	378	72	71	243	395
102	Chair	1	1	50	131	89	110	174	38	372	71	69	239	392
102	Chair	1	1	60	127	85	107	157	37	385	73	71	242	390
102	Chair	1	1	70	131	89	110	174	37	369	73	70	236	390
102	Chair	1	1	80	130	88	109	170	37	382	74	71	240	390
102	Chair	1	1	90	128	87	108	166	37	390	73	70	244	392
102	Chair	1	1	100	124	83	103	152	37	398	74	70	248	394
102	Chair	1	1	110	123	82	102	146	37	413	75	70	255	398
102	Chair	1	1	120	132	91	112	164	37	399	76	72	252	399
102	Cage	1	1	0	111	78	94	122	37	503	76	67	272	385
102	Cage	1	1	210	114	78	95	144	38	424	76	68	250	384
102	Cage	1	1	240	110	77	92	139	38	439	77	66	246	372
102	Cage	1	1	270	113	80	96	141	38	429	79	68	249	380

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
102	Cage	1	1	300	104	74	88	134	38	448	83	67	244	364
102	Cage	1	1	330	99	71	84	130	38	463	80	68	241	355
102	Cage	1	1	360	100	71	84	131	38	465	79	67	247	363
102	Cage	1	1	390	101	72	85	130	38	466	81	68	252	370
102	Cage	1	1	420	102	72	85	127	38	473	81	69	250	364
102	Cage	1	1	450	97	67	81	129	38	467	79	72	253	370
102	Cage	1	1	480	98	68	82	134	38	460	77	70	258	380
102	Cage	1	1	510	100	68	84	134	37	457	78	68	260	385
102	Cage	1	1	540	97	67	81	137	37	448	75	71	253	379
102	Cage	1	1	570	101	70	85	141	37	439	66	72	250	378
102	Cage	1	1	600	113	75	93	144	37	440	64	70	249	376
102	Cage	1	1	630	112	76	93	146	37	435	64	70	249	378
102	Cage	1	1	660	111	77	93	148	37	428	64	69	246	377
102	Cage	1	1	690	109	75	91	147	37	422	64	70	245	377
102	Cage	1	1	720	105	73	89	151	37	411	67	70	246	384
102	Cage	1	1	750	111	79	96	143	37	422	68	71	248	381
102	Cage	1	1	780	102	72	87	142	37	427	70	70	252	385
102	Cage	1	1	810	100	72	86	145	37	423	72	68	253	390
102	Cage	1	1	840	101	73	87	147	37	417	71	70	244	377
102	Cage	1	1	870	104	74	89	153	37	412	68	71	243	379
102	Cage	1	1	900	105	76	90	145	37	421	69	69	252	389
102	Cage	1	1	930	107	78	92	150	37	413	68	69	245	381
102	Cage	1	1	960	106	77	91	149	37	413	67	69	242	378
102	Cage	1	1	990	106	77	91	150	37	407	67	69	243	381
102	Cage	1	1	1020	109	79	94	149	37	413	70	68	245	381
102	Cage	1	1	1050	113	82	97	150	37	410	68	67	242	379
102	Cage	1	1	1080	108	78	92	142	37	427	71	66	249	382
102	Cage	1	1	1110	112	82	96	143	37	431	71	66	250	381
102	Cage	1	1	1140	120	85	101	140	37	417	74	69	246	382
102	Cage	1	1	1170	109	80	93	132	38	420	75	68	247	382
102	Cage	1	1	1200	109	81	94	132	38	451	78	68	255	381
102	Cage	1	1	1230	111	81	95	133	38	448	80	69	258	386
102	Cage	1	1	1260	110	81	94	131	38	445	79	69	259	389
102	Cage	1	1	1290	112	82	96	135	38	458	79	69	263	390
102	Cage	1	1	1320	119	85	101	152	38	407	76	68	247	388
102	Cage	1	1	1350	105	77	90	135	39	445	84	70	255	382
102	Cage	1	1	1380	98	70	83	128	38	472	89	69	262	381
102	Cage	1	1	1410	98	71	83	131	38	461	83	68	251	371
102	Cage	1	1	1440	97	70	82	130	38	468	85	68	258	377
102	Chair	2	4	0	122	84	103	143	38	425	76	69	254	391
102	Chair	2	4	10	142	103	123	219	38	268	63	68	204	394
102	Chair	2	4	20	131	92	112	189	37	319	73	70	220	391
102	Chair	2	4	30	129	88	109	168	37	358	76	71	235	393

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
102	Chair	2	4	40	127	86	107	161	38	376	75	69	240	392
102	Chair	2	4	50	126	86	106	159	38	382	74	70	243	393
102	Chair	2	4	60	129	91	110	182	38	352	72	69	231	390
102	Chair	2	4	70	127	89	108	178	37	360	74	70	229	382
102	Chair	2	4	80	128	88	108	173	37	368	72	69	237	391
102	Chair	2	4	90	125	86	106	161	38	380	73	70	238	387
102	Chair	2	4	100	124	85	104	151	38	398	73	70	248	393
102	Chair	2	4	110	124	84	104	149	38	409	74	70	250	391
102	Chair	2	4	120	131	92	112	166	38	402	75	70	251	396
102	Cage	2	4	0	106	75	90	122	37	504	78	65	265	375
102	Cage	2	4	210	118	82	100	155	38	394	75	69	243	387
102	Cage	2	4	240	106	75	89	142	38	423	78	67	248	382
102	Cage	2	4	270	104	75	89	136	38	445	79	66	247	372
102	Cage	2	4	300	102	73	86	138	38	438	80	67	247	373
102	Cage	2	4	330	97	70	82	133	38	450	84	67	237	354
102	Cage	2	4	360	101	73	86	134	38	453	77	67	245	364
102	Cage	2	4	390	99	70	83	127	38	465	82	68	243	356
102	Cage	2	4	420	99	71	83	129	38	463	83	69	235	346
102	Cage	2	4	450	94	66	79	124	38	483	87	68	256	368
102	Cage	2	4	480	93	65	78	124	37	489	86	68	269	384
102	Cage	2	4	510	98	69	83	129	37	475	77	68	262	381
102	Cage	2	4	540	104	71	87	131	37	470	72	69	257	376
102	Cage	2	4	570	99	68	83	124	37	488	74	71	262	375
102	Cage	2	4	600	101	70	85	127	37	486	75	70	261	375
102	Cage	2	4	630	97	67	81	129	37	475	72	70	262	380
102	Cage	2	4	660	101	70	85	126	37	481	74	70	258	373
102	Cage	2	4	690	96	67	82	127	37	476	75	71	259	375
102	Cage	2	4	720	100	70	85	126	37	485	76	70	263	378
102	Cage	2	4	750	96	68	82	128	37	473	79	68	264	384
102	Cage	2	4	780	99	70	85	127	37	478	77	68	260	377
102	Cage	2	4	810	97	69	83	127	37	480	77	69	260	375
102	Cage	2	4	840	100	72	86	132	37	463	77	68	258	380
102	Cage	2	4	870	111	78	94	133	36	457	72	68	257	381
102	Cage	2	4	900	108	76	92	137	36	448	73	67	257	384
102	Cage	2	4	930	110	78	94	128	36	473	73	67	257	375
102	Cage	2	4	960	109	78	93	123	37	487	75	68	262	375
102	Cage	2	4	990	113	81	97	122	37	494	75	69	261	373
102	Cage	2	4	1020	109	77	93	123	37	461	74	69	258	381
102	Cage	2	4	1050	111	78	95	121	37	487	76	68	267	384
102	Cage	2	4	1080	111	78	95	121	37	505	79	68	272	384
102	Cage	2	4	1110	105	74	89	117	37	497	76	67	271	385
102	Cage	2	4	1140	113	79	96	129	37	469	76	68	267	391
102	Cage	2	4	1170	113	81	96	122	37	480	77	69	272	395

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
102	Cage	2	4	1200	108	79	92	122	38	439	74	68	256	388
102	Cage	2	4	1230	110	81	94	130	38	473	78	69	271	396
102	Cage	2	4	1260	107	79	91	119	38	449	77	69	263	393
102	Cage	2	4	1290	113	83	97	126	38	488	79	69	272	392
102	Cage	2	4	1320	115	84	98	123	38	482	79	69	268	388
102	Cage	2	4	1350	114	84	98	130	38	466	82	69	260	383
102	Cage	2	4	1380	102	74	88	124	38	481	85	69	265	383
102	Cage	2	4	1410	100	72	85	134	38	456	88	67	257	382
102	Cage	2	4	1440	98	70	83	128	38	473	87	68	258	376
103	Chair	1	1	0	112	77	93	116	38	543	86	64	259	353
103	Chair	1	1	10	130	90	108	179	38	378	84	64	225	366
103	Chair	1	1	20	123	85	103	146	38	411	85	64	230	359
103	Chair	1	1	30	119	81	99	132	38	458	88	63	236	350
103	Chair	1	1	40	120	82	99	131	38	467	88	64	240	353
103	Chair	1	1	50	114	78	94	124	38	490	86	63	243	348
103	Chair	1	1	60	113	78	94	115	38	524	86	63	251	350
103	Chair	1	1	70	113	78	94	115	38	526	86	63	251	349
103	Chair	1	1	80	116	80	97	127	38	480	85	63	242	351
103	Chair	1	1	90	111	76	91	112	38	540	85	64	252	345
103	Chair	1	1	100	117	81	97	119	38	511	86	63	249	350
103	Chair	1	1	110	124	84	102	136	38	455	86	64	240	358
103	Chair	1	1	120	133	91	109	198	38	466	88	64	241	354
103	Cage	1	1	0	106	71	86	91	37	735	79	62	315	372
103	Cage	1	1	210	121	86	102	132	38	482	88	62	239	345
103	Cage	1	1	240	138	97	116	127	37	505	82	60	243	344
103	Cage	1	1	270	123	86	102	120	37	519	84	60	250	348
103	Cage	1	1	300	120	85	100	116	38	535	81	59	253	348
103	Cage	1	1	330	114	82	95	115	38	522	79	60	259	360
103	Cage	1	1	360	118	82	97	118	38	535	78	58	257	354
103	Cage	1	1	390	110	77	91	109	38	554	75	58	268	361
103	Cage	1	1	420	104	71	85	102	38	591	73	59	280	365
103	Cage	1	1	450	101	68	82	94	38	643	78	58	297	371
103	Cage	1	1	480	100	68	82	95	37	639	80	58	299	375
103	Cage	1	1	510	98	66	80	90	37	667	80	58	304	373
103	Cage	1	1	540	97	65	79	93	37	649	80	58	302	376
103	Cage	1	1	570	97	66	80	93	37	648	84	59	301	376
103	Cage	1	1	600	97	66	80	92	37	661	82	58	304	375
103	Cage	1	1	630	96	65	79	96	37	634	80	58	300	380
103	Cage	1	1	660	96	66	79	94	37	647	82	58	303	378
103	Cage	1	1	690	96	67	80	94	37	646	82	59	304	380
103	Cage	1	1	720	97	66	80	96	37	642	82	58	304	382
103	Cage	1	1	750	96	66	80	93	37	659	83	58	307	380
103	Cage	1	1	780	95	66	79	97	37	637	82	58	305	384

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
103	Cage	1	1	810	97	67	80	91	37	671	82	59	313	384
103	Cage	1	1	840	98	68	82	87	37	693	84	59	319	383
103	Cage	1	1	870	99	67	81	97	37	635	82	58	308	389
103	Cage	1	1	900	100	69	83	94	37	666	84	58	310	382
103	Cage	1	1	930	100	67	82	90	37	684	82	60	311	379
103	Cage	1	1	960	95	64	78	85	37	711	81	58	317	378
103	Cage	1	1	990	99	67	82	82	37	742	79	58	324	378
103	Cage	1	1	1020	103	72	85	85	37	799	77	58	335	377
103	Cage	1	1	1050	110	74	90	85	37	716	78	58	318	379
103	Cage	1	1	1080	99	68	82	79	37	779	80	62	331	377
103	Cage	1	1	1110	106	72	88	82	37	772	79	59	326	373
103	Cage	1	1	1140	114	77	94	98	37	619	80	58	293	374
103	Cage	1	1	1170	108	75	90	99	38	621	81	60	290	372
103	Cage	1	1	1200	108	76	90	90	38	676	81	62	304	373
103	Cage	1	1	1230	109	76	90	92	38	658	83	62	297	369
103	Cage	1	1	1260	109	77	91	89	38	685	82	62	304	370
103	Cage	1	1	1290	113	79	94	97	38	636	81	61	294	371
103	Cage	1	1	1320	116	82	97	97	39	633	81	62	289	366
103	Cage	1	1	1350	118	85	100	104	38	587	81	59	279	366
103	Cage	1	1	1380	121	85	101	99	38	621	81	59	283	361
103	Cage	1	1	1410	108	76	90	92	38	657	78	58	295	366
103	Cage	1	1	1440	111	79	93	101	38	609	81	59	285	367
103	Chair	2	4	0	115	80	96	111	38	537	85	64	258	354
103	Chair	2	4	10	142	102	121	210	38	355	81	62	217	365
103	Chair	2	4	20	123	87	104	141	38	426	86	64	235	362
103	Chair	2	4	30	121	84	101	133	38	452	86	63	237	354
103	Chair	2	4	40	116	80	96	120	38	504	86	64	245	347
103	Chair	2	4	50	119	83	99	128	38	478	86	63	242	352
103	Chair	2	4	60	113	79	94	114	38	531	85	64	251	345
103	Chair	2	4	70	114	80	95	118	38	512	85	64	248	348
103	Chair	2	4	80	113	78	94	111	38	545	85	64	256	348
103	Chair	2	4	90	114	79	95	118	38	516	85	63	250	350
103	Chair	2	4	100	110	76	92	108	38	566	86	63	261	349
103	Chair	2	4	110	114	77	94	112	38	537	86	63	258	355
103	Chair	2	4	120	121	82	100	136	38	496	85	63	249	355
103	Cage	2	4	0	106	71	86	89	37	715	79	62	314	375
103	Cage	2	4	210	118	85	100	149	38	450	82	61	241	361
103	Cage	2	4	240	113	82	96	127	38	483	86	60	244	352
103	Cage	2	4	270	123	88	104	127	38	499	81	60	247	351
103	Cage	2	4	300	122	88	103	122	38	511	81	58	246	346
103	Cage	2	4	330	108	78	91	107	38	564	83	58	261	349
103	Cage	2	4	360	105	75	88	105	38	581	82	58	269	355
103	Cage	2	4	390	102	71	84	97	38	625	81	58	283	359

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
103	Cage	2	4	420	101	70	83	104	38	585	81	58	275	361
103	Cage	2	4	450	93	62	76	93	38	649	80	58	294	365
103	Cage	2	4	480	92	61	76	86	38	702	80	58	311	372
103	Cage	2	4	510	92	61	75	87	37	696	81	58	315	379
103	Cage	2	4	540	94	62	76	85	37	707	83	58	317	378
103	Cage	2	4	570	94	63	77	85	37	703	85	57	317	378
103	Cage	2	4	600	93	63	77	89	37	673	84	58	310	378
103	Cage	2	4	630	95	64	78	91	37	659	85	58	306	378
103	Cage	2	4	660	98	66	81	91	37	663	86	58	306	377
103	Cage	2	4	690	98	67	81	89	37	674	85	57	307	374
103	Cage	2	4	720	97	66	80	89	37	676	84	57	310	377
103	Cage	2	4	750	98	67	81	92	37	658	82	58	307	380
103	Cage	2	4	780	99	68	83	86	37	701	79	58	315	377
103	Cage	2	4	810	103	70	86	84	37	718	77	57	317	375
103	Cage	2	4	840	100	69	83	83	37	730	77	57	319	375
103	Cage	2	4	870	98	68	82	85	37	769	82	58	330	377
103	Cage	2	4	900	99	69	82	81	36	825	82	61	342	377
103	Cage	2	4	930	97	67	80	75	36	864	82	65	350	377
103	Cage	2	4	960	101	70	84	77	36	864	83	65	349	376
103	Cage	2	4	990	100	70	83	81	36	798	83	62	340	382
103	Cage	2	4	1020	105	74	87	80	36	819	83	63	341	378
103	Cage	2	4	1050	103	76	86	90	36	857	83	65	348	377
103	Cage	2	4	1080	103	75	87	88	36	828	84	64	343	379
103	Cage	2	4	1110	103	74	86	90	36	796	79	59	333	375
103	Cage	2	4	1140	117	82	97	112	36	617	81	59	294	379
103	Cage	2	4	1170	111	76	92	106	37	575	81	59	277	367
103	Cage	2	4	1200	109	77	91	100	38	615	80	61	286	368
103	Cage	2	4	1230	112	79	94	108	38	578	83	61	274	363
103	Cage	2	4	1260	111	78	93	101	38	613	82	61	283	365
103	Cage	2	4	1290	108	76	90	92	38	666	81	62	294	363
103	Cage	2	4	1320	111	77	92	85	38	715	78	63	305	364
103	Cage	2	4	1350	117	82	98	105	38	581	81	59	279	369
103	Cage	2	4	1380	121	85	101	107	38	578	82	59	272	359
103	Cage	2	4	1410	117	81	97	113	38	561	78	59	270	362
103	Cage	2	4	1440	115	81	96	125	38	513	78	59	259	364
104	Chair	1	1	0	129	86	106	129	39	465	93	57	208	305
104	Chair	1	1	10	160	108	133	213	39	331	84	58	192	334
104	Chair	1	1	20	137	91	114	162	39	372	89	58	192	315
104	Chair	1	1	30	134	89	111	150	39	399	92	58	192	304
104	Chair	1	1	40	128	85	106	140	39	430	91	58	197	301
104	Chair	1	1	50	130	86	107	141	39	437	92	58	199	302
104	Chair	1	1	60	135	90	112	156	39	405	89	57	194	305
104	Chair	1	1	70	135	91	112	155	39	417	89	57	195	302

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
104	Chair	1	1	80	132	88	109	147	39	428	90	58	196	300
104	Chair	1	1	90	137	92	113	162	39	384	89	57	189	305
104	Chair	1	1	100	132	88	109	149	39	407	90	57	191	300
104	Chair	1	1	110	138	91	113	150	39	403	90	56	192	303
104	Chair	1	1	120	135	89	110	152	39	408	91	56	193	304
104	Cage	1	1	0	112	74	92	110	38	548	81	69	247	338
104	Cage	1	1	210	130	90	109	133	38	460	86	61	205	303
104	Cage	1	1	240	127	89	107	133	38	467	85	61	210	308
104	Cage	1	1	270	115	80	97	129	38	471	86	60	212	309
104	Cage	1	1	300	111	78	93	127	38	476	83	62	216	313
104	Cage	1	1	330	108	75	89	113	38	540	84	59	229	313
104	Cage	1	1	360	109	76	91	122	38	503	82	62	225	319
104	Cage	1	1	390	106	73	88	113	38	537	80	66	232	318
104	Cage	1	1	420	103	71	85	109	38	565	78	68	244	325
104	Cage	1	1	450	107	72	88	106	38	560	81	69	263	352
104	Cage	1	1	480	107	72	88	107	38	577	79	71	266	352
104	Cage	1	1	510	110	74	91	110	38	599	79	71	276	359
104	Cage	1	1	540	111	74	91	110	38	582	79	70	277	365
104	Cage	1	1	570	111	74	91	111	38	562	80	70	273	366
104	Cage	1	1	600	107	71	87	104	37	596	80	70	283	369
104	Cage	1	1	630	108	72	88	108	37	580	80	70	279	367
104	Cage	1	1	660	107	72	88	108	37	570	80	70	281	372
104	Cage	1	1	690	108	72	88	103	37	615	82	70	286	368
104	Cage	1	1	720	111	75	92	109	38	570	80	71	282	375
104	Cage	1	1	750	113	76	94	114	37	548	81	71	275	373
104	Cage	1	1	780	115	77	95	114	37	566	83	70	277	370
104	Cage	1	1	810	116	78	97	125	37	512	83	71	263	368
104	Cage	1	1	840	113	77	94	131	37	482	82	70	253	365
104	Cage	1	1	870	113	77	94	127	37	495	84	70	254	362
104	Cage	1	1	900	114	78	95	128	37	509	82	70	258	363
104	Cage	1	1	930	115	78	96	120	38	542	82	71	266	363
104	Cage	1	1	960	113	76	93	111	37	581	81	72	284	374
104	Cage	1	1	990	118	80	99	112	37	567	80	72	280	374
104	Cage	1	1	1020	118	80	98	110	37	592	80	73	288	378
104	Cage	1	1	1050	115	78	95	101	38	620	80	73	298	381
104	Cage	1	1	1080	117	80	98	107	38	604	79	74	294	382
104	Cage	1	1	1110	116	80	97	111	38	569	80	72	284	379
104	Cage	1	1	1140	111	76	93	118	38	509	79	72	258	363
104	Cage	1	1	1170	108	72	88	92	38	606	82	71	298	384
104	Cage	1	1	1200	118	80	98	135	38	478	79	72	263	382
104	Cage	1	1	1230	118	79	98	149	39	428	77	70	228	350
104	Cage	1	1	1260	118	79	98	135	39	460	76	71	231	342
104	Cage	1	1	1290	123	83	102	161	39	434	78	70	228	347

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
104	Cage	1	1	1320	126	85	104	141	39	444	78	69	227	342
104	Cage	1	1	1350	122	84	102	127	38	478	85	67	239	346
104	Cage	1	1	1380	115	78	94	118	38	525	81	68	252	350
104	Cage	1	1	1410	111	75	91	116	38	540	78	70	264	360
104	Cage	1	1	1440	110	75	90	114	38	546	77	71	260	354
104	Chair	2	4	0	138	91	114	149	39	405	80	62	204	321
104	Chair	2	4	10	164	110	136	222	39	310	83	57	193	348
104	Chair	2	4	20	146	98	121	175	39	348	87	58	189	320
104	Chair	2	4	30	141	94	117	169	39	360	89	58	188	313
104	Chair	2	4	40	134	90	111	151	39	401	88	57	195	309
104	Chair	2	4	50	131	88	108	146	39	413	86	58	198	308
104	Chair	2	4	60	136	91	113	151	39	403	84	58	200	315
104	Chair	2	4	70	145	97	120	175	39	377	86	57	196	320
104	Chair	2	4	80	158	107	131	213	39	345	86	57	183	313
104	Chair	2	4	90	158	107	131	218	39	322	86	57	180	318
104	Chair	2	4	100	156	105	129	209	39	329	85	57	183	320
104	Chair	2	4	110	145	95	119	170	39	358	92	57	187	313
104	Chair	2	4	120	145	95	118	168	39	376	90	57	189	309
104	Cage	2	4	0	112	75	92	110	38	568	80	71	262	353
104	Cage	2	4	210	131	92	111	150	39	417	82	62	207	321
104	Cage	2	4	240	118	84	100	135	38	449	86	62	208	310
104	Cage	2	4	270	118	81	98	135	38	458	82	61	216	321
104	Cage	2	4	300	109	76	91	124	39	492	79	64	225	321
104	Cage	2	4	330	107	74	88	114	38	541	77	68	237	324
104	Cage	2	4	360	107	75	89	116	38	528	78	69	249	344
104	Cage	2	4	390	106	73	87	111	38	550	80	69	265	358
104	Cage	2	4	420	108	75	89	115	38	543	80	69	261	355
104	Cage	2	4	450	105	72	87	114	38	545	79	71	272	370
104	Cage	2	4	480	106	73	88	112	38	563	77	72	278	372
104	Cage	2	4	510	107	73	88	111	38	579	77	72	283	373
104	Cage	2	4	540	106	72	87	110	38	583	78	71	283	373
104	Cage	2	4	570	107	72	88	107	38	607	77	72	289	373
104	Cage	2	4	600	107	71	88	102	38	610	77	71	296	380
104	Cage	2	4	630	108	72	88	102	38	623	80	70	293	374
104	Cage	2	4	660	113	75	92	93	38	657	83	70	302	375
104	Cage	2	4	690	109	72	89	99	38	646	78	72	302	378
104	Cage	2	4	720	111	74	91	98	37	657	82	72	300	374
104	Cage	2	4	750	111	75	92	103	37	653	83	71	297	370
104	Cage	2	4	780	109	73	90	97	37	667	80	73	305	376
104	Cage	2	4	810	111	74	91	94	37	710	83	72	308	369
104	Cage	2	4	840	109	73	90	90	37	687	81	74	314	383
104	Cage	2	4	870	111	74	91	90	37	709	82	72	317	380
104	Cage	2	4	900	113	77	94	95	37	706	82	72	316	380

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
104	Cage	2	4	930	112	76	93	97	37	681	82	73	308	378
104	Cage	2	4	960	111	75	92	96	37	711	82	74	313	375
104	Cage	2	4	990	112	76	93	97	37	701	84	72	304	367
104	Cage	2	4	1020	116	79	97	102	37	724	84	73	307	366
104	Cage	2	4	1050	113	76	94	100	37	678	83	73	306	376
104	Cage	2	4	1080	110	74	91	101	38	648	80	73	301	378
104	Cage	2	4	1110	111	73	90	92	38	687	79	73	311	378
104	Cage	2	4	1140	116	78	96	125	38	500	78	73	260	370
104	Cage	2	4	1170	116	79	96	139	38	460	79	72	242	358
104	Cage	2	4	1200	111	74	91	107	38	576	80	72	270	359
104	Cage	2	4	1230	118	80	99	145	38	474	77	72	243	354
104	Cage	2	4	1260	123	83	102	157	39	471	76	72	237	347
104	Cage	2	4	1290	119	80	99	141	39	465	76	72	232	342
104	Cage	2	4	1320	122	81	100	129	39	489	75	72	241	346
104	Cage	2	4	1350	126	85	105	131	39	466	83	67	236	346
104	Cage	2	4	1380	114	78	95	124	38	496	85	68	244	347
104	Cage	2	4	1410	112	75	91	105	38	600	81	71	265	347
104	Cage	2	4	1440	114	77	94	110	38	596	79	71	269	352
201	Chair	1	1	0	123	84	105	99	37	599	104	58	250	325
201	Chair	1	1	10	153	115	135	182	37	383	81	54	197	319
201	Chair	1	1	20	123	87	106	114	37	524	98	57	228	316
201	Chair	1	1	30	128	88	109	114	37	519	98	57	228	319
201	Chair	1	1	40	115	78	98	98	37	597	108	56	231	300
201	Chair	1	1	50	111	75	94	89	37	.	.	.	.	.
201	Chair	1	1	60	112	77	95	91	37	613	104	56	248	321
201	Chair	1	1	70	127	89	109	108	37	523	92	57	232	323
201	Chair	1	1	80	118	80	100	92	37	555	103	56	241	325
201	Chair	1	1	90	119	80	101	92	37	544	96	57	245	335
201	Chair	1	1	100	130	90	111	117	37	495	91	57	226	322
201	Chair	1	1	110	126	87	108	116	37	500	90	57	225	320
201	Chair	1	1	120	130	90	111	126	37	490	91	57	224	323
201	Cage	1	1	0	109	75	91	89	37	842	79	59	337	369
201	Cage	1	1	210	123	89	108	113	37	535	87	63	235	322
201	Cage	1	1	240	130	95	113	118	38	522	82	56	234	324
201	Cage	1	1	270	128	94	111	109	38	558	83	62	251	337
201	Cage	1	1	300	114	79	96	106	37	568	82	61	252	335
201	Cage	1	1	330	110	76	93	98	37	622	80	60	262	332
201	Cage	1	1	360	107	72	89	94	37	645	80	59	265	331
201	Cage	1	1	390	108	72	89	89	37	684	80	57	271	328
201	Cage	1	1	420	105	70	87	87	37	698	76	60	283	339
201	Cage	1	1	450	100	66	82	89	38	673	72	63	291	355
201	Cage	1	1	480	102	68	85	88	37	685	73	61	293	354
201	Cage	1	1	510	102	67	84	86	37	704	78	57	298	355

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
201	Cage	1	1	540	101	66	83	81	37	741	74	61	309	360
201	Cage	1	1	570	102	67	85	82	37	730	78	58	311	365
201	Cage	1	1	600	102	67	84	80	37	751	76	59	318	368
201	Cage	1	1	630	104	69	86	81	37	775	75	58	323	367
201	Cage	1	1	660	102	66	84	81	37	742	78	58	320	372
201	Cage	1	1	690	102	66	84	79	37	750	78	58	323	374
201	Cage	1	1	720	101	65	82	79	37	758	76	61	325	374
201	Cage	1	1	750	100	65	82	83	37	732	75	61	324	379
201	Cage	1	1	780	102	66	83	79	37	760	79	58	328	377
201	Cage	1	1	810	102	66	83	79	37	791	77	60	337	380
201	Cage	1	1	840	103	67	84	80	37	803	77	61	337	377
201	Cage	1	1	870	102	67	84	82	37	772	76	62	332	380
201	Cage	1	1	900	103	67	85	81	37	764	79	59	331	380
201	Cage	1	1	930	102	65	83	78	37	821	77	62	343	379
201	Cage	1	1	960	104	69	87	83	37	764	78	60	333	382
201	Cage	1	1	990	105	71	87	86	37	789	78	59	334	377
201	Cage	1	1	1020	106	73	87	95	37	805	77	59	337	377
201	Cage	1	1	1050	107	74	88	93	37	824	77	58	343	379
201	Cage	1	1	1080	113	80	96	92	37	760	80	60	328	376
201	Cage	1	1	1110	113	80	95	97	37	722	79	61	316	373
201	Cage	1	1	1140	107	69	87	80	37	832	77	60	333	366
201	Cage	1	1	1170	105	68	85	79	37	858	78	60	338	366
201	Cage	1	1	1200	110	79	92	99	37	788	79	60	322	364
201	Cage	1	1	1230	114	76	95	101	38	707	78	61	296	354
201	Cage	1	1	1260	110	71	90	109	38	596	81	63	260	338
201	Cage	1	1	1290	112	75	94	121	39	542	83	63	247	337
201	Cage	1	1	1320	125	89	107	135	39	500	84	63	241	342
201	Cage	1	1	1350	137	99	118	143	37	491	84	63	246	352
201	Cage	1	1	1380	121	86	104	119	37	546	83	62	252	342
201	Cage	1	1	1410	112	76	95	100	38	624	81	59	274	347
201	Cage	1	1	1440	107	69	88	89	38	673	79	54	281	343
201	Chair	3	4	0	127	90	110	125	37	477	87	56	216	315
201	Chair	3	4	10	148	112	131	187	38	378	80	53	190	309
201	Chair	3	4	20	121	88	106	127	38	457	87	57	216	320
201	Chair	3	4	30	118	83	101	120	38	485	92	57	221	319
201	Chair	3	4	40	116	82	101	113	38	519	95	56	222	310
201	Chair	3	4	50	117	83	101	120	38	478	92	57	222	322
201	Chair	3	4	60	113	80	98	112	38	523	96	56	223	309
201	Chair	3	4	70	113	79	97	106	38	555	97	56	231	312
201	Chair	3	4	80	112	78	96	105	37	560	97	55	231	311
201	Chair	3	4	90	117	83	101	119	37	498	91	56	221	315
201	Chair	3	4	100	116	82	99	116	38	496	93	55	223	318
201	Chair	3	4	110	117	82	100	117	38	503	93	56	222	314

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
201	Chair	3	4	120	123	88	106	134	38	489	90	56	218	313
201	Cage	3	4	0	105	72	88	103	37	719	79	58	302	359
201	Cage	3	4	210	125	92	110	137	39	447	82	55	214	321
201	Cage	3	4	240	131	96	114	117	37	519	88	62	244	340
201	Cage	3	4	270	117	84	101	109	37	561	87	60	245	328
201	Cage	3	4	300	114	82	98	121	37	531	82	56	239	330
201	Cage	3	4	330	113	80	97	112	37	566	83	60	248	331
201	Cage	3	4	360	107	73	90	100	37	631	84	62	262	330
201	Cage	3	4	390	105	70	87	90	38	673	84	59	272	332
201	Cage	3	4	420	106	70	88	91	37	678	83	60	273	333
201	Cage	3	4	450	98	63	79	81	38	740	76	60	301	351
201	Cage	3	4	480	97	62	79	79	37	760	76	61	311	356
201	Cage	3	4	510	100	65	82	79	37	762	76	60	312	357
201	Cage	3	4	540	98	62	79	77	37	782	79	59	320	362
201	Cage	3	4	570	96	60	77	73	37	821	79	58	325	358
201	Cage	3	4	600	97	61	78	73	37	836	78	57	325	356
201	Cage	3	4	630	102	66	83	75	37	835	78	58	326	357
201	Cage	3	4	660	97	60	77	71	37	848	78	58	332	360
201	Cage	3	4	690	97	60	78	70	37	859	79	57	336	362
201	Cage	3	4	720	98	62	79	71	37	855	79	59	338	366
201	Cage	3	4	750	98	62	80	69	37	870	79	60	341	366
201	Cage	3	4	780	99	63	80	70	37	887	79	59	342	364
201	Cage	3	4	810	98	62	79	66	37	914	79	59	347	363
201	Cage	3	4	840	99	64	81	69	37	876	80	59	344	368
201	Cage	3	4	870	105	71	88	75	37	828	79	61	334	368
201	Cage	3	4	900	101	68	85	74	37	820	79	60	335	372
201	Cage	3	4	930	100	66	83	74	37	830	79	60	336	370
201	Cage	3	4	960	102	68	86	74	37	845	78	61	338	370
201	Cage	3	4	990	104	71	87	78	37	881	78	60	342	366
201	Cage	3	4	1020	103	69	86	75	37	884	79	60	343	365
201	Cage	3	4	1050	105	74	87	82	36	934	81	59	348	361
201	Cage	3	4	1080	106	76	89	87	36	876	79	61	343	369
201	Cage	3	4	1110	106	75	88	89	37	862	80	60	340	369
201	Cage	3	4	1140	108	74	91	89	37	785	80	61	317	359
201	Cage	3	4	1170	106	70	88	93	37	754	81	61	308	357
201	Cage	3	4	1200	107	70	89	108	37	613	83	62	262	337
201	Cage	3	4	1230	103	64	83	87	38	722	81	57	269	317
201	Cage	3	4	1260	105	67	86	102	38	647	82	60	250	312
201	Cage	3	4	1290	106	69	88	100	38	663	81	61	249	307
201	Cage	3	4	1320	122	85	104	129	39	538	83	62	251	345
201	Cage	3	4	1350	116	82	99	156	37	458	83	63	233	347
201	Cage	3	4	1380	106	71	89	124	38	569	84	59	234	312
201	Cage	3	4	1410	108	72	90	110	38	624	84	59	243	309

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
201	Cage	3	4	1440	105	69	87	107	38	631	82	55	238	301
202	Chair	1	1	0	119	82	101	165	38	354	61	49	203	342
202	Chair	1	1	10	137	97	117	217	38	306	64	49	180	326
202	Chair	1	1	20	130	91	111	199	38	304	63	51	182	330
202	Chair	1	1	30	130	91	111	211	38	284	60	50	176	330
202	Chair	1	1	40	132	92	112	206	38	293	59	50	179	330
202	Chair	1	1	50	120	83	101	176	38	344	57	48	194	331
202	Chair	1	1	60	117	78	98	164	38	352	58	47	204	344
202	Chair	1	1	70	131	91	112	203	38	299	59	51	182	332
202	Chair	1	1	80	127	89	108	190	38	337	58	48	194	334
202	Chair	1	1	90	123	83	104	174	38	350	58	47	199	337
202	Chair	1	1	100	124	82	104	180	38	337	60	50	199	343
202	Chair	1	1	110	131	91	111	204	38	300	62	51	181	331
202	Chair	1	1	120	136	95	116	213	38	292	60	51	178	329
202	Cage	1	1	0	103	69	86	117	37	525	63	54	250	347
202	Cage	1	1	210	129	93	112	145	36	420	63	53	214	331
202	Cage	1	1	240	125	85	105	156	37	389	61	52	208	334
202	Cage	1	1	270	123	86	105	146	37	416	63	52	216	335
202	Cage	1	1	300	132	94	113	145	37	416	61	53	219	340
202	Cage	1	1	330	122	83	102	143	37	429	60	53	232	354
202	Cage	1	1	360	124	84	104	145	38	423	60	52	229	353
202	Cage	1	1	390	127	88	107	146	38	416	60	53	232	359
202	Cage	1	1	420	110	71	90	138	38	440	60	55	245	370
202	Cage	1	1	450	106	72	88	133	37	453	60	56	253	376
202	Cage	1	1	480	103	71	86	130	37	463	61	56	257	378
202	Cage	1	1	510	109	76	92	136	37	447	60	54	252	377
202	Cage	1	1	540	103	69	85	122	37	493	61	58	265	377
202	Cage	1	1	570	107	72	89	134	37	455	62	54	254	377
202	Cage	1	1	600	105	71	87	129	37	469	65	54	257	375
202	Cage	1	1	630	104	70	86	124	37	486	67	55	261	374
202	Cage	1	1	660	103	69	85	126	37	480	70	54	262	378
202	Cage	1	1	690	106	73	88	135	37	451	64	54	252	376
202	Cage	1	1	720	105	72	88	129	37	465	62	54	258	379
202	Cage	1	1	750	101	69	84	132	37	443	62	50	260	391
202	Cage	1	1	780	102	70	85	130	37	443	62	49	261	392
202	Cage	1	1	810	101	69	84	130	37	434	60	48	257	390
202	Cage	1	1	840	103	71	86	138	37	424	60	50	250	384
202	Cage	1	1	870	105	71	87	132	37	447	62	51	255	382
202	Cage	1	1	900	102	68	84	139	37	421	60	49	250	386
202	Cage	1	1	930	100	65	83	124	37	485	64	56	263	378
202	Cage	1	1	960	104	70	87	127	37	476	67	55	261	378
202	Cage	1	1	990	102	68	84	127	37	475	66	55	259	376
202	Cage	1	1	1020	100	66	82	126	37	486	63	56	264	379

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
202	Cage	1	1	1050	97	62	78	116	37	526	65	54	275	381
202	Cage	1	1	1080	98	63	80	108	37	574	68	55	285	378
202	Cage	1	1	1110	107	71	89	123	37	501	67	55	261	370
202	Cage	1	1	1140	105	68	85	117	37	520	67	55	261	362
202	Cage	1	1	1170	102	65	82	99	38	607	61	56	288	372
202	Cage	1	1	1200	106	70	87	114	38	542	63	56	270	368
202	Cage	1	1	1230	110	72	90	128	38	478	64	54	244	354
202	Cage	1	1	1260	108	71	88	125	38	490	63	54	249	357
202	Cage	1	1	1290	116	78	97	160	38	404	61	52	223	352
202	Cage	1	1	1320	119	82	101	146	38	416	63	55	221	344
202	Cage	1	1	1350	122	88	105	136	37	454	65	54	231	344
202	Cage	1	1	1380	115	81	97	123	36	505	63	55	244	344
202	Cage	1	1	1410	106	74	89	123	37	499	62	55	249	353
202	Cage	1	1	1440	102	70	85	119	37	520	61	55	259	359
202	Chair	3	4	0	131	92	112	210	38	281	60	50	180	341
202	Chair	3	4	10	135	95	115	210	38	316	59	48	185	330
202	Chair	3	4	20	131	90	111	201	38	303	57	48	187	339
202	Chair	3	4	30	117	79	98	171	38	342	57	46	201	343
202	Chair	3	4	40	107	70	89	148	38	.	.	.	.	.
202	Chair	3	4	50	118	80	99	172	38	372	65	49	206	339
202	Chair	3	4	60	118	80	99	168	38	350	58	47	203	343
202	Chair	3	4	70	122	82	102	183	38	323	59	49	196	344
202	Chair	3	4	80	124	85	105	182	38	340	58	48	200	343
202	Chair	3	4	90	128	88	108	196	38	313	61	51	186	333
202	Chair	3	4	100	127	86	107	197	38	311	63	52	190	341
202	Chair	3	4	110	137	95	116	225	38	272	59	49	170	326
202	Chair	3	4	120	136	95	116	216	38	281	58	51	173	328
202	Cage	3	4	0	100	68	84	107	37	576	62	56	270	358
202	Cage	3	4	210	116	81	99	154	38	391	64	54	211	337
202	Cage	3	4	240	134	99	117	151	37	403	63	53	212	335
202	Cage	3	4	270	155	117	137	152	37	397	62	53	213	339
202	Cage	3	4	300	129	92	111	144	37	422	62	53	223	343
202	Cage	3	4	330	121	85	103	134	37	458	61	53	235	348
202	Cage	3	4	360	118	81	99	134	37	462	61	53	237	350
202	Cage	3	4	390	113	77	94	128	38	483	60	55	247	356
202	Cage	3	4	420	107	73	89	127	38	484	60	55	254	367
202	Cage	3	4	450	100	67	82	120	37	509	62	57	271	380
202	Cage	3	4	480	98	65	81	118	37	515	62	57	273	381
202	Cage	3	4	510	102	70	85	118	37	518	63	57	272	379
202	Cage	3	4	540	102	70	85	118	37	519	62	58	271	377
202	Cage	3	4	570	100	68	83	112	37	541	65	57	280	381
202	Cage	3	4	600	103	71	86	124	37	489	68	54	263	377
202	Cage	3	4	630	100	69	83	122	37	501	66	56	268	378

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
202	Cage	3	4	660	97	65	80	116	37	523	63	57	273	379
202	Cage	3	4	690	95	63	78	109	37	558	63	57	284	381
202	Cage	3	4	720	97	66	80	116	37	523	67	54	273	378
202	Cage	3	4	750	98	67	81	116	37	520	66	54	272	378
202	Cage	3	4	780	97	65	80	112	37	542	65	55	277	377
202	Cage	3	4	810	98	67	82	114	37	526	66	54	275	380
202	Cage	3	4	840	97	65	80	114	37	539	67	53	282	386
202	Cage	3	4	870	96	65	79	113	37	537	70	53	277	381
202	Cage	3	4	900	99	67	82	117	37	518	65	57	272	379
202	Cage	3	4	930	99	67	83	107	37	568	72	55	287	381
202	Cage	3	4	960	99	66	82	110	37	551	70	56	282	381
202	Cage	3	4	990	96	63	79	99	37	616	63	59	299	383
202	Cage	3	4	1020	96	63	78	100	37	609	66	58	301	387
202	Cage	3	4	1050	97	64	79	97	37	631	71	56	308	389
202	Cage	3	4	1080	99	66	81	93	37	656	70	57	311	386
202	Cage	3	4	1110	101	68	84	106	37	586	70	56	291	383
202	Cage	3	4	1140	110	73	91	107	37	569	61	56	275	366
202	Cage	3	4	1170	106	70	87	100	37	605	62	57	290	375
202	Cage	3	4	1200	104	69	86	110	38	564	63	57	276	370
202	Cage	3	4	1230	110	73	90	114	38	534	60	54	261	359
202	Cage	3	4	1260	110	73	91	118	38	516	61	54	255	356
202	Cage	3	4	1290	112	75	93	117	38	525	59	54	258	357
202	Cage	3	4	1320	118	80	99	157	38	392	64	53	220	352
202	Cage	3	4	1350	120	84	103	137	37	446	64	54	231	346
202	Cage	3	4	1380	119	84	101	138	37	461	63	54	230	340
202	Cage	3	4	1410	104	71	87	122	37	513	59	55	250	349
202	Cage	3	4	1440	105	72	87	123	38	509	59	55	250	352
203	Chair	1	1	0	134	89	111	122	39	513	85	61	273	381
203	Chair	1	1	10	143	100	121	182	39	423	82	63	264	406
203	Chair	1	1	20	125	85	104	135	39	449	81	63	261	389
203	Chair	1	1	30	124	83	103	129	39	474	84	60	267	388
203	Chair	1	1	40	130	88	108	139	39	464	84	60	254	374
203	Chair	1	1	50	135	91	112	144	39	456	82	61	255	378
203	Chair	1	1	60	130	87	107	134	38	469	83	60	256	374
203	Chair	1	1	70	135	90	112	135	39	463	83	60	257	378
203	Chair	1	1	80	127	85	104	125	39	502	85	59	269	380
203	Chair	1	1	90	136	91	112	127	39	486	84	60	262	376
203	Chair	1	1	100	133	88	109	117	39	516	85	59	274	382
203	Chair	1	1	110	130	86	107	115	39	525	85	59	277	383
203	Chair	1	1	120	140	94	115	138	39	500	86	60	270	383
203	Cage	1	1	0	131	87	106	122	38	711	85	62	331	400
203	Cage	1	1	210	123	87	103	112	38	541	85	57	288	393
203	Cage	1	1	240	135	94	113	124	38	490	86	57	273	391

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
203	Cage	1	1	270	128	88	105	122	38	496	87	58	272	386
203	Cage	1	1	300	121	84	99	122	38	492	88	56	275	393
203	Cage	1	1	330	122	85	100	119	38	509	86	57	279	391
203	Cage	1	1	360	129	89	106	119	38	508	85	57	274	385
203	Cage	1	1	390	127	87	104	118	38	510	87	57	274	384
203	Cage	1	1	420	119	81	97	122	38	492	85	57	274	390
203	Cage	1	1	450	120	83	98	126	38	477	75	57	279	404
203	Cage	1	1	480	114	77	93	110	38	546	84	57	295	399
203	Cage	1	1	510	115	79	94	111	38	544	80	57	299	406
203	Cage	1	1	540	114	78	93	101	38	598	83	57	314	406
203	Cage	1	1	570	118	82	97	111	37	546	79	58	305	414
203	Cage	1	1	600	114	78	93	102	38	592	80	58	316	411
203	Cage	1	1	630	115	79	93	94	37	651	84	57	329	409
203	Cage	1	1	660	115	78	93	87	37	707	89	58	342	409
203	Cage	1	1	690	118	80	96	83	37	733	83	58	346	405
203	Cage	1	1	720	120	80	96	75	37	814	90	59	370	413
203	Cage	1	1	750	120	79	96	69	37	885	93	61	389	415
203	Cage	1	1	780	120	80	97	71	37	849	87	59	384	418
203	Cage	1	1	810	121	80	98	67	37	915	86	59	397	416
203	Cage	1	1	840	123	80	99	68	37	914	93	64	401	422
203	Cage	1	1	870	120	79	97	66	37	965	94	66	414	423
203	Cage	1	1	900	118	78	96	68	37	940	91	62	414	429
203	Cage	1	1	930	117	78	95	70	37	944	90	61	414	427
203	Cage	1	1	960	121	86	99	86	37	958	90	60	416	427
203	Cage	1	1	990	117	78	95	68	37	949	92	62	412	425
203	Cage	1	1	1020	117	82	96	81	37	984	93	64	415	420
203	Cage	1	1	1050	122	84	99	79	37	940	92	61	403	418
203	Cage	1	1	1080	122	83	99	76	37	918	91	60	398	418
203	Cage	1	1	1110	139	92	114	121	37	553	85	59	302	412
203	Cage	1	1	1140	135	90	110	151	38	418	78	60	254	394
203	Cage	1	1	1170	120	78	96	87	38	687	84	59	321	389
203	Cage	1	1	1200	129	86	105	112	38	564	82	59	295	396
203	Cage	1	1	1230	124	81	99	82	38	694	83	58	331	399
203	Cage	1	1	1260	122	79	97	79	39	700	85	58	341	409
203	Cage	1	1	1290	128	84	103	93	39	641	85	58	317	398
203	Cage	1	1	1320	128	85	104	99	39	615	86	58	314	402
203	Cage	1	1	1350	127	86	103	87	38	694	90	59	329	398
203	Cage	1	1	1380	119	78	94	82	38	659	86	58	335	413
203	Cage	1	1	1410	122	83	99	96	38	611	88	58	315	405
203	Cage	1	1	1440	123	84	100	95	38	659	90	58	313	389
203	Chair	3	4	0	133	88	110	126	39	485	82	62	265	380
203	Chair	3	4	10	156	111	131	231	39	.	.	.	.	.
203	Chair	3	4	20	153	106	128	199	38	.	.	.	.	.

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
203	Chair	3	4	30	152	106	128	209	38	.	.	.	.	.
203	Chair	3	4	40	150	105	126	201	38	358	67	66	228	381
203	Chair	3	4	50	154	105	128	197	38	378	71	64	239	388
203	Chair	3	4	60	142	97	118	169	38	398	73	63	243	385
203	Chair	3	4	70	137	93	114	154	38	434	77	61	249	378
203	Chair	3	4	80	133	91	110	148	38	437	77	62	248	376
203	Chair	3	4	90	131	88	108	131	39	460	79	61	254	375
203	Chair	3	4	100	134	90	111	134	39	454	80	61	254	378
203	Chair	3	4	110	133	90	110	136	39	450	80	61	255	381
203	Chair	3	4	120	139	94	116	147	39	447	80	62	254	380
203	Cage	3	4	0	133	87	106	101	37	721	86	62	332	398
203	Cage	3	4	210	125	87	105	125	38	490	81	58	277	396
203	Cage	3	4	240	147	103	123	128	38	481	83	58	268	387
203	Cage	3	4	270	126	86	103	124	38	486	82	57	275	394
203	Cage	3	4	300	127	88	104	125	38	492	83	57	274	392
203	Cage	3	4	330	125	87	103	122	38	500	85	57	274	388
203	Cage	3	4	360	125	87	103	121	38	495	85	57	274	390
203	Cage	3	4	390	123	84	101	122	38	493	85	57	273	389
203	Cage	3	4	420	119	82	97	119	38	504	83	57	280	395
203	Cage	3	4	450	118	82	97	122	38	491	76	57	287	410
203	Cage	3	4	480	118	82	97	117	38	512	79	56	291	407
203	Cage	3	4	510	118	82	97	109	38	550	82	56	301	406
203	Cage	3	4	540	116	80	95	116	38	522	79	57	298	413
203	Cage	3	4	570	114	78	93	99	37	613	82	58	320	410
203	Cage	3	4	600	115	78	92	90	37	677	87	58	334	408
203	Cage	3	4	630	115	80	94	95	37	640	84	58	328	412
203	Cage	3	4	660	115	78	93	87	37	704	85	58	341	409
203	Cage	3	4	690	113	75	90	74	37	820	86	59	367	406
203	Cage	3	4	720	115	77	93	71	37	868	88	60	383	413
203	Cage	3	4	750	113	75	91	68	37	924	92	65	399	417
203	Cage	3	4	780	114	75	91	64	37	977	93	68	411	416
203	Cage	3	4	810	115	77	93	67	37	953	89	62	409	420
203	Cage	3	4	840	116	77	94	67	37	950	89	62	410	422
203	Cage	3	4	870	118	79	96	68	37	947	91	63	409	422
203	Cage	3	4	900	114	76	92	64	37	984	94	67	415	420
203	Cage	3	4	930	115	77	94	67	37	960	92	64	413	423
203	Cage	3	4	960	117	80	96	69	37	952	92	62	409	421
203	Cage	3	4	990	116	79	95	68	37	954	89	60	410	421
203	Cage	3	4	1020	117	78	95	68	37	960	95	66	408	419
203	Cage	3	4	1050	118	81	96	76	37	933	92	61	403	419
203	Cage	3	4	1080	119	80	97	73	37	899	92	61	394	418
203	Cage	3	4	1110	133	87	108	93	37	735	90	62	349	413
203	Cage	3	4	1140	136	89	111	141	37	434	81	60	259	394

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
203	Cage	3	4	1170	119	77	95	85	38	698	85	58	327	393
203	Cage	3	4	1200	130	87	106	116	38	537	83	58	288	396
203	Cage	3	4	1230	118	76	94	90	38	662	84	58	325	403
203	Cage	3	4	1260	119	76	94	80	38	720	85	58	340	402
203	Cage	3	4	1290	125	81	100	77	38	745	84	58	354	413
203	Cage	3	4	1320	129	86	105	98	39	612	86	58	316	406
203	Cage	3	4	1350	130	89	107	96	38	649	88	59	319	399
203	Cage	3	4	1380	124	84	100	96	38	633	88	58	317	401
203	Cage	3	4	1410	121	82	99	108	38	576	88	58	301	398
203	Cage	3	4	1440	119	80	96	102	38	576	84	57	304	401
204	Chair	1	1	0	145	96	118	95	37	679	67	56	352	432
204	Chair	1	1	10	163	118	140	194	37	352	62	61	311	525
204	Chair	1	1	20	147	103	124	147	37	421	64	62	306	473
204	Chair	1	1	30	150	105	127	151	37	437	64	62	301	457
204	Chair	1	1	40	135	95	114	139	37	455	64	63	313	466
204	Chair	1	1	50	140	98	118	122	37	622	66	59	326	422
204	Chair	1	1	60	139	97	117	118	37	672	66	58	334	414
204	Chair	1	1	70	143	99	120	121	37	602	66	58	324	426
204	Chair	1	1	80	142	99	119	116	37	630	66	58	333	427
204	Chair	1	1	90	145	98	120	102	37	728	66	57	351	415
204	Chair	1	1	100	142	96	117	101	37	681	66	56	351	432
204	Chair	1	1	110	143	97	118	106	37	662	66	57	346	431
204	Chair	1	1	120	148	102	123	118	37	625	66	57	338	436
204	Cage	1	1	0	136	95	114	95	37	692	68	59	374	453
204	Cage	1	1	210	140	100	119	131	38	494	65	58	287	412
204	Cage	1	1	240	144	102	121	139	38	450	65	61	280	418
204	Cage	1	1	270	129	90	108	133	38	483	65	57	284	410
204	Cage	1	1	300	125	88	104	119	38	507	62	55	296	418
204	Cage	1	1	330	128	91	107	125	38	491	63	55	289	415
204	Cage	1	1	360	129	91	108	120	38	491	63	56	296	425
204	Cage	1	1	390	132	94	111	126	38	491	63	56	302	432
204	Cage	1	1	420	126	87	105	127	38	.	.	.	.	.
204	Cage	1	1	450	125	88	105	130	38	455	62	61	302	448
204	Cage	1	1	480	125	88	105	135	38	457	62	61	298	442
204	Cage	1	1	510	121	85	102	124	38	489	62	60	313	448
204	Cage	1	1	540	122	86	103	126	38	484	62	61	317	456
204	Cage	1	1	570	122	86	102	123	38	631	67	54	337	429
204	Cage	1	1	600	121	85	101	118	38	623	72	54	360	458
204	Cage	1	1	630	120	84	100	108	38	547	66	57	350	475
204	Cage	1	1	660	122	87	103	122	37	534	68	56	342	470
204	Cage	1	1	690	124	88	105	121	38	568	69	54	345	460
204	Cage	1	1	720	120	83	99	109	38	515	63	59	351	489
204	Cage	1	1	750	123	87	103	117	37	560	70	55	354	476

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
204	Cage	1	1	780	127	90	107	122	38	607	71	53	351	453
204	Cage	1	1	810	122	84	101	108	38	612	73	53	371	478
204	Cage	1	1	840	125	87	104	114	38	603	72	55	365	474
204	Cage	1	1	870	125	88	105	117	38	622	72	53	372	475
204	Cage	1	1	900	125	87	104	109	38	605	69	56	363	470
204	Cage	1	1	930	121	84	101	108	37	611	68	58	367	473
204	Cage	1	1	960	125	88	105	107	37	628	72	54	371	472
204	Cage	1	1	990	124	85	103	100	37	592	67	58	374	488
204	Cage	1	1	1020	124	86	103	97	37	598	67	59	369	480
204	Cage	1	1	1050	128	89	107	94	37	663	71	56	390	483
204	Cage	1	1	1080	131	93	111	97	37	696	70	56	395	475
204	Cage	1	1	1110	140	96	117	118	37	528	68	59	328	453
204	Cage	1	1	1140	140	99	118	107	38	573	71	57	349	464
204	Cage	1	1	1170	138	98	116	109	38	560	72	56	351	472
204	Cage	1	1	1200	144	103	123	120	38	512	70	58	320	448
204	Cage	1	1	1230	148	102	123	113	38	537	71	58	331	454
204	Cage	1	1	1260	147	101	122	120	38	519	69	59	327	456
204	Cage	1	1	1290	152	105	127	136	38	460	66	62	298	440
204	Cage	1	1	1320	149	103	125	131	38	520	67	59	303	424
204	Cage	1	1	1350	140	95	115	113	38	557	67	59	330	447
204	Cage	1	1	1380	134	92	110	101	38	542	66	60	335	457
204	Cage	1	1	1410	131	92	109	108	38	537	67	60	325	445
204	Cage	1	1	1440	131	91	108	101	38	548	68	61	330	448
204	Chair	3	4	0	148	103	124	105	37	660	67	57	335	419
204	Chair	3	4	10	166	125	146	195	37	320	64	60	286	505
204	Chair	3	4	20	153	112	132	157	37	372	63	61	306	503
204	Chair	3	4	30	152	110	131	144	37	423	64	61	311	482
204	Chair	3	4	40	145	105	124	138	37	463	65	60	326	481
204	Chair	3	4	50	143	105	123	127	37	506	65	59	340	481
204	Chair	3	4	60	141	103	121	119	38	721	66	58	316	383
204	Chair	3	4	70	147	107	126	119	38	646	65	61	329	422
204	Chair	3	4	80	150	108	127	116	37	634	65	60	335	431
204	Chair	3	4	90	145	104	123	109	37	729	64	60	330	392
204	Chair	3	4	100	145	104	123	116	37	665	65	59	335	419
204	Chair	3	4	110	144	102	121	109	37	687	65	59	338	416
204	Chair	3	4	120	143	103	122	118	37	637	65	59	331	427
204	Cage	3	4	0	140	99	118	97	37	652	69	59	366	458
204	Cage	3	4	210	135	98	115	144	38	433	65	61	270	411
204	Cage	3	4	240	137	100	118	140	38	462	66	60	275	409
204	Cage	3	4	270	134	97	114	132	38	484	64	60	280	405
204	Cage	3	4	300	131	96	112	123	38	517	66	57	285	400
204	Cage	3	4	330	133	98	114	124	38	510	66	57	284	401
204	Cage	3	4	360	131	95	111	110	38	535	66	54	296	408

Table E-1. Unadjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
204	Cage	3	4	390	130	94	110	118	38	513	67	56	304	427
204	Cage	3	4	420	124	88	104	109	38	554	67	59	341	461
204	Cage	3	4	450	127	91	107	107	38	523	66	60	344	476
204	Cage	3	4	480	127	91	108	104	38	586	66	59	351	462
204	Cage	3	4	510	127	90	107	97	38	661	68	58	375	467
204	Cage	3	4	540	130	90	109	87	37	775	71	56	406	463
204	Cage	3	4	570	131	93	111	94	37	670	68	57	384	475
204	Cage	3	4	600	132	94	111	100	37	643	67	59	374	472
204	Cage	3	4	630	130	91	109	89	37	683	67	58	398	486
204	Cage	3	4	660	127	89	106	87	37	742	67	58	415	486
204	Cage	3	4	690	131	95	112	98	37	653	67	58	384	480
204	Cage	3	4	720	137	98	116	94	37	643	67	59	382	480
204	Cage	3	4	750	136	97	115	102	37	611	68	60	362	466
204	Cage	3	4	780	126	88	106	84	37	638	67	58	401	506
204	Cage	3	4	810	124	90	106	96	37	709	72	59	398	475
204	Cage	3	4	840	128	92	109	86	37	752	70	59	410	476
204	Cage	3	4	870	128	92	109	86	37	828	71	60	426	470
204	Cage	3	4	900	127	91	109	79	37	862	71	60	439	473
204	Cage	3	4	930	131	94	112	79	37	879	71	60	445	475
204	Cage	3	4	960	129	91	109	76	37	862	71	59	447	483
204	Cage	3	4	990	126	89	106	74	36	710	74	59	416	498
204	Cage	3	4	1020	147	108	126	81	36	733	71	61	384	452
204	Cage	3	4	1050	159	116	136	92	36	617	69	59	354	455
204	Cage	3	4	1080	152	109	129	83	36	715	71	58	391	466
204	Cage	3	4	1110	142	102	121	110	37	619	69	58	348	448
204	Cage	3	4	1140	137	101	118	109	37	603	68	59	339	442
204	Cage	3	4	1170	141	104	122	113	37	578	67	60	335	444
204	Cage	3	4	1200	144	105	123	120	37	517	67	62	319	445
204	Cage	3	4	1230	144	105	124	126	38	504	66	61	311	440
204	Cage	3	4	1260	146	106	125	128	38	493	66	61	305	436
204	Cage	3	4	1290	150	108	128	136	38	488	65	61	298	429
204	Cage	3	4	1320	148	108	127	138	38	464	64	61	295	434
204	Cage	3	4	1350	141	103	121	125	38	505	66	60	307	433
204	Cage	3	4	1380	136	99	116	111	38	557	67	57	320	434
204	Cage	3	4	1410	134	99	115	116	38	572	66	56	313	419
204	Cage	3	4	1440	131	96	111	105	38	552	67	57	319	433

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
101	Chair	1	1	10	7.1	5.0	6.6	58.1	-0.1	-63.9	-2.9	-0.7	-8.8	17.2
101	Chair	1	1	20	4.4	2.1	3.6	17.2	-0.0	-40.2	-2.3	-0.2	-7.1	7.6
101	Chair	1	1	30	5.6	2.6	4.2	14.5	0.1	-33.1	-2.2	-0.1	-9.0	1.0
101	Chair	1	1	40	4.5	2.4	3.5	12.6	0.1	-32.7	-2.3	-0.2	-8.5	1.5
101	Chair	1	1	50	3.5	1.5	2.3	11.9	0.1	-28.0	-3.2	-0.3	-9.4	-2.2
101	Chair	1	1	60	3.5	2.2	2.6	16.5	0.1	-34.6	-4.3	-0.5	-11.6	-2.4
101	Chair	1	1	70	4.9	3.9	4.4	27.3	0.1	-48.8	-5.6	-0.8	-14.5	-0.2
101	Chair	1	1	80	-0.2	-0.0	-0.4	6.6	-0.0	-19.5	-3.1	-0.2	-6.2	-1.0
101	Chair	1	1	90	5.0	2.8	3.6	12.0	-0.0	-24.9	-3.9	-0.3	-8.7	-2.5
101	Chair	1	1	100	2.7	1.3	1.5	2.6	-0.0	-7.3	-2.8	-0.2	-8.1	-9.7
101	Chair	1	1	110	6.6	4.9	5.8	26.3	-0.0	-52.0	-4.3	-1.1	-14.7	1.3
101	Chair	1	1	120	19.5	14.2	17.0	65.7	-0.1	-78.6	-6.6	-2.1	-21.2	6.1
101	Cage	1	1	210	-4.1	-1.4	-0.9	46.3	1.1	-296.9	-15.1	-1.5	-70.9	-6.1
101	Cage	1	1	240	-12.4	-8.6	-9.0	36.9	1.0	-259.4	-7.7	-1.2	-69.7	-19.1
101	Cage	1	1	270	4.3	3.7	5.4	23.9	1.1	-222.7	-5.3	0.0	-58.4	-16.2
101	Cage	1	1	300	-5.5	-3.7	-3.5	23.2	0.9	-224.3	-3.0	-0.5	-58.0	-15.0
101	Cage	1	1	330	-16.3	-13.3	-14.1	17.1	0.8	-194.8	-0.8	-1.4	-58.1	-24.6
101	Cage	1	1	360	-14.0	-10.9	-11.7	6.1	0.7	-133.5	2.5	-0.2	-40.3	-19.5
101	Cage	1	1	390	-14.5	-12.5	-12.5	20.0	0.6	-197.7	-1.0	-0.9	-52.2	-15.0
101	Cage	1	1	420	-18.6	-16.9	-17.4	-4.3	0.7	-56.5	1.1	1.1	-14.5	-8.2
101	Cage	1	1	450	-20.6	-18.8	-19.6	-16.8	0.6	49.4	-1.7	1.4	2.1	-15.2
101	Cage	1	1	480	-22.4	-20.7	-21.7	-20.0	0.5	40.6	0.7	0.8	4.9	-9.3
101	Cage	1	1	510	-23.8	-21.0	-22.4	-15.5	0.2	35.0	3.6	0.8	4.6	-4.6
101	Cage	1	1	540	-19.6	-17.1	-18.3	-17.3	0.3	22.0	4.3	1.2	8.8	0.1
101	Cage	1	1	570	-22.5	-20.0	-21.5	-22.5	0.3	58.1	5.8	0.6	13.8	-2.8
101	Cage	1	1	600	-18.6	-16.9	-18.0	-24.4	0.1	77.3	5.2	0.8	17.4	-3.1
101	Cage	1	1	630	-21.3	-18.6	-20.1	-21.9	0.3	66.8	5.7	1.1	22.2	4.9
101	Cage	1	1	660	-19.8	-18.1	-19.3	-26.7	0.2	105.5	5.5	1.0	27.6	1.8
101	Cage	1	1	690	-21.8	-18.6	-20.4	-22.4	0.1	75.6	8.2	0.7	24.7	6.5
101	Cage	1	1	720	-15.9	-14.2	-15.2	-27.1	-0.1	106.4	9.5	0.8	28.6	2.9
101	Cage	1	1	750	-16.7	-14.6	-15.7	-26.7	0.0	112.2	7.9	1.1	32.7	6.6
101	Cage	1	1	780	-18.5	-15.1	-16.6	-23.3	0.0	72.4	8.9	1.1	30.3	13.8
101	Cage	1	1	810	-17.4	-15.2	-16.3	-27.3	-0.1	131.8	8.9	1.3	37.5	7.9
101	Cage	1	1	840	-14.6	-12.9	-13.8	-30.9	-0.1	166.4	13.0	0.9	38.8	1.2
101	Cage	1	1	870	-17.5	-15.1	-16.5	-29.5	-0.2	162.7	11.1	1.1	43.3	7.0
101	Cage	1	1	900	-17.5	-15.2	-16.3	-28.6	-0.1	154.0	9.4	2.0	45.6	11.9
101	Cage	1	1	930	-16.4	-13.8	-14.9	-33.3	-0.3	216.0	12.8	1.4	48.7	0.9
101	Cage	1	1	960	-13.0	-9.8	-10.5	-27.3	-0.4	199.0	12.3	1.1	45.1	3.1
101	Cage	1	1	990	-9.4	-7.5	-7.6	-28.7	-0.4	249.8	11.3	1.6	54.4	1.5
101	Cage	1	1	1020	-11.6	-8.4	-9.0	-25.3	-0.5	247.5	10.5	1.2	57.5	5.7
101	Cage	1	1	1050	-12.0	-8.2	-9.8	-22.2	-0.6	260.1	11.9	1.0	60.9	6.4
101	Cage	1	1	1080	-6.5	-2.9	-5.0	-17.9	-0.4	211.7	9.2	0.9	44.8	-0.5
101	Cage	1	1	1110	-8.5	-3.7	-6.5	-23.1	-0.3	245.3	10.8	0.9	53.9	0.8

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
101	Cage	1	1	1140	2.7	1.9	3.1	-15.5	-0.1	39.6	8.6	0.5	22.2	15.4
101	Cage	1	1	1170	2.7	2.4	3.5	5.8	0.2	-123.3	1.0	-0.1	-14.5	14.6
101	Cage	1	1	1200	3.3	3.0	3.8	6.0	0.6	-118.5	-1.1	0.1	-17.8	8.9
101	Cage	1	1	1230	4.7	4.4	5.6	31.8	1.0	-226.4	-7.9	-0.9	-44.3	7.9
101	Cage	1	1	1260	5.6	4.4	6.0	21.2	1.2	-201.9	-4.0	-0.6	-36.2	9.5
101	Cage	1	1	1290	0.3	0.4	1.2	19.3	1.3	-203.6	-5.2	-1.0	-38.7	7.2
101	Cage	1	1	1320	9.2	5.2	8.2	29.1	1.4	-232.9	-7.6	-1.7	-47.0	5.6
101	Cage	1	1	1350	7.4	4.5	7.0	3.4	1.3	-123.8	0.1	0.3	-25.8	-2.1
101	Cage	1	1	1380	-4.1	-7.2	-5.1	9.4	0.8	-127.0	-3.3	-0.4	-28.4	-1.2
101	Cage	1	1	1410	-8.7	-10.1	-9.2	-6.4	0.8	-48.1	0.3	0.3	-10.8	-2.6
101	Cage	1	1	1440	-5.5	-6.8	-5.4	20.0	0.8	-181.2	0.2	-0.0	-36.8	2.4
101	Chair	2	4	10	12.7	12.9	13.3	79.8	0.1	-114.9	-4.0	-1.5	-28.6	9.4
101	Chair	2	4	20	10.6	10.4	11.0	44.7	0.1	-96.5	-4.2	-0.8	-19.3	14.9
101	Chair	2	4	30	6.3	7.9	7.4	36.4	0.2	-86.7	-5.8	-0.3	-20.5	7.7
101	Chair	2	4	40	2.9	4.8	4.0	17.0	0.2	-48.9	-2.7	-0.1	-10.5	5.3
101	Chair	2	4	50	5.5	5.9	5.7	18.0	0.1	-51.3	-3.0	-0.2	-13.4	1.7
101	Chair	2	4	60	3.2	4.0	3.4	17.8	0.2	-50.6	-3.4	-0.1	-13.5	1.2
101	Chair	2	4	70	4.8	4.8	4.8	28.5	0.2	-67.2	-6.0	-0.3	-17.4	2.9
101	Chair	2	4	80	6.2	6.6	6.4	19.6	0.1	-56.2	-3.4	-0.0	-14.4	2.6
101	Chair	2	4	90	4.7	5.1	5.0	30.2	0.1	-71.9	-5.7	-0.2	-17.0	5.9
101	Chair	2	4	100	3.1	4.0	3.5	17.3	0.2	-47.2	-3.4	0.0	-9.6	5.9
101	Chair	2	4	110	8.1	8.2	8.4	43.3	0.1	-89.4	-6.6	-0.6	-20.4	9.3
101	Chair	2	4	120	7.6	7.6	7.9	40.4	0.1	-83.4	-6.2	-0.5	-19.6	7.5
101	Cage	2	4	210	16.9	17.1	18.8	94.2	1.4	-520.0	-27.8	-2.1	-115.6	5.3
101	Cage	2	4	240	10.2	11.8	12.6	57.3	1.3	-435.5	-18.6	-1.6	-97.0	-3.0
101	Cage	2	4	270	-6.0	-0.9	-2.1	39.4	0.9	-366.4	-9.8	-0.7	-85.8	-12.3
101	Cage	2	4	300	-1.2	1.2	1.1	52.4	1.0	-406.5	-10.5	-1.5	-97.0	-14.3
101	Cage	2	4	330	-7.9	-3.4	-4.7	41.3	0.9	-374.9	-6.8	-1.1	-87.5	-12.3
101	Cage	2	4	360	-9.1	-4.9	-6.6	20.9	1.2	-241.9	-2.5	0.4	-60.1	-16.3
101	Cage	2	4	390	-12.4	-8.7	-10.0	15.8	1.1	-234.7	-0.5	0.0	-59.6	-17.4
101	Cage	2	4	420	-13.8	-10.6	-12.0	9.5	1.1	-187.4	-0.9	0.9	-44.8	-11.8
101	Cage	2	4	450	-15.3	-11.5	-13.2	5.3	1.0	-159.0	-1.1	0.9	-38.3	-10.9
101	Cage	2	4	480	-15.9	-12.8	-14.6	-4.0	0.7	-79.7	-2.9	0.8	-19.0	-6.6
101	Cage	2	4	510	-15.5	-13.1	-14.6	-7.4	0.6	-50.9	-2.7	0.9	-12.0	-5.2
101	Cage	2	4	540	-16.0	-13.9	-15.5	-9.7	0.4	-13.6	-1.3	0.8	-4.3	-4.4
101	Cage	2	4	570	-12.3	-10.8	-12.0	-11.0	0.3	-0.2	0.2	1.0	0.2	-2.5
101	Cage	2	4	600	-11.8	-9.2	-10.7	-6.5	0.3	-33.5	0.7	0.7	0.9	6.1
101	Cage	2	4	630	-6.6	-5.5	-6.2	-7.9	0.1	-16.7	0.6	1.1	4.0	5.7
101	Cage	2	4	660	-5.9	-4.3	-5.0	-8.5	0.1	-27.0	2.4	1.1	5.8	9.5
101	Cage	2	4	690	0.5	0.7	0.6	-10.6	0.1	-9.9	2.1	0.8	6.6	6.6
101	Cage	2	4	720	-3.8	-2.6	-3.1	-9.3	-0.0	-1.5	5.0	1.7	11.4	10.9
101	Cage	2	4	750	-2.9	-1.6	-2.1	-11.0	-0.2	25.7	2.8	0.4	12.3	5.6
101	Cage	2	4	780	-1.9	-0.4	-1.2	-10.4	-0.2	27.6	5.5	1.2	15.2	8.5

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
101	Cage	2	4	810	-2.8	-0.6	-1.7	-9.6	-0.3	19.8	4.7	1.0	15.4	10.6
101	Cage	2	4	840	-0.3	1.5	0.9	-10.6	-0.2	70.9	4.3	0.8	19.0	4.0
101	Cage	2	4	870	-4.7	-2.2	-3.5	-13.3	-0.5	76.6	6.7	1.0	22.7	6.2
101	Cage	2	4	900	-4.1	-1.3	-2.5	-10.9	-0.8	121.2	8.4	1.8	28.1	3.1
101	Cage	2	4	930	-3.5	-0.5	-1.6	-12.8	-0.8	137.2	7.0	1.2	32.6	4.9
101	Cage	2	4	960	-1.4	2.3	0.7	-6.8	-0.8	164.7	8.1	1.8	37.2	4.0
101	Cage	2	4	990	1.0	4.8	3.2	-2.6	-0.9	154.9	7.8	0.9	38.1	7.6
101	Cage	2	4	1020	2.3	6.8	4.6	-0.6	-0.7	87.7	7.4	0.7	30.8	14.8
101	Cage	2	4	1050	4.1	9.5	6.3	3.2	-0.5	122.8	7.8	1.0	33.7	8.8
101	Cage	2	4	1080	2.0	6.3	4.1	-3.8	-0.4	150.6	6.5	0.4	35.6	6.0
101	Cage	2	4	1110	4.7	7.8	6.3	-4.3	-0.4	87.6	6.4	0.5	24.6	8.1
101	Cage	2	4	1140	10.1	12.1	10.8	-2.3	-0.2	63.9	5.6	0.6	17.7	4.8
101	Cage	2	4	1170	14.7	14.5	15.1	-3.3	-0.1	-36.2	5.0	0.8	0.9	8.0
101	Cage	2	4	1200	10.4	11.6	11.5	-6.2	0.2	-26.4	-0.9	0.3	-1.9	2.9
101	Cage	2	4	1230	11.1	11.7	12.2	15.9	0.5	-233.5	0.6	-0.3	-34.9	17.5
101	Cage	2	4	1260	14.7	14.3	14.9	11.0	0.8	-191.3	-5.5	-0.2	-34.3	7.2
101	Cage	2	4	1290	17.2	16.1	17.1	22.7	1.1	-268.8	-5.4	-0.8	-51.8	5.6
101	Cage	2	4	1320	12.9	12.7	13.1	19.7	1.3	-252.3	-4.6	-0.6	-49.0	3.5
101	Cage	2	4	1350	24.1	20.4	22.8	31.8	1.5	-323.2	-5.4	-0.9	-62.6	6.5
101	Cage	2	4	1380	11.4	11.5	12.2	20.6	1.2	-271.5	-5.7	0.2	-49.4	7.5
101	Cage	2	4	1410	6.6	6.3	6.9	31.0	1.0	-305.5	-5.8	-0.4	-65.2	-1.8
101	Cage	2	4	1440	9.0	7.9	8.7	33.3	1.1	-300.3	-8.9	-0.4	-68.6	-7.6
102	Chair	1	1	10	10.5	10.7	11.0	31.0	-0.1	-74.4	-5.9	-2.1	-23.2	4.1
102	Chair	1	1	20	-1.5	-3.4	-2.7	-3.2	-0.1	-12.7	2.0	1.3	-1.0	4.7
102	Chair	1	1	30	-1.9	-5.0	-3.5	-16.2	-0.2	9.9	-0.5	1.8	7.9	7.2
102	Chair	1	1	40	-3.3	-6.3	-4.7	-23.6	-0.2	25.8	-0.6	1.4	13.2	7.4
102	Chair	1	1	50	-0.7	-2.2	-1.4	-8.7	-0.2	20.1	-1.8	-0.2	9.1	4.0
102	Chair	1	1	60	-4.3	-5.9	-5.0	-25.7	-0.3	33.4	0.4	1.3	12.4	2.4
102	Chair	1	1	70	-0.8	-2.1	-1.3	-8.6	-0.3	17.3	0.2	0.8	6.8	2.0
102	Chair	1	1	80	-1.8	-3.2	-2.4	-12.2	-0.3	29.4	0.6	0.9	10.8	2.1
102	Chair	1	1	90	-3.9	-3.9	-4.0	-16.3	-0.4	38.0	0.2	-0.1	14.7	3.9
102	Chair	1	1	100	-7.7	-7.9	-8.2	-30.7	-0.4	46.1	0.8	-0.1	18.8	6.3
102	Chair	1	1	110	-8.6	-9.3	-9.1	-36.2	-0.3	60.5	2.1	0.2	25.6	10.1
102	Chair	1	1	120	0.4	0.1	0.2	-18.8	-0.4	46.7	2.5	2.1	22.1	11.1
102	Cage	1	1	210	3.4	-0.9	1.7	22.2	0.5	-79.3	-0.3	1.1	-22.1	-1.1
102	Cage	1	1	240	-1.2	-1.0	-1.2	16.7	0.6	-64.0	0.6	-0.6	-25.9	-13.6
102	Cage	1	1	270	1.8	1.9	2.1	19.2	0.7	-74.1	2.3	1.5	-22.8	-4.8
102	Cage	1	1	300	-6.6	-4.3	-5.6	12.5	0.7	-55.2	6.7	0.8	-28.1	-20.9
102	Cage	1	1	330	-11.5	-7.3	-10.0	8.3	0.9	-39.6	3.3	1.3	-30.2	-30.2
102	Cage	1	1	360	-10.5	-7.4	-9.6	8.9	0.9	-38.5	2.3	-0.0	-24.7	-22.3
102	Cage	1	1	390	-9.7	-6.3	-8.5	7.7	0.9	-37.2	4.2	1.1	-19.2	-14.8
102	Cage	1	1	420	-9.2	-6.5	-8.5	4.6	0.8	-30.2	4.2	1.9	-21.5	-20.9
102	Cage	1	1	450	-13.9	-11.1	-12.6	7.4	0.6	-35.9	2.4	4.9	-19.0	-15.4

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
102	Cage	1	1	480	-12.4	-10.6	-11.2	12.0	0.3	-42.6	0.2	3.5	-14.1	-5.3
102	Cage	1	1	510	-10.9	-10.2	-10.0	11.8	0.1	-46.3	1.9	1.4	-11.7	-0.1
102	Cage	1	1	540	-13.5	-11.7	-12.1	14.6	0.0	-54.8	-1.7	4.0	-18.2	-6.5
102	Cage	1	1	570	-9.7	-8.4	-8.4	18.9	-0.1	-64.5	-10.0	5.5	-21.7	-7.4
102	Cage	1	1	600	2.2	-3.3	-0.4	21.8	-0.4	-63.2	-12.8	3.2	-22.2	-8.8
102	Cage	1	1	630	0.8	-2.8	-0.7	23.6	-0.4	-68.2	-12.6	3.0	-22.6	-7.2
102	Cage	1	1	660	0.5	-1.6	-0.3	26.3	-0.4	-75.2	-12.8	2.7	-25.4	-8.5
102	Cage	1	1	690	-2.1	-3.1	-2.3	25.0	-0.2	-81.5	-11.9	3.4	-26.7	-7.7
102	Cage	1	1	720	-5.9	-5.0	-4.8	28.6	-0.1	-92.2	-9.0	3.6	-25.5	-0.8
102	Cage	1	1	750	0.7	0.2	1.9	20.9	-0.3	-80.6	-8.8	4.7	-23.9	-3.8
102	Cage	1	1	780	-8.7	-6.8	-6.5	20.4	-0.4	-76.5	-6.5	3.3	-20.1	0.2
102	Cage	1	1	810	-10.8	-6.9	-7.6	22.5	-0.5	-80.5	-4.1	1.8	-18.3	5.0
102	Cage	1	1	840	-10.2	-5.5	-6.8	24.9	-0.4	-85.8	-5.4	3.4	-28.1	-7.9
102	Cage	1	1	870	-6.5	-4.3	-4.7	30.7	-0.4	-91.3	-8.4	4.1	-28.8	-6.3
102	Cage	1	1	900	-6.2	-2.8	-3.7	22.8	-0.5	-81.6	-7.2	2.5	-19.5	3.5
102	Cage	1	1	930	-3.6	-0.3	-1.4	28.0	-0.5	-89.6	-8.3	2.3	-26.9	-4.0
102	Cage	1	1	960	-4.6	-1.1	-2.6	26.8	-0.3	-89.9	-9.0	2.8	-29.2	-7.6
102	Cage	1	1	990	-5.0	-1.7	-3.1	28.2	-0.3	-96.0	-9.5	2.1	-28.5	-3.7
102	Cage	1	1	1020	-1.4	0.7	0.1	26.7	-0.3	-90.6	-6.8	1.0	-27.1	-4.1
102	Cage	1	1	1050	2.3	3.5	3.3	28.4	-0.3	-92.7	-8.1	-0.0	-29.2	-6.3
102	Cage	1	1	1080	-2.6	-0.4	-1.2	20.2	-0.3	-75.8	-4.9	-0.5	-22.3	-3.3
102	Cage	1	1	1110	1.4	3.2	2.7	20.7	-0.3	-72.6	-5.7	-0.8	-21.7	-3.6
102	Cage	1	1	1140	9.0	6.2	7.6	17.6	-0.1	-86.2	-2.4	2.6	-25.5	-2.9
102	Cage	1	1	1170	-2.0	1.4	-0.4	9.5	0.4	-82.7	-1.2	1.5	-24.5	-2.7
102	Cage	1	1	1200	-1.7	3.0	0.1	10.1	0.6	-51.9	2.1	1.7	-16.5	-4.2
102	Cage	1	1	1230	0.4	2.4	1.0	10.6	0.9	-54.9	3.5	2.4	-13.9	1.1
102	Cage	1	1	1260	-1.0	2.5	0.2	8.8	1.0	-57.7	2.8	2.3	-13.1	3.7
102	Cage	1	1	1290	1.6	3.8	2.0	13.3	1.0	-44.9	2.4	1.9	-9.1	4.8
102	Cage	1	1	1320	7.8	6.4	7.2	30.0	1.1	-96.4	-0.0	1.7	-24.7	3.2
102	Cage	1	1	1350	-5.4	-1.8	-3.5	13.3	1.4	-58.0	8.1	3.0	-16.8	-2.7
102	Cage	1	1	1380	-12.4	-8.8	-11.0	5.6	1.1	-31.5	12.2	2.5	-9.9	-3.7
102	Cage	1	1	1410	-12.7	-7.7	-10.5	9.0	1.1	-42.0	6.4	1.9	-20.5	-14.5
102	Cage	1	1	1440	-13.3	-8.9	-11.4	8.0	0.8	-34.9	8.7	1.6	-13.9	-8.0
102	Chair	2	4	10	19.8	18.8	19.6	76.1	-0.2	-156.5	-12.7	-0.8	-50.6	2.5
102	Chair	2	4	20	9.4	7.7	8.5	46.0	-0.4	-105.5	-3.3	1.4	-34.1	-0.3
102	Chair	2	4	30	7.2	3.7	5.6	25.0	-0.3	-66.2	-0.2	2.5	-19.0	2.3
102	Chair	2	4	40	5.5	2.2	3.9	17.7	-0.3	-48.8	-1.3	0.3	-14.4	0.6
102	Chair	2	4	50	4.4	1.8	3.3	15.8	-0.3	-42.6	-1.5	0.7	-11.4	2.3
102	Chair	2	4	60	7.5	6.7	7.1	39.0	-0.3	-72.9	-3.7	0.4	-23.7	-1.2
102	Chair	2	4	70	5.7	5.0	5.2	35.4	-0.4	-64.2	-2.3	0.8	-25.8	-9.5
102	Chair	2	4	80	5.8	4.3	5.1	30.3	-0.4	-56.2	-4.1	0.3	-17.8	-0.4
102	Chair	2	4	90	3.3	1.9	2.5	17.6	-0.3	-44.5	-2.6	1.0	-16.0	-4.0
102	Chair	2	4	100	2.0	0.4	1.3	8.3	-0.2	-26.9	-2.4	0.9	-6.7	2.0

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
102	Chair	2	4	110	1.9	0.3	1.1	5.9	-0.2	-15.8	-1.6	1.6	-4.5	0.3
102	Chair	2	4	120	9.7	8.2	9.0	23.1	-0.2	-23.1	-1.0	1.4	-3.6	5.2
102	Cage	2	4	210	11.9	7.4	10.3	33.1	0.6	-110.0	-3.0	3.1	-22.5	11.7
102	Cage	2	4	240	-0.7	-0.3	-0.2	20.6	0.6	-81.2	0.3	2.1	-16.9	6.8
102	Cage	2	4	270	-1.9	0.3	-0.8	14.7	0.7	-59.1	1.2	0.5	-17.8	-3.8
102	Cage	2	4	300	-4.4	-1.9	-3.5	16.1	0.7	-65.6	2.4	2.0	-18.3	-2.2
102	Cage	2	4	330	-9.1	-5.0	-7.5	11.3	0.8	-53.8	5.8	1.5	-27.8	-21.4
102	Cage	2	4	360	-5.2	-2.0	-4.0	12.6	0.9	-51.3	-1.5	1.3	-20.5	-11.3
102	Cage	2	4	390	-7.8	-4.9	-7.1	5.0	0.8	-39.4	4.5	2.8	-22.6	-19.2
102	Cage	2	4	420	-7.8	-4.1	-6.7	7.0	0.9	-40.9	5.1	3.5	-29.7	-29.1
102	Cage	2	4	450	-12.3	-8.6	-10.6	2.8	0.8	-21.5	9.1	3.0	-9.6	-7.4
102	Cage	2	4	480	-13.7	-10.0	-11.4	2.0	0.2	-14.8	8.0	2.5	3.4	8.6
102	Cage	2	4	510	-8.3	-5.8	-6.4	7.0	-0.1	-29.2	-0.8	2.8	-2.9	5.5
102	Cage	2	4	540	-2.6	-3.5	-2.7	9.5	-0.4	-34.3	-5.9	3.7	-7.9	0.4
102	Cage	2	4	570	-7.3	-7.2	-6.8	2.1	-0.5	-16.1	-3.8	5.4	-3.4	-0.5
102	Cage	2	4	600	-5.9	-5.3	-4.9	5.4	-0.5	-18.3	-3.4	4.5	-3.9	-0.1
102	Cage	2	4	630	-9.8	-7.8	-8.2	7.6	-0.6	-29.3	-5.8	4.8	-3.5	4.8
102	Cage	2	4	660	-4.9	-4.8	-4.2	4.6	-0.7	-22.8	-3.6	4.9	-6.8	-2.4
102	Cage	2	4	690	-9.9	-7.4	-7.9	5.3	-0.6	-28.0	-2.6	5.9	-6.5	-0.1
102	Cage	2	4	720	-6.0	-4.8	-4.4	4.6	-0.6	-19.3	-2.0	4.7	-2.4	2.4
102	Cage	2	4	750	-10.7	-6.8	-7.9	6.2	-0.7	-31.0	1.0	2.8	-1.3	8.6
102	Cage	2	4	780	-7.2	-4.5	-5.2	5.6	-0.5	-26.2	-0.9	3.0	-4.7	1.9
102	Cage	2	4	810	-9.1	-6.2	-6.9	5.7	-0.5	-24.0	-1.4	4.0	-5.4	-0.1
102	Cage	2	4	840	-6.1	-2.8	-3.7	10.3	-0.5	-40.8	-0.5	2.7	-6.8	4.9
102	Cage	2	4	870	4.3	2.7	4.2	11.4	-0.8	-46.8	-6.0	2.3	-7.7	5.8
102	Cage	2	4	900	1.5	1.2	2.0	15.9	-1.0	-56.2	-5.4	1.1	-8.6	8.6
102	Cage	2	4	930	3.3	3.4	3.9	6.6	-0.8	-31.2	-4.7	1.7	-7.7	-0.5
102	Cage	2	4	960	2.7	2.7	3.5	1.7	-0.8	-16.6	-3.1	3.0	-3.7	-0.3
102	Cage	2	4	990	6.8	6.2	7.5	0.1	-0.8	-10.4	-3.3	3.1	-3.8	-2.8
102	Cage	2	4	1020	3.0	2.7	3.8	1.3	-0.7	-42.6	-4.2	3.2	-7.1	5.7
102	Cage	2	4	1050	4.1	3.5	4.9	-0.8	-0.7	-17.1	-2.0	2.9	2.1	8.7
102	Cage	2	4	1080	5.0	3.3	5.3	-0.9	-0.6	1.3	0.6	2.8	6.8	8.1
102	Cage	2	4	1110	-1.7	-0.4	-0.6	-4.9	-0.6	-6.6	-1.8	2.1	5.7	9.1
102	Cage	2	4	1140	6.8	4.3	6.0	7.4	-0.4	-34.9	-2.4	2.6	1.7	15.2
102	Cage	2	4	1170	6.2	6.5	6.6	0.8	-0.1	-24.0	-1.2	3.9	7.3	19.1
102	Cage	2	4	1200	1.6	4.2	2.7	0.8	0.3	-64.9	-3.6	2.9	-9.0	12.3
102	Cage	2	4	1230	3.6	6.5	4.7	8.3	0.5	-30.6	0.3	3.3	5.8	20.1
102	Cage	2	4	1260	0.6	3.7	1.8	-2.7	0.7	-55.0	-1.2	3.3	-2.3	18.0
102	Cage	2	4	1290	6.7	8.6	7.2	4.0	0.9	-15.8	0.6	3.3	7.1	16.1
102	Cage	2	4	1320	8.4	9.4	8.4	1.9	0.9	-22.2	0.9	3.3	3.1	13.0
102	Cage	2	4	1350	7.6	9.2	8.4	8.9	1.2	-37.9	3.9	3.1	-4.8	7.3
102	Cage	2	4	1380	-4.1	-0.4	-1.9	2.9	0.9	-22.6	7.0	3.5	-0.1	7.2
102	Cage	2	4	1410	-6.6	-3.0	-4.5	12.3	0.7	-48.1	10.1	1.9	-7.9	6.2

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
102	Cage	2	4	1440	-8.6	-4.9	-7.1	6.4	0.8	-31.0	8.7	2.6	-7.0	0.6
103	Chair	1	1	10	17.0	13.4	15.1	62.6	-0.3	-164.8	-1.8	0.3	-33.7	13.4
103	Chair	1	1	20	11.0	7.7	9.2	29.4	-0.3	-131.5	-0.5	0.4	-28.9	6.2
103	Chair	1	1	30	6.6	4.1	5.2	15.3	-0.2	-85.3	1.9	-0.2	-22.7	-2.9
103	Chair	1	1	40	7.0	5.1	5.7	14.4	-0.1	-75.7	1.7	-0.1	-19.1	-0.5
103	Chair	1	1	50	1.4	1.5	1.0	7.9	-0.1	-52.5	-0.3	-0.3	-16.2	-4.6
103	Chair	1	1	60	0.2	1.1	0.3	-1.3	-0.1	-19.0	0.4	-0.2	-7.3	-2.9
103	Chair	1	1	70	0.8	0.8	0.3	-1.4	-0.1	-16.6	-0.1	-0.3	-7.7	-4.5
103	Chair	1	1	80	4.0	3.2	3.1	10.1	-0.0	-62.4	-0.6	-0.3	-17.1	-2.2
103	Chair	1	1	90	-1.7	-1.2	-2.0	-4.6	-0.0	-2.5	-1.3	0.3	-6.8	-8.2
103	Chair	1	1	100	4.9	3.7	3.8	2.2	-0.0	-31.5	-0.0	-0.4	-10.2	-3.4
103	Chair	1	1	110	11.9	6.7	8.7	19.9	0.0	-88.2	0.6	0.1	-19.2	5.4
103	Chair	1	1	120	20.2	14.4	16.0	81.7	0.1	-77.1	2.5	0.2	-18.1	1.3
103	Cage	1	1	210	15.6	15.2	16.5	40.8	1.0	-252.5	8.7	-0.4	-76.3	-26.2
103	Cage	1	1	240	32.2	26.4	30.3	35.2	0.5	-230.2	3.0	-2.8	-72.4	-27.6
103	Cage	1	1	270	16.9	15.7	16.6	28.9	0.7	-216.3	4.7	-1.9	-65.3	-23.5
103	Cage	1	1	300	13.7	14.2	14.2	24.9	1.0	-200.3	1.5	-3.2	-61.9	-23.5
103	Cage	1	1	330	8.2	11.4	9.5	23.2	0.9	-212.8	-0.0	-2.5	-56.5	-12.0
103	Cage	1	1	360	12.0	10.9	11.5	26.2	0.8	-200.3	-1.5	-4.0	-58.3	-18.1
103	Cage	1	1	390	4.3	6.1	5.0	18.0	1.0	-181.1	-4.2	-4.2	-47.2	-10.7
103	Cage	1	1	420	-1.6	0.6	-0.6	10.7	1.2	-143.8	-6.4	-3.8	-35.1	-6.6
103	Cage	1	1	450	-5.3	-2.8	-3.8	2.4	0.9	-92.2	-0.9	-4.3	-18.4	-0.8
103	Cage	1	1	480	-6.2	-3.2	-4.0	3.1	0.4	-95.7	1.4	-4.7	-16.0	3.6
103	Cage	1	1	510	-7.6	-4.7	-5.6	-1.0	0.2	-68.0	1.2	-4.0	-11.5	0.9
103	Cage	1	1	540	-9.0	-6.2	-7.1	1.8	0.1	-85.6	0.5	-4.2	-13.3	4.6
103	Cage	1	1	570	-8.7	-4.5	-5.9	1.8	0.1	-86.9	4.6	-3.0	-13.7	4.2
103	Cage	1	1	600	-8.5	-4.5	-5.9	0.5	0.1	-73.5	2.6	-4.9	-10.7	3.5
103	Cage	1	1	630	-10.1	-5.4	-7.0	4.3	0.2	-100.9	0.5	-4.0	-14.6	8.1
103	Cage	1	1	660	-9.9	-5.0	-6.8	2.6	0.3	-87.9	2.9	-4.7	-12.4	6.1
103	Cage	1	1	690	-9.6	-4.1	-6.0	2.8	0.4	-89.2	2.9	-3.7	-11.4	8.3
103	Cage	1	1	720	-9.1	-4.4	-6.0	4.3	0.3	-92.7	3.2	-4.1	-10.8	9.9
103	Cage	1	1	750	-9.4	-4.6	-6.1	1.0	0.3	-76.3	3.9	-4.2	-7.8	8.1
103	Cage	1	1	780	-10.6	-4.7	-6.7	5.6	0.2	-97.6	3.2	-4.4	-10.3	12.8
103	Cage	1	1	810	-8.4	-3.8	-5.4	-0.3	0.1	-64.2	3.2	-3.6	-1.8	12.8
103	Cage	1	1	840	-7.4	-2.8	-4.2	-4.1	0.1	-42.0	5.1	-3.4	3.6	11.8
103	Cage	1	1	870	-7.2	-3.4	-4.5	5.3	0.2	-99.5	3.3	-4.2	-6.6	17.6
103	Cage	1	1	900	-5.8	-2.0	-2.9	2.3	0.1	-68.6	5.0	-4.0	-5.1	9.9
103	Cage	1	1	930	-6.3	-3.4	-4.2	-2.0	0.0	-51.1	2.8	-2.6	-3.7	7.1
103	Cage	1	1	960	-10.4	-6.4	-7.7	-6.5	0.2	-23.7	1.6	-4.9	1.5	6.3
103	Cage	1	1	990	-7.0	-3.3	-4.1	-9.2	0.1	6.9	-0.4	-4.8	8.7	6.7
103	Cage	1	1	1020	-3.1	0.9	-0.8	-6.6	-0.0	64.6	-1.7	-4.7	20.3	4.8
103	Cage	1	1	1050	3.8	3.0	4.6	-6.1	0.0	-18.9	-0.7	-4.4	2.5	7.0
103	Cage	1	1	1080	-6.9	-2.6	-3.7	-12.7	-0.1	44.0	1.2	-0.4	15.6	5.1

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
103	Cage	1	1	1110	0.4	1.8	2.1	-9.7	0.0	37.3	-0.3	-3.7	10.9	1.7
103	Cage	1	1	1140	7.7	6.2	7.9	6.7	0.4	-116.0	1.2	-4.1	-22.1	2.8
103	Cage	1	1	1170	2.4	4.3	4.0	7.4	1.0	-113.9	1.9	-2.1	-24.7	0.2
103	Cage	1	1	1200	2.5	5.3	4.5	-1.4	1.3	-58.4	1.6	-0.4	-11.1	1.0
103	Cage	1	1	1230	3.3	5.2	4.6	0.6	1.5	-76.7	3.7	-0.8	-18.0	-2.6
103	Cage	1	1	1260	3.6	6.0	5.2	-3.0	1.6	-50.1	2.9	-0.9	-10.7	-1.4
103	Cage	1	1	1290	7.1	7.8	7.9	5.7	1.6	-99.2	2.3	-1.1	-21.5	-0.2
103	Cage	1	1	1320	10.3	11.2	11.3	5.3	1.8	-101.9	1.8	-0.4	-26.3	-6.0
103	Cage	1	1	1350	12.5	14.0	14.0	12.1	1.5	-147.8	1.8	-3.9	-36.2	-5.2
103	Cage	1	1	1380	14.7	14.6	15.4	7.2	1.2	-113.5	2.3	-3.2	-31.8	-10.6
103	Cage	1	1	1410	2.4	5.8	4.2	0.7	1.1	-78.0	-1.2	-4.5	-20.1	-5.5
103	Cage	1	1	1440	5.5	8.5	7.4	9.9	1.1	-126.1	1.5	-3.4	-30.2	-4.4
103	Chair	2	4	10	27.4	22.7	25.4	99.2	0.0	-182.1	-3.8	-1.8	-40.7	10.8
103	Chair	2	4	20	8.6	7.3	8.1	30.1	-0.1	-111.1	0.7	0.1	-22.7	7.4
103	Chair	2	4	30	6.5	4.1	5.2	22.2	-0.1	-85.7	1.1	-0.4	-21.4	-0.7
103	Chair	2	4	40	0.9	0.2	0.6	8.9	-0.0	-33.4	1.6	0.0	-13.0	-7.6
103	Chair	2	4	50	4.0	3.2	3.7	16.9	0.0	-59.5	1.4	-0.3	-16.0	-2.3
103	Chair	2	4	60	-2.0	-0.7	-1.3	2.6	0.0	-6.2	0.3	0.1	-7.5	-9.1
103	Chair	2	4	70	-0.7	0.4	-0.2	6.8	0.0	-24.9	-0.1	0.1	-10.0	-5.7
103	Chair	2	4	80	-2.3	-1.6	-1.9	-0.2	0.0	7.9	0.5	0.1	-2.3	-6.1
103	Chair	2	4	90	-0.8	-0.4	-0.7	6.6	0.0	-21.4	0.0	-0.4	-7.7	-3.9
103	Chair	2	4	100	-4.4	-3.3	-3.9	-2.9	0.1	28.4	0.7	-0.1	2.8	-5.7
103	Chair	2	4	110	-1.1	-2.7	-2.0	1.1	0.0	0.1	1.6	-0.4	0.3	1.1
103	Chair	2	4	120	6.4	2.5	4.1	24.9	0.0	-40.9	0.6	-0.8	-9.6	0.5
103	Cage	2	4	210	12.1	13.1	13.1	60.1	1.4	-264.5	3.4	-0.4	-72.4	-13.6
103	Cage	2	4	240	7.4	10.9	9.7	38.4	1.3	-232.0	6.7	-2.1	-69.7	-22.1
103	Cage	2	4	270	17.4	16.4	17.3	38.2	1.1	-215.8	2.3	-1.4	-67.1	-23.3
103	Cage	2	4	300	16.8	16.1	16.7	32.6	0.7	-204.1	1.6	-3.2	-68.0	-28.9
103	Cage	2	4	330	2.7	6.3	4.3	17.6	0.9	-150.6	3.6	-3.3	-52.7	-25.7
103	Cage	2	4	360	-0.5	3.1	1.1	16.1	1.1	-133.3	3.1	-3.2	-44.4	-20.0
103	Cage	2	4	390	-3.1	-0.3	-2.2	7.7	1.2	-90.1	1.6	-3.8	-31.2	-15.7
103	Cage	2	4	420	-4.4	-1.5	-3.2	15.1	1.2	-129.6	2.2	-3.5	-39.1	-14.0
103	Cage	2	4	450	-12.7	-9.1	-10.9	3.8	1.1	-65.2	0.9	-3.2	-19.8	-9.2
103	Cage	2	4	480	-13.3	-10.0	-10.9	-3.6	0.7	-12.2	0.8	-3.7	-2.5	-2.8
103	Cage	2	4	510	-13.6	-10.3	-11.2	-2.2	0.3	-19.0	2.1	-3.6	1.5	4.8
103	Cage	2	4	540	-11.9	-9.7	-10.3	-3.9	-0.2	-7.8	4.4	-4.0	3.3	3.2
103	Cage	2	4	570	-11.3	-8.9	-9.6	-3.7	-0.3	-11.2	6.0	-4.2	2.7	3.1
103	Cage	2	4	600	-12.1	-8.8	-9.8	0.2	-0.2	-41.4	5.4	-4.0	-4.3	3.3
103	Cage	2	4	630	-10.8	-7.2	-8.4	2.1	-0.1	-55.6	6.2	-3.9	-7.7	3.1
103	Cage	2	4	660	-7.9	-5.2	-5.9	1.7	-0.1	-51.9	6.9	-3.6	-7.6	2.3
103	Cage	2	4	690	-7.2	-4.7	-5.2	0.2	-0.1	-40.5	5.8	-4.5	-6.9	-0.2
103	Cage	2	4	720	-8.3	-5.3	-6.0	-0.1	-0.1	-38.2	5.0	-4.3	-4.0	2.8
103	Cage	2	4	750	-7.5	-4.2	-5.0	2.6	0.0	-56.3	3.1	-3.3	-6.6	5.2

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
103	Cage	2	4	780	-6.3	-3.4	-3.6	-3.1	0.1	-14.0	-0.1	-4.0	0.8	2.3
103	Cage	2	4	810	-2.9	-1.5	-0.6	-5.2	0.0	3.3	-1.8	-4.3	3.3	0.2
103	Cage	2	4	840	-5.9	-3.0	-3.1	-5.6	-0.1	15.1	-1.7	-4.6	5.3	0.0
103	Cage	2	4	870	-7.1	-3.5	-4.7	-4.3	-0.3	54.0	3.2	-3.3	15.8	2.7
103	Cage	2	4	900	-6.8	-2.7	-4.4	-8.2	-0.4	110.3	2.6	-1.0	27.9	2.4
103	Cage	2	4	930	-8.2	-4.7	-6.0	-14.4	-0.4	149.1	3.1	3.1	36.0	2.4
103	Cage	2	4	960	-4.4	-1.2	-2.6	-11.8	-0.5	149.3	3.9	3.3	35.0	1.5
103	Cage	2	4	990	-5.9	-1.6	-3.1	-7.9	-0.5	83.3	4.5	0.7	26.2	7.5
103	Cage	2	4	1020	-1.1	2.1	1.0	-8.9	-0.6	103.9	4.2	1.6	26.9	3.1
103	Cage	2	4	1050	-2.3	4.3	0.0	0.6	-0.7	142.7	4.5	3.6	34.4	2.4
103	Cage	2	4	1080	-2.4	3.2	0.1	-1.4	-0.5	113.7	4.8	2.3	29.3	4.2
103	Cage	2	4	1110	-3.0	2.9	-0.5	0.7	-0.6	81.4	0.4	-2.3	19.4	0.8
103	Cage	2	4	1140	11.6	10.3	11.0	23.3	-0.4	-97.9	2.4	-2.9	-20.1	4.9
103	Cage	2	4	1170	5.0	4.8	5.1	16.9	0.4	-139.2	1.9	-2.7	-37.4	-7.8
103	Cage	2	4	1200	3.0	5.3	4.3	10.7	1.1	-99.2	1.0	-1.0	-27.7	-6.9
103	Cage	2	4	1230	6.4	7.7	7.1	18.5	1.3	-137.0	3.9	-0.4	-40.4	-11.3
103	Cage	2	4	1260	5.7	7.0	6.3	11.8	1.5	-101.6	3.0	-0.9	-30.8	-10.0
103	Cage	2	4	1290	2.9	4.2	3.3	2.4	1.5	-48.4	1.8	0.1	-19.9	-11.6
103	Cage	2	4	1320	4.9	5.3	5.1	-4.6	1.6	-0.0	-0.5	1.4	-8.5	-10.9
103	Cage	2	4	1350	11.5	10.3	11.4	16.2	1.4	-133.8	2.1	-3.1	-34.7	-5.9
103	Cage	2	4	1380	15.7	13.5	14.7	17.7	1.3	-137.1	2.9	-2.3	-42.3	-15.2
103	Cage	2	4	1410	11.8	9.8	10.6	23.8	1.0	-153.2	-1.2	-2.9	-43.9	-12.5
103	Cage	2	4	1440	9.5	9.6	9.3	35.5	1.4	-201.7	-0.6	-2.1	-54.8	-10.5
104	Chair	1	1	10	31.3	22.2	26.5	83.2	-0.1	-133.4	-9.0	1.0	-15.8	28.8
104	Chair	1	1	20	8.0	5.6	7.2	32.5	-0.1	-92.8	-3.6	0.5	-15.9	9.6
104	Chair	1	1	30	5.0	3.4	4.3	21.1	-0.0	-65.5	-0.9	0.1	-15.4	-0.7
104	Chair	1	1	40	-0.8	-0.3	-0.6	10.6	-0.1	-34.9	-1.2	0.3	-10.8	-4.5
104	Chair	1	1	50	1.2	0.2	0.6	11.5	-0.1	-28.0	-1.2	0.5	-8.5	-3.0
104	Chair	1	1	60	6.0	4.7	5.2	26.4	-0.1	-60.2	-3.8	-0.4	-13.9	0.1
104	Chair	1	1	70	6.3	5.0	5.6	25.4	0.0	-47.4	-3.7	-0.3	-12.5	-2.6
104	Chair	1	1	80	3.3	2.3	2.6	17.3	0.1	-37.1	-2.7	0.1	-11.9	-5.2
104	Chair	1	1	90	8.0	5.7	6.8	32.6	0.1	-80.4	-4.2	-0.5	-18.7	-0.1
104	Chair	1	1	100	3.5	2.0	2.5	19.8	0.1	-58.2	-2.7	-0.3	-16.9	-5.4
104	Chair	1	1	110	9.7	5.5	7.0	20.3	0.1	-62.2	-2.7	-1.5	-15.6	-2.0
104	Chair	1	1	120	5.8	2.8	3.8	22.4	0.1	-57.0	-1.6	-1.2	-14.2	-1.6
104	Cage	1	1	210	17.7	15.3	17.1	22.9	-0.1	-88.0	4.7	-7.3	-42.4	-34.9
104	Cage	1	1	240	15.7	14.6	15.6	22.8	0.3	-81.1	3.9	-7.6	-37.9	-30.2
104	Cage	1	1	270	3.7	5.9	4.9	18.8	0.6	-77.2	5.1	-8.5	-35.9	-29.0
104	Cage	1	1	300	-1.1	3.8	1.3	17.0	0.5	-71.9	1.5	-7.2	-31.9	-24.9
104	Cage	1	1	330	-4.0	0.3	-2.6	3.1	0.4	-8.0	2.2	-9.4	-18.4	-24.6
104	Cage	1	1	360	-2.3	2.0	-0.5	11.7	0.5	-45.3	0.8	-6.4	-22.2	-19.2
104	Cage	1	1	390	-5.8	-1.5	-4.1	2.8	0.6	-11.0	-1.4	-3.1	-15.0	-19.8
104	Cage	1	1	420	-8.4	-3.4	-6.3	-1.3	0.5	17.0	-3.5	-0.5	-3.5	-12.3

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
104	Cage	1	1	450	-4.9	-2.0	-4.1	-3.7	0.1	12.1	-0.6	0.6	15.4	14.5
104	Cage	1	1	480	-4.9	-2.6	-4.2	-2.6	-0.1	29.0	-2.1	2.5	18.6	14.0
104	Cage	1	1	510	-1.4	-0.1	-1.2	0.1	-0.2	50.9	-1.9	2.1	28.1	21.5
104	Cage	1	1	540	-1.1	-0.7	-1.0	0.3	-0.3	33.8	-2.2	1.4	29.3	26.9
104	Cage	1	1	570	-1.1	-0.7	-1.1	1.2	-0.3	13.9	-1.6	1.0	25.3	28.4
104	Cage	1	1	600	-4.6	-3.7	-4.4	-5.7	-0.4	48.6	-1.4	1.4	35.4	30.9
104	Cage	1	1	630	-4.2	-2.7	-3.4	-1.7	-0.4	32.5	-1.4	1.7	31.2	29.6
104	Cage	1	1	660	-4.4	-2.7	-3.5	-2.4	-0.4	22.5	-0.9	1.3	33.1	34.5
104	Cage	1	1	690	-3.4	-2.8	-3.5	-7.3	-0.3	66.8	0.9	1.1	38.2	29.8
104	Cage	1	1	720	-0.7	0.8	0.4	-1.3	-0.3	21.9	-1.4	2.4	34.4	36.7
104	Cage	1	1	750	0.9	1.8	2.0	4.0	-0.4	0.4	-0.6	2.1	27.8	34.8
104	Cage	1	1	780	3.1	2.6	2.9	3.9	-0.4	18.0	1.6	1.6	29.4	31.7
104	Cage	1	1	810	4.5	4.0	5.1	14.9	-0.4	-36.2	1.9	1.8	15.4	30.7
104	Cage	1	1	840	1.1	2.5	2.7	20.6	-0.4	-66.0	1.0	0.8	5.7	27.4
104	Cage	1	1	870	1.5	2.3	2.5	17.0	-0.4	-52.4	2.2	1.3	6.3	24.6
104	Cage	1	1	900	2.7	3.2	3.5	17.6	-0.3	-38.5	0.5	1.6	10.6	25.7
104	Cage	1	1	930	3.4	3.9	4.2	10.1	-0.3	-5.7	0.9	2.0	18.5	24.9
104	Cage	1	1	960	0.9	1.2	1.2	0.9	-0.3	33.6	-0.6	3.4	36.1	35.8
104	Cage	1	1	990	6.6	5.9	6.8	1.6	-0.4	18.9	-1.2	3.1	32.1	35.8
104	Cage	1	1	1020	6.0	5.2	5.7	0.0	-0.4	44.4	-1.2	4.0	40.7	40.2
104	Cage	1	1	1050	3.0	3.1	3.2	-8.9	-0.3	72.0	-1.6	4.4	50.3	43.2
104	Cage	1	1	1080	5.7	5.6	5.9	-2.8	-0.2	56.1	-2.8	5.1	46.8	43.8
104	Cage	1	1	1110	4.6	5.4	5.2	0.7	-0.0	20.7	-1.8	3.3	36.7	41.5
104	Cage	1	1	1140	-0.3	1.7	1.3	8.3	0.1	-39.2	-2.4	3.4	11.0	25.2
104	Cage	1	1	1170	-3.7	-2.0	-3.4	-18.3	0.3	58.6	0.7	2.5	50.5	45.7
104	Cage	1	1	1200	6.5	5.3	6.3	25.0	0.5	-69.8	-2.5	3.3	15.2	44.6
104	Cage	1	1	1230	6.1	4.7	6.1	38.8	0.7	-119.5	-4.1	1.7	-19.5	11.8
104	Cage	1	1	1260	6.1	4.8	5.9	25.3	1.1	-88.1	-5.0	2.4	-16.2	4.5
104	Cage	1	1	1290	11.6	8.5	10.7	51.3	1.2	-113.9	-3.8	0.9	-19.7	9.3
104	Cage	1	1	1320	13.9	10.1	12.2	30.6	1.4	-103.7	-3.8	0.1	-20.2	4.4
104	Cage	1	1	1350	10.5	9.2	10.2	17.3	0.7	-69.6	3.2	-2.1	-8.3	8.5
104	Cage	1	1	1380	2.7	3.3	2.6	7.8	0.5	-22.4	-0.4	-0.5	5.0	12.4
104	Cage	1	1	1410	-1.1	0.9	-0.6	6.1	0.5	-8.0	-3.8	0.7	16.1	22.2
104	Cage	1	1	1440	-2.2	0.2	-1.5	4.1	0.4	-1.8	-4.2	2.4	12.6	16.0
104	Chair	2	4	10	25.2	18.4	22.0	73.1	0.1	-94.8	3.4	-4.5	-11.0	26.8
104	Chair	2	4	20	7.9	6.1	7.0	26.1	0.2	-57.6	6.9	-3.5	-14.9	-0.7
104	Chair	2	4	30	2.3	2.5	2.7	20.4	0.2	-45.0	8.8	-4.3	-15.9	-7.8
104	Chair	2	4	40	-4.4	-1.2	-2.9	1.8	0.2	-4.0	8.1	-4.8	-8.2	-12.0
104	Chair	2	4	50	-7.2	-3.8	-5.5	-2.5	0.2	8.2	6.7	-4.2	-5.7	-12.6
104	Chair	2	4	60	-1.9	-0.1	-1.0	2.2	0.2	-1.7	4.4	-3.7	-3.9	-6.1
104	Chair	2	4	70	6.3	5.7	5.8	26.1	0.1	-28.3	6.6	-5.1	-7.4	-1.0
104	Chair	2	4	80	19.5	15.7	17.4	63.9	0.2	-60.5	5.9	-5.1	-20.3	-7.8
104	Chair	2	4	90	19.7	15.7	17.6	69.6	0.2	-83.2	6.5	-5.0	-23.6	-2.8

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
104	Chair	2	4	100	17.4	13.7	15.2	60.6	0.2	-76.0	5.7	-4.7	-21.0	-1.2
104	Chair	2	4	110	6.9	3.5	4.9	21.5	0.2	-47.2	12.2	-5.1	-16.4	-7.9
104	Chair	2	4	120	6.3	3.4	4.6	19.2	0.1	-29.7	10.6	-4.9	-14.4	-11.6
104	Cage	2	4	210	18.8	16.8	18.7	39.6	0.8	-151.4	2.3	-8.9	-55.2	-31.7
104	Cage	2	4	240	5.1	8.7	7.6	25.1	0.2	-118.9	5.6	-9.2	-54.5	-42.6
104	Cage	2	4	270	5.5	6.7	6.1	24.8	0.5	-109.8	2.0	-10.1	-45.8	-32.0
104	Cage	2	4	300	-2.9	1.7	-0.9	13.5	0.6	-75.9	-0.5	-7.3	-37.2	-31.4
104	Cage	2	4	330	-5.7	-0.7	-3.8	3.5	0.6	-27.5	-2.6	-3.5	-24.6	-28.7
104	Cage	2	4	360	-5.2	-0.2	-3.4	5.5	0.5	-39.5	-2.0	-2.5	-13.0	-8.5
104	Cage	2	4	390	-6.2	-1.9	-4.8	0.4	0.5	-17.7	0.5	-2.4	2.6	5.2
104	Cage	2	4	420	-4.8	-0.2	-3.2	4.3	0.5	-24.9	-0.2	-2.1	-1.3	2.6
104	Cage	2	4	450	-7.7	-2.4	-5.1	3.5	0.3	-22.6	-1.3	-0.3	10.2	17.1
104	Cage	2	4	480	-6.3	-2.0	-4.4	2.1	0.0	-5.3	-2.7	0.4	16.3	19.7
104	Cage	2	4	510	-5.3	-1.9	-3.8	0.8	-0.1	11.3	-3.4	0.7	20.9	20.7
104	Cage	2	4	540	-5.9	-2.8	-4.9	-0.0	-0.2	15.4	-2.0	-0.3	20.9	19.9
104	Cage	2	4	570	-5.0	-2.4	-4.1	-3.5	-0.2	38.6	-2.7	0.3	27.1	20.8
104	Cage	2	4	600	-5.1	-3.3	-4.5	-8.3	-0.3	41.5	-2.5	0.0	33.7	27.6
104	Cage	2	4	630	-4.6	-3.0	-4.2	-8.4	-0.4	55.4	0.3	-0.9	31.1	21.1
104	Cage	2	4	660	0.5	0.0	0.0	-16.8	-0.3	89.3	2.9	-1.1	40.1	22.8
104	Cage	2	4	690	-3.8	-2.8	-3.4	-11.7	-0.4	77.5	-1.6	0.7	39.7	25.3
104	Cage	2	4	720	-1.3	-0.5	-0.8	-11.8	-0.5	88.6	1.8	0.5	38.2	21.0
104	Cage	2	4	750	-0.9	0.7	-0.0	-7.4	-0.6	84.7	3.4	0.1	34.6	17.8
104	Cage	2	4	780	-3.6	-1.5	-2.5	-13.4	-0.6	98.7	0.4	2.1	42.7	23.7
104	Cage	2	4	810	-1.8	-0.4	-1.0	-16.6	-0.7	141.5	2.7	0.2	45.6	16.2
104	Cage	2	4	840	-3.1	-1.4	-2.3	-19.8	-0.7	119.0	0.6	2.4	52.0	30.3
104	Cage	2	4	870	-1.5	-0.3	-1.1	-20.4	-0.7	140.9	2.0	0.7	54.7	27.5
104	Cage	2	4	900	0.5	2.3	1.4	-15.6	-0.7	137.9	2.3	0.9	54.4	27.9
104	Cage	2	4	930	-0.2	1.0	0.4	-13.8	-0.7	113.2	1.9	1.4	46.4	25.2
104	Cage	2	4	960	-1.1	0.3	-0.2	-14.6	-0.6	143.1	2.5	2.4	50.8	22.7
104	Cage	2	4	990	-0.2	1.5	0.8	-13.3	-0.7	133.0	3.6	1.1	42.2	14.5
104	Cage	2	4	1020	4.0	4.3	4.4	-8.4	-0.6	155.9	4.1	1.7	45.2	13.3
104	Cage	2	4	1050	0.7	1.2	1.3	-10.1	-0.5	109.7	2.6	2.0	43.8	23.6
104	Cage	2	4	1080	-2.4	-0.5	-1.4	-9.6	-0.3	79.7	-0.0	2.1	38.7	25.0
104	Cage	2	4	1110	-1.9	-1.8	-2.1	-18.3	-0.2	118.6	-0.5	1.7	48.4	25.6
104	Cage	2	4	1140	3.1	3.4	3.9	14.6	-0.0	-67.9	-2.4	1.7	-1.8	17.3
104	Cage	2	4	1170	3.7	3.8	4.3	28.4	0.3	-107.9	-1.3	0.6	-20.4	5.0
104	Cage	2	4	1200	-1.4	-0.3	-1.3	-3.7	0.5	7.7	-0.4	0.4	8.2	6.3
104	Cage	2	4	1230	6.0	5.6	6.4	34.7	0.5	-93.7	-2.8	0.6	-19.3	1.8
104	Cage	2	4	1260	10.5	8.0	10.0	46.5	0.8	-97.1	-4.1	1.1	-24.7	-5.5
104	Cage	2	4	1290	7.0	5.1	6.6	30.3	1.0	-103.1	-3.5	0.2	-29.9	-11.0
104	Cage	2	4	1320	9.8	6.7	8.2	18.6	1.1	-78.8	-4.8	0.8	-21.3	-7.0
104	Cage	2	4	1350	13.7	10.4	12.6	21.1	0.7	-101.7	3.3	-4.0	-26.1	-6.3
104	Cage	2	4	1380	1.8	2.8	2.4	13.4	0.4	-71.9	4.6	-3.5	-18.3	-5.7

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
104	Cage	2	4	1410	-0.6	-0.1	-1.3	-5.1	0.4	31.5	0.9	-0.6	3.3	-5.8
104	Cage	2	4	1440	1.6	2.3	1.3	-0.2	0.3	28.2	-0.5	-0.5	6.7	-0.2
201	Chair	1	1	10	30.1	30.9	30.3	83.3	-0.2	-215.8	-23.1	-3.5	-52.9	-5.7
201	Chair	1	1	20	0.2	2.1	1.3	14.9	-0.1	-74.8	-6.8	-0.4	-22.2	-9.2
201	Chair	1	1	30	4.4	4.1	4.2	15.4	-0.0	-79.7	-6.6	-0.4	-21.9	-6.0
201	Chair	1	1	40	-8.6	-6.7	-7.5	-1.5	0.1	-1.6	3.5	-1.3	-18.9	-25.2
201	Chair	1	1	50	-12.2	-9.6	-11.0	-10.3	0.1	.	.	.	.	.
201	Chair	1	1	60	-11.6	-7.9	-10.1	-7.9	0.1	13.9	-0.8	-1.9	-1.8	-3.9
201	Chair	1	1	70	3.5	4.3	3.8	8.7	0.0	-76.4	-12.5	-0.4	-18.1	-1.8
201	Chair	1	1	80	-5.7	-4.3	-5.1	-7.2	0.1	-44.0	-1.7	-2.0	-9.4	0.1
201	Chair	1	1	90	-4.6	-3.9	-4.3	-7.0	0.2	-54.9	-8.2	-0.3	-5.2	9.9
201	Chair	1	1	100	6.5	5.4	5.7	17.6	0.2	-103.8	-13.6	-0.8	-24.7	-2.7
201	Chair	1	1	110	3.3	2.3	2.5	16.6	0.3	-98.6	-14.4	-0.4	-24.9	-5.0
201	Chair	1	1	120	6.6	6.0	5.7	26.7	0.4	-108.6	-13.0	-0.9	-25.8	-2.1
201	Cage	1	1	210	14.0	14.4	16.8	23.5	0.4	-307.0	8.5	3.8	-102.3	-47.7
201	Cage	1	1	240	21.1	19.6	22.6	28.9	0.7	-319.6	3.0	-3.2	-103.0	-44.9
201	Cage	1	1	270	19.0	19.2	20.6	19.7	0.8	-283.4	4.4	3.6	-85.9	-32.6
201	Cage	1	1	300	4.4	4.3	5.6	17.0	0.4	-273.5	3.1	2.4	-84.9	-34.4
201	Cage	1	1	330	0.8	0.7	2.0	8.7	0.4	-219.9	1.7	1.3	-75.4	-37.0
201	Cage	1	1	360	-2.8	-2.9	-1.9	5.0	0.4	-197.2	1.8	0.2	-71.6	-38.4
201	Cage	1	1	390	-1.9	-2.5	-1.3	-0.7	0.4	-157.5	1.1	-2.2	-66.3	-41.3
201	Cage	1	1	420	-4.4	-5.0	-4.0	-2.6	0.6	-143.6	-2.4	0.9	-54.1	-30.3
201	Cage	1	1	450	-9.8	-9.3	-9.0	-0.0	0.8	-169.0	-6.3	3.9	-45.9	-14.2
201	Cage	1	1	480	-7.7	-6.8	-6.2	-1.1	0.5	-156.8	-5.3	2.3	-44.3	-15.3
201	Cage	1	1	510	-7.8	-7.5	-6.8	-3.4	0.2	-137.5	-0.4	-1.6	-39.3	-14.4
201	Cage	1	1	540	-8.0	-8.9	-7.7	-8.0	0.0	-100.7	-4.3	2.6	-27.5	-9.3
201	Cage	1	1	570	-7.0	-7.5	-6.3	-7.0	-0.1	-111.9	-0.9	-0.7	-25.5	-4.1
201	Cage	1	1	600	-7.2	-8.4	-6.9	-9.3	-0.2	-90.3	-2.2	0.4	-18.6	-1.4
201	Cage	1	1	630	-5.2	-5.5	-4.5	-7.9	0.0	-66.7	-3.5	-0.3	-14.3	-2.2
201	Cage	1	1	660	-7.4	-8.7	-7.3	-8.3	-0.1	-100.0	-0.3	-0.8	-17.2	2.7
201	Cage	1	1	690	-6.9	-8.5	-6.8	-10.5	-0.0	-91.5	-0.4	-0.6	-14.5	4.5
201	Cage	1	1	720	-8.5	-10.1	-8.7	-10.2	0.0	-83.3	-3.1	2.2	-11.8	4.9
201	Cage	1	1	750	-9.2	-9.7	-8.6	-6.7	0.3	-109.8	-3.2	2.3	-12.9	10.2
201	Cage	1	1	780	-7.7	-8.8	-7.3	-10.6	-0.1	-81.6	0.5	-1.0	-9.2	7.4
201	Cage	1	1	810	-7.4	-9.4	-7.4	-10.2	-0.1	-50.7	-1.2	1.7	-0.1	10.6
201	Cage	1	1	840	-6.8	-8.2	-6.7	-9.3	0.1	-38.3	-1.5	2.4	-0.1	7.9
201	Cage	1	1	870	-7.0	-8.3	-6.7	-7.6	0.1	-69.8	-3.0	2.9	-4.8	10.5
201	Cage	1	1	900	-6.3	-7.7	-6.0	-8.6	-0.2	-77.4	0.1	0.6	-6.4	10.3
201	Cage	1	1	930	-7.6	-9.8	-8.0	-11.7	-0.1	-20.7	-1.3	2.7	5.7	9.4
201	Cage	1	1	960	-5.1	-5.9	-4.2	-6.5	-0.0	-77.6	-0.3	1.3	-4.1	13.2
201	Cage	1	1	990	-4.6	-4.1	-4.0	-2.9	-0.2	-53.1	-0.4	0.6	-3.2	7.5
201	Cage	1	1	1020	-3.7	-2.0	-3.5	5.6	-0.2	-37.0	-2.0	0.2	0.1	7.8
201	Cage	1	1	1050	-2.1	-1.0	-2.3	3.5	-0.1	-18.0	-2.1	-0.4	6.2	10.0

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
201	Cage	1	1	1080	4.1	5.2	5.0	2.4	-0.3	-81.3	1.0	1.5	-9.3	7.2
201	Cage	1	1	1110	3.2	4.6	4.4	7.4	-0.2	-120.0	-0.0	2.6	-21.1	4.2
201	Cage	1	1	1140	-2.6	-6.3	-4.2	-9.4	0.3	-9.6	-1.9	1.2	-3.7	-2.9
201	Cage	1	1	1170	-4.4	-7.3	-6.2	-10.2	0.6	16.5	-0.6	1.5	1.5	-3.0
201	Cage	1	1	1200	1.0	3.8	1.6	9.8	0.5	-54.2	0.2	0.9	-15.4	-5.1
201	Cage	1	1	1230	4.5	1.2	4.4	12.2	0.9	-134.3	-0.7	2.4	-40.9	-15.3
201	Cage	1	1	1260	0.2	-4.1	-0.5	19.8	1.5	-246.1	2.6	4.0	-77.2	-31.2
201	Cage	1	1	1290	3.1	0.5	3.4	32.0	1.7	-299.8	4.8	4.4	-90.0	-32.2
201	Cage	1	1	1320	15.3	14.3	16.7	45.8	2.0	-341.9	5.8	4.3	-95.6	-27.2
201	Cage	1	1	1350	27.4	23.8	27.2	53.9	0.4	-351.1	5.2	4.4	-91.0	-17.2
201	Cage	1	1	1380	11.9	10.8	13.2	30.0	0.3	-295.8	4.0	3.7	-85.3	-27.1
201	Cage	1	1	1410	2.8	1.4	3.8	10.9	0.8	-217.6	2.7	0.7	-63.2	-21.8
201	Cage	1	1	1440	-1.9	-6.1	-3.1	0.0	1.0	-168.2	-0.1	-4.4	-55.8	-26.4
201	Chair	3	4	10	20.4	21.4	20.9	62.2	0.5	-99.1	-6.8	-2.2	-26.5	-5.9
201	Chair	3	4	20	-6.3	-2.3	-4.1	1.8	0.6	-20.1	0.2	1.5	-0.5	4.9
201	Chair	3	4	30	-9.9	-7.2	-8.5	-5.5	0.7	7.9	5.8	1.8	4.8	3.6
201	Chair	3	4	40	-11.1	-7.8	-9.4	-12.0	0.6	41.2	8.7	0.8	6.0	-5.5
201	Chair	3	4	50	-10.5	-7.0	-8.7	-5.3	0.5	0.8	4.9	1.0	5.7	7.1
201	Chair	3	4	60	-14.0	-10.1	-12.2	-13.4	0.5	45.3	9.6	0.2	6.9	-5.6
201	Chair	3	4	70	-14.9	-10.9	-12.9	-19.5	0.5	77.1	10.2	0.4	14.7	-3.1
201	Chair	3	4	80	-15.5	-11.9	-13.7	-20.3	0.4	82.1	10.0	-0.3	15.1	-3.8
201	Chair	3	4	90	-10.6	-6.7	-8.8	-6.6	0.4	20.8	4.2	-0.1	4.8	-0.5
201	Chair	3	4	100	-11.9	-8.4	-10.4	-9.2	0.5	18.7	6.2	-0.5	6.5	2.6
201	Chair	3	4	110	-10.8	-8.2	-9.7	-7.9	0.5	25.9	6.1	-0.1	6.1	-0.7
201	Chair	3	4	120	-4.8	-2.6	-4.0	9.2	0.6	11.7	3.3	0.0	1.7	-2.4
201	Cage	3	4	210	19.6	20.0	22.1	33.4	1.8	-271.9	3.2	-3.7	-87.3	-37.8
201	Cage	3	4	240	25.4	24.3	26.8	14.3	0.4	-200.3	9.5	3.6	-57.4	-18.7
201	Cage	3	4	270	11.9	12.1	13.9	5.8	0.4	-158.1	7.9	1.5	-56.3	-30.3
201	Cage	3	4	300	8.5	10.3	11.0	18.0	0.5	-188.4	3.1	-2.3	-62.4	-28.7
201	Cage	3	4	330	7.7	8.0	9.3	8.9	0.2	-152.8	4.6	2.1	-53.4	-27.3
201	Cage	3	4	360	2.0	0.7	2.0	-3.3	0.7	-87.6	5.1	3.3	-40.2	-28.7
201	Cage	3	4	390	-0.2	-2.5	-0.9	-12.7	1.0	-45.6	5.3	1.1	-29.8	-26.7
201	Cage	3	4	420	0.6	-1.6	0.1	-12.2	0.7	-41.3	4.3	1.8	-28.4	-26.0
201	Cage	3	4	450	-7.2	-9.5	-8.3	-21.9	0.8	20.8	-2.3	1.7	-0.5	-8.0
201	Cage	3	4	480	-8.1	-9.9	-8.8	-24.2	0.7	41.1	-3.0	2.3	8.7	-2.4
201	Cage	3	4	510	-5.5	-7.4	-5.7	-24.5	0.4	43.0	-2.7	1.3	9.7	-1.6
201	Cage	3	4	540	-7.8	-10.5	-8.9	-26.5	0.1	63.1	-0.1	0.3	18.0	3.5
201	Cage	3	4	570	-8.9	-12.2	-10.6	-29.9	0.1	102.4	0.4	-0.1	22.9	-0.3
201	Cage	3	4	600	-8.1	-11.2	-9.5	-30.3	0.1	116.9	-0.4	-0.9	23.6	-2.4
201	Cage	3	4	630	-3.7	-6.5	-4.8	-28.0	0.1	116.5	-0.3	-0.7	24.0	-1.6
201	Cage	3	4	660	-8.4	-12.5	-10.2	-32.0	0.2	129.5	-0.2	0.0	29.7	1.5
201	Cage	3	4	690	-8.1	-11.8	-9.8	-33.2	0.2	139.8	0.4	-1.1	33.9	3.7
201	Cage	3	4	720	-7.5	-10.5	-8.6	-32.3	0.2	135.6	0.4	0.8	36.0	7.3

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
201	Cage	3	4	750	-6.8	-10.1	-8.0	-34.3	0.1	151.4	0.1	1.4	38.8	6.9
201	Cage	3	4	780	-6.2	-8.7	-7.1	-33.3	0.1	167.8	0.0	1.0	40.5	5.3
201	Cage	3	4	810	-7.0	-10.1	-8.3	-37.6	0.1	195.3	0.6	0.9	45.3	4.5
201	Cage	3	4	840	-6.3	-8.0	-6.7	-34.4	0.0	157.3	1.5	0.5	42.4	9.7
201	Cage	3	4	870	-0.1	-0.9	0.8	-27.8	-0.1	109.3	0.1	2.4	32.5	9.7
201	Cage	3	4	900	-4.1	-4.5	-2.9	-28.7	-0.0	101.1	-0.0	2.2	33.6	13.3
201	Cage	3	4	930	-5.8	-6.4	-4.7	-28.8	-0.2	110.9	-0.1	2.1	34.3	10.9
201	Cage	3	4	960	-3.0	-3.8	-1.9	-29.0	-0.1	126.0	-0.4	2.5	36.7	10.8
201	Cage	3	4	990	-1.5	-1.0	-0.9	-25.0	-0.2	162.1	-0.3	2.2	40.4	7.0
201	Cage	3	4	1020	-2.1	-2.8	-1.5	-28.6	-0.2	165.4	0.3	1.9	40.9	6.8
201	Cage	3	4	1050	-0.6	2.2	-0.2	-20.9	-0.5	214.7	2.0	1.0	46.2	2.1
201	Cage	3	4	1080	1.2	3.8	1.5	-16.5	-0.4	157.5	0.7	2.5	41.6	10.2
201	Cage	3	4	1110	0.3	3.4	0.9	-13.7	-0.3	142.8	1.4	1.9	38.0	9.9
201	Cage	3	4	1140	2.6	2.3	3.3	-14.2	-0.2	66.2	1.1	2.6	14.8	0.0
201	Cage	3	4	1170	0.5	-1.6	0.7	-10.6	0.2	34.6	2.1	3.2	6.6	-1.4
201	Cage	3	4	1200	1.6	-1.7	1.7	4.5	0.7	-106.0	4.8	4.1	-40.1	-21.5
201	Cage	3	4	1230	-2.7	-8.0	-4.1	-15.9	1.3	2.8	2.2	-1.5	-33.3	-41.3
201	Cage	3	4	1260	-0.8	-4.6	-1.5	-1.6	1.5	-71.8	3.2	1.9	-52.2	-46.9
201	Cage	3	4	1290	0.8	-3.3	0.0	-3.2	1.7	-56.2	2.5	3.0	-52.5	-51.6
201	Cage	3	4	1320	16.4	13.4	16.5	26.2	1.7	-181.3	4.5	3.4	-50.7	-14.1
201	Cage	3	4	1350	10.6	10.0	11.9	52.6	0.5	-261.3	4.3	4.6	-69.0	-12.1
201	Cage	3	4	1380	0.7	-0.6	1.1	21.1	0.9	-150.3	5.4	0.4	-67.6	-46.5
201	Cage	3	4	1410	3.0	0.1	2.2	6.6	1.2	-94.8	5.6	1.1	-59.2	-49.6
201	Cage	3	4	1440	-0.0	-2.7	-0.3	3.5	1.3	-87.5	3.8	-3.3	-64.0	-57.4
202	Chair	1	1	10	17.6	15.2	16.3	52.4	0.4	-48.5	3.3	-0.3	-22.7	-15.7
202	Chair	1	1	20	11.1	9.2	9.9	34.4	0.2	-49.9	1.4	1.3	-21.2	-12.3
202	Chair	1	1	30	11.3	8.9	10.1	46.1	0.1	-70.2	-1.6	1.0	-27.4	-12.1
202	Chair	1	1	40	12.6	10.6	11.5	41.2	0.1	-61.3	-2.5	0.6	-24.3	-11.5
202	Chair	1	1	50	0.6	1.0	0.6	11.8	0.1	-9.9	-4.2	-1.2	-8.9	-10.7
202	Chair	1	1	60	-2.1	-3.4	-2.8	-0.2	-0.1	-2.3	-2.9	-2.1	0.8	2.1
202	Chair	1	1	70	11.8	9.2	10.8	38.3	-0.1	-54.8	-1.6	1.6	-21.4	-9.8
202	Chair	1	1	80	8.1	6.8	7.6	25.7	0.0	-16.7	-3.5	-1.2	-9.4	-8.1
202	Chair	1	1	90	3.6	1.6	2.8	9.4	-0.0	-3.9	-3.5	-2.5	-3.5	-4.6
202	Chair	1	1	100	4.5	0.5	2.7	15.6	-0.0	-17.3	-0.8	0.6	-4.1	1.0
202	Chair	1	1	110	12.1	8.8	10.5	39.3	0.0	-54.1	0.4	1.7	-21.5	-10.4
202	Chair	1	1	120	17.1	13.3	15.1	48.6	0.1	-61.7	-1.2	1.7	-25.3	-13.1
202	Cage	1	1	210	25.7	24.1	26.1	28.3	-0.6	-105.1	0.4	-1.5	-35.3	-15.4
202	Cage	1	1	240	22.0	16.4	19.5	38.9	-0.4	-136.2	-1.8	-2.2	-41.8	-13.0
202	Cage	1	1	270	20.3	17.1	18.8	28.7	0.3	-109.0	0.1	-2.1	-33.9	-12.0
202	Cage	1	1	300	28.7	24.9	27.2	28.1	0.4	-108.8	-1.5	-1.7	-30.4	-6.6
202	Cage	1	1	330	18.6	14.2	16.1	25.9	0.4	-95.6	-2.4	-1.3	-17.8	7.4
202	Cage	1	1	360	21.0	15.4	17.9	28.1	0.4	-102.0	-2.8	-2.0	-20.5	6.1
202	Cage	1	1	390	24.2	18.7	21.0	28.7	0.5	-109.0	-2.9	-1.6	-18.0	12.6

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
202	Cage	1	1	420	7.2	2.3	4.0	21.1	0.4	-84.5	-2.3	1.1	-4.4	23.1
202	Cage	1	1	450	2.7	2.6	2.2	16.3	0.3	-72.4	-2.8	1.6	3.3	29.6
202	Cage	1	1	480	-0.0	1.6	0.2	13.2	0.2	-61.6	-1.6	1.3	7.7	31.6
202	Cage	1	1	510	6.2	6.6	6.1	18.9	-0.0	-78.0	-2.9	-0.2	1.9	30.0
202	Cage	1	1	540	0.2	-0.1	-0.8	4.9	-0.3	-31.9	-1.4	4.1	15.0	30.5
202	Cage	1	1	570	3.3	3.2	2.9	16.5	-0.3	-69.6	-0.8	-0.6	4.1	29.8
202	Cage	1	1	600	2.1	2.1	1.7	11.8	-0.2	-56.1	2.0	-0.3	7.2	28.6
202	Cage	1	1	630	0.8	1.0	0.4	7.1	-0.5	-38.6	4.3	1.0	11.2	27.7
202	Cage	1	1	660	-0.6	0.5	-0.9	8.4	-0.5	-44.7	6.9	0.1	11.8	30.8
202	Cage	1	1	690	2.4	3.6	2.6	17.3	-0.5	-74.3	1.5	-0.6	2.3	28.9
202	Cage	1	1	720	1.8	3.2	2.2	12.2	-0.3	-60.1	-0.3	-0.4	8.5	32.2
202	Cage	1	1	750	-2.2	-0.0	-1.9	15.0	-0.4	-81.6	-0.8	-4.3	10.4	44.6
202	Cage	1	1	780	-1.0	1.1	-0.5	13.2	-0.5	-81.7	-0.8	-4.9	11.2	45.6
202	Cage	1	1	810	-1.7	-0.1	-1.5	13.0	-0.4	-90.9	-2.2	-5.8	7.4	43.7
202	Cage	1	1	840	-0.5	2.0	0.5	20.6	-0.3	-100.5	-2.6	-3.9	0.0	37.1
202	Cage	1	1	870	1.6	1.9	1.3	14.7	-0.3	-77.9	-1.0	-3.4	5.5	35.2
202	Cage	1	1	900	-1.2	-0.5	-1.4	21.4	-0.4	-104.3	-2.8	-5.1	0.1	38.8
202	Cage	1	1	930	-2.7	-3.6	-3.0	7.2	-0.3	-39.6	1.2	1.8	13.4	31.5
202	Cage	1	1	960	0.9	0.7	1.0	9.7	-0.4	-49.1	4.5	0.3	10.8	31.2
202	Cage	1	1	990	-1.3	-1.1	-1.5	9.6	-0.4	-50.0	3.3	1.1	9.4	29.4
202	Cage	1	1	1020	-3.2	-3.2	-3.5	8.5	-0.4	-38.9	0.1	1.8	14.0	32.4
202	Cage	1	1	1050	-6.5	-6.8	-7.4	-0.8	-0.3	1.4	2.9	0.1	25.4	33.8
202	Cage	1	1	1080	-4.8	-5.6	-5.9	-8.9	-0.4	48.6	5.8	0.2	35.0	30.9
202	Cage	1	1	1110	4.2	1.6	3.0	5.8	-0.3	-24.3	4.6	1.0	11.2	23.2
202	Cage	1	1	1140	1.3	-1.4	-0.3	-0.3	0.1	-4.5	4.3	0.7	10.9	15.7
202	Cage	1	1	1170	-1.6	-3.6	-4.0	-17.8	0.5	82.0	-1.6	2.0	38.5	24.8
202	Cage	1	1	1200	2.3	0.7	0.9	-3.1	0.6	17.3	0.8	1.4	20.2	21.4
202	Cage	1	1	1230	6.4	3.0	4.4	10.4	0.9	-46.6	1.4	-0.4	-5.8	7.2
202	Cage	1	1	1260	4.4	1.6	2.5	7.3	1.2	-35.2	0.1	-0.1	-0.9	10.0
202	Cage	1	1	1290	12.9	9.3	11.3	42.6	1.2	-121.3	-1.2	-1.9	-26.8	5.4
202	Cage	1	1	1320	15.6	12.8	14.8	28.8	1.1	-109.3	0.1	0.6	-28.3	-2.8
202	Cage	1	1	1350	19.0	18.7	19.7	18.4	-0.2	-71.1	2.3	-0.0	-18.2	-2.8
202	Cage	1	1	1380	11.7	11.8	11.7	5.9	-0.6	-20.3	0.5	0.6	-5.9	-2.9
202	Cage	1	1	1410	2.9	4.6	3.4	6.2	-0.0	-26.2	-0.6	0.7	-0.7	6.5
202	Cage	1	1	1440	-1.6	1.0	-1.0	1.4	0.3	-4.6	-1.4	1.0	8.9	12.6
202	Chair	3	4	10	3.9	2.7	3.5	0.8	0.2	35.2	-1.4	-2.3	5.0	-10.9
202	Chair	3	4	20	0.1	-2.0	-0.9	-8.9	0.2	22.1	-2.5	-2.0	6.4	-1.1
202	Chair	3	4	30	-13.9	-13.0	-13.4	-38.3	0.0	60.8	-2.9	-4.3	20.3	2.9
202	Chair	3	4	40	-23.3	-22.2	-22.9	-61.8	-0.2	.	.	.	.	.
202	Chair	3	4	50	-12.5	-12.0	-12.1	-37.9	-0.2	90.9	4.8	-1.5	25.3	-1.4
202	Chair	3	4	60	-13.1	-11.6	-12.2	-41.4	-0.2	68.9	-1.5	-3.8	22.5	2.7
202	Chair	3	4	70	-8.8	-9.7	-9.2	-27.1	-0.2	41.9	-1.2	-1.8	15.2	3.8
202	Chair	3	4	80	-6.7	-7.0	-6.7	-27.7	-0.2	58.8	-1.5	-2.8	19.4	2.2

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
202	Chair	3	4	90	-2.9	-4.1	-3.3	-14.0	-0.1	32.5	1.5	0.8	6.1	-7.2
202	Chair	3	4	100	-3.3	-5.7	-4.4	-12.4	-0.2	29.9	2.6	1.1	9.7	0.7
202	Chair	3	4	110	6.2	2.8	4.7	15.6	-0.0	-8.5	-0.8	-1.9	-10.1	-14.2
202	Chair	3	4	120	5.0	2.5	4.0	6.5	0.1	-0.1	-1.8	0.1	-6.8	-12.6
202	Cage	3	4	210	15.6	13.2	15.4	47.4	0.5	-185.7	1.4	-2.0	-59.3	-21.0
202	Cage	3	4	240	34.0	31.3	33.5	43.5	0.1	-173.5	0.9	-3.0	-57.5	-23.0
202	Cage	3	4	270	55.1	49.1	53.2	44.6	-0.1	-178.9	0.1	-3.3	-56.6	-19.5
202	Cage	3	4	300	28.8	24.5	27.4	36.6	0.0	-154.0	-0.7	-2.7	-47.2	-15.1
202	Cage	3	4	330	21.1	17.3	19.5	27.2	0.1	-117.9	-0.9	-3.0	-35.0	-10.4
202	Cage	3	4	360	17.6	13.2	15.0	26.5	0.0	-114.3	-1.6	-3.1	-32.7	-8.1
202	Cage	3	4	390	12.7	9.8	10.7	20.5	0.5	-93.7	-2.6	-1.1	-23.2	-2.1
202	Cage	3	4	420	7.1	5.0	5.0	19.7	0.6	-92.4	-1.9	-0.8	-15.5	8.5
202	Cage	3	4	450	-0.3	-1.1	-1.4	12.8	0.4	-67.1	-0.3	1.2	1.0	22.3
202	Cage	3	4	480	-2.0	-2.4	-3.0	11.0	0.3	-60.9	-0.4	1.3	3.4	23.3
202	Cage	3	4	510	1.6	2.3	1.4	10.7	0.2	-58.4	0.6	0.6	2.2	20.7
202	Cage	3	4	540	1.5	2.2	1.4	10.4	-0.2	-57.6	-0.1	1.8	0.9	18.7
202	Cage	3	4	570	-0.4	0.6	-0.5	5.1	-0.1	-35.3	2.5	0.7	9.7	22.7
202	Cage	3	4	600	2.9	3.0	2.6	16.7	-0.3	-86.9	6.1	-2.2	-6.7	18.6
202	Cage	3	4	630	-0.1	0.9	-0.2	14.4	-0.4	-75.2	3.7	0.1	-2.2	20.5
202	Cage	3	4	660	-3.5	-3.0	-3.9	9.1	-0.1	-52.9	0.4	1.3	3.5	20.5
202	Cage	3	4	690	-5.4	-4.8	-5.7	1.6	-0.1	-18.0	1.1	1.4	14.2	22.7
202	Cage	3	4	720	-3.7	-2.0	-3.5	9.0	-0.2	-52.9	4.5	-1.8	3.1	20.0
202	Cage	3	4	750	-2.7	-1.0	-2.3	9.4	-0.2	-56.2	3.3	-2.3	2.3	20.1
202	Cage	3	4	780	-3.0	-2.3	-3.3	4.6	-0.2	-33.8	2.9	-1.2	7.3	19.0
202	Cage	3	4	810	-1.9	-0.4	-1.5	7.4	-0.2	-50.3	3.6	-2.4	5.4	22.3
202	Cage	3	4	840	-3.8	-2.7	-4.0	6.6	-0.2	-37.7	4.9	-3.3	12.4	28.0
202	Cage	3	4	870	-4.6	-2.5	-4.0	6.0	-0.4	-39.2	7.9	-3.5	6.9	22.5
202	Cage	3	4	900	-1.6	-0.9	-1.2	9.7	-0.4	-58.6	2.7	0.9	2.2	21.4
202	Cage	3	4	930	-1.4	-0.8	-1.0	-0.2	-0.3	-8.0	9.7	-0.6	16.9	23.2
202	Cage	3	4	960	-1.5	-1.3	-1.4	2.8	-0.3	-25.2	7.5	0.1	12.3	22.8
202	Cage	3	4	990	-4.0	-5.0	-5.0	-8.3	-0.2	39.7	1.1	3.0	29.2	25.1
202	Cage	3	4	1020	-4.7	-4.6	-5.3	-6.8	-0.2	32.9	3.2	1.7	31.0	29.4
202	Cage	3	4	1050	-3.6	-3.7	-4.1	-10.4	-0.4	55.1	8.3	-0.3	37.7	30.9
202	Cage	3	4	1080	-1.8	-2.0	-2.3	-14.5	-0.4	80.0	7.2	0.8	41.2	28.2
202	Cage	3	4	1110	0.9	0.6	0.5	-0.8	-0.3	10.1	7.6	-0.4	21.5	24.7
202	Cage	3	4	1140	9.5	5.6	7.9	0.3	-0.2	-6.8	-1.8	-0.2	4.7	7.9
202	Cage	3	4	1170	5.5	2.0	3.2	-7.0	0.4	29.0	-0.5	1.0	20.3	17.3
202	Cage	3	4	1200	4.1	1.7	2.7	2.7	0.5	-12.7	0.2	0.7	6.4	12.2
202	Cage	3	4	1230	9.3	5.2	6.9	6.8	0.8	-42.8	-1.9	-1.6	-8.7	1.5
202	Cage	3	4	1260	9.4	5.6	7.1	11.2	1.2	-60.8	-1.8	-2.0	-15.2	-1.8
202	Cage	3	4	1290	11.3	7.0	9.0	10.1	1.2	-51.1	-3.2	-1.7	-12.1	-0.8
202	Cage	3	4	1320	17.8	12.8	15.8	49.4	1.3	-184.1	1.8	-3.1	-50.0	-6.3
202	Cage	3	4	1350	19.9	16.8	19.1	29.7	0.4	-129.8	1.3	-1.6	-38.9	-11.7

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
202	Cage	3	4	1380	18.7	16.3	18.0	30.8	-0.4	-115.4	0.6	-1.8	-39.5	-17.8
202	Cage	3	4	1410	3.8	3.5	3.1	15.0	0.4	-63.7	-3.0	-0.9	-20.3	-8.5
202	Cage	3	4	1440	4.3	4.4	3.7	16.0	0.7	-66.9	-3.5	-0.9	-19.6	-6.3
203	Chair	1	1	10	8.2	10.6	10.1	60.3	-0.1	-89.4	-2.8	2.3	-8.9	24.6
203	Chair	1	1	20	-9.1	-4.3	-6.6	12.7	-0.1	-63.8	-3.2	1.9	-12.1	8.3
203	Chair	1	1	30	-10.5	-5.9	-8.3	6.6	-0.1	-38.9	-1.0	-0.6	-5.8	7.2
203	Chair	1	1	40	-4.8	-1.2	-2.8	17.0	-0.1	-48.9	-1.0	-0.4	-18.2	-6.9
203	Chair	1	1	50	0.4	1.6	1.1	21.5	-0.1	-56.6	-2.5	-0.3	-18.0	-3.2
203	Chair	1	1	60	-4.9	-2.3	-3.6	12.3	-0.1	-43.2	-1.6	-0.5	-17.1	-7.4
203	Chair	1	1	70	0.8	1.1	0.9	13.2	-0.1	-49.7	-1.6	-0.9	-16.1	-3.0
203	Chair	1	1	80	-7.6	-4.1	-6.6	2.9	-0.1	-10.3	0.1	-1.4	-4.0	-1.4
203	Chair	1	1	90	1.1	1.4	1.3	5.3	-0.1	-26.8	-0.8	-0.7	-10.7	-4.7
203	Chair	1	1	100	-1.2	-1.4	-1.6	-5.4	-0.1	3.0	0.8	-2.2	1.3	0.8
203	Chair	1	1	110	-4.2	-2.7	-3.9	-7.5	-0.0	12.7	0.7	-1.8	4.5	1.6
203	Chair	1	1	120	5.5	4.6	4.5	15.6	-0.0	-12.8	1.2	-1.1	-2.3	2.0
203	Cage	1	1	210	-8.1	0.1	-3.0	-10.2	0.2	-170.0	-0.3	-4.4	-42.8	-7.5
203	Cage	1	1	240	4.1	7.5	7.0	2.0	0.2	-220.8	0.3	-4.3	-57.9	-9.6
203	Cage	1	1	270	-2.8	1.8	-0.3	-0.1	0.5	-214.9	2.1	-3.9	-59.1	-13.7
203	Cage	1	1	300	-9.5	-2.5	-6.4	0.2	0.7	-219.3	2.7	-5.4	-55.5	-7.1
203	Cage	1	1	330	-9.5	-1.8	-5.7	-3.0	0.5	-201.6	0.3	-4.8	-52.0	-9.0
203	Cage	1	1	360	-2.2	2.0	0.2	-2.4	0.5	-202.7	-0.2	-4.5	-57.1	-15.7
203	Cage	1	1	390	-3.6	0.4	-1.6	-4.0	0.5	-200.6	2.0	-4.4	-56.7	-16.1
203	Cage	1	1	420	-11.8	-5.8	-9.1	0.3	0.5	-218.8	-0.4	-4.3	-57.4	-10.1
203	Cage	1	1	450	-11.4	-3.6	-7.4	4.3	0.5	-233.9	-10.4	-4.6	-52.2	3.7
203	Cage	1	1	480	-17.0	-9.2	-13.2	-11.7	0.4	-164.8	-1.7	-5.0	-36.4	-1.2
203	Cage	1	1	510	-16.2	-7.2	-11.6	-11.0	0.2	-167.1	-4.9	-5.1	-31.9	5.9
203	Cage	1	1	540	-16.7	-8.6	-13.1	-20.5	0.0	-112.6	-2.2	-4.9	-17.3	5.9
203	Cage	1	1	570	-12.9	-4.2	-8.4	-10.8	-0.0	-164.8	-6.0	-4.2	-25.8	13.7
203	Cage	1	1	600	-16.8	-8.6	-12.8	-20.2	0.0	-118.9	-4.9	-4.2	-15.2	10.3
203	Cage	1	1	630	-15.9	-8.0	-12.5	-27.7	-0.1	-60.0	-1.1	-4.3	-2.4	9.0
203	Cage	1	1	660	-16.2	-8.5	-13.0	-35.0	-0.4	-4.2	3.4	-4.0	11.2	8.5
203	Cage	1	1	690	-13.0	-6.1	-9.9	-38.3	-0.3	22.6	-2.5	-4.1	14.7	5.3
203	Cage	1	1	720	-11.3	-7.0	-9.4	-46.9	-0.4	103.3	5.0	-2.2	38.9	12.6
203	Cage	1	1	750	-11.3	-8.0	-10.1	-52.4	-0.5	174.7	7.4	-0.6	58.3	15.1
203	Cage	1	1	780	-11.3	-6.7	-8.8	-50.3	-0.4	138.1	1.6	-2.6	53.2	18.3
203	Cage	1	1	810	-9.8	-6.2	-7.7	-54.4	-0.4	204.6	0.9	-2.6	66.3	16.2
203	Cage	1	1	840	-8.2	-6.2	-7.0	-54.0	-0.5	203.4	7.6	2.3	70.5	21.5
203	Cage	1	1	870	-11.0	-7.3	-8.6	-56.2	-0.7	254.0	8.4	4.1	83.3	22.5
203	Cage	1	1	900	-13.2	-8.2	-10.2	-54.1	-0.5	229.2	6.1	0.7	83.3	28.7
203	Cage	1	1	930	-14.5	-8.4	-10.9	-51.7	-0.5	233.6	5.1	-0.7	82.9	27.2
203	Cage	1	1	960	-10.2	-0.4	-6.6	-35.4	-0.4	247.6	5.0	-1.5	85.5	26.9
203	Cage	1	1	990	-13.7	-8.4	-10.6	-53.5	-0.5	238.1	6.7	0.1	81.4	25.0
203	Cage	1	1	1020	-14.0	-4.3	-10.0	-40.3	-0.7	273.0	7.8	2.1	84.2	19.8

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
203	Cage	1	1	1050	-9.1	-2.7	-6.4	-42.7	-0.5	229.4	6.9	-0.6	72.2	18.0
203	Cage	1	1	1080	-8.6	-3.7	-6.4	-45.6	-0.6	207.0	5.2	-1.4	67.3	17.6
203	Cage	1	1	1110	7.9	5.6	7.8	-0.3	-0.3	-157.9	-0.3	-2.5	-28.7	12.1
203	Cage	1	1	1140	3.8	3.0	4.5	29.0	0.3	-293.2	-7.3	-1.8	-77.4	-6.6
203	Cage	1	1	1170	-10.9	-8.4	-10.0	-34.8	0.6	-24.2	-1.2	-3.1	-10.0	-10.8
203	Cage	1	1	1200	-2.5	-0.5	-0.8	-9.6	0.6	-146.4	-3.4	-3.1	-36.3	-4.0
203	Cage	1	1	1230	-7.5	-5.5	-6.6	-39.9	0.9	-17.0	-2.3	-3.3	-0.2	-1.7
203	Cage	1	1	1260	-8.9	-7.2	-8.5	-42.4	1.0	-10.8	-0.2	-3.3	9.6	8.5
203	Cage	1	1	1290	-2.8	-2.4	-2.5	-28.7	1.1	-69.9	-0.3	-4.1	-13.8	-1.9
203	Cage	1	1	1320	-3.3	-2.0	-2.1	-22.9	1.2	-96.0	0.5	-3.8	-17.3	1.5
203	Cage	1	1	1350	-4.1	-1.0	-2.4	-34.3	0.5	-17.1	4.6	-3.1	-1.5	-1.8
203	Cage	1	1	1380	-11.9	-8.3	-11.8	-39.8	0.2	-51.9	0.5	-3.6	3.6	12.7
203	Cage	1	1	1410	-8.6	-3.8	-6.6	-25.5	0.3	-100.1	2.5	-4.0	-15.7	4.6
203	Cage	1	1	1440	-8.0	-2.7	-5.9	-27.1	0.1	-51.9	4.8	-4.0	-18.2	-11.1
203	Chair	3	4	10	22.4	23.0	21.9	105.0	-0.0	.	.	.	.	.
203	Chair	3	4	20	19.4	17.7	18.7	73.0	-0.1	.	.	.	.	.
203	Chair	3	4	30	19.0	17.8	18.6	83.9	-0.2	.	.	.	.	.
203	Chair	3	4	40	17.0	16.3	16.9	75.5	-0.2	-127.0	-15.7	3.9	-37.0	0.3
203	Chair	3	4	50	20.8	16.8	18.9	71.5	-0.2	-106.4	-11.7	2.7	-25.8	8.0
203	Chair	3	4	60	8.3	8.6	8.9	43.7	-0.1	-86.9	-9.6	1.0	-22.0	4.7
203	Chair	3	4	70	3.7	4.5	4.4	28.5	-0.1	-50.7	-5.4	-0.5	-16.1	-2.3
203	Chair	3	4	80	-0.9	2.4	0.9	22.1	-0.1	-48.1	-5.0	-0.1	-16.3	-4.3
203	Chair	3	4	90	-2.8	-0.8	-1.4	5.5	-0.1	-24.5	-3.8	-0.5	-10.4	-5.1
203	Chair	3	4	100	0.4	1.3	1.2	7.9	-0.0	-31.2	-2.9	-0.4	-10.6	-2.7
203	Chair	3	4	110	-0.7	1.1	0.5	9.9	0.0	-34.7	-2.5	-0.3	-9.4	0.6
203	Chair	3	4	120	5.9	6.0	6.3	21.1	0.0	-37.5	-2.2	-0.1	-10.5	0.1
203	Cage	3	4	210	-7.8	0.7	-1.8	24.5	1.1	-230.9	-5.4	-3.7	-55.4	-1.7
203	Cage	3	4	240	14.3	16.0	16.6	27.0	0.6	-239.3	-3.3	-3.4	-64.3	-11.1
203	Cage	3	4	270	-6.9	-0.7	-3.9	23.0	0.9	-234.7	-4.2	-4.8	-57.8	-3.9
203	Cage	3	4	300	-6.3	1.8	-2.1	24.1	0.9	-229.2	-3.3	-4.4	-58.0	-6.1
203	Cage	3	4	330	-7.9	0.8	-3.2	21.1	0.9	-220.8	-1.6	-4.7	-58.0	-9.6
203	Cage	3	4	360	-8.4	0.4	-3.3	20.7	0.9	-225.8	-1.4	-4.5	-58.3	-8.1
203	Cage	3	4	390	-10.4	-2.2	-5.8	21.3	0.9	-227.9	-1.4	-4.4	-59.1	-8.4
203	Cage	3	4	420	-13.9	-4.2	-9.1	18.6	0.7	-216.8	-3.5	-4.9	-51.8	-2.4
203	Cage	3	4	450	-14.9	-4.6	-9.6	21.6	0.7	-229.9	-10.2	-4.3	-45.2	12.0
203	Cage	3	4	480	-15.1	-4.8	-9.7	16.9	0.7	-209.1	-7.6	-5.6	-41.3	9.2
203	Cage	3	4	510	-15.0	-4.4	-9.5	8.8	0.5	-170.4	-3.9	-5.7	-31.4	8.2
203	Cage	3	4	540	-17.2	-6.1	-11.1	15.2	0.4	-198.9	-7.4	-4.3	-34.2	15.3
203	Cage	3	4	570	-18.6	-8.6	-13.6	-1.2	0.3	-107.6	-4.0	-3.9	-12.6	12.0
203	Cage	3	4	600	-18.3	-8.8	-14.0	-10.4	0.1	-44.1	0.6	-3.5	2.0	9.8
203	Cage	3	4	630	-17.8	-6.9	-12.0	-5.6	0.1	-81.1	-2.1	-3.7	-4.1	13.8
203	Cage	3	4	660	-18.1	-8.1	-13.0	-13.2	0.1	-16.5	-1.3	-3.4	9.0	11.0
203	Cage	3	4	690	-19.9	-11.6	-16.2	-26.3	0.0	99.7	-0.1	-2.6	34.3	8.2

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
203	Cage	3	4	720	-17.7	-9.4	-13.4	-29.7	-0.1	147.4	2.2	-1.2	50.6	14.9
203	Cage	3	4	750	-19.6	-11.5	-15.5	-33.1	-0.1	203.1	5.3	3.3	66.6	19.0
203	Cage	3	4	780	-19.1	-11.1	-15.1	-36.4	-0.3	256.2	7.1	6.6	78.4	18.7
203	Cage	3	4	810	-17.7	-9.4	-13.1	-34.1	-0.1	232.6	2.9	-0.0	76.3	21.9
203	Cage	3	4	840	-17.5	-9.1	-12.8	-33.5	-0.0	229.3	3.3	-0.1	77.4	24.0
203	Cage	3	4	870	-15.3	-7.5	-10.8	-33.0	-0.1	226.1	5.1	1.5	76.8	24.4
203	Cage	3	4	900	-18.7	-10.1	-14.0	-36.1	-0.3	263.2	7.5	5.3	82.9	21.8
203	Cage	3	4	930	-18.0	-9.2	-12.9	-33.3	-0.4	239.2	6.2	2.6	80.5	24.8
203	Cage	3	4	960	-15.7	-7.0	-10.6	-32.1	-0.2	230.9	5.4	0.8	76.9	23.5
203	Cage	3	4	990	-16.7	-8.0	-11.4	-32.3	-0.2	233.5	2.7	-2.0	77.5	23.2
203	Cage	3	4	1020	-16.1	-8.7	-11.7	-33.0	-0.3	239.7	9.0	4.8	76.2	21.1
203	Cage	3	4	1050	-15.1	-5.8	-10.2	-24.7	-0.3	212.6	5.7	-0.8	70.7	21.3
203	Cage	3	4	1080	-14.0	-6.8	-9.7	-27.2	-0.3	177.8	5.7	-0.8	62.0	20.7
203	Cage	3	4	1110	-0.5	0.7	1.1	-7.5	-0.3	14.1	4.0	-0.0	16.6	15.7
203	Cage	3	4	1140	2.8	2.8	4.3	40.5	0.3	-287.2	-5.7	-2.0	-73.5	-3.5
203	Cage	3	4	1170	-14.1	-9.5	-11.9	-16.0	0.7	-22.5	-0.9	-3.3	-5.4	-4.8
203	Cage	3	4	1200	-3.4	0.2	-0.3	15.5	0.9	-183.9	-3.3	-3.4	-44.5	-1.5
203	Cage	3	4	1230	-15.1	-10.4	-12.5	-10.8	1.2	-58.4	-2.3	-3.6	-6.8	4.7
203	Cage	3	4	1260	-13.9	-10.1	-12.2	-20.9	1.3	-0.8	-1.2	-3.5	7.8	4.6
203	Cage	3	4	1290	-8.1	-5.6	-6.8	-23.7	1.3	24.1	-1.8	-3.1	22.1	15.0
203	Cage	3	4	1320	-3.6	-1.0	-1.6	-2.5	1.4	-108.8	0.0	-3.8	-16.4	8.1
203	Cage	3	4	1350	-2.7	2.2	0.8	-4.6	0.9	-71.7	1.7	-3.0	-13.6	0.8
203	Cage	3	4	1380	-9.1	-2.9	-6.1	-4.6	0.5	-88.1	1.5	-3.5	-15.5	3.2
203	Cage	3	4	1410	-11.7	-4.2	-7.7	7.4	0.8	-144.3	2.1	-3.9	-31.3	0.7
203	Cage	3	4	1440	-13.6	-6.8	-10.8	1.9	1.0	-144.7	-2.4	-4.1	-28.8	2.8
204	Chair	1	1	10	18.1	21.5	22.0	98.9	0.0	-327.5	-5.3	4.8	-41.1	92.4
204	Chair	1	1	20	2.3	6.1	6.0	52.2	-0.0	-257.9	-2.9	5.9	-45.6	40.6
204	Chair	1	1	30	5.5	8.1	8.6	56.0	-0.0	-241.9	-3.4	5.6	-51.0	24.3
204	Chair	1	1	40	-9.5	-1.9	-4.2	44.3	-0.0	-224.5	-2.4	6.5	-38.5	33.5
204	Chair	1	1	50	-4.5	1.6	-0.7	27.2	-0.0	-57.6	-1.1	2.7	-26.0	-10.6
204	Chair	1	1	60	-5.3	0.6	-1.4	23.0	-0.0	-7.3	-0.8	1.6	-18.2	-18.8
204	Chair	1	1	70	-2.0	2.8	1.2	26.8	0.0	-77.4	-1.4	1.4	-27.9	-6.9
204	Chair	1	1	80	-2.2	2.8	1.0	21.7	0.0	-49.4	-1.3	1.2	-19.2	-6.0
204	Chair	1	1	90	0.0	1.7	1.2	7.0	-0.0	48.9	-1.0	0.4	-1.0	-17.1
204	Chair	1	1	100	-2.9	-0.2	-1.4	5.9	-0.0	2.1	-1.2	-0.3	-0.7	-0.9
204	Chair	1	1	110	-2.0	0.7	-0.5	10.9	-0.1	-17.0	-0.6	0.5	-5.9	-1.0
204	Chair	1	1	120	3.3	5.4	4.8	23.3	-0.1	-54.0	-1.0	0.7	-13.7	3.6
204	Cage	1	1	210	4.5	5.2	4.7	36.0	0.5	-198.2	-3.2	-0.4	-86.9	-40.4
204	Cage	1	1	240	7.6	7.1	7.3	44.5	0.8	-241.7	-3.6	1.9	-94.0	-34.5
204	Cage	1	1	270	-7.3	-4.4	-5.9	37.8	0.6	-208.6	-3.5	-1.5	-90.0	-42.9
204	Cage	1	1	300	-11.3	-6.2	-9.7	24.0	0.7	-185.1	-5.9	-3.4	-77.2	-34.5
204	Cage	1	1	330	-8.1	-3.5	-6.6	30.2	0.9	-200.8	-5.6	-3.7	-84.4	-38.4
204	Cage	1	1	360	-6.8	-3.3	-6.1	24.8	0.8	-200.8	-5.2	-2.7	-77.2	-27.9

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
204	Cage	1	1	390	-4.0	-0.3	-2.8	31.6	0.8	-201.0	-5.0	-2.4	-71.6	-20.5
204	Cage	1	1	420	-10.1	-7.2	-8.9	31.7	0.9	.	.	.	.	.
204	Cage	1	1	450	-11.3	-6.6	-9.3	35.1	1.0	-237.2	-6.5	2.3	-71.5	-4.4
204	Cage	1	1	480	-10.9	-6.2	-8.6	40.4	0.8	-235.2	-6.5	2.8	-75.2	-10.9
204	Cage	1	1	510	-14.6	-9.1	-11.9	28.8	0.7	-203.0	-6.4	1.5	-61.0	-4.9
204	Cage	1	1	540	-14.5	-8.1	-11.3	31.6	0.6	-207.5	-6.1	2.0	-56.7	3.4
204	Cage	1	1	570	-14.2	-8.7	-11.7	28.3	0.5	-61.1	-1.6	-4.2	-36.8	-23.7
204	Cage	1	1	600	-15.2	-9.7	-12.6	22.9	0.6	-68.3	3.5	-4.7	-14.0	5.6
204	Cage	1	1	630	-15.7	-10.4	-14.2	13.3	0.4	-144.8	-2.0	-1.8	-23.3	22.5
204	Cage	1	1	660	-14.1	-7.9	-11.1	27.4	0.2	-158.0	-0.6	-2.6	-31.5	17.3
204	Cage	1	1	690	-11.5	-6.6	-9.2	26.5	0.5	-123.9	1.3	-4.2	-29.0	7.4
204	Cage	1	1	720	-16.4	-11.4	-14.9	14.5	0.3	-177.0	-5.1	0.4	-23.1	36.5
204	Cage	1	1	750	-13.0	-8.0	-10.7	22.1	0.1	-131.3	1.5	-3.1	-19.6	23.0
204	Cage	1	1	780	-9.3	-4.8	-6.9	27.1	0.3	-84.6	2.9	-5.4	-23.1	0.3
204	Cage	1	1	810	-14.1	-10.1	-13.0	12.7	0.4	-79.4	5.3	-5.5	-2.5	24.7
204	Cage	1	1	840	-10.8	-7.2	-9.4	18.9	0.4	-88.7	3.6	-4.0	-8.4	21.3
204	Cage	1	1	870	-10.7	-7.0	-8.9	22.4	0.4	-69.4	3.9	-5.4	-1.9	22.2
204	Cage	1	1	900	-11.0	-7.1	-9.5	14.3	0.3	-86.5	1.2	-3.0	-11.0	16.8
204	Cage	1	1	930	-15.5	-10.1	-13.3	12.7	0.1	-81.0	-0.6	-1.0	-6.7	20.4
204	Cage	1	1	960	-11.0	-6.7	-9.0	12.2	0.1	-64.2	3.7	-4.8	-2.8	18.9
204	Cage	1	1	990	-12.2	-9.1	-11.4	4.9	-0.0	-99.9	-0.7	-0.6	-0.1	35.2
204	Cage	1	1	1020	-11.7	-8.5	-10.6	2.5	-0.2	-93.9	-1.0	-0.1	-5.1	26.8
204	Cage	1	1	1050	-8.5	-5.8	-7.4	-1.0	-0.3	-28.7	2.6	-2.5	16.7	30.0
204	Cage	1	1	1080	-4.7	-1.8	-3.2	2.1	0.2	4.3	1.4	-2.4	20.9	22.5
204	Cage	1	1	1110	4.3	1.9	3.0	23.2	0.2	-163.5	-0.3	0.7	-45.8	-0.1
204	Cage	1	1	1140	4.3	4.4	4.1	12.6	0.3	-119.1	2.7	-1.3	-24.5	11.2
204	Cage	1	1	1170	1.7	3.2	2.3	13.7	0.5	-131.6	4.0	-2.2	-22.6	18.8
204	Cage	1	1	1200	8.3	8.2	8.6	24.8	0.6	-179.8	2.1	-0.2	-54.1	-5.0
204	Cage	1	1	1230	11.5	7.7	9.3	18.3	0.6	-155.1	2.6	-0.5	-42.6	1.5
204	Cage	1	1	1260	11.3	6.4	8.5	25.2	0.8	-173.2	1.3	0.0	-47.1	3.3
204	Cage	1	1	1290	16.0	10.0	12.8	41.4	1.2	-231.4	-2.6	3.2	-75.4	-12.6
204	Cage	1	1	1320	12.8	8.6	10.8	36.3	1.2	-171.4	-1.5	0.8	-71.0	-28.9
204	Cage	1	1	1350	4.3	0.5	1.4	18.0	0.6	-134.9	-0.7	0.7	-43.8	-6.2
204	Cage	1	1	1380	-2.3	-2.3	-3.5	6.6	0.6	-149.4	-1.7	1.6	-38.6	4.4
204	Cage	1	1	1410	-5.2	-2.6	-4.9	13.1	0.4	-154.6	-1.5	1.7	-48.9	-7.7
204	Cage	1	1	1440	-5.1	-4.0	-5.8	6.4	0.5	-143.8	-0.6	2.7	-43.9	-5.2
204	Chair	3	4	10	18.9	21.4	22.1	90.7	0.0	-340.1	-2.7	2.3	-49.4	85.5
204	Chair	3	4	20	5.3	8.6	8.5	52.4	-0.0	-287.7	-3.6	3.4	-29.2	83.8
204	Chair	3	4	30	4.0	7.1	6.9	39.1	0.0	-237.0	-2.5	3.9	-24.1	63.0
204	Chair	3	4	40	-2.9	2.0	0.8	33.2	0.0	-197.1	-2.1	2.4	-9.3	61.7
204	Chair	3	4	50	-4.5	1.7	-0.3	21.8	0.0	-153.4	-1.4	2.1	5.3	61.8
204	Chair	3	4	60	-6.2	-0.2	-2.2	13.7	0.1	61.2	-0.7	0.5	-18.6	-36.9
204	Chair	3	4	70	-1.1	3.9	2.3	13.9	0.1	-13.8	-1.9	3.5	-5.8	2.1

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
204	Chair	3	4	80	2.4	4.6	3.9	11.2	-0.0	-26.1	-1.9	3.1	-0.3	11.8
204	Chair	3	4	90	-2.2	0.5	-0.8	4.0	-0.0	69.1	-2.4	2.4	-5.3	-27.4
204	Chair	3	4	100	-2.9	1.1	-0.5	11.0	-0.1	5.6	-1.8	1.7	-0.5	-0.5
204	Chair	3	4	110	-3.9	-1.2	-2.2	4.3	-0.1	27.2	-1.8	1.5	2.6	-3.8
204	Chair	3	4	120	-4.3	-0.7	-2.0	13.5	-0.1	-22.4	-1.5	1.7	-3.7	7.3
204	Cage	3	4	210	-5.1	-1.6	-3.2	47.5	1.0	-218.9	-4.6	2.7	-96.2	-47.1
204	Cage	3	4	240	-2.4	1.0	-0.5	42.9	1.0	-190.3	-3.3	1.3	-90.7	-49.3
204	Cage	3	4	270	-5.6	-1.8	-4.1	35.2	1.1	-168.2	-4.7	1.0	-86.2	-53.3
204	Cage	3	4	300	-8.9	-3.3	-6.6	26.7	1.2	-134.9	-2.9	-1.7	-81.1	-58.2
204	Cage	3	4	330	-6.4	-0.7	-3.8	27.8	1.2	-141.9	-3.3	-1.3	-82.1	-57.2
204	Cage	3	4	360	-8.3	-3.9	-6.8	12.9	1.2	-116.8	-3.1	-4.2	-69.9	-50.0
204	Cage	3	4	390	-9.4	-4.8	-7.9	21.8	1.3	-138.8	-2.6	-2.2	-61.8	-30.8
204	Cage	3	4	420	-15.4	-11.2	-14.2	12.5	1.3	-98.2	-2.4	0.5	-25.4	3.5
204	Cage	3	4	450	-12.6	-8.4	-10.9	9.9	1.3	-129.3	-3.6	1.7	-22.3	18.3
204	Cage	3	4	480	-12.4	-8.1	-10.4	6.8	1.2	-65.7	-3.2	0.8	-14.9	4.3
204	Cage	3	4	510	-12.8	-9.6	-11.5	0.5	0.9	8.8	-1.1	-0.4	9.2	8.9
204	Cage	3	4	540	-9.9	-8.8	-9.6	-9.6	0.8	122.6	1.9	-2.2	39.6	5.5
204	Cage	3	4	570	-8.4	-5.7	-7.2	-2.7	0.8	18.0	-0.8	-1.5	18.1	16.8
204	Cage	3	4	600	-7.6	-5.2	-6.9	3.6	0.6	-9.5	-1.8	0.0	7.6	14.2
204	Cage	3	4	630	-10.1	-8.2	-9.6	-7.3	0.5	31.2	-1.7	-0.3	32.3	28.5
204	Cage	3	4	660	-12.6	-9.8	-11.8	-10.1	0.5	90.3	-2.6	-0.7	49.3	28.3
204	Cage	3	4	690	-8.7	-4.1	-6.4	0.8	0.5	1.3	-1.9	-0.3	18.1	22.1
204	Cage	3	4	720	-2.8	-1.0	-1.9	-2.4	0.4	-9.4	-2.0	-0.0	15.6	22.6
204	Cage	3	4	750	-3.6	-2.4	-2.9	5.4	0.3	-41.3	-1.2	1.6	-4.4	8.3
204	Cage	3	4	780	-13.3	-10.7	-12.5	-12.7	0.2	-14.1	-2.4	-0.8	35.1	48.0
204	Cage	3	4	810	-15.3	-9.3	-12.0	-0.3	0.3	57.4	3.1	0.9	31.5	17.4
204	Cage	3	4	840	-11.8	-6.9	-9.1	-10.2	0.2	99.8	1.0	0.2	43.5	17.9
204	Cage	3	4	870	-11.3	-6.9	-8.8	-11.0	0.2	175.5	2.1	1.2	60.4	12.5
204	Cage	3	4	900	-12.3	-7.7	-9.6	-17.8	0.2	210.2	1.4	1.4	72.8	15.4
204	Cage	3	4	930	-8.8	-5.5	-6.5	-17.6	0.3	226.8	1.9	1.2	78.4	16.9
204	Cage	3	4	960	-10.5	-8.2	-8.8	-20.3	0.2	209.9	1.5	0.1	81.0	25.5
204	Cage	3	4	990	-14.2	-10.1	-11.8	-23.2	-0.2	58.2	4.4	0.4	49.6	40.3
204	Cage	3	4	1020	7.1	9.1	8.1	-15.4	-0.5	81.0	2.2	2.2	17.5	-6.0
204	Cage	3	4	1050	18.8	16.7	17.4	-4.2	-0.6	-35.5	-0.2	0.7	-12.0	-2.8
204	Cage	3	4	1080	12.4	10.1	11.2	-13.8	-0.4	63.1	1.4	-0.5	24.9	7.8
204	Cage	3	4	1110	2.2	3.0	2.9	13.2	0.0	-33.0	-0.4	-0.6	-17.6	-10.3
204	Cage	3	4	1140	-2.6	1.4	-0.5	12.3	0.3	-49.6	-1.5	0.1	-26.8	-15.9
204	Cage	3	4	1170	1.7	5.2	3.5	16.3	0.5	-74.2	-1.7	1.1	-31.2	-13.5
204	Cage	3	4	1200	4.1	5.6	4.8	22.9	0.8	-135.1	-2.5	3.2	-47.5	-13.2
204	Cage	3	4	1230	4.8	5.9	5.5	29.3	1.1	-148.3	-3.3	2.6	-55.2	-17.8
204	Cage	3	4	1260	6.7	7.3	7.1	31.6	1.3	-159.6	-3.3	2.7	-61.1	-22.0
204	Cage	3	4	1290	10.4	9.1	9.6	38.9	1.6	-164.4	-4.1	2.8	-67.9	-28.9
204	Cage	3	4	1320	8.7	8.6	8.7	41.8	1.7	-188.1	-4.9	2.3	-71.3	-23.8

Table E-1. Baseline-Adjusted Individual Animal CV and ECG Data

Animal	Type	Dose	Day	Time	SBP	DBP	MAP	HR	BT	RR	PR	QRS	QT	QTcB
204	Cage	3	4	1350	1.7	4.3	2.5	27.8	1.0	-147.0	-2.9	0.9	-59.6	-24.3
204	Cage	3	4	1380	-3.4	0.1	-2.4	13.9	0.9	-95.5	-2.1	-1.9	-46.0	-23.9
204	Cage	3	4	1410	-6.1	0.1	-3.5	19.5	1.0	-79.9	-2.9	-2.7	-53.1	-39.2
204	Cage	3	4	1440	-8.9	-2.8	-6.7	8.2	1.1	-99.7	-2.2	-2.1	-46.6	-24.8

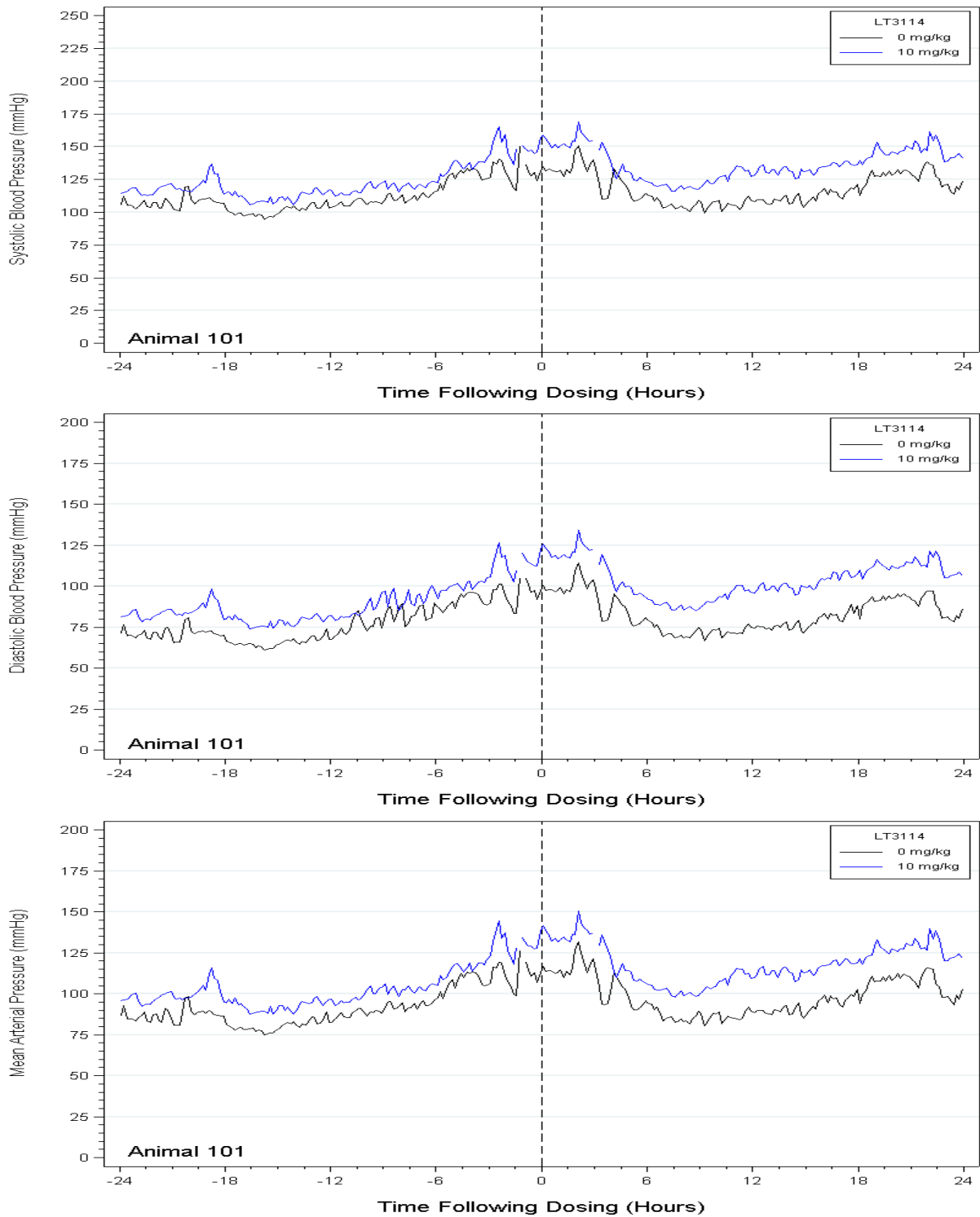


Figure E-1. Animal 101, Systemic Blood Pressure Parameters

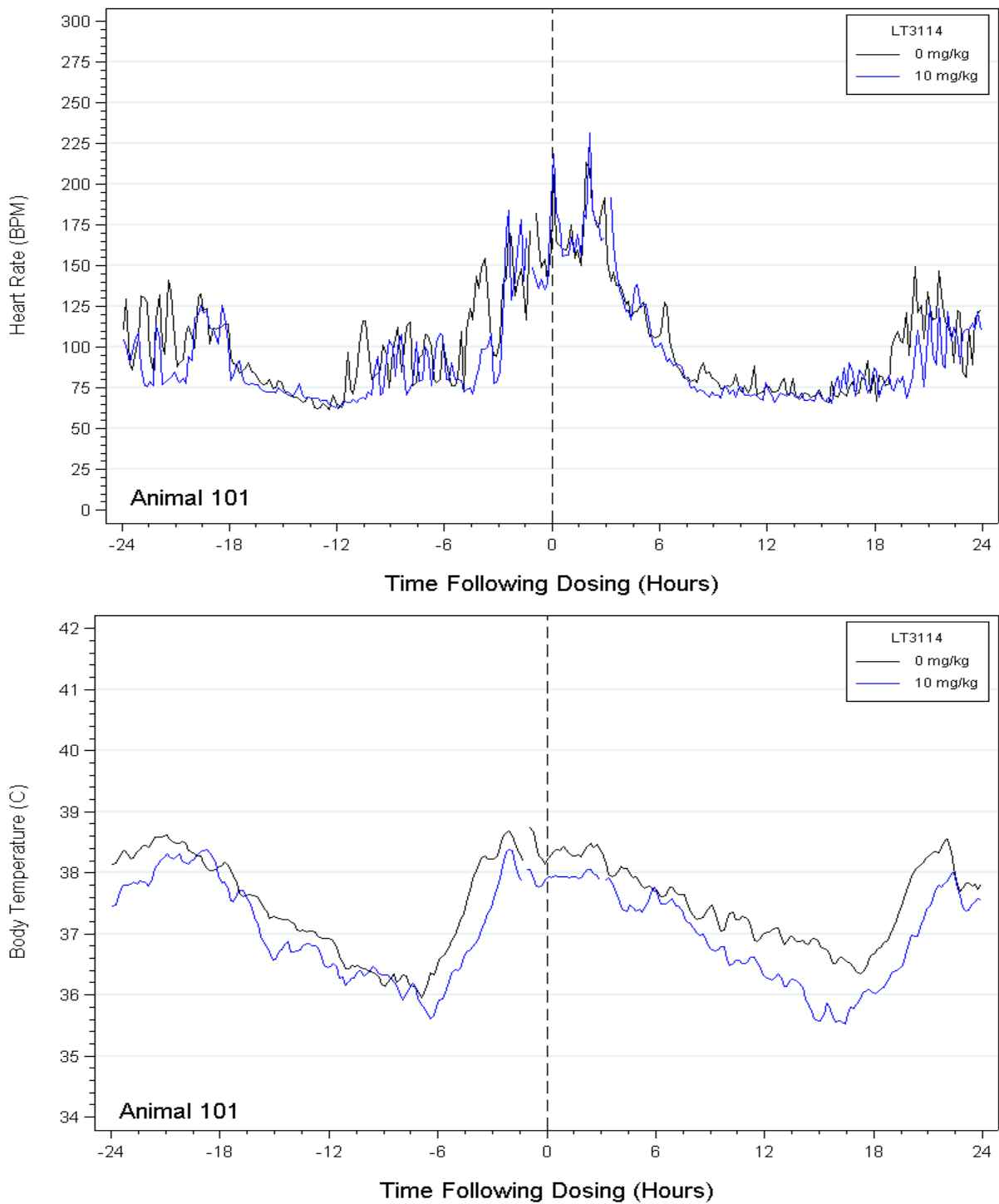


Figure E-2. Animal 101, Heart Rate and Body Temperature Parameters

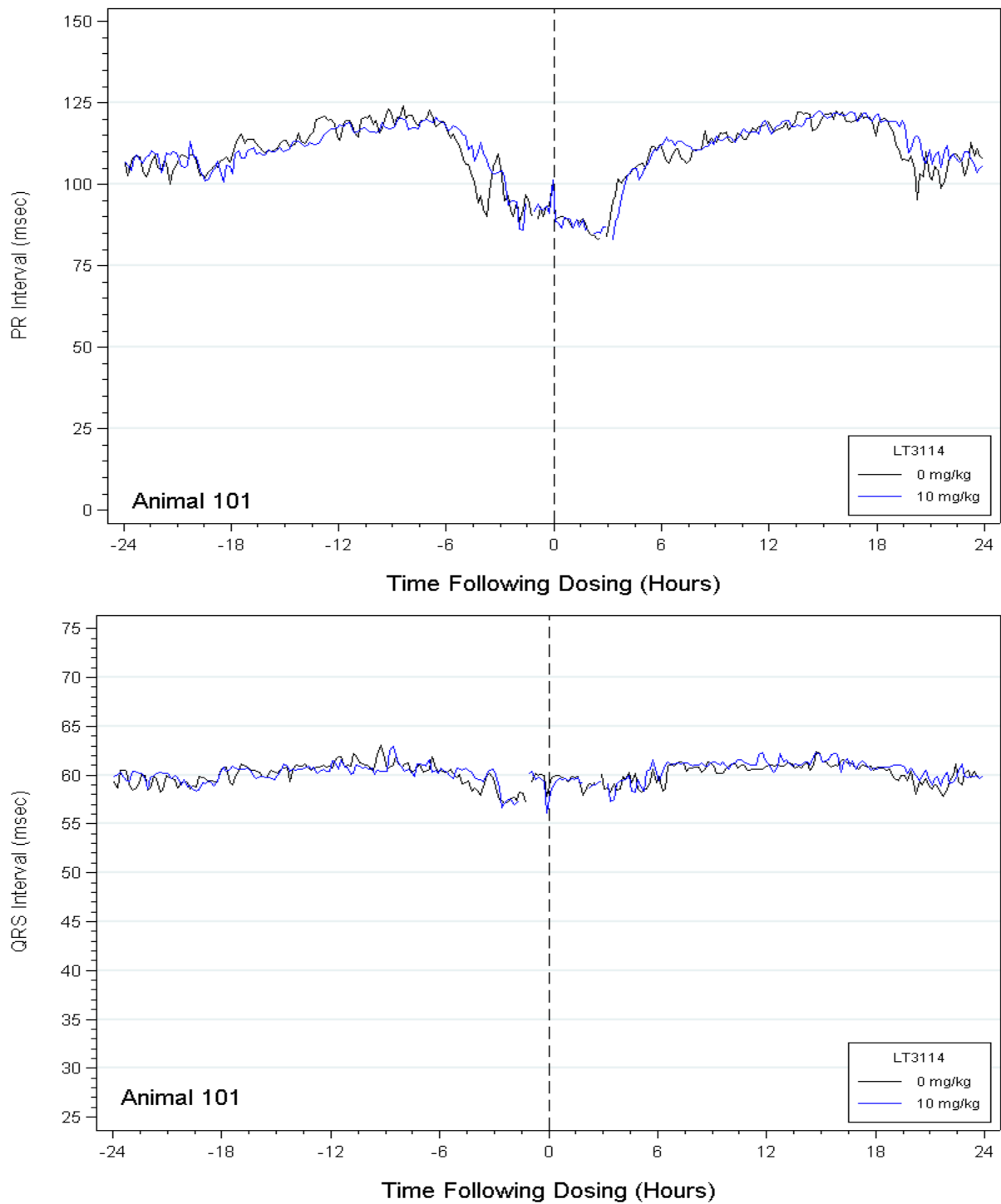


Figure E-3. Animal 101, PR and QRS Interval Parameters

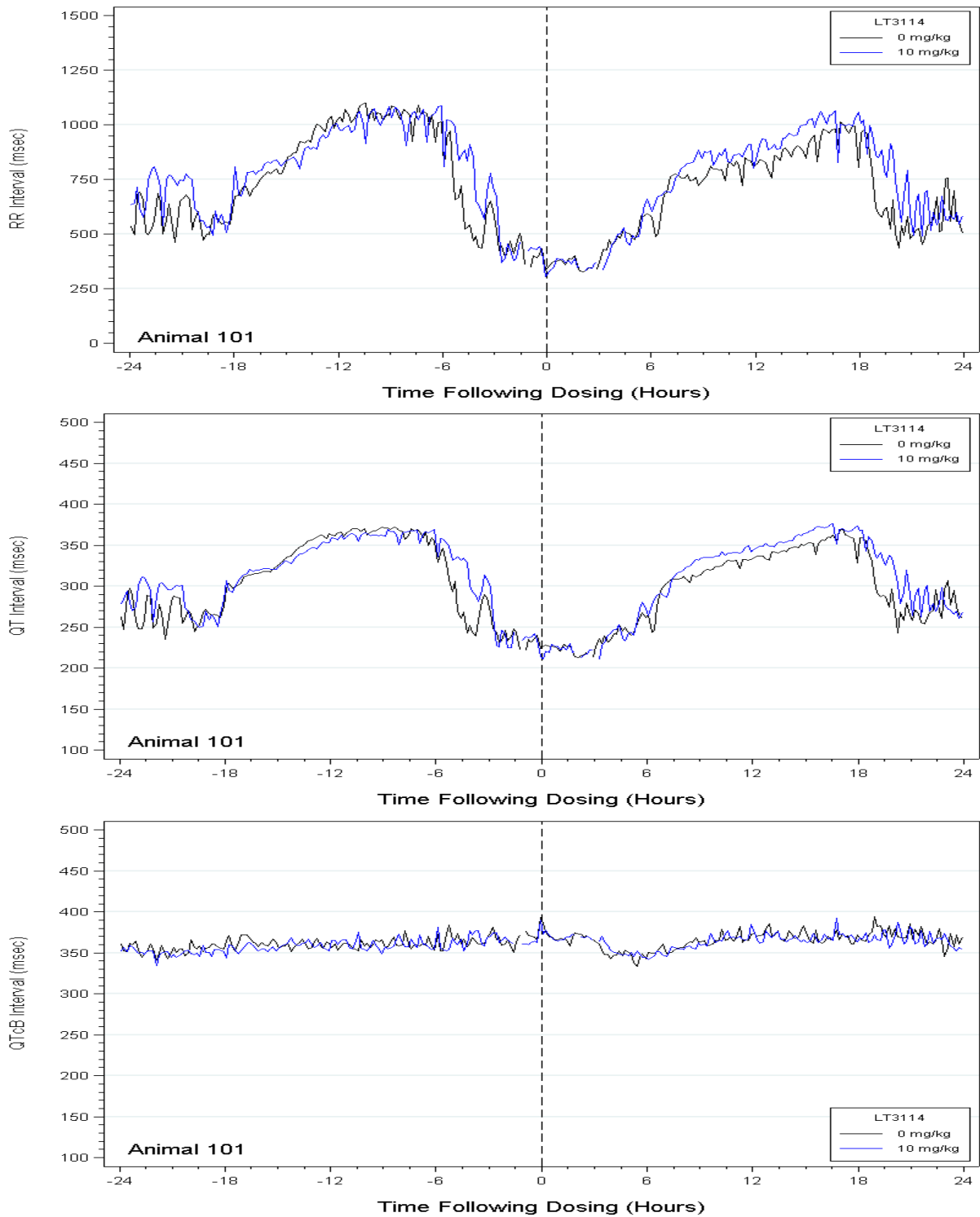


Figure E-4. Animal 101, RR, QT, and QTcB Interval Parameters

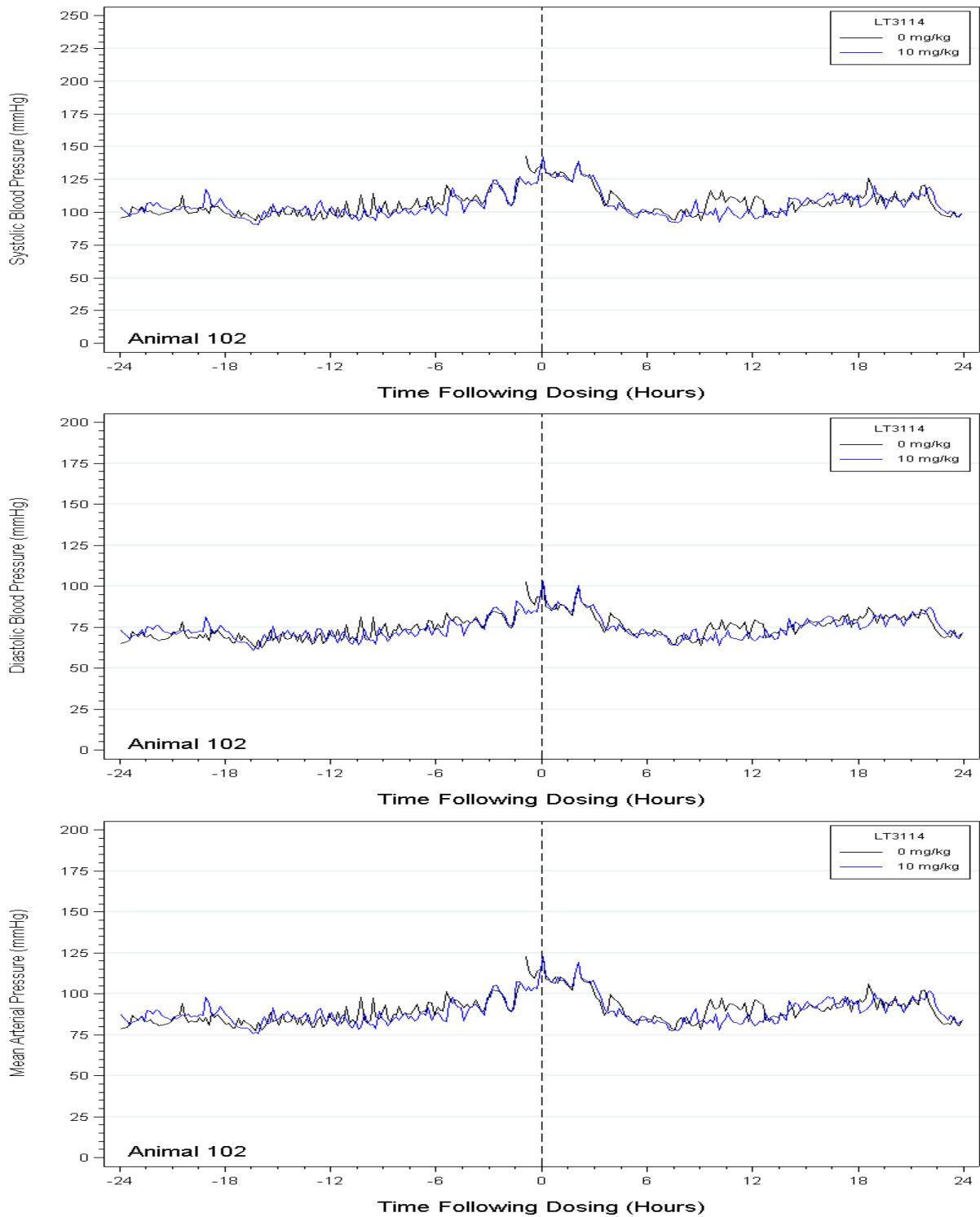


Figure E-5. Animal 102, Systemic Blood Pressure Parameters

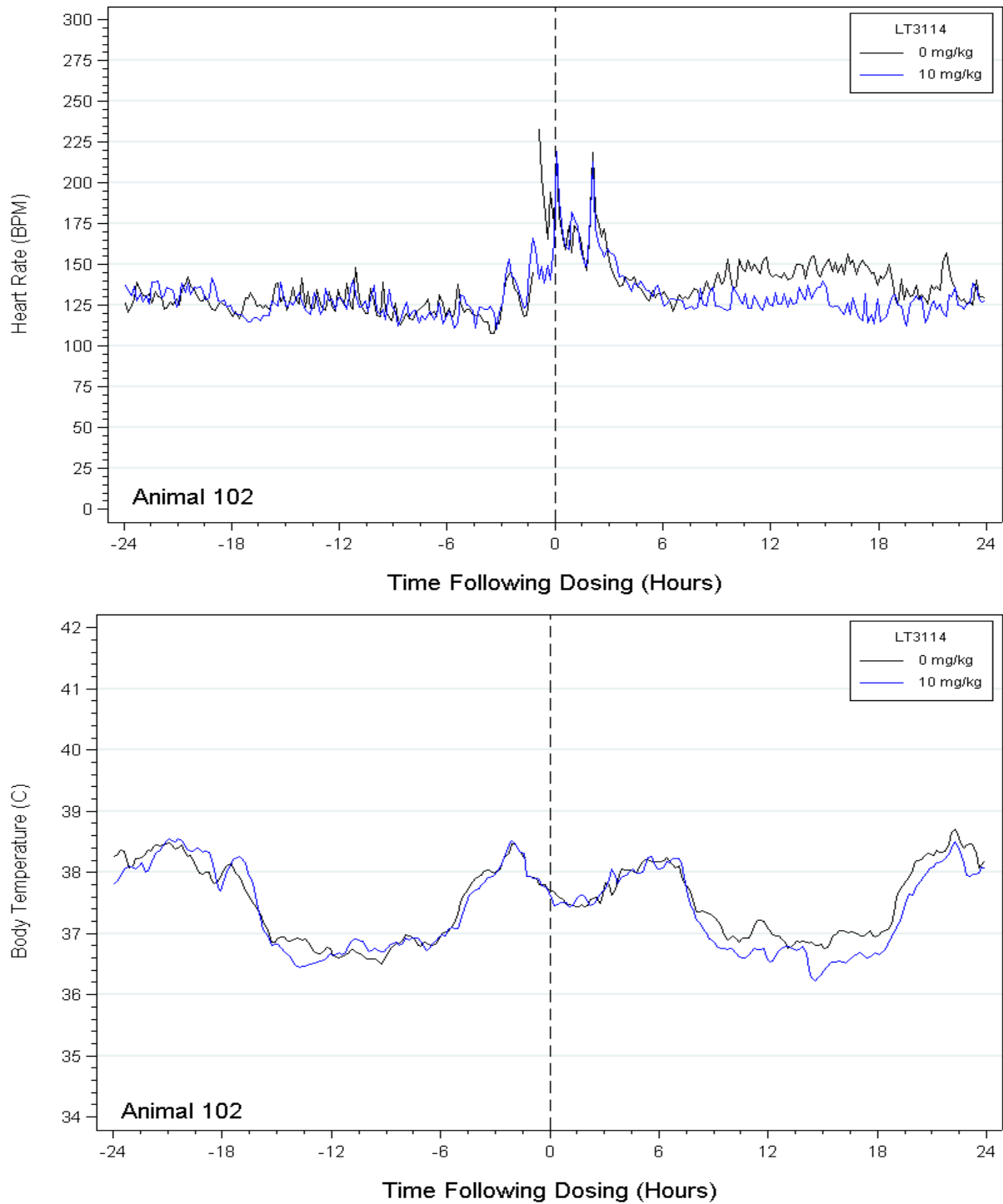


Figure E-6. Animal 102, Heart Rate and Body Temperature Parameters

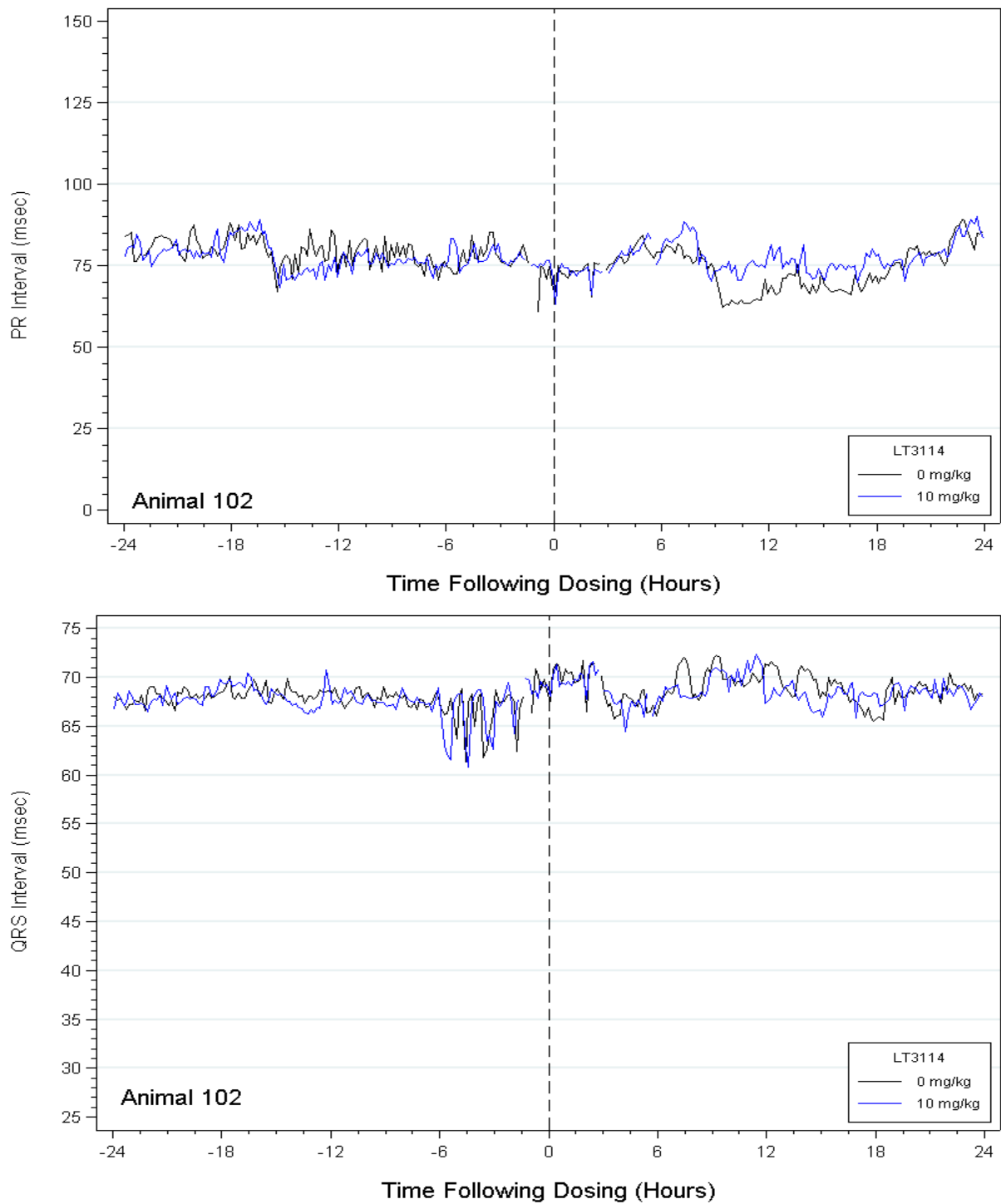


Figure E-7. Animal 102, PR and QRS Interval Parameters

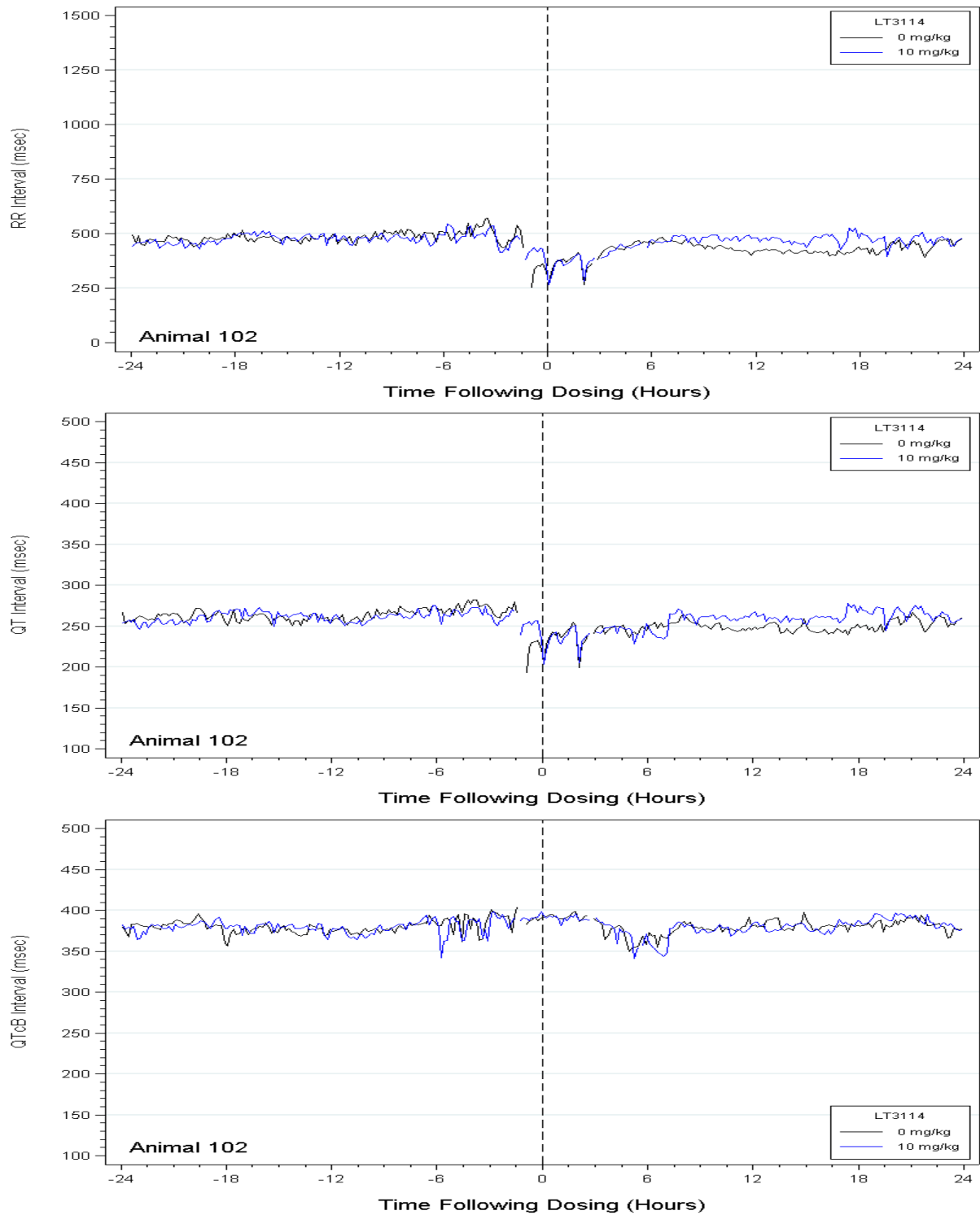


Figure E-8. Animal 102, RR, QT, and QTcB Interval Parameters

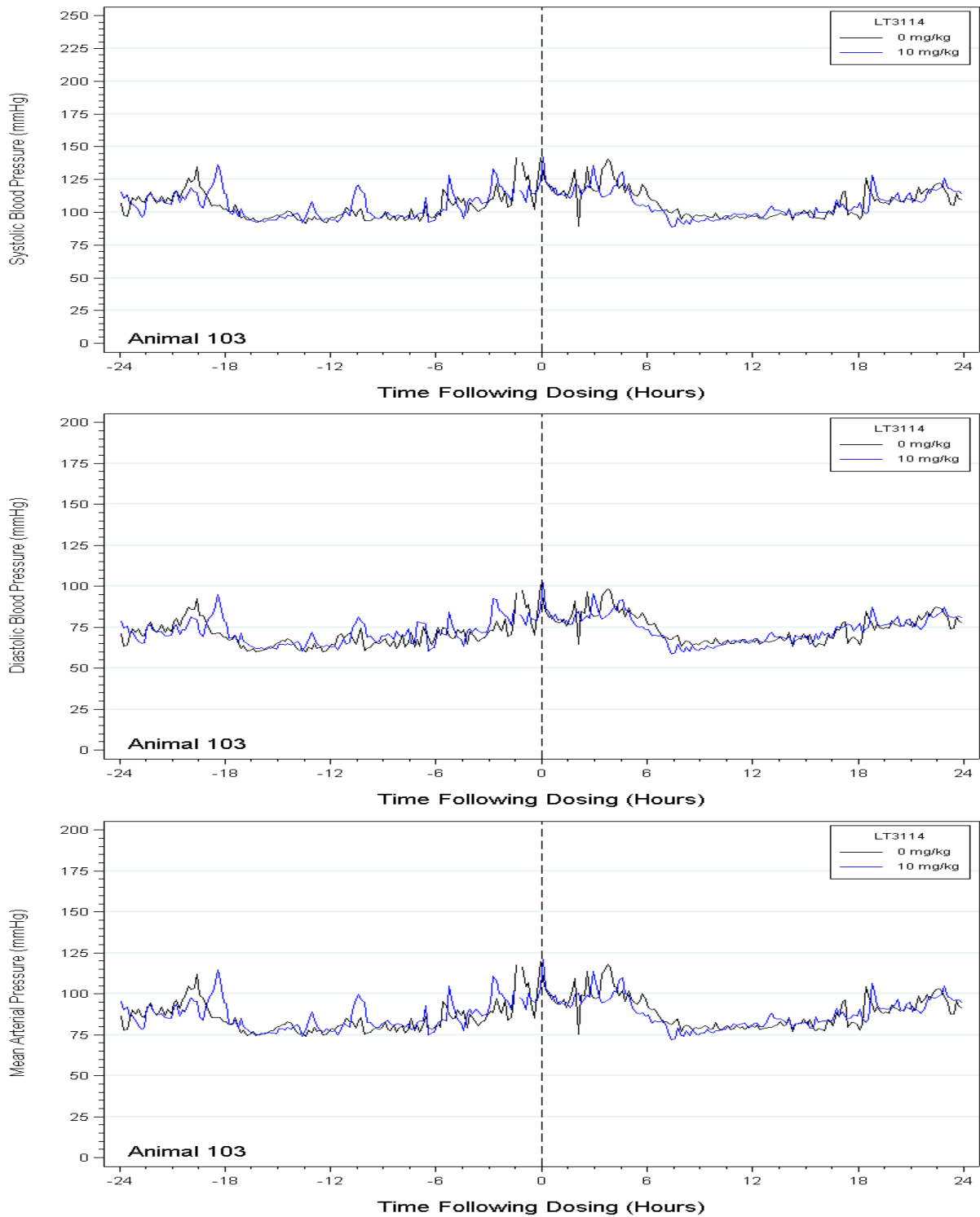


Figure E-9. Animal 103, Systemic Blood Pressure Parameters

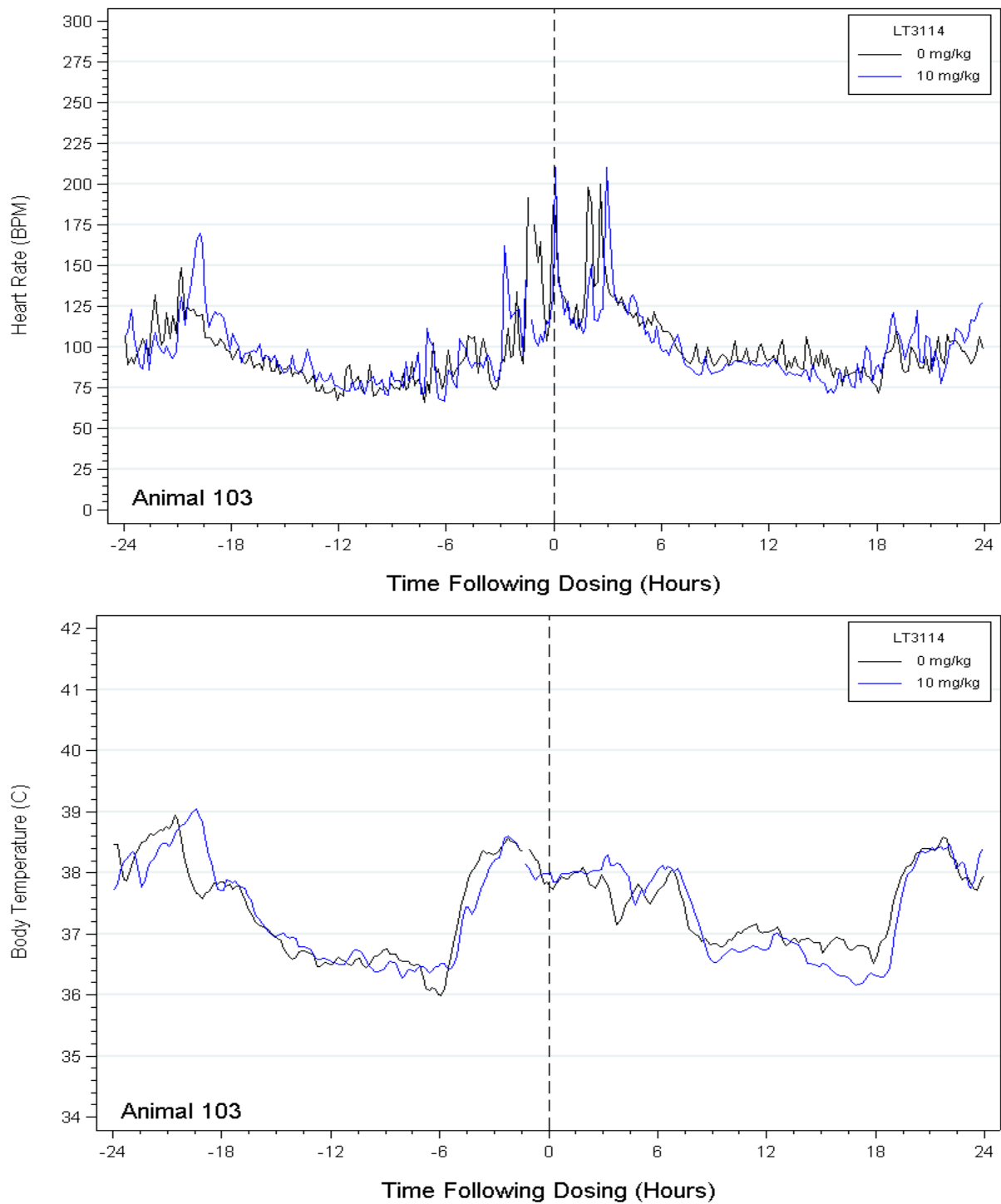


Figure E-10. Animal 103, Heart Rate and Body Temperature Parameters

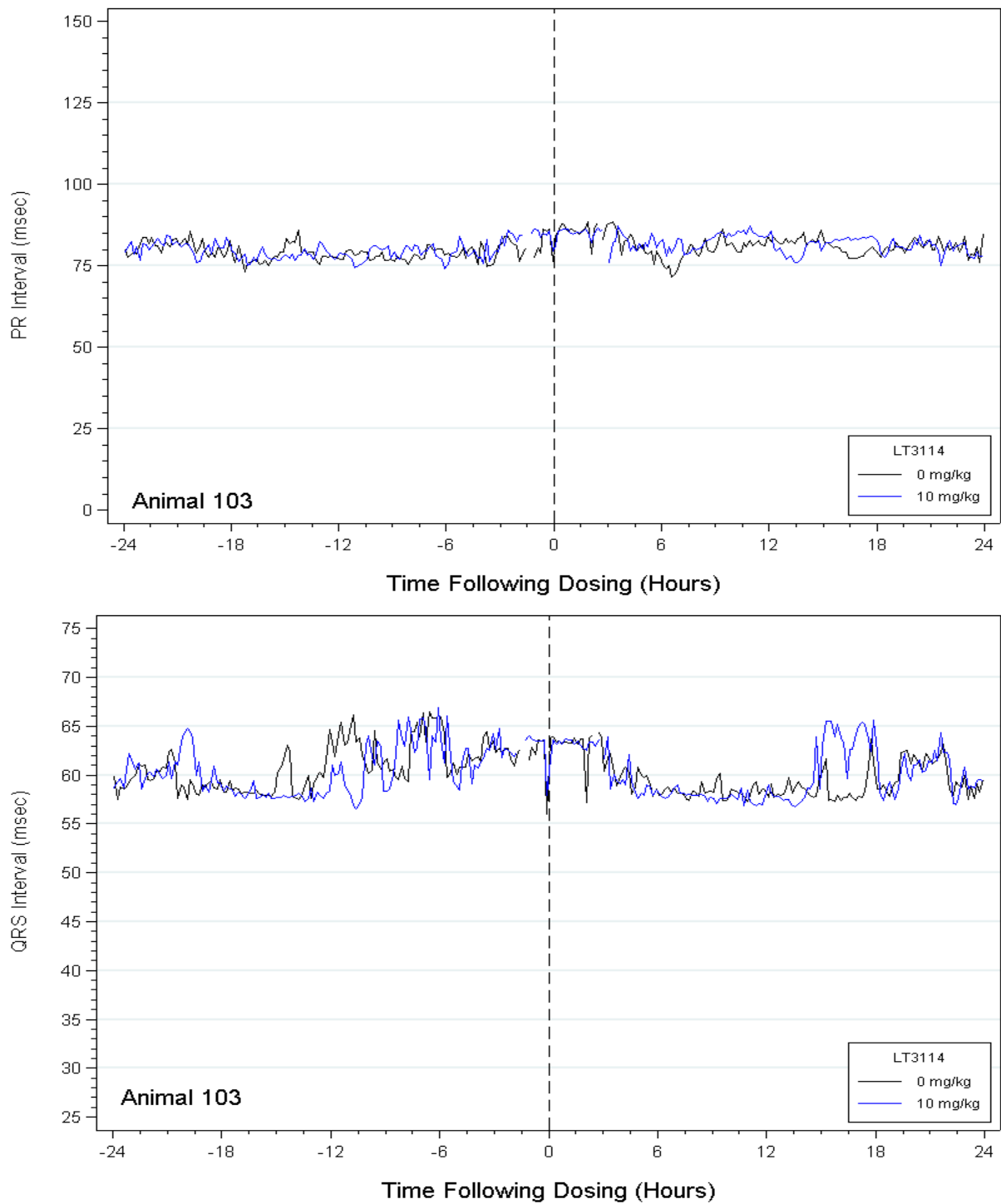


Figure E-11. Animal 103, PR and QRS Interval Parameters

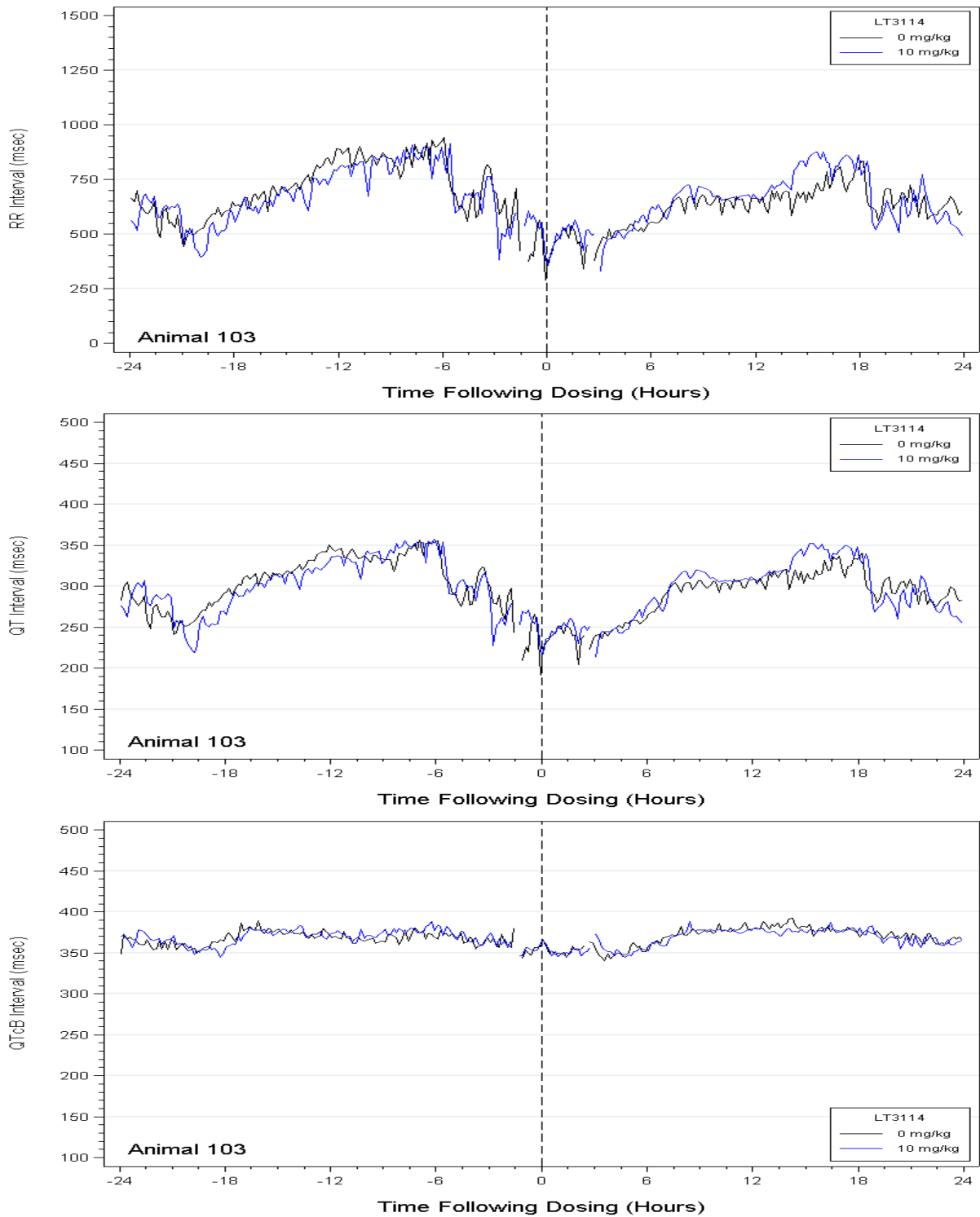


Figure E-12. Animal 103, RR, QT, and QTcB Interval Parameters

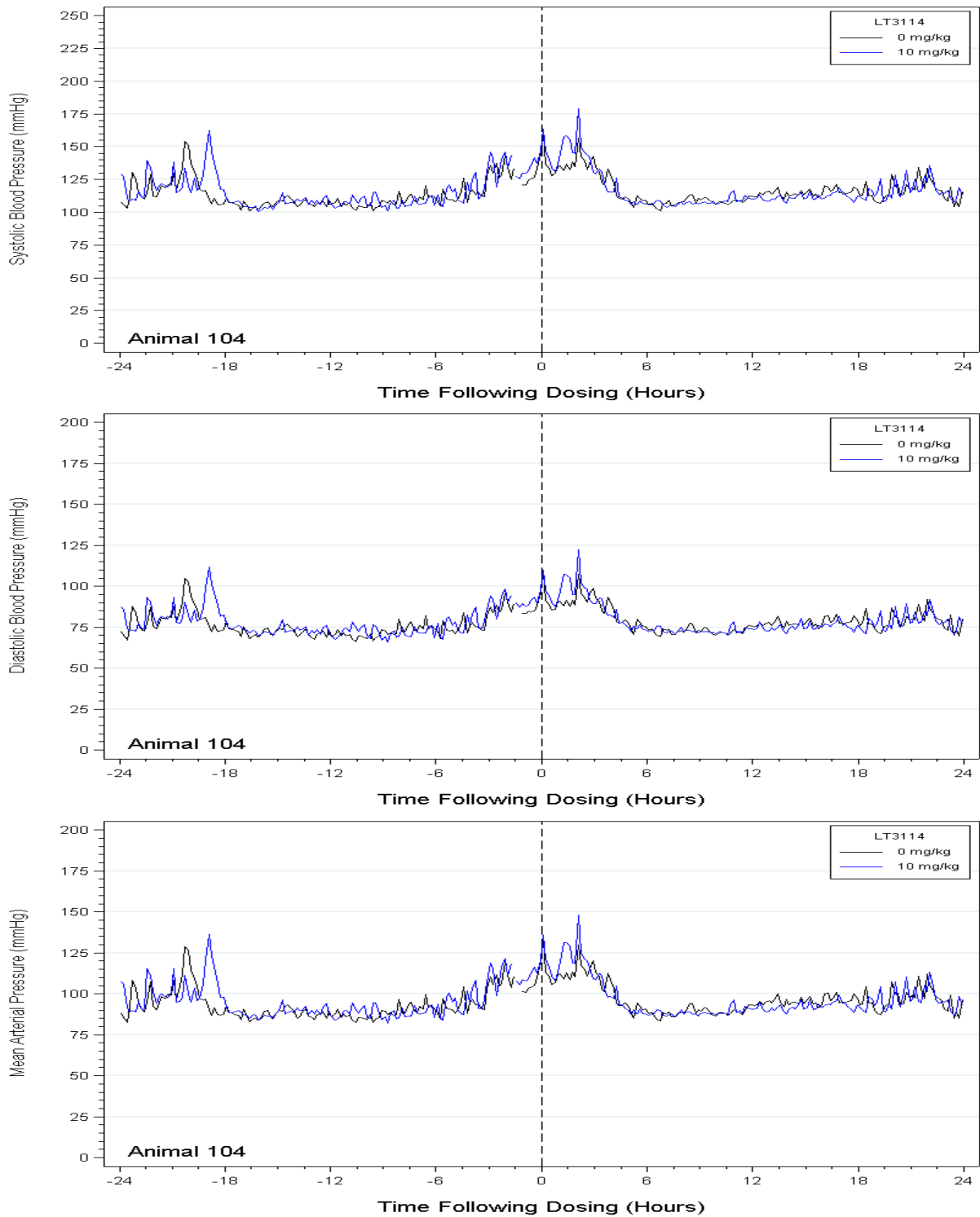


Figure E-13. Animal 104, Systemic Blood Pressure Parameters

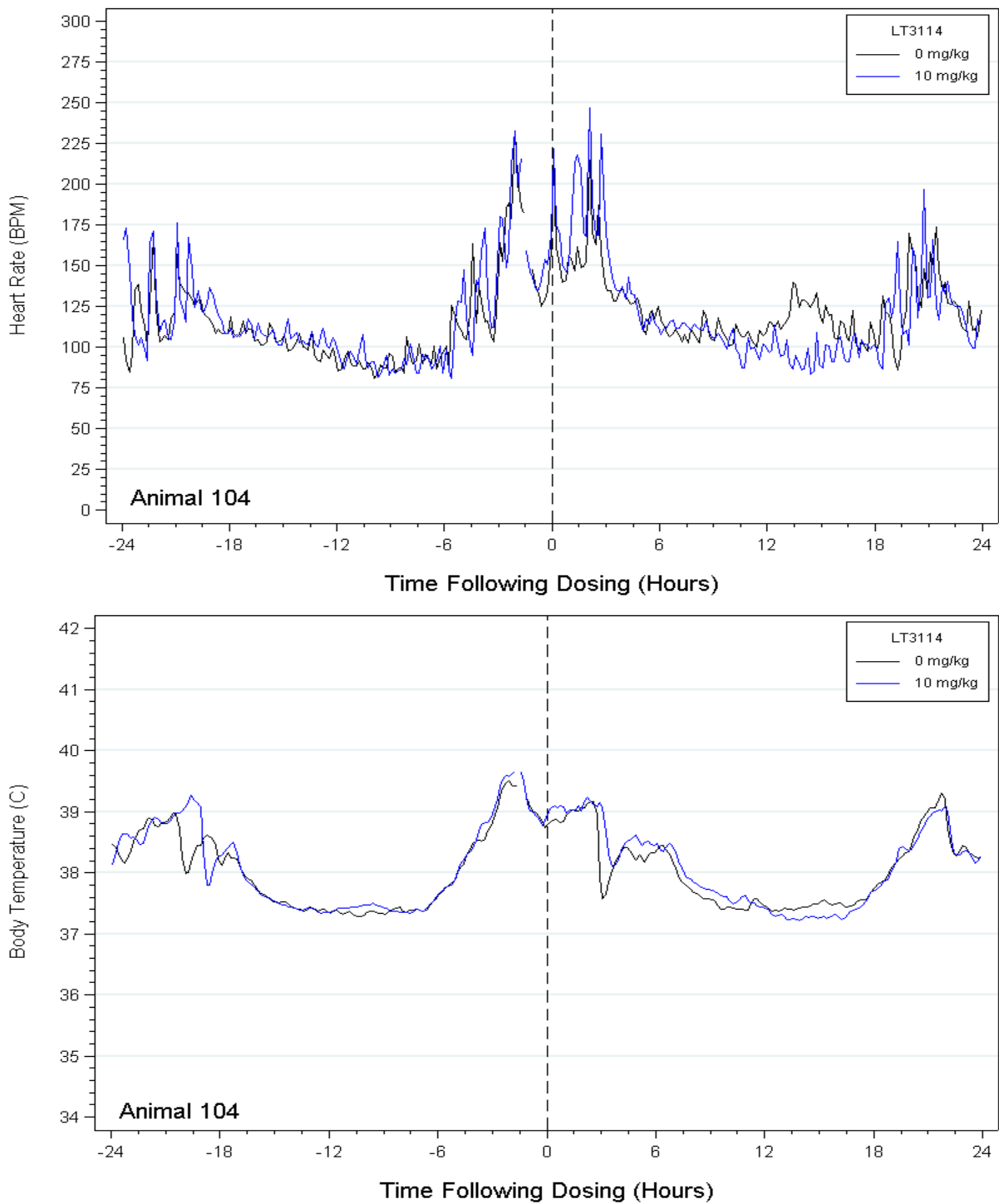


Figure E-14. Animal 104, Heart Rate and Body Temperature Parameters

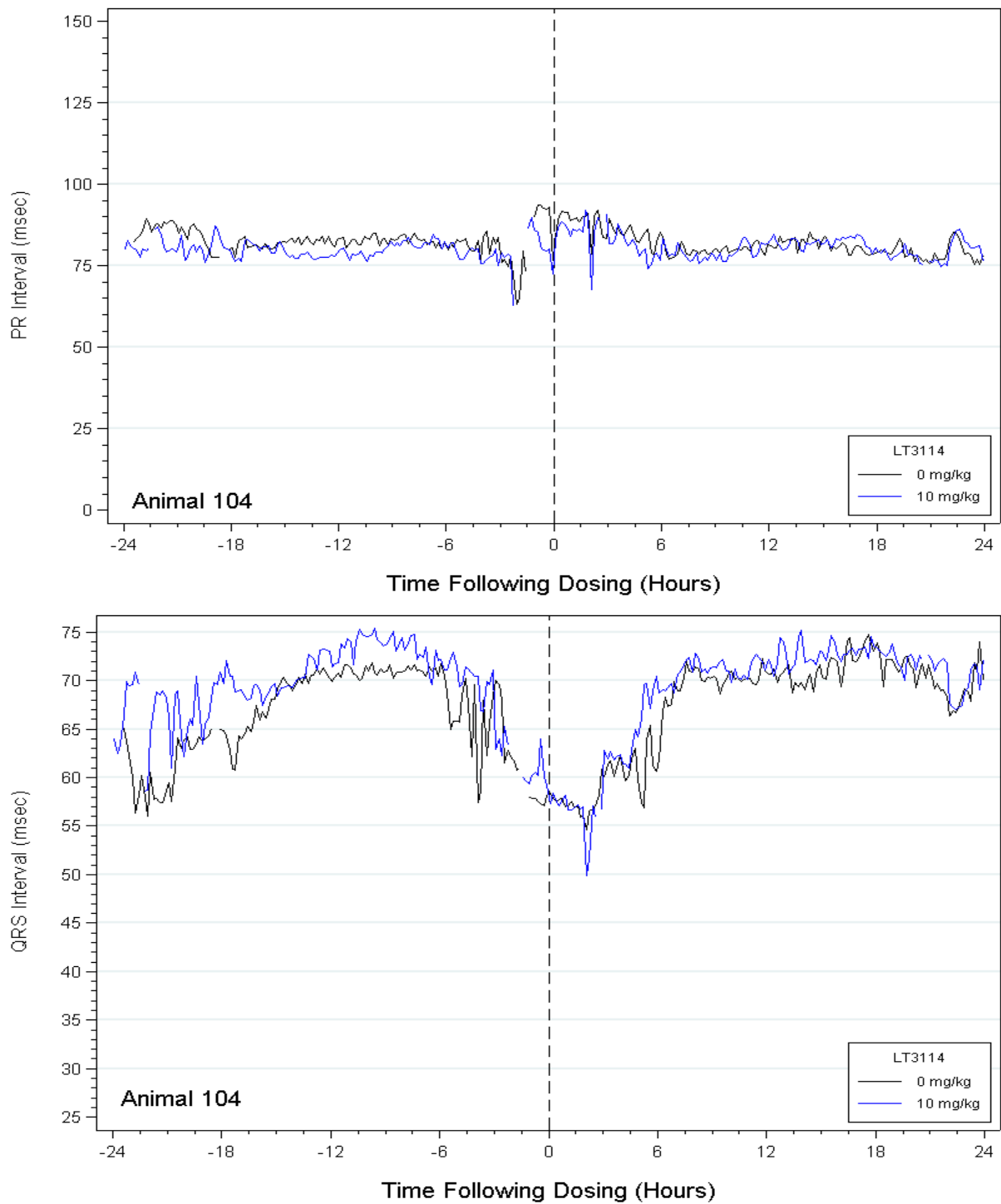


Figure E-15. Animal 104, PR and QRS Interval Parameters

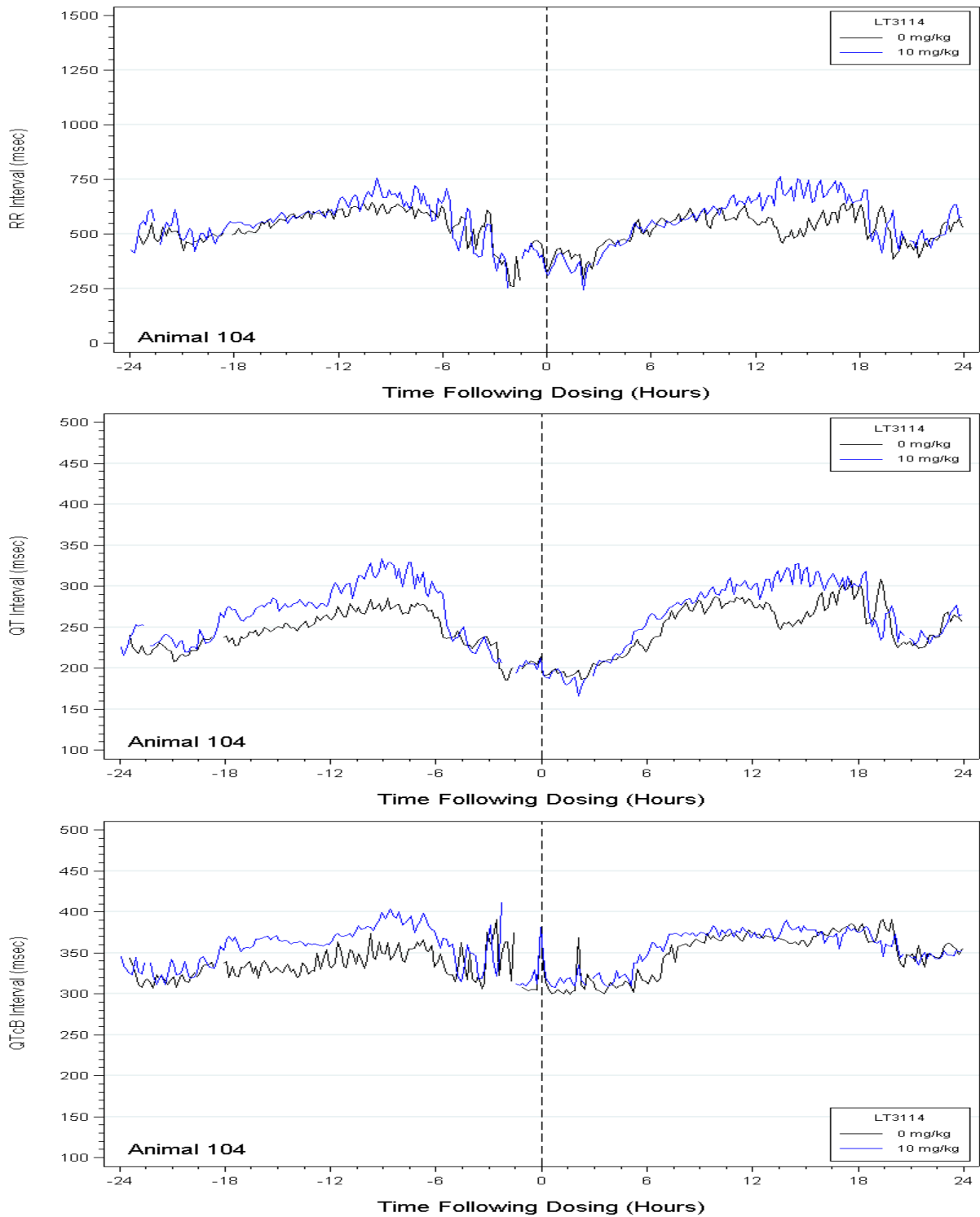


Figure E-16. Animal 104, RR, QT, and QTcB Interval Parameters

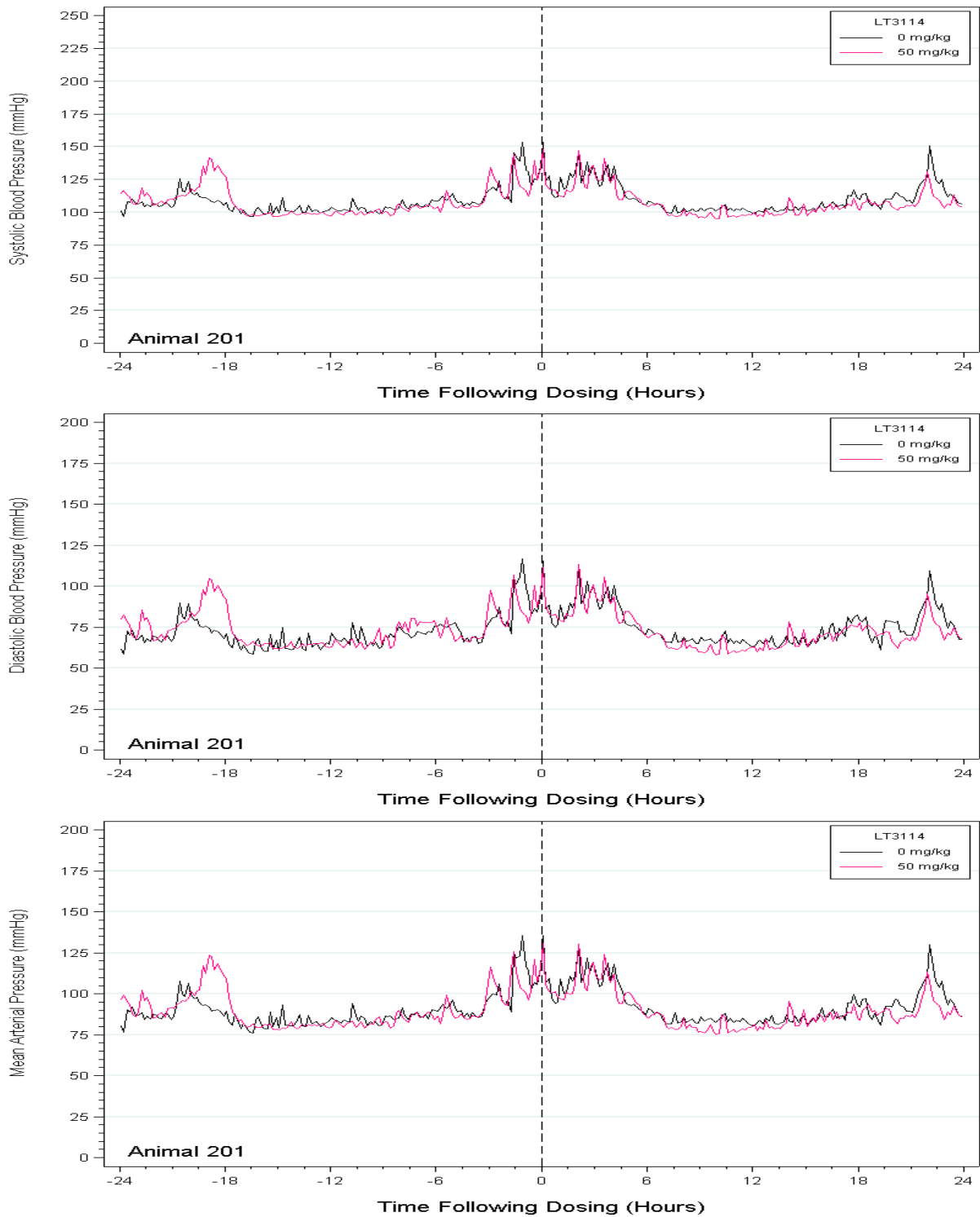


Figure E-17. Animal 201, Systemic Blood Pressure Parameters

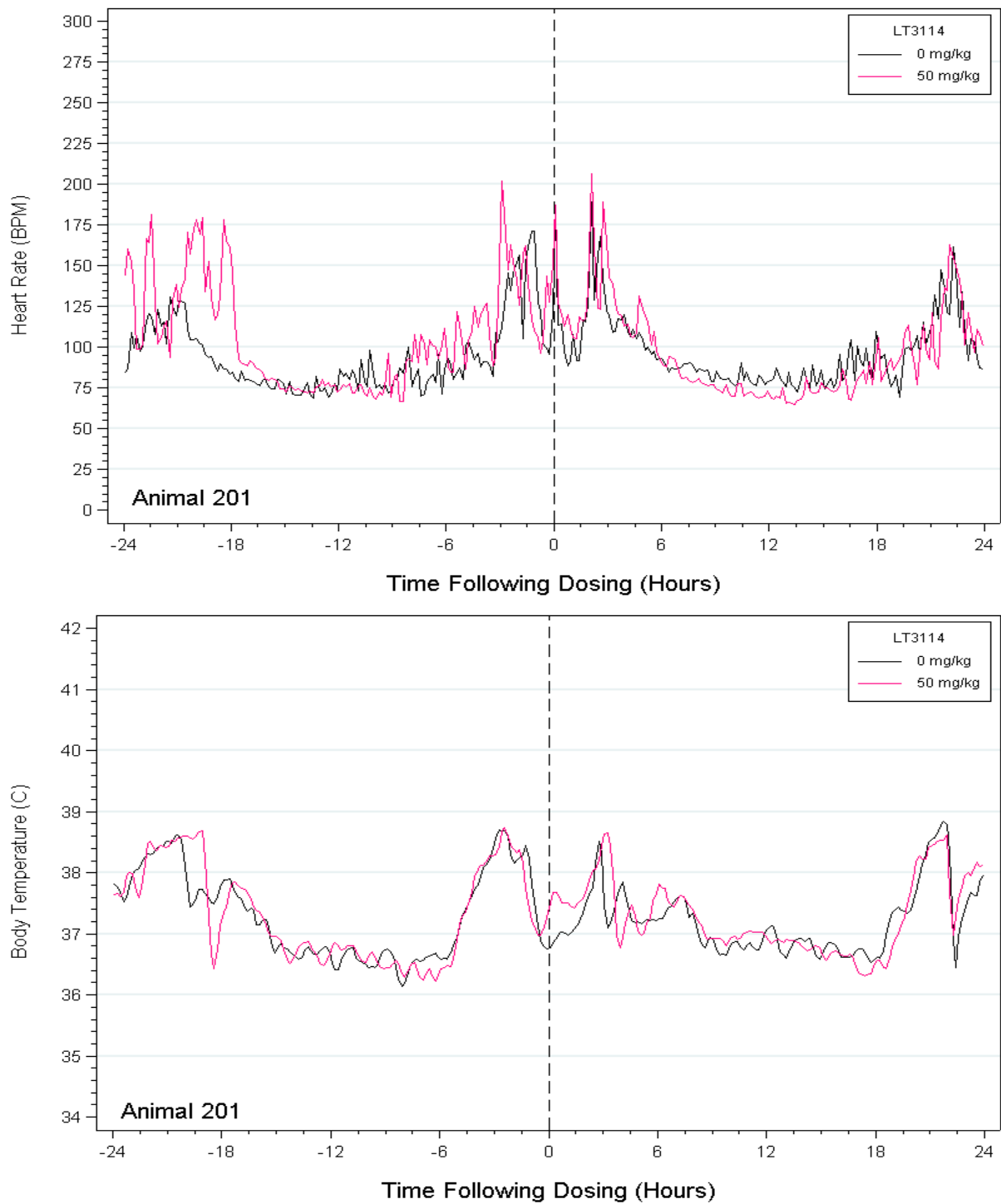


Figure E-18. Animal 201, Heart Rate and Body Temperature Parameters

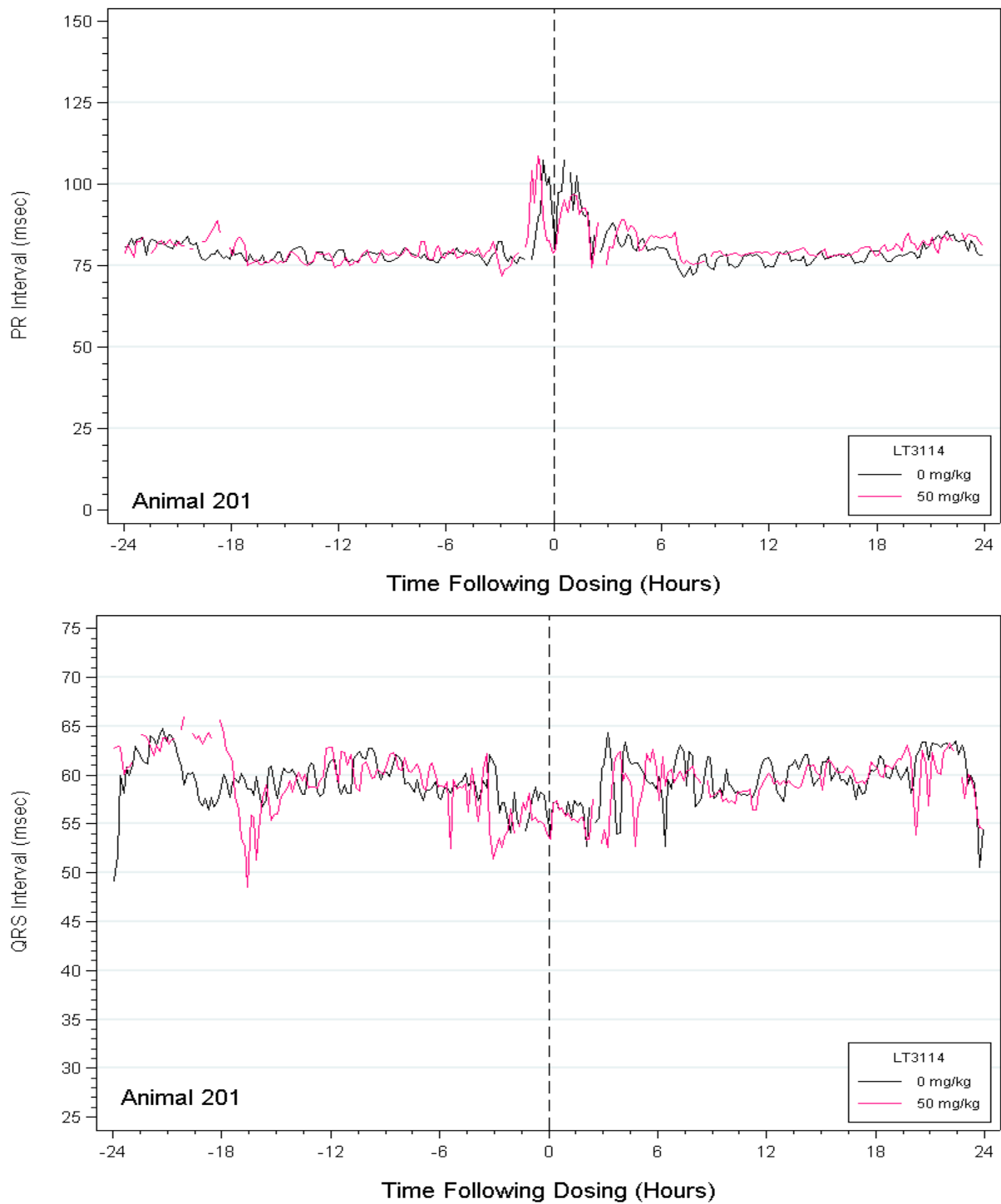


Figure E-19. Animal 201, PR and QRS Interval Parameters

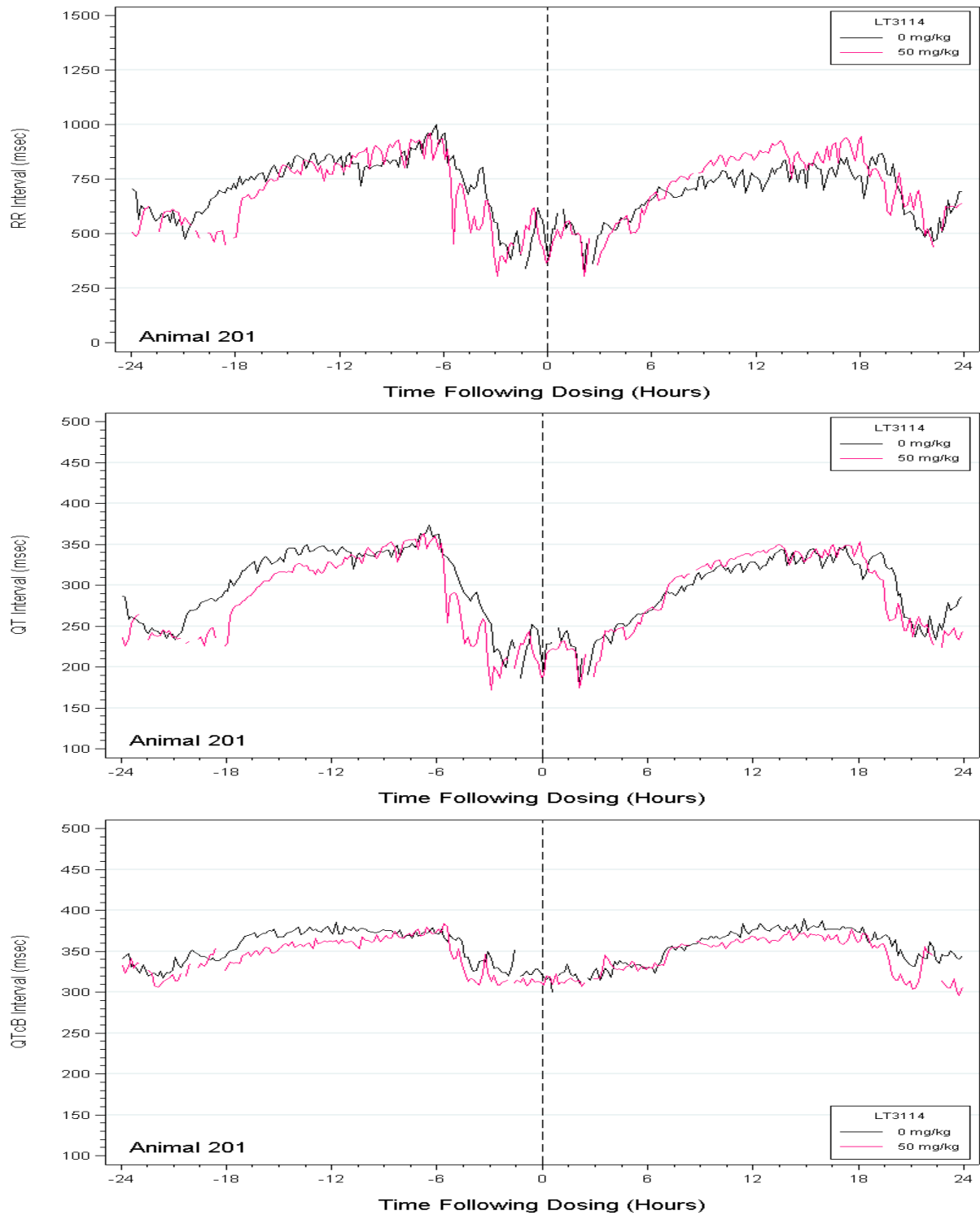


Figure E-20. Animal 201, RR, QT, and QTcB Interval Parameters

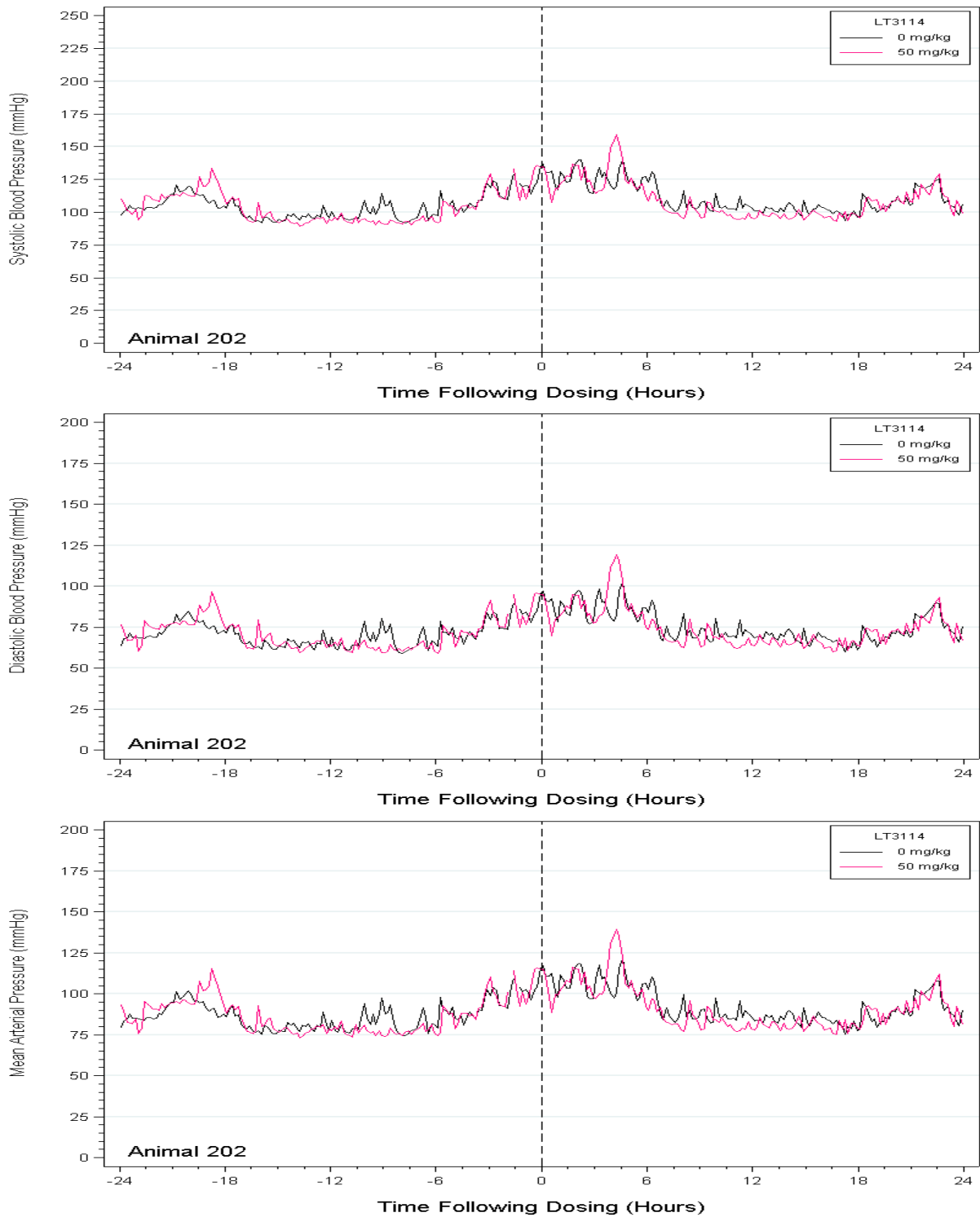


Figure E-21. Animal 202, Systemic Blood Pressure Parameters

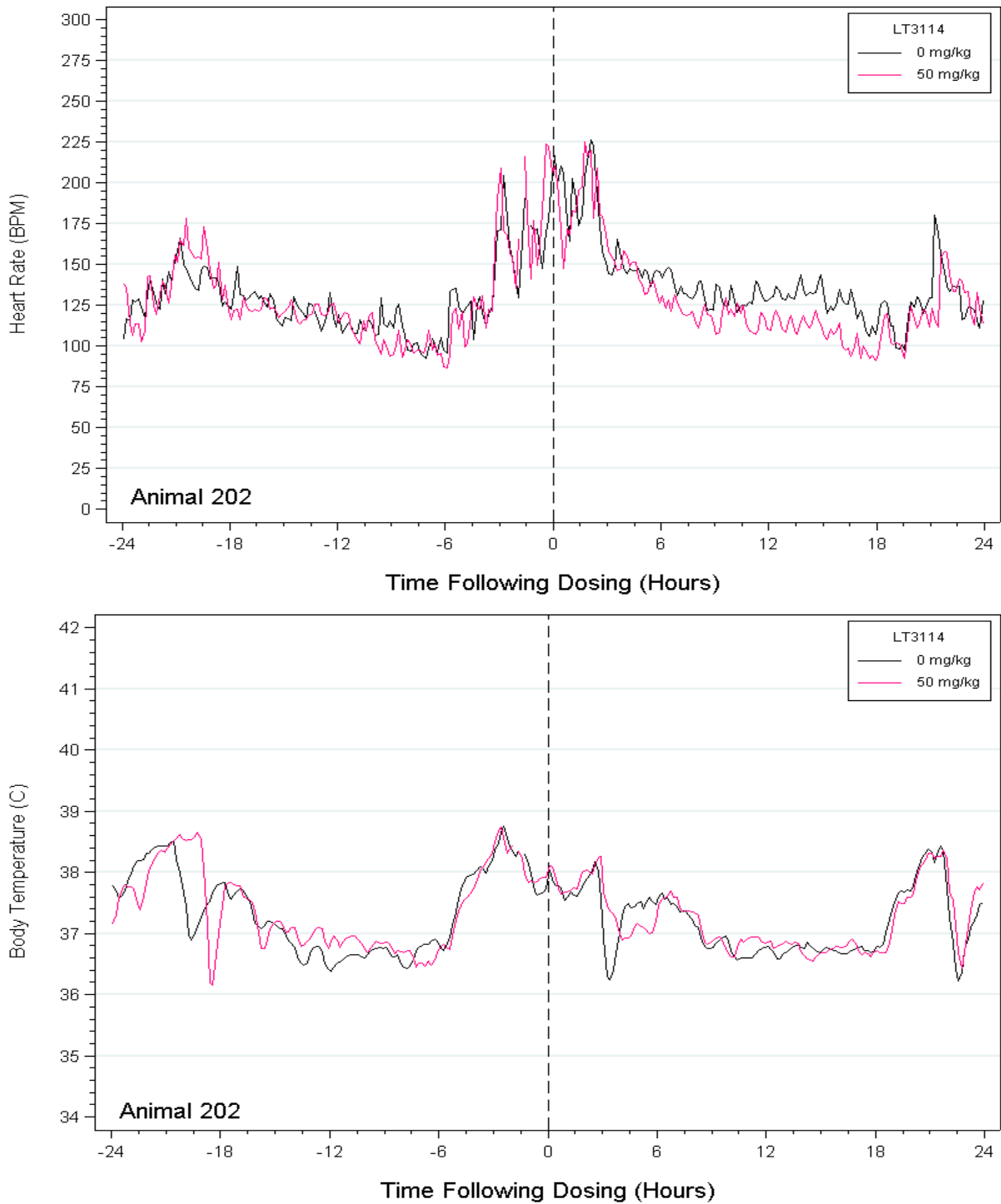


Figure E-22. Animal 202, Heart Rate and Body Temperature Parameters

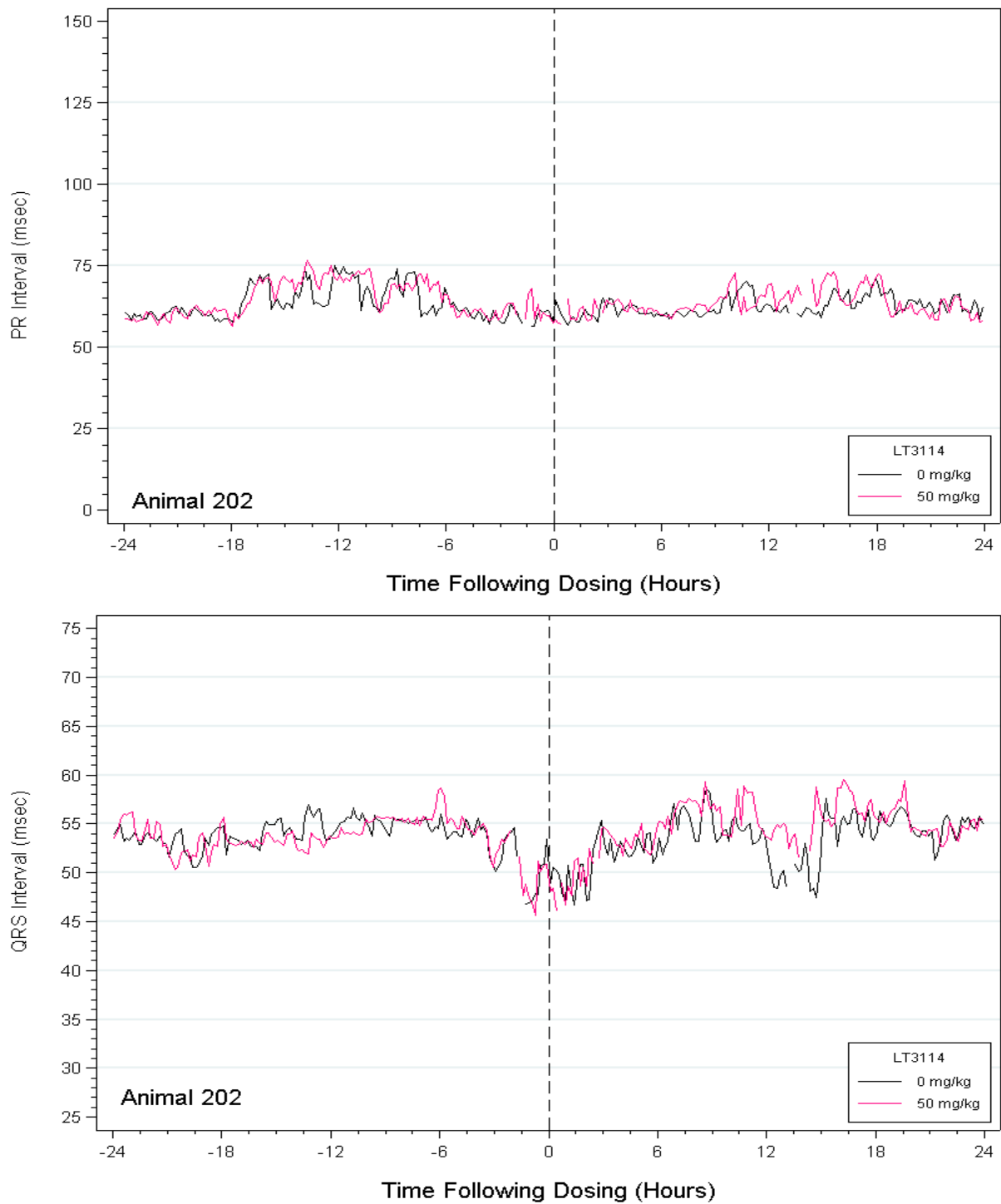


Figure E-23. Animal 202, PR and QRS Interval Parameters

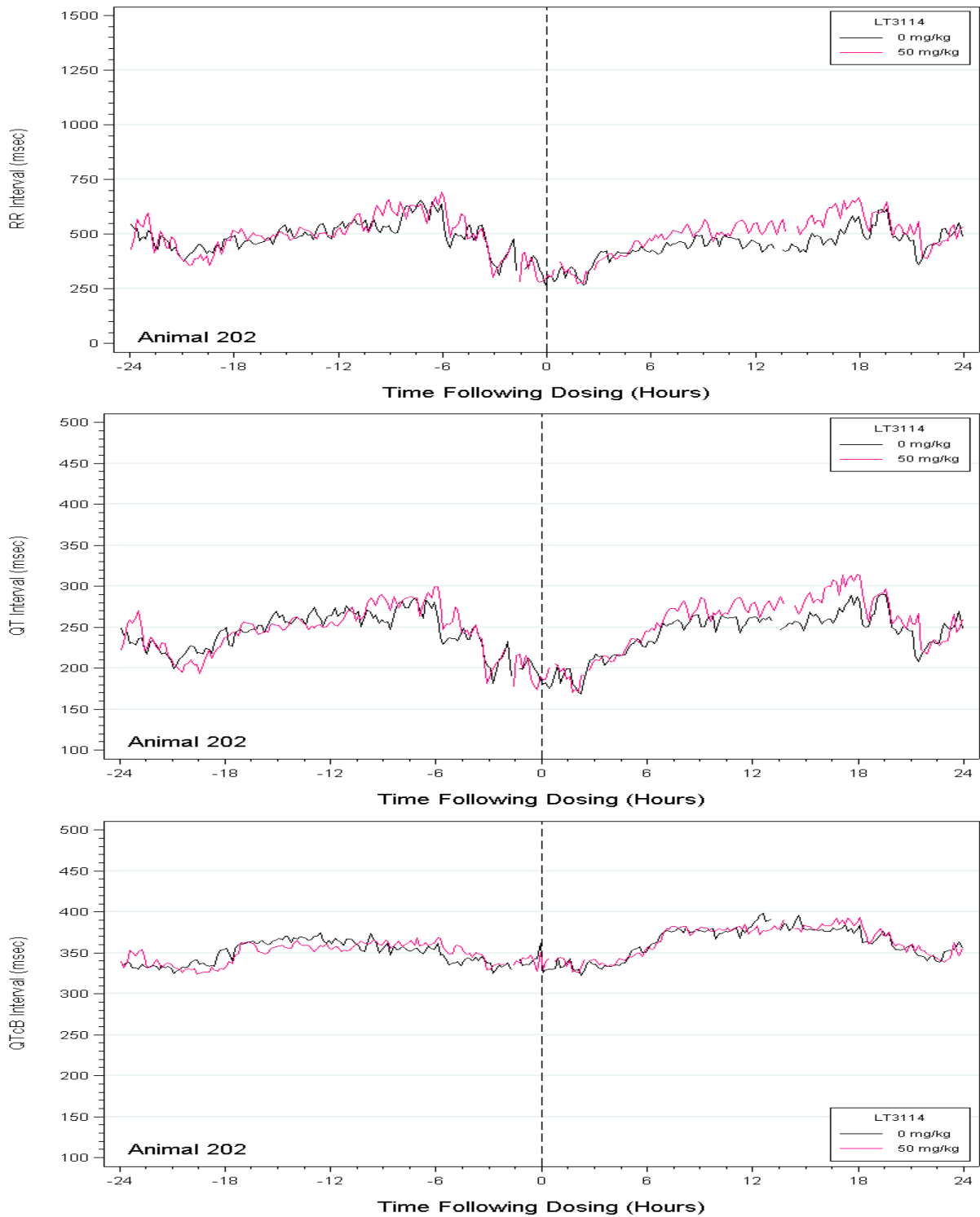


Figure E-24. Animal 202, RR, QT, and QTcB Interval Parameters

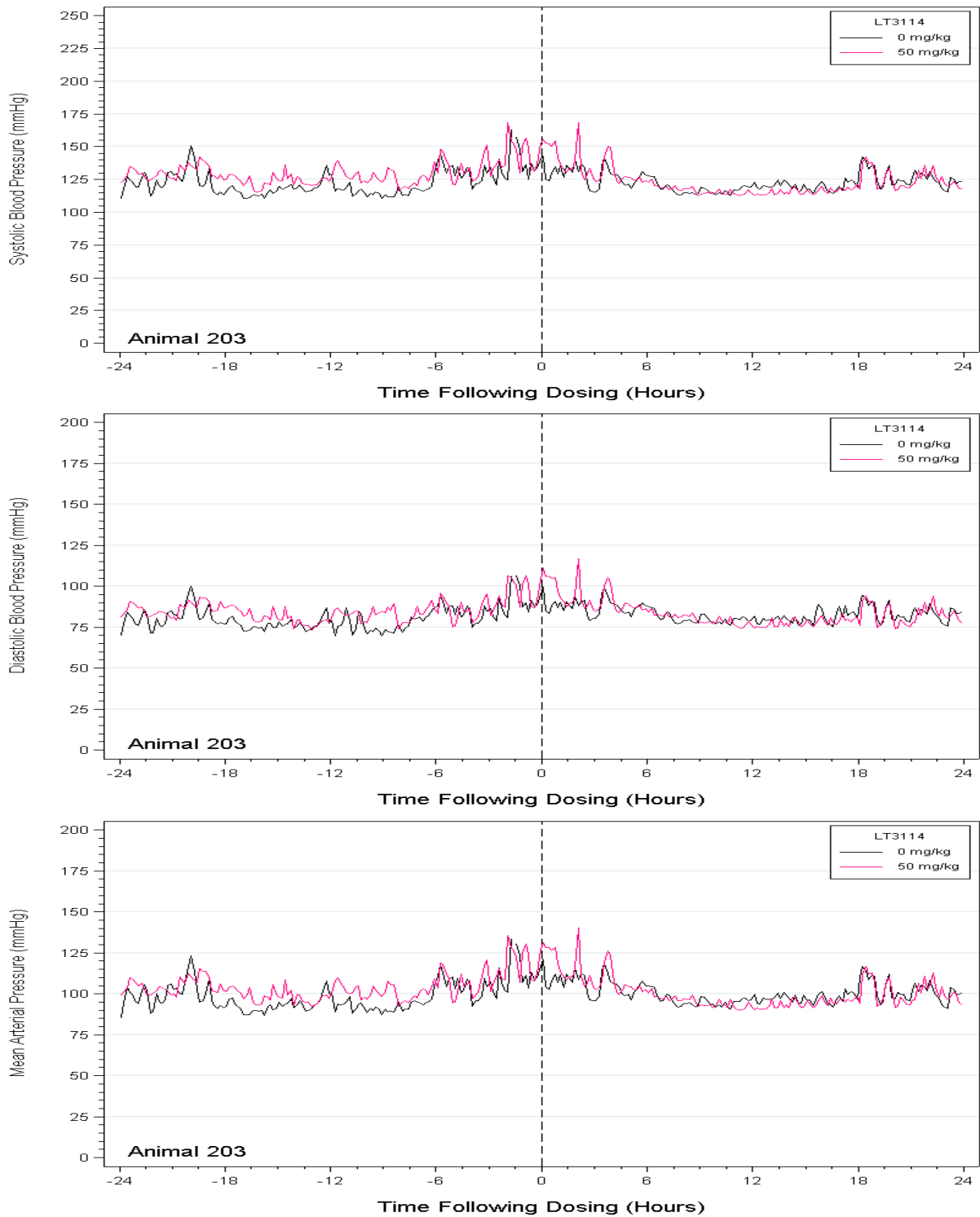


Figure E-25. Animal 203, Systemic Blood Pressure Parameters

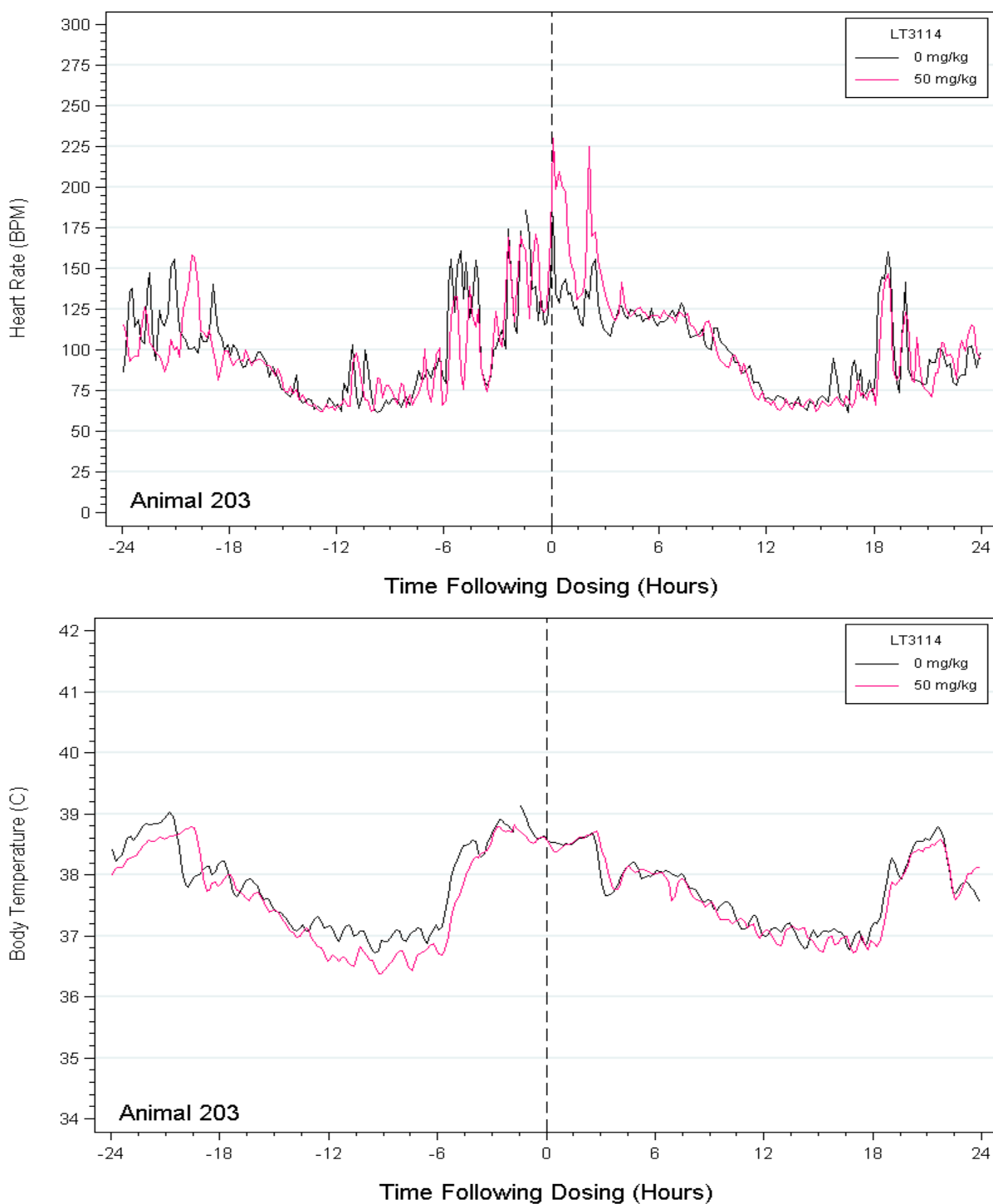


Figure E-26. Animal 203, Heart Rate and Body Temperature Parameters

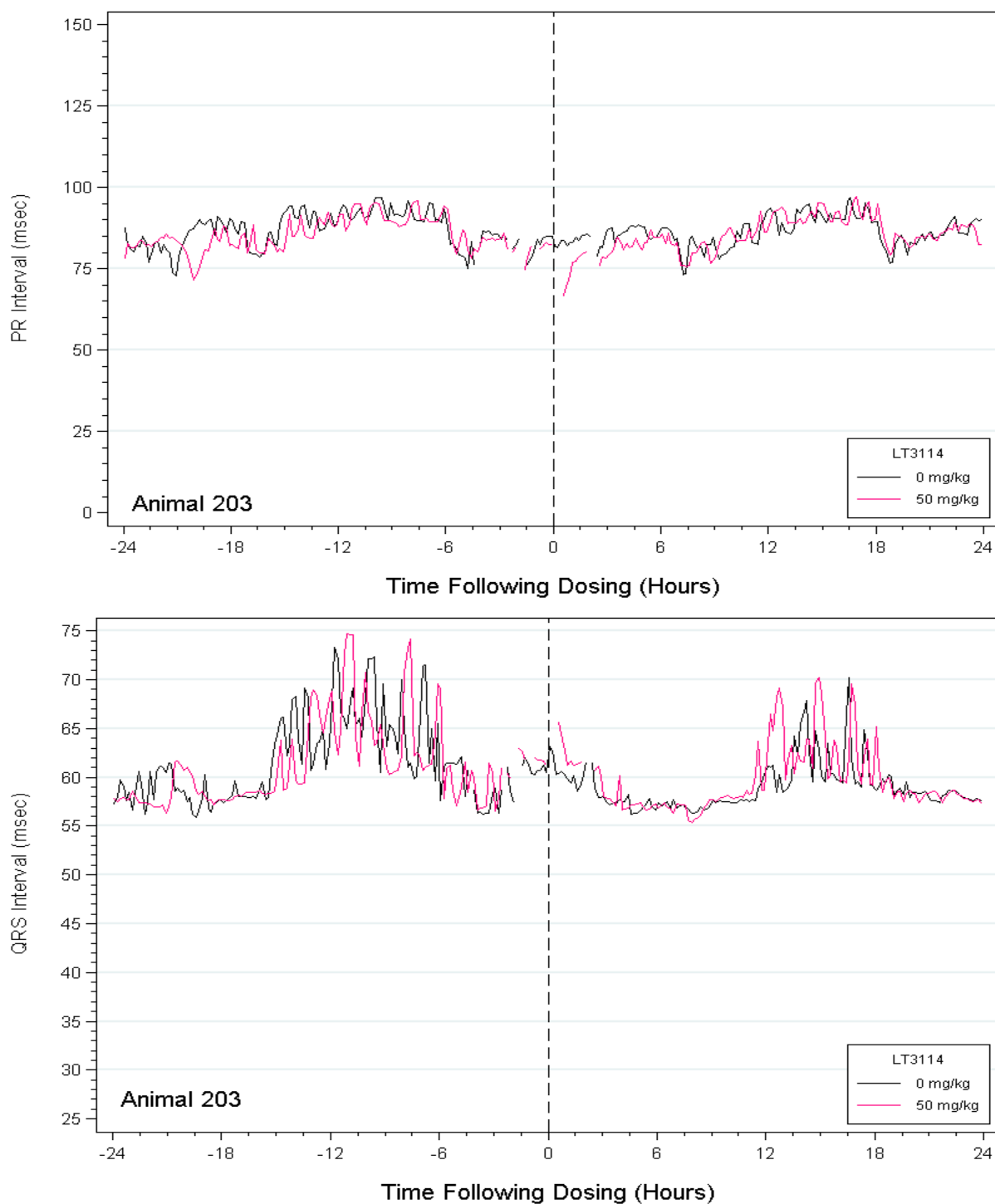


Figure E-27. Animal 203, PR and QRS Interval Parameters

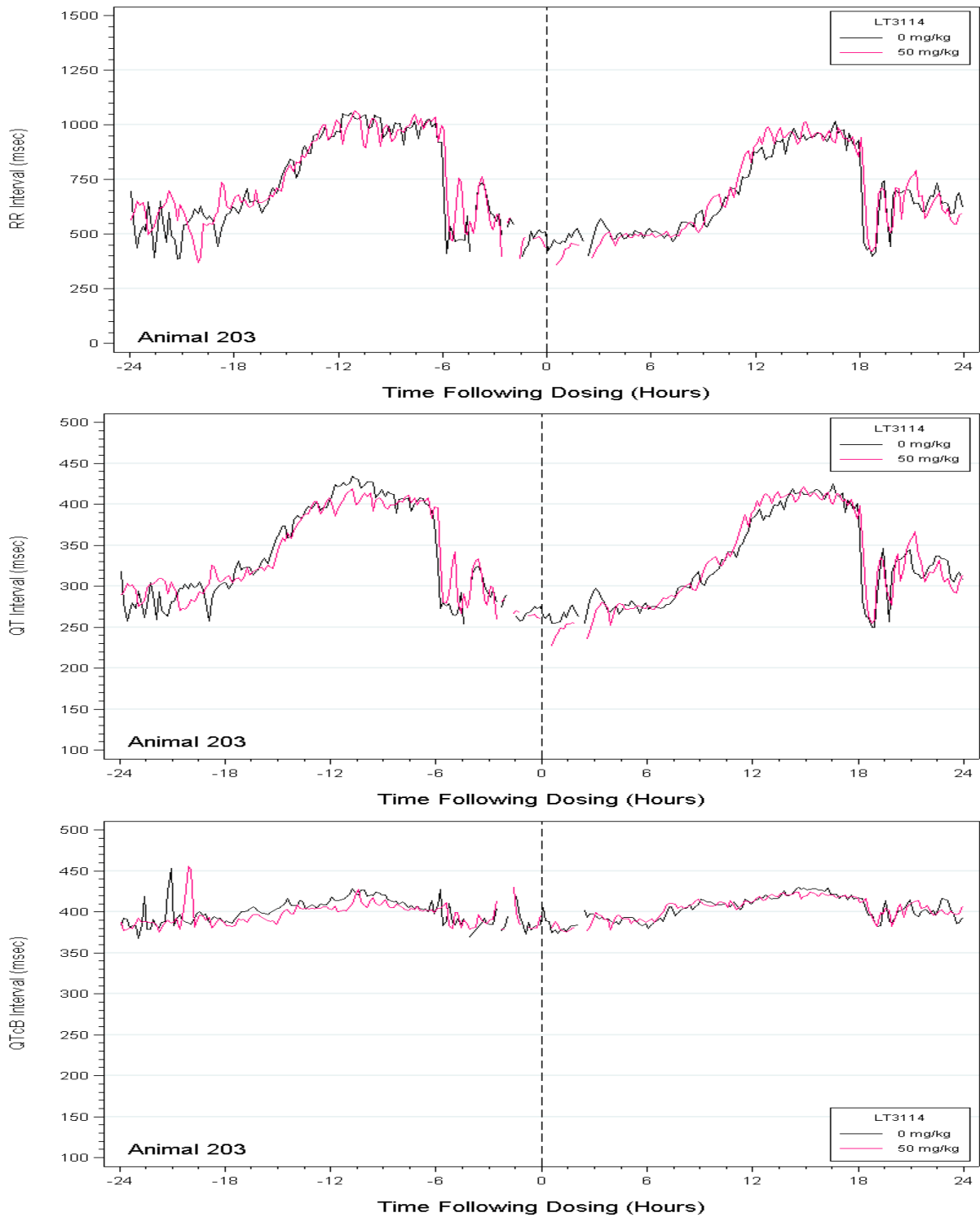


Figure E-28. Animal 203, RR, QT, and QTcB Interval Parameters

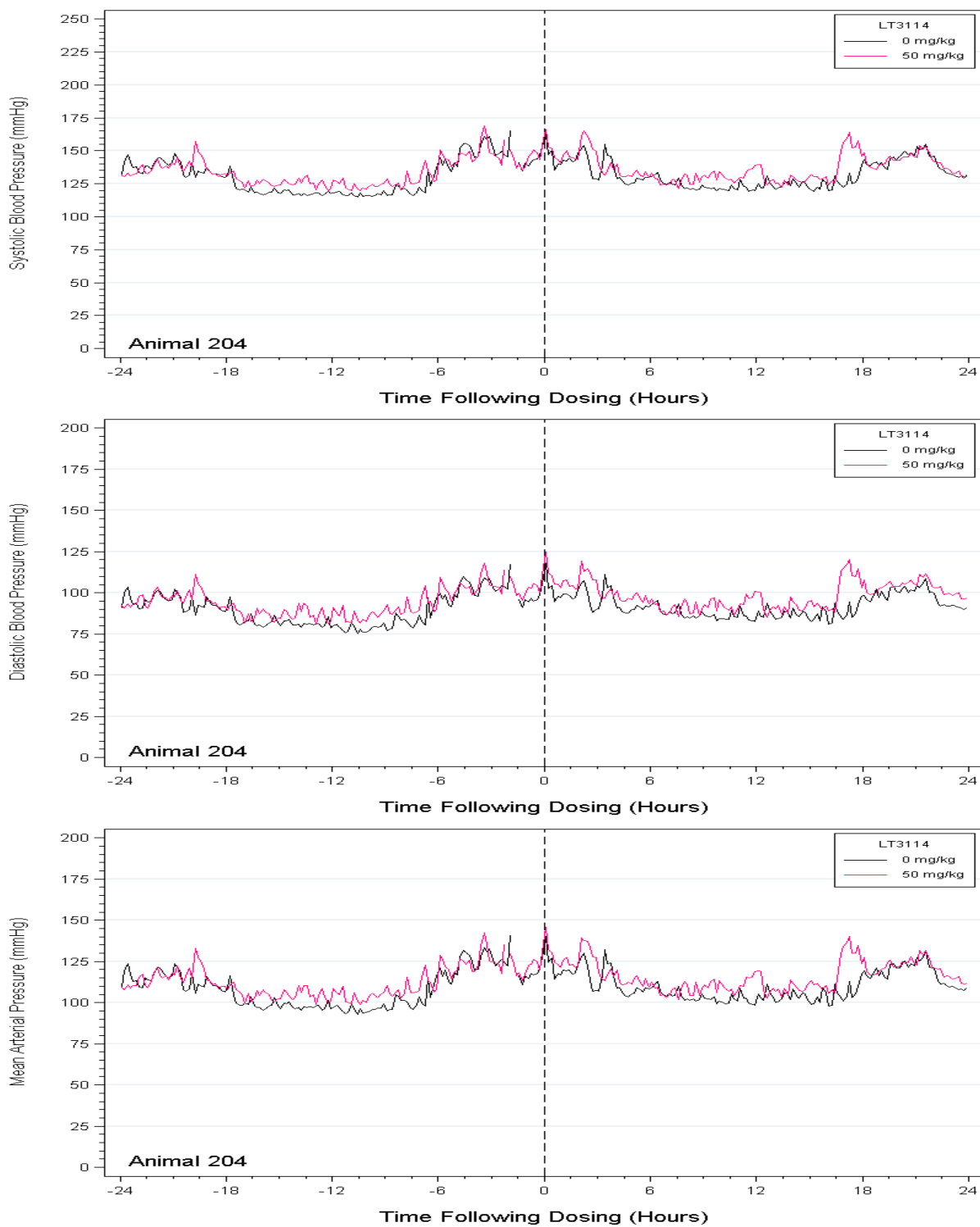


Figure E-29. Animal 204, Systemic Blood Pressure Parameters

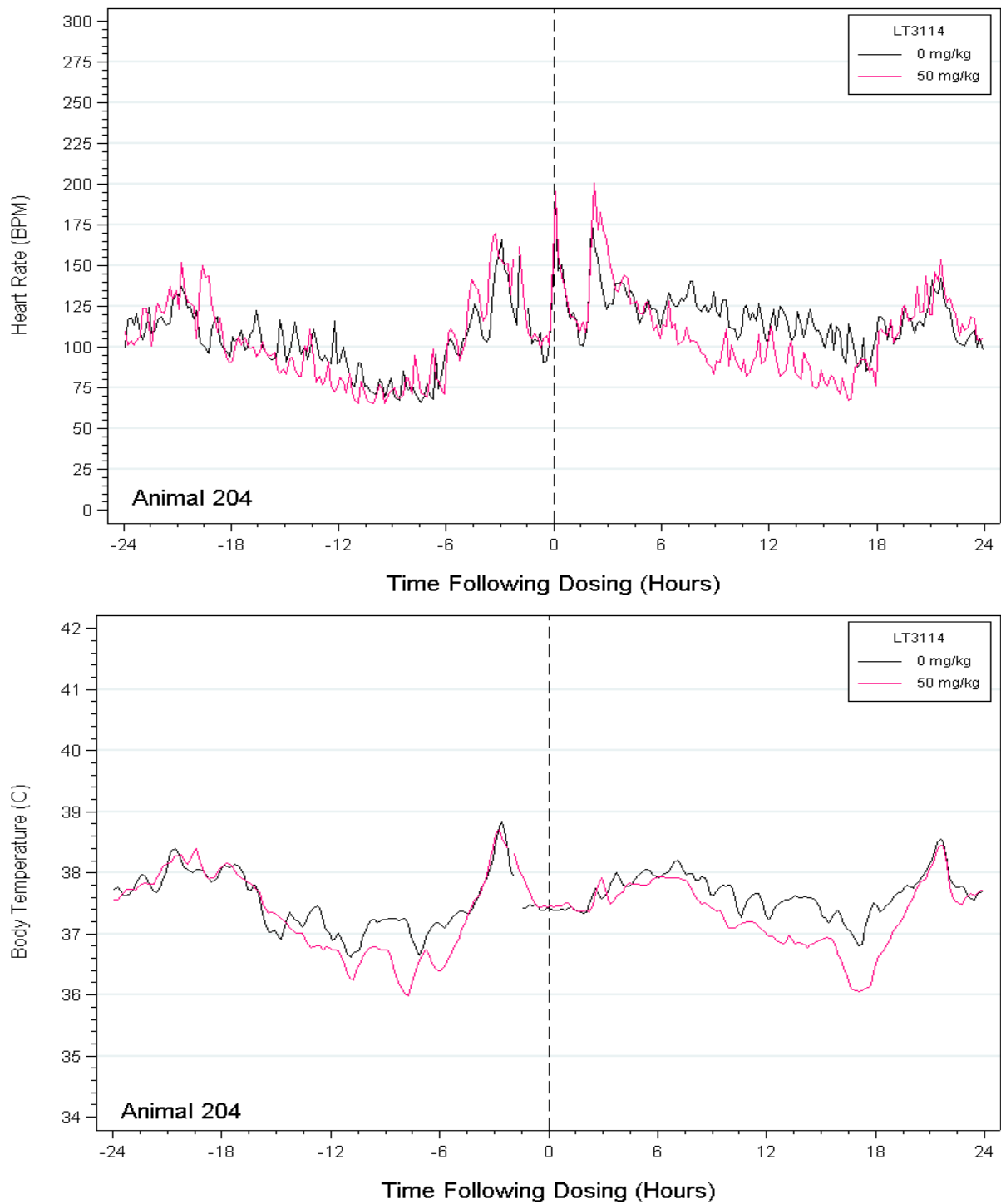


Figure E-30. Animal 204, Heart Rate and Body Temperature Parameters

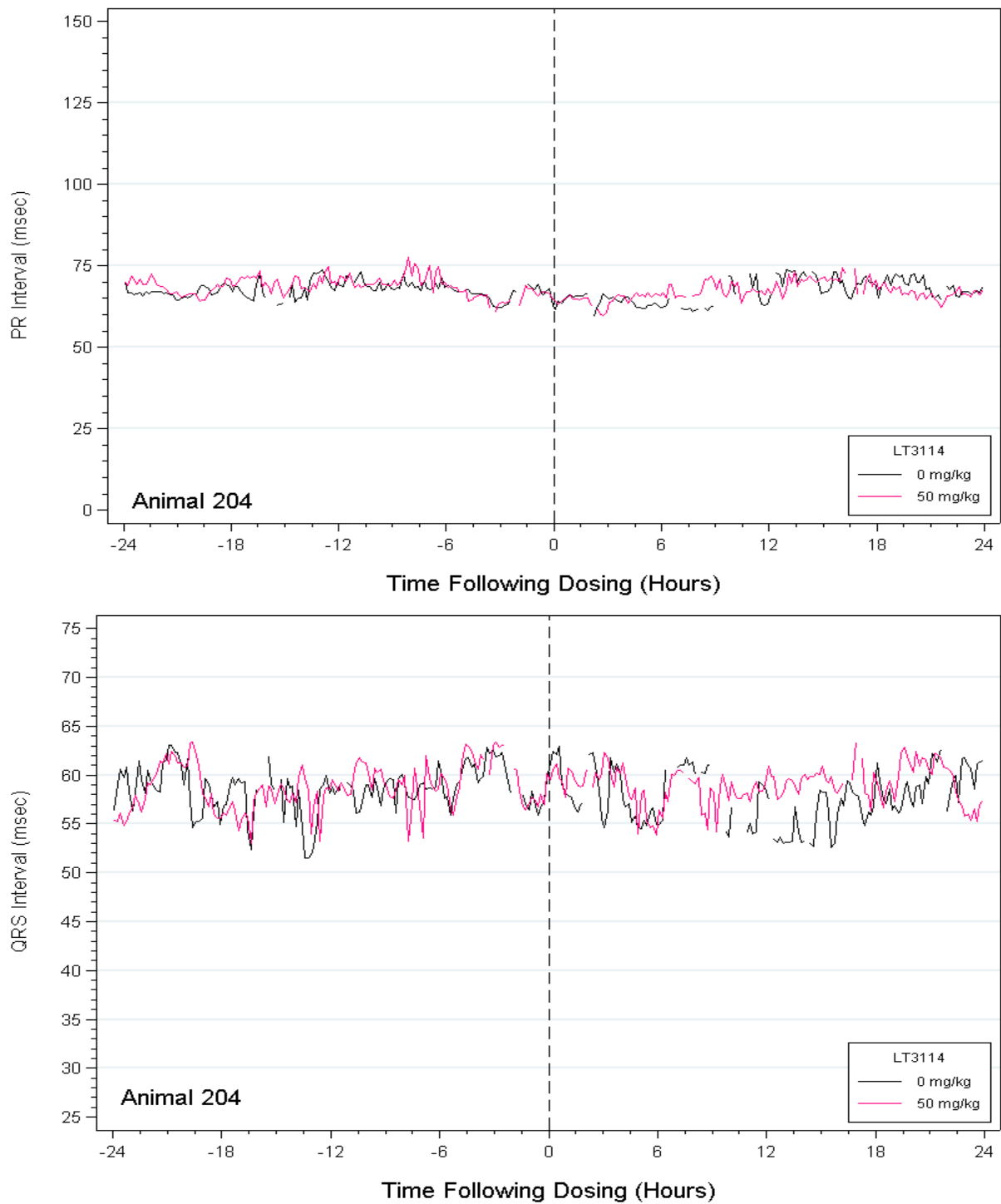


Figure E-31. Animal 204, PR and QRS Interval Parameters

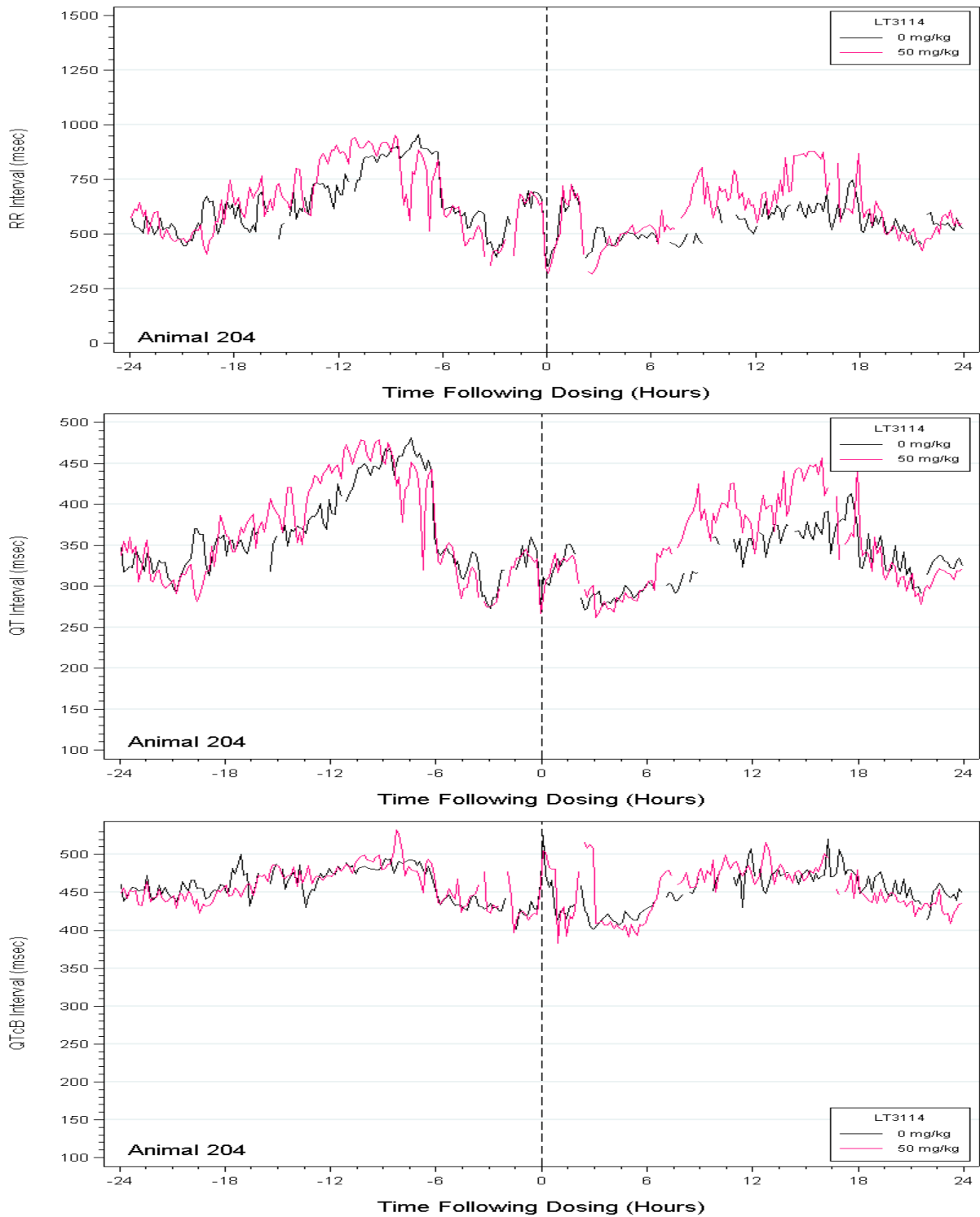


Figure E-32. Animal 204, RR, QT, and QTcB Interval Parameters

APPENDIX F

QUALITATIVE EVALUATION OF ECG WAVEFORMS

Robert L. Hamlin, D.V.M., Ph.D., D.A.C.V.I.M.1520 Grenoble Rd.
Columbus, Ohio 43221
614-486-7542
Fax: 614-486-7545

November 20, 2014

Dr. Craig R. Hassler
Battelle Memorial Institute
505 King Avenue
Columbus, Ohio 43201**RE: Qualitative ECG Evaluation for Monkeys on Battelle Study Number 52447**

With few exceptions when muscle tremor of motion artifacts occurs, all ECGs are of excellent quality for interpretation. As usual with serial ECGs, there are frequent changes in both heart rates and in configurations of component of deflections (in particular ST-T complexes).

Monkey 101 (vehicle, Day 1) had changes in heart rate accompanied by dramatic changes in configurations of ST-T not related to the test article.

Monkey 102 (vehicle, Day 1) had either incomplete Right Bundle Branch Block (RBBB) or left anterior hemiblock, but no ECG changes related to the test article.

Monkey 103 (vehicle, Day 1) had occasional junctional premature depolarizations; this does not indicate a drug effect. On Day 4, following the low dose, this animal displayed the premature depolarizations possibly more prevalently. It is impossible to say if this was drug related since the beats occurred so infrequently.

Monkey 104 showed no ECG changes of significance.

Monkey 201 (vehicle, Day 1) had remarkable ventilatory artifacts producing what appeared to be electrical alternans, but is most likely artifact.

Monkey 202 through 204 showed no ECG changes of significance.

There were no ECG findings attributable to the test article at either dose level for this study.

Sincerely,



Robert L. Hamlin, D.V.M., Ph.D., D.A.C.V.I.M.
(Cardiology/Internal Medicine)11-26-14

Date

APPENDIX G

DETERMINATION OF LT3114 IN MONKEY K₂EDTA PLASMA SAMPLES BY ELISA



SAMPLE ANALYSIS REPORT

Determination of LT3114 in Monkey K₂EDTA Plasma Samples by ELISA

QPS Project Number: 344-1411
Client Project Number: 1259SL35.001

Sponsor
Lpath Incorporated
4025 Sorrento Valley Blvd
San Diego, CA 92121

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QPS, LLC
Delaware Technology Park
3 Innovation Way, Suite 240
Newark, DE 19711
WEB: www.qps-usa.com
TEL: (302) 369-5601
FAX: (302) 369-5602

1259SL35.001

SAMPLE ANALYSIS REPORT**REPORT TITLE:**Determination of LT3114 in Monkey K₂EDTA Plasma Samples by ELISA**QPS PROJECT NUMBER:**

344-1411

CONTRIBUTING SCIENTISTS:

Melissa Calvarese, B.S.

TESTING FACILITY:

QPS, LLC

Delaware Technology Park

3 Innovation Way, Suite 240

Newark, DE 19711

BIOANALYTICAL STUDY DATES:

Sept 24, 2014 to Oct 7, 2014

SPONSOR:

Lpath Incorporated

4025 Sorrento Valley Blvd.

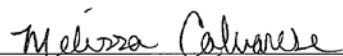
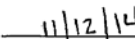
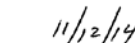
San Diego, CA 92121

SPONSOR REPRESENTATIVE:

Gary Woodnutt

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1. SIGNATURE PAGE**WRITTEN AND REVIEWED BY:**Melissa Calvarese
Principal Investigator
QPS, LLC
Date**APPROVED BY:**Chad Asher, B.S.
Group Leader and Head of PK Bioanalysis
QPS, LLC
Date**Confidential**

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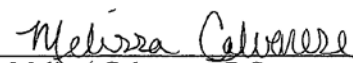
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2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58; and the FDA *Guidance for Industry: Bioanalytical Method Validation*, May 2001.

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.



Melissa Calvarese, B.S.
Principal Investigator
QPS, LLC

11/12/14
Date

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3. QUALITY ASSURANCE STATEMENT

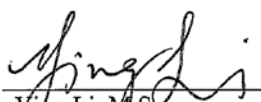
QPS 344-1411

Protocol Title: A Single Slow Bolus Injection Study of LT3114 in Cynomolgus Monkeys

This QPS study has been audited and found to accurately reflect the requirements set forth by the 344-1403 methods and standard operating procedures and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor	Audit Dates	Date Reported to Principal Investigator and Management	Date Reported to Test Facility Quality Assurance, Study Director and Management
Protocol Review/ Draft Study Plan	S. Ma	26 Sep 2014	26 Sep 2014	7 Oct 2014
Study Inspection/ Instrument Setup/ Analysis	Y. Li	7 Oct 2014	8 Oct 2014	20 Oct 2014
Study Records/ Draft Report	Y. Li	14 Oct 2014	14 Oct 2014	20 Oct 2014

The undersigned has assured that the QPS final report accurately reflects the source data.



Ying Li, M.S.
QPS Quality Assurance

11-11-2014

Date

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4. PROTOCOL TITLE

A Single Slow Bolus Injection Study of LT3114 in Cynomolgus Monkeys

5. SUMMARY

On 24 Sep 2014, a total of 16 monkey K₂EDTA plasma samples were received frozen on dry ice from Battelle, Columbus, Ohio. The samples were stored in a -70°C freezer until analysis. The sample receipt and sample analysis information is summarized in Tables 1 and 2, respectively. The samples were analyzed using the method entitled "Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA" which was validated at QPS, LLC. The details of the assay procedure and validation data are listed and summarized in the assay validation report (QPS 344-1403).

LT3114 was obtained from Lpath Inc and was subsequently used in preparing standard and quality control (QC) solutions in this study.

6. OBJECTIVE

To determine the concentrations of LT3114 in monkey plasma samples obtained from Lpath pre-clinical Protocol 1259SL35.001.

7. ANALYTE INFORMATION

Reference Standard	
Name:	LT3114
Concentration:	10.0 ^a mg/mL
Potency:	9.8mg/mL
Supplied Form	Lpath
Lot Number:	103948
Expiration Date:	8/30/2015 ^b
Storage Conditions:	-70°C
^a With potency applied, concentration will be used at 9.8mg/mL.	
^b May be updated in study record if additional stability is performed	

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8. VALIDATION SUMMARY FOR LT3114

Report Title	Determination of LT3114 in Monkey K ₂ EDTA Plasma by ELISA
Report Number	QPS 344-1403
Analyte Name	LT3114
Sample Volume:	5 μ L (Before 100x dilution, duplicate measurement)
Analytical Method Type	ELISA
Sample Processing Method	None
Calibration Range:	100 – 2000 ng/mL
QC Concentrations	100, 300, 750, 1500, and 2000 ng/mL
QC Intra-batch Precision (%CV)	1.9% - 5.3%
QC Intra-batch Accuracy (%Diff)	-3.0% - 13.0%
QC Inter-batch Precision (%CV)	5.4% - 12.9%
QC Inter-batch Accuracy (%Diff)	-3.3% - 4.0%
Room temp Stability in Monkey K ₂ EDTA Plasma	24 Hours at Room Temperature
Freeze/thaw Stability in Monkey K ₂ EDTA Plasma	5 Cycles at -70°C and -20°C/ Room Temperature
Long-term Storage Stability in Monkey K ₂ EDTA Plasma	92 Days at -20°C and -70°C
Dilution Linearity	900,000 ng/mL diluted 1, 100, 500, 1000, and 5000 -fold
Selectivity (10 lots, spiked 100, 2000 ng/mL)	100% lots tested within 100 $\pm$ 25% Recovery

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9. ABBREVIATIONS AND DEFINITIONS

This section provides abbreviations and definitions of terms and concepts that may be commonly used throughout this report.

AQL	Above Quantifiable Limit
BQL	Below Quantifiable Limit
Exc	Sample excluded according to QPS Validation Protocol 344-1403
LLOQ	Lower Limit of Quantification
NC	Not Calculated
NA	Not Applicable
NR	Not Reported
OD	Optical Density
QC	Quality Control
r²	Coefficient of Determination
SD	Standard Deviation
ULOQ	Upper Limit of Quantification
Precision	$\% \text{Coefficient of Variation (\%CV)} = 100 \times \frac{\text{Standard Deviation}}{\text{Mean Concentration}}$
Accuracy	$(\% \text{Diff}) = 100 \times \frac{\text{Mean Found Concentration} - \text{Nominal Concentration}}{\text{Nominal Concentration}}$

Quality control samples at three concentration levels (low, mid, and high) were prepared.

The concentrations of the calibration standards, and the QCs in monkey plasma were determined based on the actual concentrations of the stock solutions.

All statistics (e.g., Mean, S.D., %CV, %Diff) found in the data tables are calculated by Watson™ LIMS or based on the “precision as displayed” option of Microsoft® Excel.

10. STUDY INFORMATION**10.1. Sample Collection**

Plasma was collected from two dose groups on 15 Sept 2014 and 18 Sept 2014 into tubes containing K₂EDTA as the anticoagulant at the following time points: approximately 2 hours after the start of dosing, on Days 1 and 4. Plasma samples were stored at approximately -70°C at Battelle until shipment frozen on dry ice to QPS.

The PK samples were to be obtained according to protocol as follows:

2 timepoints x 4subjects x 2 groups = 16 samples

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10.2. Sample Collection Dates

Samples were collected from 15 Sept 2014 through 18 Sept 2014.

10.3. Sample Receipt Dates and Sample Inventory

Sample receipt date: 24 Sept 2014.

10.4. Sample Analysis Dates

Analysis of samples began on 6 Oct 2014 and concluded on 7 Oct 2014 (Table 2).

10.5. Sample Storage Conditions

Samples were received at QPS frozen on dry ice and upon arrival they were stored at -70°C. At most 22 days transpired between sample collection and analysis for all the samples. The currently validated stability for QPS method 344-1403 is sufficient to cover the study samples.

10.6. Sample Disposition

Sample disposition will be governed by the most current version of SOP LP-001, "Sample Chain of Custody."

11. ANALYTICAL INFORMATION**11.1. Analytical Method Validation**

This analysis was carried out according to the method outlined in QPS Project Number 344-1403 entitled "Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA"

11.2. Regression Analysis

Calculations were done using "Logistic, autoestimate" regression using Watson™ LIMS (v. 6.4.0.04, Thermo Electron Corporation, USA)

12. ACCEPTANCE CRITERIA

Acceptance criteria for calibration standards and QC's are based upon QPS 344-1403 validation report and QPS SOP TLM-002. A calibration standard is accepted if its back-calculated concentration does not deviate more than $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) from nominal concentration.

For a run to be accepted, a minimum of 75% of the total number of calibration standards in the calibration range (between the LLOQ and ULOQ of the run) should be within $100 \pm 20\%$ ($100 \pm 25\%$ at LLOQ and ULOQ standards) of their nominal value.

For QCs and study samples, each sample will be assayed twice and the average value will be reported if the % CV from the two measurements is $\leq 25\%$. Otherwise, the mean value from

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the two measurements will not be reported. At least 2/3 of the back calculated QC concentrations should be within $100 \pm 20\%$ of their nominal value, 50% of QCs at each concentration should not deviate more than $\pm 20\%$ from their nominal value.

13. REPEAT/REASSAY CRITERIA

Samples with concentrations above the quantifiable limit (AQL) were reassayed using an appropriate dilution factor. All reported reassay sample results were determined using the decision tree found in the current version of QPS SOP TLM-002.

14. RESULTS

14.1. Method Performance

Standard curve regression parameters for LT3114 in monkey plasma are summarized in Table 3. Back-calculated concentrations for LT3114 standards are listed in Table 4.

Back calculated concentrations for QCs in monkey plasma are listed in Table 5. The %CV of the QCs for LT3114 at 300, 750, and 1500 ng/mL ranged from 1.4% to 7.5% while %Diff ranged from -2.7% to 13.3%.

All 16 samples were analyzed in 2 runs.

14.2. Summary of Repeat/Reassay Analysis

Eight samples were reassayed due to AQL results. Reassay sample concentrations are listed in Table 6.

15. STUDY SPECIFIC DEVIATIONS OR OBSERVATIONS

There were no protocol deviations or study observations affecting the analytical results

16. CONCLUSION

LT3114 in monkey K₂EDTA plasma were analyzed using a validated method. Concentrations of LT3114 in individual monkey plasma samples are summarized in Table 7. All concentrations are expressed in terms of ng/mL. Sample concentrations that were calculated below 100 ng/mL for LT3114 are reported as below quantifiable limit (BQL).

17. REFERENCES

Study Protocol

“A Single Slow Bolus Injection Study of LT3114 in Cynomolgus Monkeys”, 3 Sept 2014
signed by Michael A. Hawk.

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Standard Operating Procedure:

QPS LP-001: "Sample Chain of Custody"

QPS LP-004: "Analytical Standards and Quality Controls"

QPS LP-007: "Analytical Reference Compounds"

QPS LP-034: "Receipt and Handling of Biological Matrices"

QPS GP-014: "Archives"

QPS TLM-001: "Validation of Quantitative Ligand Binding Assays"

QPS TLM-002: "Sample Analysis and Reanalysis for Ligand Binding Assays"

Study Notebooks

QPS 344-1411

Validation Notebook(s) and Report(s)QPS 344-1403: "Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA"**18. ARCHIVE STATEMENT**

All raw data (including electronic Softmax Pro files), a copy of this bioanalytical report and required supporting information for this study will be held under the control of QPS, LLC, 3 Innovation Way, Suite 240, Newark, DE 19711, USA. Study records will be archived according to the most current version of SOP GN-014, "Archives."

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19. TABLES

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Table 1 **Sample Receipt Tracking Log**

Date Received	Matrix	Original	Total
9/24/2014	Monkey Plasma	16	16
Total:			16

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Table 2 **Sample Analysis Tracking Log**

Curve Number	Assay Date	Run Description	Run Acceptance ^a
1	6-Oct-14	141006-3441411-MKPL-MAC-1A-001	Accepted
2	7-Oct-14	141007-3441411-MKPL-MAC-1A-002	Accepted

a: Run acceptance based on QPS protocol 344-1403

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Table 3 **LT3114 Standard Curve Regression Parameters**

Assay Date	Curve Number	Min	Max	Slope	Ed50	R-Squared	LLOQ	ULOQ
Oct-14	1	0.100803756	4.246368666	1.370921394	3739.797432994	0.9891	100	2000
Oct-14	2	0.065679799	6.692723438	1.137616938	9634.050295509	0.9992	100	2000

$$Y_{\text{P}} = (\text{Min} - \text{Max}) / (1 + (\text{Conc.} / \text{Ed50})^{**} \text{Slope}) + \text{Max}$$

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Table 4 **Back-Calculated Concentrations (ng/mL) of LT3114 Calibration Standards in Monkey K₂EDTA Plasma**

Assay Date	Curve Number	Concentration (ng/mL)							
		100	200	400	800	1000	1400	1800	2000
6-Oct-14	1	85.7	220	433	793	960	1330	2010	1740
		98.6	227	461	834	996	1390	1880	Exc
7-Oct-14	2	100	207	419	793	993	1390	1810	2000
		104	211	426	808	956	1380	1860	1990
n		4	4	4	4	4	4	4	3
Mean		97.1	216	435	807	976	1370	1890	1910
S.D.		7.92	9.00	18.4	19.3	21.2	28.7	85.2	147
%CV		8.2	4.2	4.2	2.4	2.2	2.1	4.5	7.7
%Diff		-2.9	8.0	8.8	0.9	-2.4	-2.1	5.0	-4.5

Exc: Excluded according to QPS Protocol 344-1403

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Table 5 **Back-Calculated Concentrations (ng/mL) of LT3114 Quality Control
Samples in Monkey K₂EDTA Plasma**

Assay Date	Curve Number	Concentration (ng/ml)		
		300	750	1500
6-Oct-14	1	336	798	1440
		334	779	1440
7-Oct-14	2	314	775	1440
		375	794	1500
n		4	4	4
Mean		340	787	1460
S.D.		25.5	11.2	30.0
%CV		7.5	1.4	2.1
%Diff		13.3	4.9	-2.7

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Table 6 Reassay/Repeat Information Table

Subject	Treatment	Time	Original Conc. ng/mL	Original Curve Number	Reason for Reassay	Reassay Conc. ng/mL	Reassay Curve Number	Reported Conc. ng/mL	Reason for Reported Conc.
101	10mg/kg	Day 4 2h	AQL>(40000)	1	1	313000	2	313000	1
102	10mg/kg	Day 4 2h	AQL>(40000)	1	1	332000	2	332000	1
103	10mg/kg	Day 4 2h	AQL>(40000)	1	1	371000	2	371000	1
104	10mg/kg	Day 4 2h	AQL>(40000)	1	1	291000	2	291000	1
201	50mg/kg	Day 4 2h	AQL>(200000)	1	1	1650000	2	1650000	1
202	50mg/kg	Day 4 2h	AQL>(200000)	1	1	1590000	2	1590000	1
203	50mg/kg	Day 4 2h	AQL>(200000)	1	1	1450000	2	1450000	1
204	50mg/kg	Day 4 2h	AQL>(200000)	1	1	2210000	2	2210000	1

AQL: Above Quantifiable Limit

Reasons For Reassay:

1). AQL

Reasons For Reported Conc:

1). Concentration reported as per QPS SOP TLM-002

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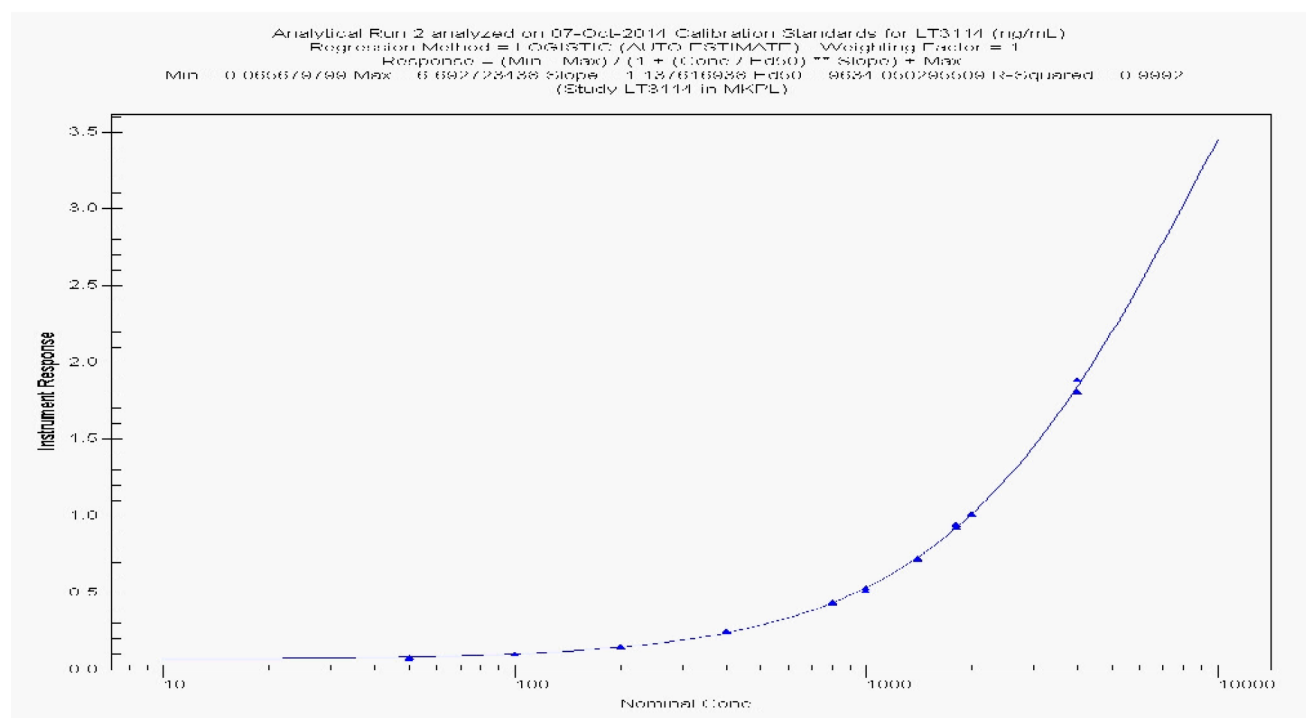
Table 7 **Concentration (ng/mL) of LT3114 in Individual Monkey K₂EDTA Plasma Samples**

Subject	Treatment Description	Day Nominal	Hour Nominal	Concentration (ng/mL)
101	10mg/kg	1	2	BQL<(100)
101	10mg/kg	4	2	313000
102	10mg/kg	1	2	BQL<(100)
102	10mg/kg	4	2	332000
103	10mg/kg	1	2	BQL<(100)
103	10mg/kg	4	2	371000
104	10mg/kg	1	2	BQL<(100)
104	10mg/kg	4	2	291000
201	50mg/kg	1	2	BQL<(100)
201	50mg/kg	4	2	1650000
202	50mg/kg	1	2	BQL<(100)
202	50mg/kg	4	2	1590000
203	50mg/kg	1	2	BQL<(100)
203	50mg/kg	4	2	1450000
204	50mg/kg	1	2	BQL<(100)
204	50mg/kg	4	2	2210000

BQL: Below Quantifiable Limit

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20. FIGURES**Figure 1 Typical Standard Curve for LT3114 ELISA in Monkey K₂EDTA Plasma**

Run ID: 141007-3441411-MKPL-MAC-1A-002

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Appendix A Certificate of Analysis of LT3114

Lpath Inc.		Certificate of Analysis	
TITLE	Preliminary Lpathomab Reference Standard 10 mg/ml , 0.5 mL		REVISION No.
			1.0.0
	PAGE	1 of 2	

Name	Part Number		Lot Number	
Lpathomab DS	TBD		Gallus-MO 103948	
Container/Closure	Fill Volume	Label Strength	Manufacturer	Filler
2-mL Cryovial	0.5 mL	10 mg /mL	Gallus-MO St. Louis, MO	Eurofins Lancaster Laboratories Lancaster, PA
Formulation Buffer			Storage Temperature	
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0			≤-60°C	
Production Date	Retest Date	Use Approved Until Last Day of	Project /Use	
JUN 2014	JUN 2015	AUG 2015	Preliminary Product Reference Standard for QC Testing	

Attribute	Test	Specification	Results
Identity	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution	Conforms
	pH	5.0 ± 0.2	5.1
	CGE (Non-reduced)	Major band at 150 kDa	Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Conforms
	ICE		Comparable to previous reference standard
		Minor Peak at pI ~8.3	15%
		Major Peak at pI ~8.4	60%
		Minor Peak at pI ~8.6	13%
		Faint Peak at pI ~8.7	4%
Potency	Concentration by OD 280 nm	10.0 ± 0.5 mg/mL	9.8 mg/mL
	CGE, non-reduced	Monomer + Non-glycosylated Monomer ≥ 95%	98.8%
		Non-glycosylated Monomer Report %	2.6%
		No other impurity > 3%	Conforms

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Lpath Inc.		Certificate of Analysis	
TITLE	Preliminary Lpathomab Reference Standard 10 mg/ml , 0.5 mL		REVISION No. 1.0.0
			PAGE 2 of 2

Attribute	Test	Specification		Results
Purity	COE, reduced	HC + LC	≥ 95%	98.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No other impurity > 3%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 92%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No individual impurity > 3%	None detected

JUDGMENT	
Based on a comprehensive review of the manufacturing and testing records, I approve this reference standard for use in GMP/GLP conforming chemical, biochemical and in-vitro testing of Lpathomab samples For QC Testing Only. Not for Preclinical or Clinical Trial Use. Quality Assurance: <u></u> Date: August 21, 2014 Acting Head of QA, Frieder K. Hofmann, PhD	

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ELISA Validation Report

Determination of LT3114 in Monkey K₂EDTA Plasma By ELISA

QPS Project Number: 344-1403
Client Project Number: NA

Sponsor
Lpath Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121

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QPS, LLC
Delaware Technology Park
3 Innovation Way, Suite 240
Newark, DE 19711
WEB: www.qps-usa.com
TEL: (302) 369-5601
FAX: (302) 369-5602

ELISA VALIDATION REPORT**REPORT TITLE:**

Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA

QPS PROJECT NUMBER:

344-1403

VALIDATION INITIATED:

9 Jul 2014

CONTRIBUTING SCIENTISTS:

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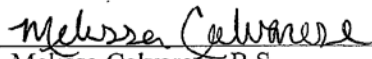
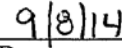
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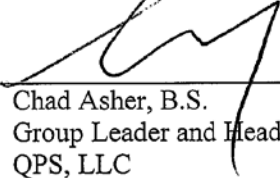
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1. SIGNATURE PAGE**WRITTEN BY:**Melissa Calvarese, B.S.
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Date

APPROVED BY:Chad Asher, B.S.
Group Leader and Head of PK Bioanalysis
QPS, LLC

Date

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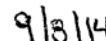
2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58; the Organization for Economic Co-operation and Development (OECD) Principles of Good Laboratory Practice [C (97) 186/Final]; the Japanese Ministry of Health, Labor, and Welfare (MHLW); Ordinance No. 21, March 26, 1997; and the FDA Guidance for Industry: Bioanalytical Method Validation, Sept 2013.

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.



Melissa Calvarese, B.S.
Principal Investigator
QPS, LLC



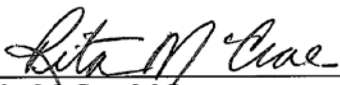
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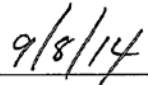
3. QUALITY ASSURANCE STATEMENT**QPS 344-1403****Protocol Title:** Determination of LT3114 in Monkey K₂EDTA Plasma by ELISA

This QPS study has been audited and found to accurately reflect the requirements set forth by the 344-1403 validation protocol and standard operating procedures, and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor	Audit Dates	Date Audit Findings Reported to QPS Principal Investigator and Management
Draft Protocol	Y Li	2 Jul 2014	2 Jul 2014
Study Records/ Draft Report	R. McCrae	13,14 Aug 2014	14 Aug 2014



Rita McCrae M.S.
Quality Assurance Auditor
QPS, LLC



Date

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4. VALIDATION SUMMARY FOR LT3114

An immunoassay method for the determination of LT3114 in monkey K₂EDTA plasma was developed and validated. Experiments were conducted at QPS to evaluate the precision and accuracy of the assay in monkey plasma.

The ELISA assay utilized a sandwich format, in which, microtiter plates are pre-coated with LPA-BSA Conjugate. The plate is washed. Samples (including standards and QCs) are added and incubated on the plate. LT3114 is detected by HRP conjugated goat anti-human IgG monkey adsorbed. TMB is used as a substrate for the HRP. The reaction is stopped with 1N HCl.

Assay range, intra-batch and inter-batch precision / accuracy data, along with the stability, dilution linearity, and matrix evaluation data are provided in the following summary table. Additional long term frozen stability will be evaluated at a later date, and will be added as an addendum.

Report Title	Determination of LT3114 in Monkey K ₂ EDTA Plasma by ELISA
Report Number	QPS 344-1403
Analyte Name	LT3114
Sample Volume:	5µL (Before 100x dilution, duplicate measurement)
Analytical Method Type	ELISA
Calibration Range:	100 – 2000 ng/mL
QC Concentrations	100, 300, 750, 1500, and 2000 ng/mL
QC Intra-batch Precision (%CV)	1.9% - 5.3%
QC Intra-batch Accuracy (%RE)	-3.0% - 13.0
QC Inter-batch Precision (%CV)	5.4% - 12.9%
QC Inter-batch Accuracy (%RE)	-3.3% - 4.0%
Room temp Stability in Monkey K ₂ EDTA Plasma	24 Hours at Room Temperature
Freeze/thaw Stability in Monkey K ₂ EDTA Plasma	5 Cycles at -70°C and -20°C/ Room Temperature
Long-term Storage Stability in Monkey K ₂ EDTA Plasma	30 Days at -20°C and -70°C
Dilution Linearity	900,000 ng/mL diluted 1, 100, 500, 1000, and 5000 -fold
Selectivity (10 lots, spiked 100, 2000 ng/mL)	100% lots tested within 100±25% Recovery

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5. OBJECTIVE

The purpose of this study was to validate an ELISA to determine the concentrations of LT3114 in monkey K₂EDTA plasma samples collected during a nonclinical study.

6. ABBREVIATIONS AND DEFINITIONS

This section provides abbreviations and definitions of terms and concepts that may be commonly used throughout this report.

AQL	Above Quantifiable Limit
BQL	Below Quantifiable Limit
Exc	Sample excluded according to QPS Validation Protocol 344-1403
LLOQ	Lower Limit of Quantification
LLOQ QC	Lower Limit of Quantification Quality Control
MRD	Minimum required Dilution
NC	Not Calculated
NA	Not Applicable
NR	Not Reported
OD	Optical Density
QC	Quality Control
r²	Coefficient of Determination
SD	Standard Deviation
SOP	Standard Operating Procedure
ULOQ	Upper Limit of Quantification
ULOQ QC	Upper Limit of Quantification Quality Control

$$\text{Precision} \quad \% \text{Coefficient of Variation (\%CV)} = 100 \times \frac{\text{Standard Deviation}}{\text{Mean Concentration}}$$

$$\text{Accuracy} \quad (\% \text{RE}) = 100 \times \frac{\text{Mean Found Concentration} - \text{Nominal Concentration}}{\text{Nominal Concentration}}$$

Calibration standard and Quality control samples at five concentration levels were prepared in matrix and were determined based upon the actual concentrations of the stock solutions.

All statistics in the data tables are calculated by Watson™ LIMS or based on the “precision as displayed” option of Microsoft® Excel (e.g., Mean, S.D., %CV, %Diff).

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7. ANALYTE INFORMATION

Name: LT3114
Concentration: 20.14^a mg/mL
Purity: 96%
Supplied Form: Lpath
Lot Number: LP121416
Expiration Date: 4/30/16 (assigned per QPS SOP LP-007)^b
Storage Conditions: -70°C

^aWith purity applied, concentration will be used at 19.33mg/mL.

^bNo supporting documentation available for stability or retest date of compound at current storage condition.

8. ACCEPTANCE CRITERIA

The validation acceptance criteria and the statistical data will be determined at a minimum to QPS' SOPs TLM-001 Validation of Binding Assays and QPS validation protocol 344-1403. Additional experiments (if any) not governed under QPS' SOPs will follow the instructions provided by the sponsor.

Six successful, complete precision and accuracy runs (out of which three are consecutive runs) are required to validate a method. The evaluation of an intra-batch validation run includes a minimum of six replicates of QCs at each concentration and a minimum of five QC concentrations (including LLOQ and ULOQ). The evaluation of inter-batch validation runs (minimum of six runs) includes a minimum of three replicates of QCs at each concentration per run. Each run must include a minimum of five QC concentrations.

Method Acceptance: A method is accepted if it successfully fulfills the following:

- Intra-batch Accuracy and Precision acceptance criteria: For each intra-batch validation run, the mean of six or more replicates at each concentration of the QC samples is not allowed to deviate more than $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ QCs) from its nominal value. In addition, the intra-batch precision at each concentration must not exceed 20% (25% at LLOQ and ULOQ QCs) when calculated as the %CV.
- Inter-batch Accuracy and Precision acceptance criteria: The mean value at each QC concentration must not exceed $\pm 20.0\%$ ($\pm 25.0\%$ at LLOQ and ULOQ) from its nominal value. In addition, the inter-batch precision at each concentration must not exceed 20% (25% at LLOQ and ULOQ QCs) when calculated as the %CV.
- Dilution test acceptance criteria: The mean of the back calculated concentration (after the dilution factor has been applied) must be within

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100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered acceptable.

- Assay selectivity acceptance criteria: At least 80% of the lots spiked with LT3114 should have recovery between 75-125%.
- Room temperature and freeze/thaw stability acceptance criteria: The mean back calculated concentration must be within 100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered stable.
- Frozen stability samples acceptance criteria: The back calculated concentration is compared to the nominal value of the QC samples, which was confirmed in the initial analysis as properly prepared. The mean value for each concentration level must be within 100±20% of the nominal value, and the %CV of the calculated concentrations must not exceed 20% to be considered stable.

Run Acceptance criteria:

- For QCs, validation samples and future study samples, each replicate will be measured twice and the average will be reported if the %CV from the two measurements is $\leq 25\%$. Otherwise, the value from the two measurements will not be reported.
- Production runs or sample analysis runs are accepted based on the performance of calibration standards and QCs. Only standards back-calculated to over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) different from their nominal concentrations will be excluded from the regression analysis. If more than one standard back calculates to over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ), exclusion should be carried out one standard at a time to see if the other standards would fall within $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) of the nominal value after the exclusion.

If both of the calibration standards at the lowest (or highest) concentration of the curve are excluded from the regression analysis due to various reasons (e.g., not enough sensitivity at the lower end, curve bend at the higher end, QCs or standards at the specific concentration fail to meet acceptance criteria), the LLOQ (or ULOQ) shall be moved up (or down) to the next standard concentration in sequence (also known as the truncation of the standard curve).

Finally, a minimum of six standard concentrations must be within the assay range. Excluded calibration standards that fall outside of the calibration range shall not be used to evaluate the acceptability of the run. All excluded

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calibration standards over $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ) within the calibration range should be reported (e.g., as Exc) and should be included when evaluating the acceptability of the run.

Anchor points may be included or excluded to facilitate curve fitting based on scientific judgment.

At least 2/3 of the back calculated QC concentrations must be within $100 \pm 20\%$ ($100 \pm 25\%$ at the LLOQ and ULOQ QCs) of their nominal value, 50% of QC at each concentration must not deviate more than $\pm 20\%$ ($\pm 25\%$ at LLOQ and ULOQ QCs) from its nominal value.

- Special case of run acceptance for accuracy and precisions run:
The precision and accuracy runs are accepted solely based on the performance of the calibration curve. A minimum of 75% of the total number of calibration standards in the calibration range (between the LLOQ and ULOQ of the run) should be within $100 \pm 20\%$ ($100 \pm 25\%$ at LLOQ and ULOQ standards) of their nominal value, and a minimum of six unique standard concentrations must be within the assay range. The calibration curve must contain at least one calibration standard at both the LLOQ and ULOQ of the range.

9. MATERIALS

9.1. Sample-Handling Equipment

- 96-well microtiter plates (Costar Cat #2592)
- 96-well format micro titer/dilution tubes
- 25-100 mL Reagent reservoir
- Plate sealers
- Polypropylene plastic tubes: various volumes
- Pipette tips for single and multichannel pipettes
- Graduated cylinder: various volumes
- 8-Channel StatMatic II microplate dispenser

9.2. Analytical Equipment

Spectramax Microplate Reader (Molecular Devices).

10. METHODS

10.1. Preparation of Secondary Stock Solutions:

Two sets of stock solutions were prepared separately for standards and QCs. The prepared stock solutions were used immediately to prepare subsequent working standard and QC solutions and the remaining portion was discarded.

10.2. Preparation of Working Standard Solutions in Monkey K₂EDTA Plasma

Appropriate amount of stock was combined with monkey plasma according to the following table to make the calibration standard working solutions. Each concentration of the standards

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contains at least 95% (v/v) monkey plasma. The prepared working standard solutions were aliquoted for single use and were stored at -70°C. The blank pooled monkey plasma served as the zero standard (Blank).

Concentration (ng/mL)	Vol. of Stock 4000 ng/mL (μL)	Vol. of MKPL (μL)	Total Volume (μL)
0	0	200	200
50	5	395	400
100	5	195	200
200	10	190	200
400	20	180	200
800	40	160	200
1000	50	150	200
1400	70	130	200
1800	90	110	200
2000	100	100	200
4000	200	0	200

10.3. Preparation of Working QC Solutions in Monkey K₂EDTA Plasma

Appropriate amount of stock was combined with monkey plasma according to the following table to make the QC working solutions. Each concentration of the QCs contains at least 95% (v/v) monkey plasma. The prepared QC solutions were aliquoted for single use and were stored at -70°C.

Concentration (ng/mL)	Vol. of Stock 4000 ng/mL (μL)	Vol. of MKPL (μL)	Total Volume (μL)
100	5.0	195.0	200
300	75.0	925.0	1000
750	37.5	162.5	200
1500	375	625.0	1000
2000	100	100.0	200

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10.4. Analytical Method

<u>Parameter</u>	<u>Value of Parameter</u>
General:	
Analyte	LT3114
Matrix	Monkey K ₂ EDTA Plasma
Calibration standard concentrations	50(AP), 100, 200, 400, 800, 1000, 1400, 1800, 2000 and 4000(AP) ng/mL
Quality Control Concentrations	100, 300, 750, 1500, and 2000 ng/mL
Minimum required dilution	100x in Dilipidized Bovine Serum
Sample volume	5 µL (before dilution, 2 wells) 100 µL/well (after dilution)
ELISA procedure summary	Microtiter plates are pre-coated with LPA-BSA. After washing, samples (including standards and QCs) are added and incubated on the plate. LT3114 is detected by horseradish peroxidase conjugated goat anti-human IgG H and L, Monkey Adsorbed Ab. TMB is used as a substrate for the HRP. The reaction is stopped with 1N HCl. The color intensity is proportional to the quantity of LT3114.
Data Acquisition:	
Equipment	Molecular Devices SpectramaxPLUS ³⁸⁴
Program	Softmax Pro Version 3.1.2
Reading mode	Endpoint
Wavelength	450 nm (reading), 630 nm (reference)
Calibration	Auto-calibration on
Blanking	No pre-read, Plate blank not used
Mixing	Off
Analysis and Reporting:	
Program	Watson™ LIMS Version 7.4.1
Regression type	4-parameter (auto-estimate)
Weighting:	1
Anchoring Point	Use as needed.
Precision for Reporting:	3 significant digits

11. RESULTS AND DISCUSSION

Table 1 lists all the validation runs performed.

11.1. Regression Analysis

Calculations were done using 4-parameter (auto-estimate) regression using Watson™ LIMS (v. 7.4.1, Thermo Electron Corporation, USA) Table 2 lists the back-calculated calibration standard concentrations for LT3114 and Table 3 lists the standard curve parameters for

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LT3114. The results show a 4-Parameter fit from 100-2000 ng/mL for LT3114. Figure 1 shows a typical calibration curve of LT3114 in monkey plasma.

11.2. Precision and Accuracy

Precision and accuracy of the method were determined by analyzing QC samples at a minimum of five different concentrations within the standard curve range.

Precision was determined by the coefficient of variation (%CV) and accuracy was determined by percent recovery (%RE). The acceptable maximum %CV for a valid assay is 25% at all concentrations. The acceptable accuracy (%RE) for a valid assay is within $100 \pm 20\%$ (25% at LLOQ and ULOQ QCs) of the nominal concentration.

The intra-batch precision and accuracy within an analytical run were determined by analyzing six replicates for each QC sample including the ULOQ and the LLOQ QC from the same run. The results for intra-batch precision and accuracy (Table 4) were acceptable.

The inter-batch precision and accuracy of the assay over different analytical runs were determined by analyzing QC samples from six validation runs with three replicates in each run. The results for inter-batch precision and accuracy (Table 5A) were acceptable. In addition the results for QCs from all runs are summarized in Table 5B.

11.3. Lower Limit of Quantification

The lower limit of quantification (LLOQ) of the assay was 100 ng/mL for LT3114. For sample analysis, concentrations found below the LLOQ after the minimum required dilution will be reported as 'BQL', if applicable, since the precision and accuracy of the assay were determined to that limit.

11.4. Stability

11.4.1. Room Temperature Stability

Room temperature stability of LT3114 in monkey plasma was performed to evaluate LT3114 stability in monkey plasma for the time that the samples would be at room temperature during sample handling and processing. Stability of the QC samples was determined at two concentrations (low and high QCs) in minimum three replicates at both 6 and 24 hours. The results (Table 6) indicate that LT3114 in monkey plasma is stable at room temperature for at least 24 hours.

11.4.2. Freeze/Thaw Stability

The stability of LT3114 in monkey plasma after five freeze/thaw cycles was determined. QC samples at two concentrations (Low and high QCs) were frozen at -20°C or -70°C (for a minimum of 24 hours for cycle 1 and a minimum of 12 hours for later cycles) and thawed at room temperature. This freeze/thaw cycle was repeated five times. After the completion of the fifth freeze/thaw cycle, the samples were analyzed and compared to the nominal value. During initial testing of the freeze thaw samples the -20°C Low QC recovered lower than expected, the test was repeated at n=6, and was found to recover within the acceptable range.

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The results (Table 7A and Table 7B) indicate LT3114 is stable for five freeze/thaw cycles in monkey plasma under the described conditions.

11.5. Long-Term Frozen Storage Stability

LT3114 quality control samples in monkey plasma were stored at -70°C and -20°C for 30 days before being analyzed against a freshly prepared calibration standard curve. The results (Table 8) indicate LT3114 is stable for at least 30 days in monkey plasma at -70°C and -20°C. The frozen storage stability of LT3114 in monkey plasma at -70°C and -20°C at longer periods will be assessed at a later date and an addendum to the report will be provided.

11.6. Dilution Linearity and Hook Effect

The purpose of the dilution test was to ensure that samples could be diluted with monkey plasma without affecting the final calculated concentration. Monkey plasma samples prepared at one concentration 900,000 ng/mL were diluted with monkey plasma at dilution factors of 1, 100, 500, 1000, and 5000 in three replicates at each level. The results were corrected with the dilution factor and compared to the nominal concentration. The results in Table 9 show that the difference between the mean of the adjusted concentration (found concentration multiplied by dilution factor) and the nominal concentration of LT3114 was within the acceptable range. There was no hook effect observed.

11.7. Assay selectivity in Monkey K₂EDTA Plasma

Ten individual lots of monkey plasma were spiked with or without LT3114 and measured for LT3114 concentration. Table 10 indicates that 100% of the tested lots had a recovery between 75-125%, which is acceptable.

12. STUDY SPECIFIC DEVIATIONS OR OBSERVATIONS

DEV-14-376: In step 8 "Plate Reading" the Analytical Procedure specified Lm2 as 570nm, however the plates were all read using 630nm as Lm 2. The method development for this project was consistent with the 630nm wavelength used during validation. Throughout the study analyst and PI incorrectly initial dated that the 570nm was used as the Lm2. There is no impact on study data and a note to the record was added the study notebook for clarification.

13. CONCLUSION

An ELISA method (QPS method 344-1403) has been validated for the quantitation of LT3114 in monkey K₂EDTA plasma from 100 – 2000 ng/mL. With a validated dilution factor of 5000, this assay is suitable for measuring LT3114 in monkey K₂EDTA plasma from 100 – 10000000 ng/mL.

14. REFERENCES

Method Validation Protocol:

QPS Project No. 344-1403

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SOPs:

QPS LP-004: “Analytical Standards and Quality Controls”

QPS LP-007: “Analytical Reference Compounds”

QPS LP-034: “Receipt and Handling of Biological Matrices”

QPS TLM-001: “Validation Quantitative of Ligand Binding Assays”

Notebooks:

QPS validation notebook 344-1403

LT3114 assay notebook

15. ARCHIVE STATEMENT

All raw data (including electronic SoftmaxPro files), a copy of this bioanalytical report and required supporting information for this study will be held under the control of QPS, LLC, 3 Innovation Way, Suite 240, Newark DE, 19711, USA. Study records will be archived according to the most current version of SOP GN-014, “Archive”

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16. TABLES

Table 1 Validation Run Log

Curve Number	Analysis Date	Run Description	Accepted/ Rejected ^a
1	9-Jul-14	140709-3441403-MKPL-MAC-1A-001	Accepted
2	9-Jul-14	140709-3441403-MKPL-MAC-2B-002	Accepted
3	9-Jul-14	140709-3441403-MKPL-GP-1A-003	Accepted
4	9-Jul-14	140709-3441403-MKPL-GP-2B-004	Accepted
5	10-Jul-14	140710-3441403-MKPL-EEQ-1A-005	Accepted
6	10-Jul-14	140710-3441403-MKPL-EEQ-2B-006	Accepted
7	11-Jul-14	140711-3441403-MKPL-MAC-1A-007	Accepted
8	11-Jul-14	140711-3441403-MKPL-MAC-2B-008	Accepted
9	16-Jul-14	140716-3441403-MKPL-MAC-1A-009	Accepted
10	17-Jul-14	140717-3441403-MKPL-MAC-1A-010	Accepted
11	7-Aug-14	VOID ^b	NA
12	7-Aug-14	140807-3441403-MKPL-MAC-1A-012	Accepted

a: Acceptance based on QPS Validation Protocol 344-1403

b: Run voided due to planning error, processed as run 012

**Table 2 Back-Calculated Calibration Standard Concentrations of LT3114
Monkey K₂EDTA Plasma**

Analysis Date	Curve Number	Concentrations (ng/mL)							
		100	200	400	800	1000	1400	1800	2000
9-Jul-14	1	105	204	387	788	968	1380	1760	2060
		100	205	385	774	1090	1450	1760	1970
9-Jul-14	2	98.7	197	383	798	1010	1420	1780	2010
		98.6	202	392	800	997	1430	1780	1980
9-Jul-14	3	94.0	190	392	831	1090	1430	1730	1980
		96.2	185	376	814	1050	1460	1700	1950
9-Jul-14	4	98.8	206	396	810	978	1410	1800	1990
		96.8	203	385	801	1040	1390	1790	2000
10-Jul-14	5	107	204	407	806	1010	1410	1780	2030
		Exc	199	400	778	998	1400	1780	1990
10-Jul-14	6	97.6	211	409	792	1010	1390	1800	2030
		99.8	203	404	796	1000	1390	1780	2020
11-Jul-14	7	99.4	200	408	794	1010	1400	1850	1990
		100	191	405	779	1020	1360	1830	1960
11-Jul-14	8	104	197	402	800	1040	1410	1830	1980
		98.0	192	386	791	991	1400	1780	2000
16-Jul-14	9	96.8	221	423	787	970	1420	1820	2020
		87.7	212	421	787	967	1440	1750	2020
17-Jul-14	10	103	192	416	895	931	1400	1770	2090
		110	177	425	777	968	1370	1810	2000
7-Aug-14	12	97.9	201	408	789	1010	1400	1770	2040
		96.8	205	397	802	1010	1360	1820	2000
n		21	22	22	22	22	22	22	22
Mean		99.3	200	400	800	1010	1410	1790	2010
S.D.		4.70	9.58	13.8	25.1	38.5	26.5	34.6	32.6
%CV		4.7	4.8	3.5	3.1	3.8	1.9	1.9	1.6
%RE		-0.7	0.0	0.0	0.0	1.0	0.7	-0.6	0.5

Exc: Excluded according to QPS Validation Protocol 344-1403

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Table 3 LT3114 Standard Curve Regression Parameters

Analysis Date	Curve Number	Min	Max	Slope	Ed50	R-Squared	LLOQ	ULOQ	Regression Footnote(s)
9-Jul-14	1	0.077196810	9.200081202	1.073971400	10571.368443463	0.9991	100	2000	1
9-Jul-14	2	0.063511273	38.477247641	0.935907175	68685.749423884	0.9998	100	2000	1
9-Jul-14	3	0.064120884	4879.830532427	0.993463228	9874195.481421820	0.9985	100	2000	1
9-Jul-14	4	0.059013501	34184.680661393	1.010770010	60369913.375127300	0.9997	100	2000	1
10-Jul-14	5	0.071299796	15.840082125	1.068198645	23422.851722878	0.9997	100	2000	1
10-Jul-14	6	0.086739283	14.801481479	1.073822412	23017.011802598	0.9999	100	2000	1
11-Jul-14	7	0.079744960	15.359883908	0.999031598	21081.313630949	0.9996	100	2000	1
11-Jul-14	8	0.072687304	14.789715323	0.979564736	19582.881777004	0.9998	100	2000	1
16-Jul-14	9	0.095728082	7.747141688	1.146963989	6431.959550719	0.9996	100	2000	1
17-Jul-14	10	0.340156524	6.546377670	1.143375025	3878.050907112	0.9985	100	2000	1
7-Aug-14	12	0.176919055	7.278381753	1.069744952	5673.028396259	0.9996	100	2000	1

Regression Footnote(s):

1) Resp. = (Min - Max) / (1 + (Conc. / Ed50) ** Slope) + Max

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Table 4 Intra-Batch Precision and Accuracy of LT3114 in Monkey K₂EDTA Plasma

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	114	319	841	1530	2080
		109	316	752	1530	2000
		112	365	779	1470	1860
		112	332	759	1480	1880
		112	334	769	1490	1950
		119	341	786	1530	1890
n		6	6	6	6	6
Intrarun Mean		113	335	781	1510	1940
Intrarun SD		3.35	17.7	31.9	28.1	84.5
Intrarun %CV		3.0	5.3	4.1	1.9	4.4
Intrarun %RE		13.0	11.7	4.1	0.7	-3.0

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Table 5A Inter-Batch Precision and Accuracy of LT3114 in Monkey K₂EDTA Plasma

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	114	319	841	1530	2080
		109	316	752	1530	2000
		112	365	779	1470	1860
		112	332	759	1480	1880
		112	334	769	1490	1950
		119	341	786	1530	1890
9-Jul-14	2	97.5	267	735	1410	1910
		95.2	269	711	1400	1890
		94.6	278	714	1460	1980
9-Jul-14	3	89.4	270	719	1400	1860
		76.1	245	652	1230	1720
		84.0	252	713	1370	1860
9-Jul-14	4	98.6	383	810	1440	1970
		90.7	327	705	1430	2040
		96.8	293	764	1470	1940
10-Jul-14	5	110	329	843	1580	2140
		96.6	294	765	1380	1920
		NR	309	814	1440	2030
10-Jul-14	6	119	337	851	1550	2140
		106	304	766	1390	1930
		130	389	855	1510	2150
n		20	21	21	21	21
Mean		103	312	767	1450	1960
S.D.		13.3	40.2	54.9	78.6	109
%CV		12.9	12.9	7.2	5.4	5.6
%RE		3.0	4.0	2.3	-3.3	-2.0

NR: Not Repotable %CV>25%

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Table 5B Inter-Batch Precision and Accuracy of LT3114 in Monkey K2EDTA

Analysis Date	Curve Number	Concentrations (ng/mL)				
		100	300	750	1500	2000
9-Jul-14	1	114	319	841	1530	2080
		109	316	752	1530	2000
		112	365	779	1470	1860
		112	332	759	1480	1880
		112	334	769	1490	1950
		119	341	786	1530	1890
9-Jul-14	2	97.5	267	735	1410	1910
		95.2	269	711	1400	1890
		94.6	278	714	1460	1980
9-Jul-14	3	89.4	270	719	1400	1860
		76.1	245	652	1230	1720
		84.0	252	713	1370	1860
9-Jul-14	4	98.6	383	810	1440	1970
		90.7	327	705	1430	2040
		96.8	293	764	1470	1940
10-Jul-14	5	110	329	843	1580	2140
		96.6	294	765	1380	1920
		NR	309	814	1440	2030
10-Jul-14	6	119	337	851	1550	2140
		106	304	766	1390	1930
		130	389	855	1510	2150
11-Jul-14	7		373	769	1430	
			320	829	1610	
11-Jul-14	8		327	823	1580	
			296	795	1550	
16-Jul-14	9		323	709	1370	
			308	694	1370	
17-Jul-14	10		313	720	1500	
			339	767	1550	
7-Aug-14	12		282	667	1280	
			302	774	1570	
n		20	31	31	31	21
Mean		103	314	763	1460	1960
S.D.		13.3	35.8	54.3	89.7	109
%CV		12.9	11.4	7.1	6.1	5.6
%RE		3.0	4.7	1.7	-2.7	-2.0

NR: Not Repotable %CV>25%

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Table 6 Room Temperature Stability of LT3114 in Monkey K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300	1500
	Lot: EEQ140708-12	Lot: EEQ140708-14
6 hours	280	1380
	276	1300
	277	1280
n	3	3
Mean	278	1320
SD	2.08	52.9
%CV	0.7	4.0
%RE	-7.3	-12.0
24 hour	268	1180
	263	1280
	253	1230
n	3	3
Mean	261	1230
SD	7.64	50.0
%CV	2.9	4.1
%RE	-13.0	-18.0

Run ID: 140716-3441403-MKPL-MAC-1A-009

Table 7A Freeze/Thaw Stability of LT3114 in Monkey K₂EDTA Plasma

	Nominal Concentration (ng/mL)	
	300	1500
	Lot: EEQ140708-12	Lot: EEQ140708-14
-70°C	271	1250
5 Cycles	264	1250
	262	1220
n	3	3
Mean	266	1240
SD	4.73	17.3
%CV	1.8	1.4
%RE	-11.3	-17.3
-20°C	280	1140
5 Cycles	246	1200
	257	1150
n	3	3
Mean	261	1160
SD	17.3	32.1
%CV	6.6	2.8
%RE	-13.0	-22.7
Run ID: 140716-3441403-MKPL-MAC-1A-009		

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**Table 7B Repeat of -20°C Low QC Freeze/Thaw Stability of LT3114 in
Monkey K₂EDTA Plasma**

Nominal Concentration (ng/mL)	
1500	
Lot: EEQ140708-14	
5 Cycles	1360
	1290
	NR
	1380
	1460
	1390
n	5
Mean	1380
SD	61.1
%CV	4.4
%RE	-8.0

NR: Not Reportable %CV>25%

Run ID: 140717-3441403-MKPL-MAC-1A-010

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**Table 8 30 Day Long Term Frozen Stability of LT3114 in Monkey
K₂EDTA Plasma**

	Nominal Concentration (ng/mL)	
	300 Lot: EEQ140708-12	1500 Lot: EEQ140708-14
-20°C	265	1200
	260	1230
	263	1210
n	3	3
Mean	263	1210
SD	2.52	15.3
%CV	1.0	1.3
%RE	-12.3	-19.3
-70°C	267	1290
	279	1290
	289	1230
n	3	3
Mean	278	1270
SD	11.0	34.6
%CV	4.0	2.7
%RE	-7.3	-15.3

Run ID: 140807-3441403-MKPL-MAC-1A-012

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Table 9 Dilution Test of LT3114 in Monkey K2EDTA Plasma

Nominal Concentration (ng/mL)	Dilution Factor	LT3114 (ng/mL)
900000	1	AQL
		AQL
		AQL
		n 3
		Mean NC
		SD NC
	100	%CV NC
		%RE NC
		AQL
		AQL
		AQL
		n 3
		Mean NC
		SD NC
		%CV NC
		%RE NC
	500	1010000
		1000000
		1010000
		n 3
		Mean 1010000
		SD 5770
	1000	%CV 0.6
		%RE 12.2
		984000
		1050000
		1040000
		n 3
		Mean 1020000
		SD 35600
		%CV 3.5
		%RE 13.3
	5000	1070000
		973000
		991000
		n 3
		Mean 1010000
		SD 51600
		%CV 5.1
		%RE 12.2
	Cumulative n 9	
	Cumulative Mean 1010000	
	Cumulative SD 32500	
	Cumulative %CV 3.2	
	Cumulative %RE 12.2	

Run ID: 140711-3441403-MKPL-MAC-1A-007

AQL: Above Limit of Quantitation (>2000 ng/mL)

NC: Not calculated

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Table 10 Assay Selectivity of LT3114 in Monkey K₂EDTA Plasma

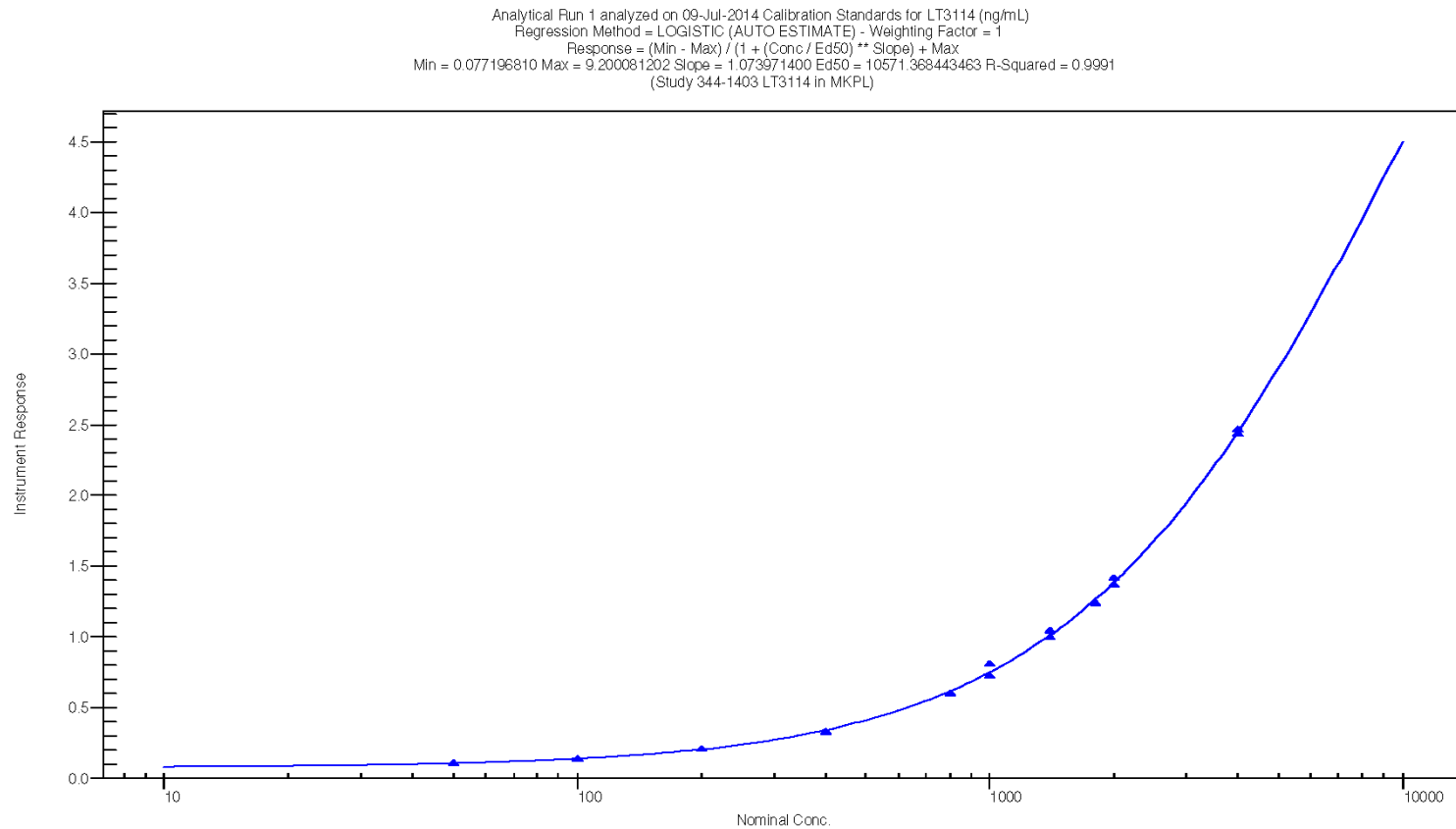
Individual Monkey Plasma	LT3114 Concentration (ng/mL)				
	Blank (0 ng/mL)	Spike-in LLOQ QC (100 ng/mL)		Spike-in ULOQ QC (2000 ng/mL)	
Lot #	Found	% Recovery of Spiked Lot		Found	% Recovery of Spiked Lot
CYN141483	BQL	115	115.0	2490	124.5
CYN141484	BQL	121	121.0	2330	116.5
CYN141485	BQL	117	117.0	2110	105.5
CYN141486	BQL	120	120.0	2200	110.0
CYN141487	BQL	120	120.0	2160	108.0
CYN141488	BQL	120	120.0	2180	109.0
CYN141489	BQL	123	123.0	2210	110.5
CYN141490	BQL	120	120.0	2210	110.5
CYN141491	BQL	122	122.0	2160	108.0
CYN141492	BQL	115	115.0	2100	105.0
n		10		10	
Mean		119		2220	
SD		2.75		116	
%CV		2.3		5.2	
%RE		19.0		11.0	

Run ID: 140711-3441403-MKPL-MAC-2B-008

BQL: Below Limit of Quantitation (<100 ng/mL)

% Recovery: (Found Concentration / Nominal of the Spiked) X 100%

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17. FIGURES**Figure 1 Typical Standard Curve for LT3114 ELISA in Monkey K₂EDTA Plasma**

Run ID:140807-3441403-MKPL-MAC-1A-001

QPS Report 344-1403

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18. APPENDICES

Appendix A Certificate of Analysis of LT3114

		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	1 of 5

Name	Part Number	Lot Number
Humanized Lpathomab	LT3114	LP121416
Formulation Buffer		Storage Temperature
PBS (Mediatech #21-040, Lot# 21040283)		2 to 8 °C
Production Date	Retest Date	Project /Use
5-10-2013	N/A	TBD

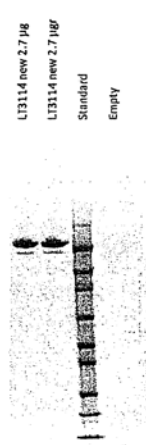
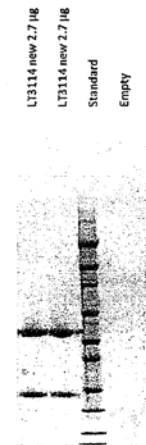
Summary
<u>Production location, date, and method</u> The antibody was produced from stable CHO clone (191+192) 3114 3B8/1B4/2F1 in 4x 3-L shake flasks at Lpath. Takedown on d10, May 10, 2013. Pooled, clarified harvest was stored at -20 °C. Lot LP121416 consists of 3x 1-L bottles of supe which cracked during storage. Lot LP121417 consists of 1x 1-L bottle of supe which did not crack during storage. Purified lots LP121416 and LP121417 were pooled to create lot LP121419. <u>Purification location, date and method</u> Antibody purification was initiated on August 21, 2013 at Lpath using affinity and AEX chromatography (MabSelect Protein A and Q membrane Sartorius #92IEXQ42D4-OOA, respectively). Following concentration, the antibody was dialyzed against 3x 2-L PBS (Mediatech) and passed through a 0.22 um filter flask unit (Corning). <u>Concentration by A₂₈₀ and extinction coefficient</u> The final sample is 21.5 mg/mL as determined by UV spectroscopy using a theoretical extinction coefficient of 1.4 mL/mg. <u>Total amount of Ab (mg)</u> 1400 mg <u>Appearance details:</u> Clear, slightly yellow solution

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Pages 1-5
Rcm 5/2/14

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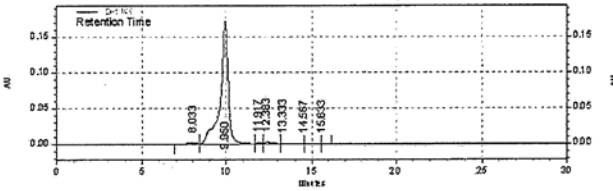
		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	2 of 5

Attribute	Test	Specifications	Results
Identity	Appearance	Particulate free, clear solution Colorless or slightly yellow	Particulate free, clear solution Slightly yellow
	SDS-PAGE (Non-Reduced)	Report Results	Major band at 165 kDa
	SDS-PAGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Two pronounced bands; One at ~55 kDa; One at ~ 28 kDa
	<i>Insert images of SDS-PAGE</i>	Non-Reduced LI3114 new 2.7 µg LI3114 new 2.7 µg Standard Empty  Reduced LI3114 new 2.7 µg LI3114 new 2.7 µg Standard Empty 	
	Isoelectric Focusing (by IEF Gel)	Report Results	Not done

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Lpath INCORPORATED		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	3 of 5

Attribute	Test	Specifications	Results																																													
Potency	Protein Concentration (OD-280)	Report Results	20.14 mg/mL																																													
	LPA Binding Direct ELISA (EC 50)	Report Results	25.05 ng/mL																																													
	Bioactivity Assay (IL-6 and EDG2)	Positive Result	Does not stimulate IL-6 EDG2 IC50 = 13.65 µg/mL																																													
Purity	Insert image of SE-HPLC <table><caption>Det 166 Results</caption><thead><tr><th>Time</th><th>Area</th><th>Area %</th><th>Height</th><th>Height %</th></tr></thead><tbody><tr><td>8.033</td><td>68097</td><td>1.14</td><td>1413</td><td>0.78</td></tr><tr><td>9.930</td><td>5707101</td><td>95.95</td><td>173377</td><td>96.24</td></tr><tr><td>11.917</td><td>35431</td><td>0.60</td><td>1744</td><td>0.97</td></tr><tr><td>12.383</td><td>97022</td><td>1.63</td><td>2645</td><td>1.47</td></tr><tr><td>13.333</td><td>31081</td><td>0.52</td><td>668</td><td>0.37</td></tr><tr><td>14.567</td><td>8116</td><td>0.14</td><td>234</td><td>0.13</td></tr><tr><td>15.633</td><td>1015</td><td>0.02</td><td>62</td><td>0.03</td></tr><tr><td>Totals</td><td>5947863</td><td>100.00</td><td>180143</td><td>100.00</td></tr></tbody></table>			Time	Area	Area %	Height	Height %	8.033	68097	1.14	1413	0.78	9.930	5707101	95.95	173377	96.24	11.917	35431	0.60	1744	0.97	12.383	97022	1.63	2645	1.47	13.333	31081	0.52	668	0.37	14.567	8116	0.14	234	0.13	15.633	1015	0.02	62	0.03	Totals	5947863	100.00	180143	100.00
Time	Area	Area %	Height	Height %																																												
8.033	68097	1.14	1413	0.78																																												
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15.633	1015	0.02	62	0.03																																												
Totals	5947863	100.00	180143	100.00																																												
SE-HPLC	Purity [% as IgG monomer]	≥ 90%	96%																																													
	Aggregates [as % of total peak area]	≤ 6%	1%																																													
	Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%																																													

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					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	4 of 5

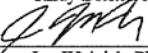
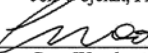
Attribute	Test		Specifications	Results
SDS-PAGE reduced	Purity [%]		≥ 90%	≥ 90%
	50 kDa Band		~65%	~67%
	25 kDa Band		~35%	~33%
	Other Impurities [%]		No individual impurity ≥ 3%	No individual impurity ≥ 3%
	Endotoxin		≤ 10.0 EU/mg of protein	0.06 EU/mg
	Bioburden		≤ 10 CFU/mL	Not done

Comments																																
Source Data: <table border="1"> <thead> <tr> <th></th> <th>Lpath notebook</th> <th>Pages</th> </tr> </thead> <tbody> <tr> <td>Production</td> <td>277</td> <td>149</td> </tr> <tr> <td>Purification</td> <td>296</td> <td>11,18,25</td> </tr> <tr> <td>UV A280</td> <td>296</td> <td>48</td> </tr> <tr> <td>SDS-PAGE</td> <td>296</td> <td>26</td> </tr> <tr> <td>LPA binding ELISA</td> <td>221</td> <td>21-22</td> </tr> <tr> <td>IL-6</td> <td>237</td> <td>48-49</td> </tr> <tr> <td>EDG2</td> <td>237</td> <td>51</td> </tr> <tr> <td>SE-HPLC</td> <td>296</td> <td>33-34</td> </tr> <tr> <td>Endotoxin</td> <td>298</td> <td>13</td> </tr> </tbody> </table> Additional Comments on Sample Quality:				Lpath notebook	Pages	Production	277	149	Purification	296	11,18,25	UV A280	296	48	SDS-PAGE	296	26	LPA binding ELISA	221	21-22	IL-6	237	48-49	EDG2	237	51	SE-HPLC	296	33-34	Endotoxin	298	13
	Lpath notebook	Pages																														
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					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	5 of 5

JUDGMENT	
Approved for testing in humans ☐ - Accepted ☐ - Rejected For Research Only Not For Use in Humans Prepared by: <u></u> Date: <u>12-6</u> 2013 Cindy Dickerson, Ph.D. Reviewed by: <u></u> Date: <u>12/6/</u> 2013 Jon Wojciak, Ph.D. Approved by: <u></u> Date: <u>3/26/2013</u> Gary Woodnutt, Ph.D.	

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